UMC 0.11um LOGIC/MIXED_MODE Al Advanced Enhancement Standard Performance 2.5V IO, Operated at 1.2V, MOSFET SPICE Model Document

Version 0.5 Phase 1
2016/09/07

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1 Introduction

1.1 Revision History

Table 1-1 Revision history

Version	Date	Author	Remark
0.2_p1	2011/03/12	YZ Wang (NMOS) Laney Fan (PMOS)	Original release.
0.3_p1	2011/09/02	YZ Wang	Revised the wmin from 0.16um to 0.35um and target values.
0.3_p2	2012/01/17	YZ Wang	Revised the section 1.2 general information in the document.
0.4_p1	2012/5/15	Ho Cheng Hu	Add bpw NMOS model card
0.5_p1	2016/09/07	Denny Hung	1. Add jtsswgs parameter which value align with jtsswgd.

1.2 General Information

This document serves as a reference to the design of products that will be manufactured by UMC using the following process.

> 0.11um Logic and Mixed-Mode 1P8M Metal Metal Capacitor Al Advanced Enhancement Process

For more information about this process and technology, please see the electrical design rule listed below.

> G-02-LOGIC/MIXED_MODE11-1P8M-MMC/AL/L110AE-EDR

The creation, verification, and usage of the compact model of the following devices are presented in this document.

> 1.2V n-ch MOSFET
> 1.2V bpw n-ch MOSFET
> 1.2V p-ch MOSFET

The following compact model is chosen for the modeling of these devices.

> BSIM3V3.24

For detailed descriptions of these compact model equations and parameters, please refer to this material.

> BSIM3V3.24 MOSFET Model User Manual

In BSIM3V3.24, there are two types of models that can be used. One is a binning model, which means the whole device dimension range is chopped into several "bins" and different model parameter sets are provided for each bin. The other one is a global model, which means one scalable parameter set is provided and it can cover the whole device geometry range. Each of these two approaches has advantages over the other.

UMC provides global models for their many benefits and the model accuracy is guaranteed by a quantitative report of model fitting error against the silicon data, which can be found in the following sections.

Note: Three choices of thick oxide IO devices are available, 2.5V, 3.3V and HG 3.3V, which cannot coexist on the same chip. If any design requirement is necessary, please discuss in advance with UMC through CE department.

1.3 Model Usage

The compact model is provided in various formats. It is guaranteed that the simulation results of this model from different simulators show differences less than 0.5% and 1% for DC and AC simulations respectively. It should be noted that the model has been verified on different simulators of the specific versions listed below and the simulation results from other versions of the simulators might be different.

Synopsys HSPICE release 2014.9SP2
Cadence Spectre version 13.1.0.066
Mentor Graphics ELDO version 2013.1

To cope with the impact of process variations on circuit performance, worst-case models are provided along with the typical case and they are listed below. Please note that 'fast' here means lower threshold voltage, higher leakage and driving current, and 'slow' means higher threshold voltage, lower leakage and driving current.

TT: Typical n-ch MOSFET and Typical p-ch MOSFET
FF: Fast n-ch MOSFET and Fast p-ch MOSFET
SS: Slow n-ch MOSFET and Slow p-ch MOSFET
FNSP: Fast n-ch MOSFET and Slow p-ch MOSFET
SNFP: Slow n-ch MOSFET and Fast p-ch MOSFET

Follow these steps to invoke the model file during simulation.

For Synopsys HSPICE
- Copy ModelFileName.mdl and ModelFileName.lib into your simulation directory.
- Add the following statement in the input file to the simulator for 90% dimension shrinkage.
.Option scale=0.9
- Add the following statement in the input file to the simulator.
.lib "./ModelFileName.lib.eldo" CaseName
, where CaseName can be tt, ff, ss, fnsp, or snfp.
- Define device condition by 'Mxx' statement;

Example:
M1 Vdd Vgg Vss Vbb DeviceName W=WDES L=LDES

For Cadence Spectre
- Copy ModelFileName.mdl.scs and ModelFileName.lib.scs into your simulation directory.
- Add the following statement in the input file to the simulator for 90% dimension shrinkage.
SetOption1 options scale=0.9
- Add the following statement in the input file to the simulator.
include "./ModelFileName.lib.scs" section=CaseName
, where CaseName can be tt, ff, ss, fnsp, or snfp.
- Define device condition by 'Mxx' statement;

Example:
M1 (Vdd Vgg Vss Vbb) DeviceName W=WDES L=LDES

For Mentor Graphics Eldo
- Copy ModelFileName.mdl.eldo and ModelFileName.lib.eldo into your simulation
directory.
- Add the following statement in the input file to the simulator for 90% dimension shrinkage.
.Option scale=0.9
- Add the following statement in the input file to the simulator.
.lib "./ModelFileName.lib.eldo" CaseName
, where CaseName can be tt, ff, ss, fnsp, or snfp.
- Define device condition by 'Mxx' statement;

Example:
M1 Vdd Vgg Vss Vbb DeviceName W=WDES L=LDES

ModelFileName for 1.2V MOSFET: l110ae_25ud12_v051
DeviceName for 1.2V n-ch MOSFET: n_25ud12_l110ae
DeviceName for 1.2V bpw n-ch MOSFET: n_bpw_25ud12_l110ae
DeviceName for 1.2V p-ch MOSFET: p_25ud12_l110ae

As stated above, one global model is provided and this model is valid within the following geometry, voltage, and temperature ranges. Device behavior beyond these ranges may not be well described by the model and is not guaranteed.

Geometry range (After 90% shrink):

For 1.2V n-ch and bpw n-ch MOSFET:

0.234 um <=L_{DES} <=45 um
0.35 um <= W_{DES} <=90 um

For 1.2V p-ch MOSFET:

0.234 um <=L_{DES} <=45 um
0.35 um <= W_{DES} <=90 um

Adopt multi-finger structures if WDES > 9 um.
0.318 um ≦ SA,SB ≦ 1.98 um

Note: SAREF = SBREF = 1.98 um is the reference dimension of the STI-stress (LOD) effect.

Voltage range:

For 1.2V n-ch and bpw n-ch MOSFET:

0V <= V_{GS} <= 1.2V(*1.1)
0V <= V_{DS} <= 1.2V(*1.1)
-1.2V <= V_{BS} <= 0V

For 1.2V p-ch MOSFET:

0V => V_{GS} => -1.2V(*1.1)
0V => V_{DS} => -1.2V(*1.1)
1.2V => V_{BS} => 0V

Temperature range:

-50 C ~ +125 C

2 1.2V N-ch MOSFET

2.1 Silicon Wafer for Modeling

The information about the silicon wafer used for characterization and model parameter extraction is summarized in the table below.

Table 2-1 Test wafer information for NMOS

	MOSFET DC model	MOSFET AC Model	Junction DC Model	Junction AC Model
Mask ID	BSW0QA	BSW0QA	BSW0QA	BSW0QA
Lot ID	D5FHR.11	D5FHR.11	D5FHR.11	D5FHR.11
Wafer Number	11	11	11	11

2.2 Electrical Parameters

The important electrical parameters for the characterization of this device are summarized in the table below.

Linear and saturation threshold voltage is extracted at
ID =300nA*WDES*0.9/ (LDES*0.9+12nm) for NMOS and
ID =-70nA*WDES*0.9/ (LDES*0.9+12nm) for PMOS, respectively.
The dimension is after 90% shrink and 12nm sizing channel length back.

All parameters are given for T=25 C, V_{BS}=0 V unless specified otherwise.

Table 2-2 Electrical parameters for NMOS

All parameters are given for T=25 ^{o}C , V_{BS}=0 V unless specified otherwise.

Target & Centered Value are given for SA=2.2u SB=2.2u

Measured & Extracted Value are given for SA=2.2u SB=2.2u

Parameter	Condition	Unit	Target	Measured Value	Extracted Value	Centered Value
Long channel device (W/L= 10 μ m / 10 μ m)						
V_{TLIN}	V_{DS} = 0.1 V	V	0.512	0.506	0.516	0.512
I_{ON}	V_{DS}=V_{GS}= 1.2 V	μA/ μm	3.87	3.85	3.86	3.87
Wide and short channel device (W/L= 10 μ m / 0.246 μ m)						
V_{TLIN}	V_{DS} = 0.1 V	V	0.5	0.497	0.498	0.500
V_{TSAT}	V_{DS} = 1.2 V	V	0.450	0.448	0.455	0.450
I_{ON}	V_{DS}=V_{GS}= 1.2 V	μA/μm	154	155.30	156.04	154.00
I_{OFF}	V_{DS}= 1.2 V, V_{GS}=0 V	A/μm	--	5.4E-13	2.9E-12	3.4E-12*
Narrow and long channel device (W/L= 0.35 μ m / 10 μ m)						
V_{TLIN}	V_{DS} = 0.1 V	V	0.512	0.508	0.505	0.512
V_{TSAT}	V_{DS}= 1.2 V	V	0.508	0.503	0.502	0.510
I_{ON}	V_{DS}=V_{GS}= 1.2 V	μA/μm	3.35	3.35	3.39	3.35
Narrow and short channel device (W/L= 0.35 μm / 0.246 μm)						
V_{TLIN}	V_{DS} = 0.1 V	V	0.486	0.478	0.489	0.486
V_{TSAT}	V_{DS}= 1.2 V	V	0.437	0.427	0.438	0.436
I_{ON}	V_{DS}=V_{GS}= 1.2 V	μA/μm	178.06	178.75	178.79	178.05
I_{OFF}	V_{DS}= 1.2 V, V_{GS}=0 V	A/μm	--	6.5E-12	4.1E-12	4.3E-12 *
Capacitance						
CJ	V_J = 0 V	F/m^2	--	7.812E-04	7.812E-04	7.812E-04
CJSW	V_J = 0 V	F/m	--	1.125E-10	1.125E-10	1.125E-10
CJSWG	V_J = 0 V	F/m	--	2.250E-10	2.155E-10	2.155E-10
C_{OVL}	V_{GS} = -0.5 V	F/m	--	2.65E-10	2.55E-10	2.55E-10

*gmindc=1e-15

2.3 MOSFET DC Model

2.3.1 Test devices

The geometry of devices that were used for MOSFET DC model parameter extraction is summarized below.

Table 2-3 Geometry of NMOS devices measured for DC parameter extraction

W\L	10	1	0.8	0.6	0.507	0.435	0.417	0.399	0.318	0.3	0.282	0.264	0.246
10	X	X	X	X	X	X	X	X	X	X	X	X	X
1											X	X	X
0.8	X	X		X	X		X		X		X	X	X
0.6				X			X		X				X
0.43	X	X			X		X		X		X	X	X
0.35	X			X	X		X		X				X

2.3.2 Measurement conditions

The drain current was measured at various bias conditions summarized in the table below:

Table 2-4 Test conditions for IV curves of NMOS

I_D vs. V_{GS}

	Start	Stop	Step
V_{DS} (V)	0.1	1.2	1.1
V_{GS} (V)	-0.5	1.35	0.025
V_{BS} (V)	0	-1.2	-0.3

I_D vs. V_{DS}

	Start	Stop	Step
V_{DS} (V)	0	1.35	0.025
V_{GS} (V)	0.7	1.2	0.125
V_{BS} (V)	0	-1.2	-1.2

2.3.3 Model verification plots

The simulation results obtained from the extracted model were plotted together with the measurement results for comparison. The L and W in the plots refer to physical dimension (LDES *0.9+12nm and WDES *0.9) respectively.

I_D vs. V_{DS} plots

The following plots show measured and simulated output I-V curves for selected devices at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

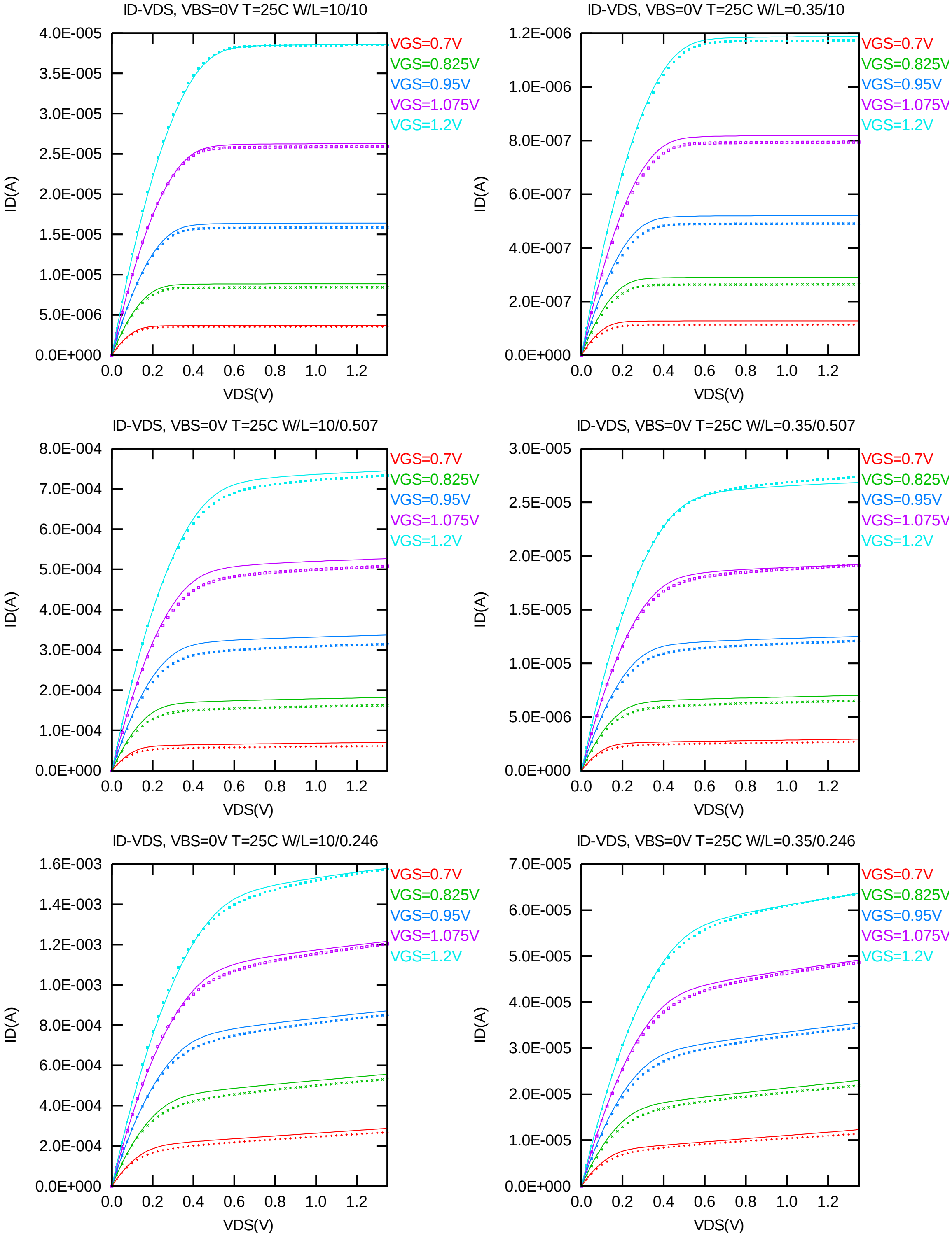

Figure 2-1 I_D vs. V_{DS} at V_{BS} = 0 V for NMOS

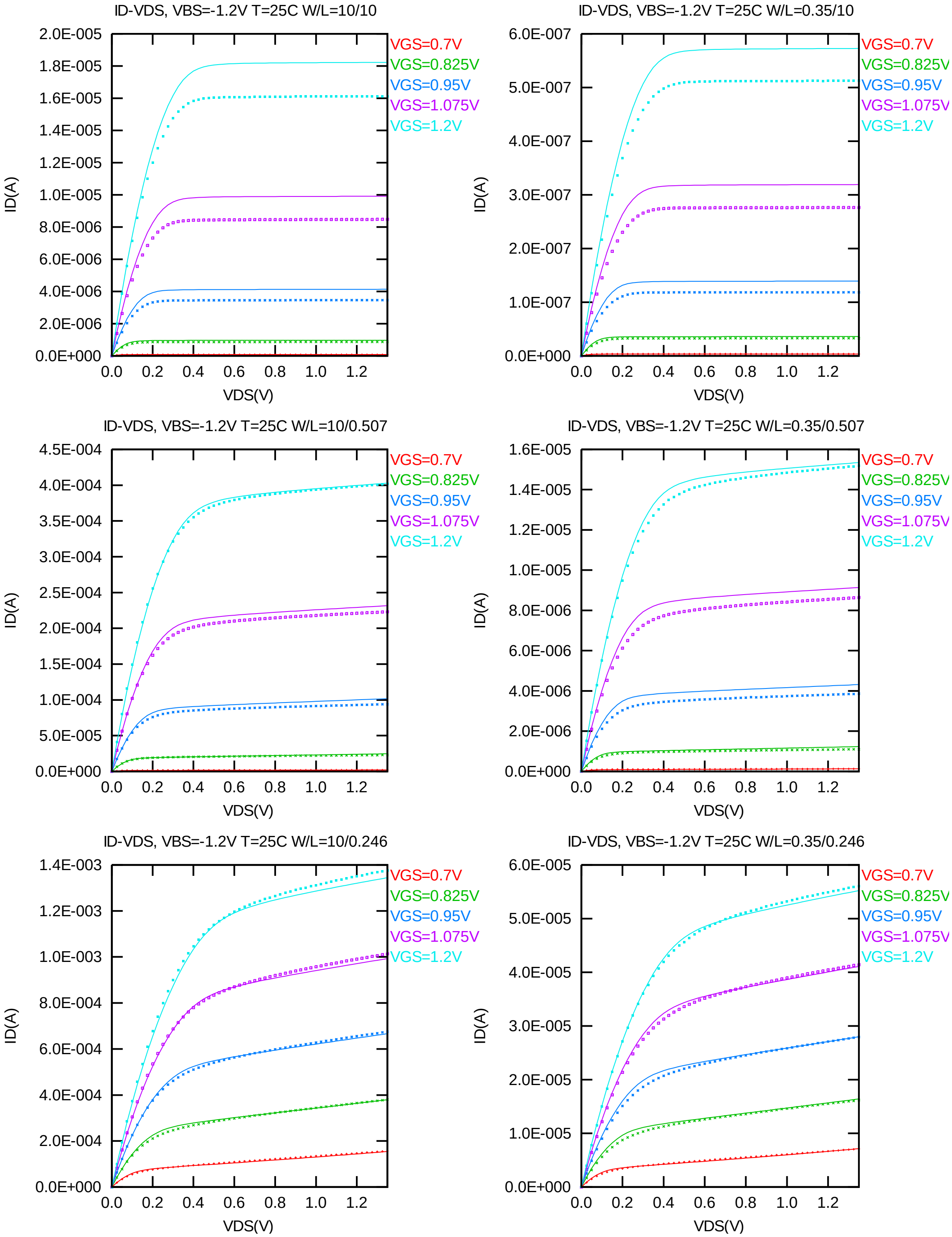

Figure 2-2 I_D vs. V_{DS} at V_{BS} = -1.2V for NMOS

G_{DS} vs. V_{DS} plots

The following plots show measured and simulated output I-V curves for selected devices at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

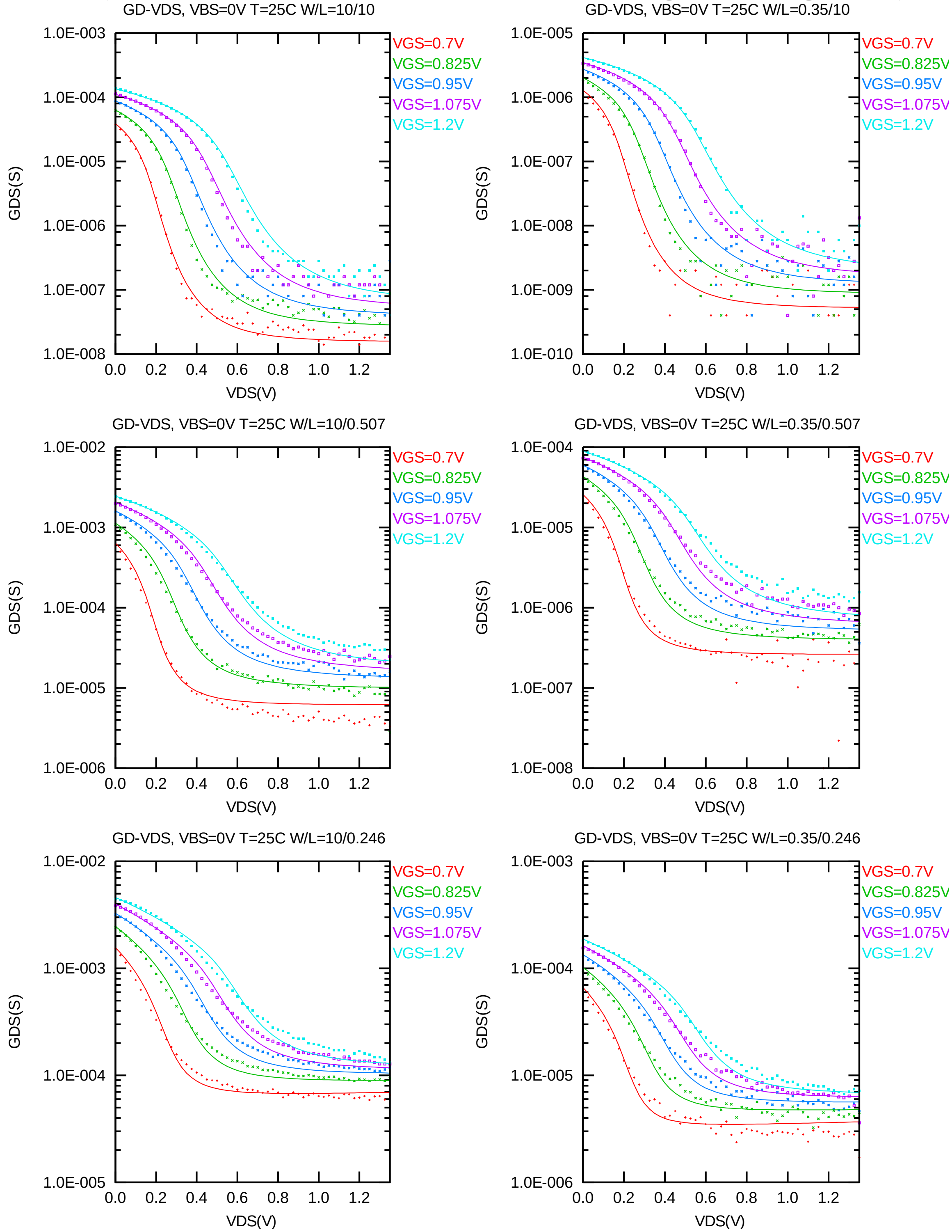

Figure 2-3 G_{DS} vs. V_{DS} at V_{BS} = 0 V for NMOS

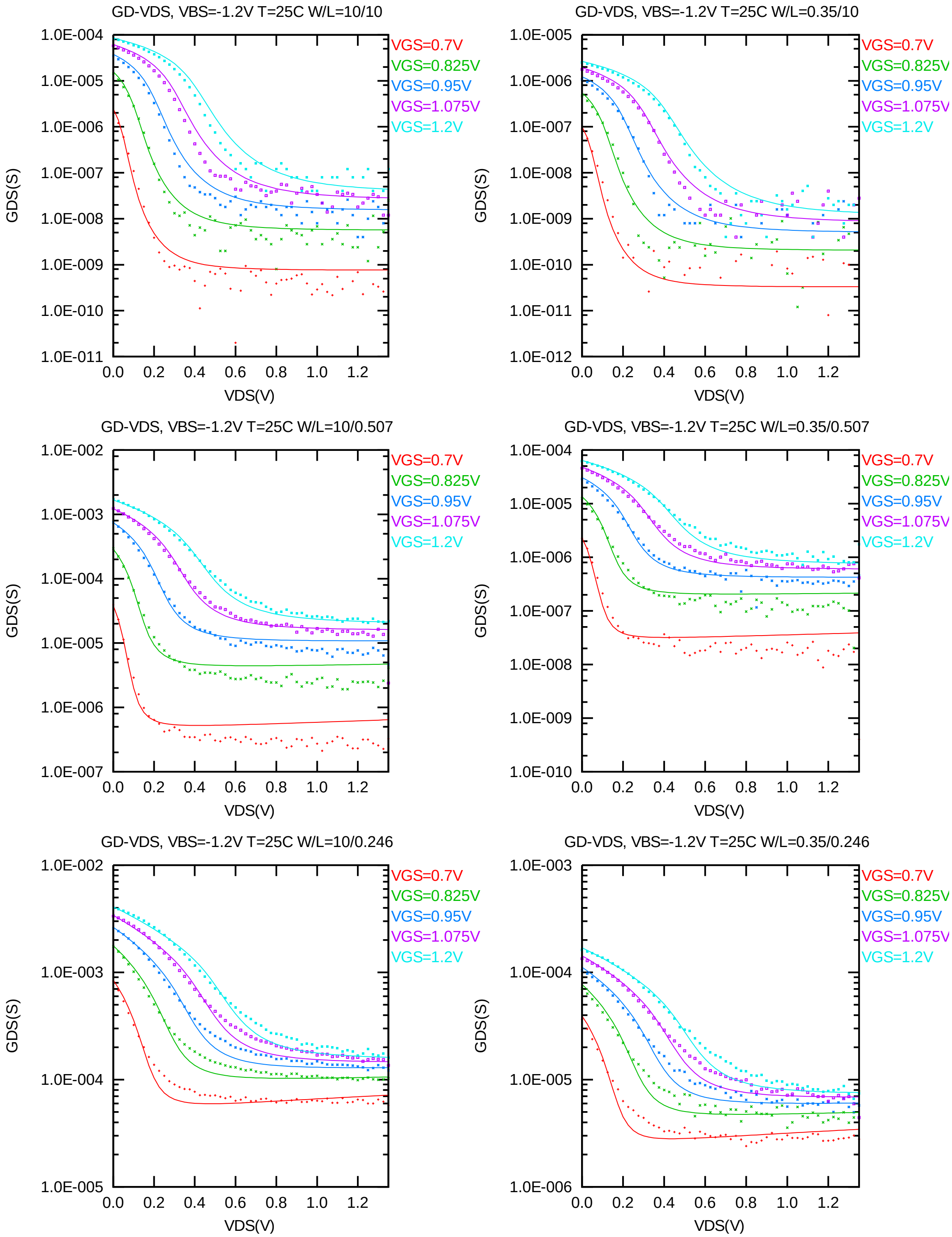

Figure 2-4 G_{DS} vs. V_{DS} at V_{BS} = -1.2V for NMOS

I_D vs. V_{GS} plots (linear scale)

The following plots show measured and simulated output I-V curves for selected devices at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

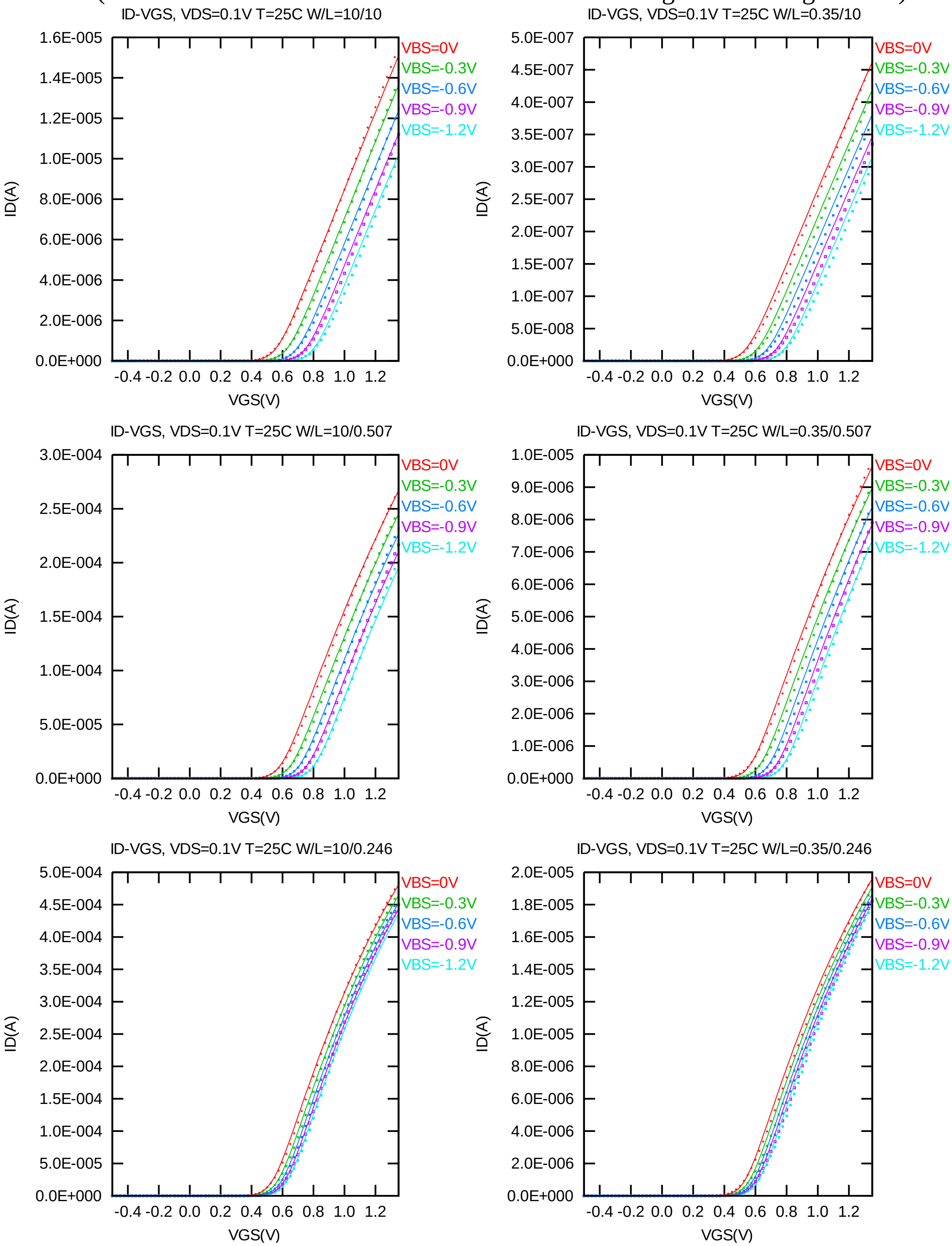

Figure 2-5 I_D vs. V_{GS} at V_{DS} = 0.1 V for NMOS

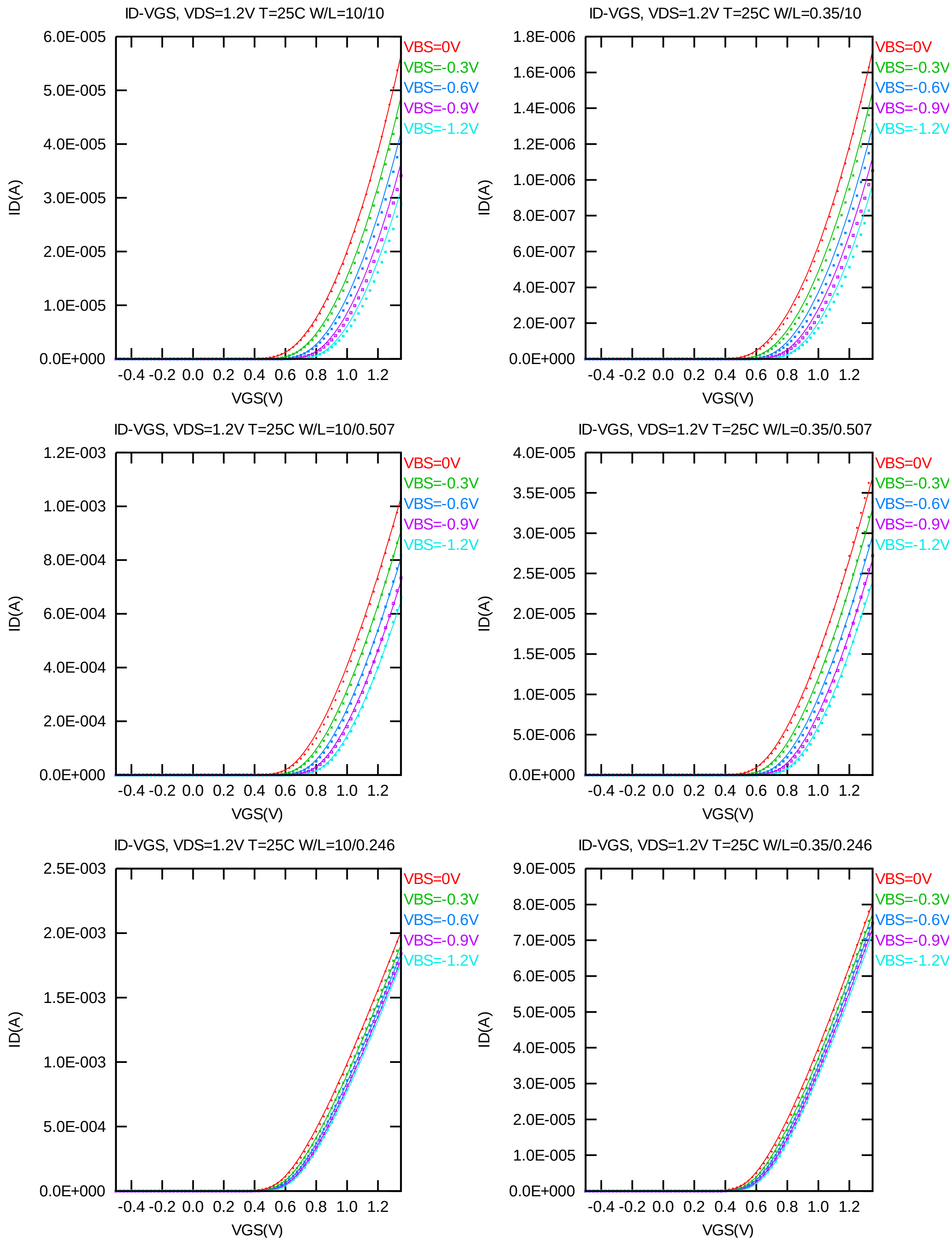

Figure 2-6 I_D vs. V_{GS} at V_{DS} = 1.2V for NMOS

I_D vs. V_{GS} plots (logarithmic scale)

The following plots show measured and simulated output I-V curves for selected devices at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

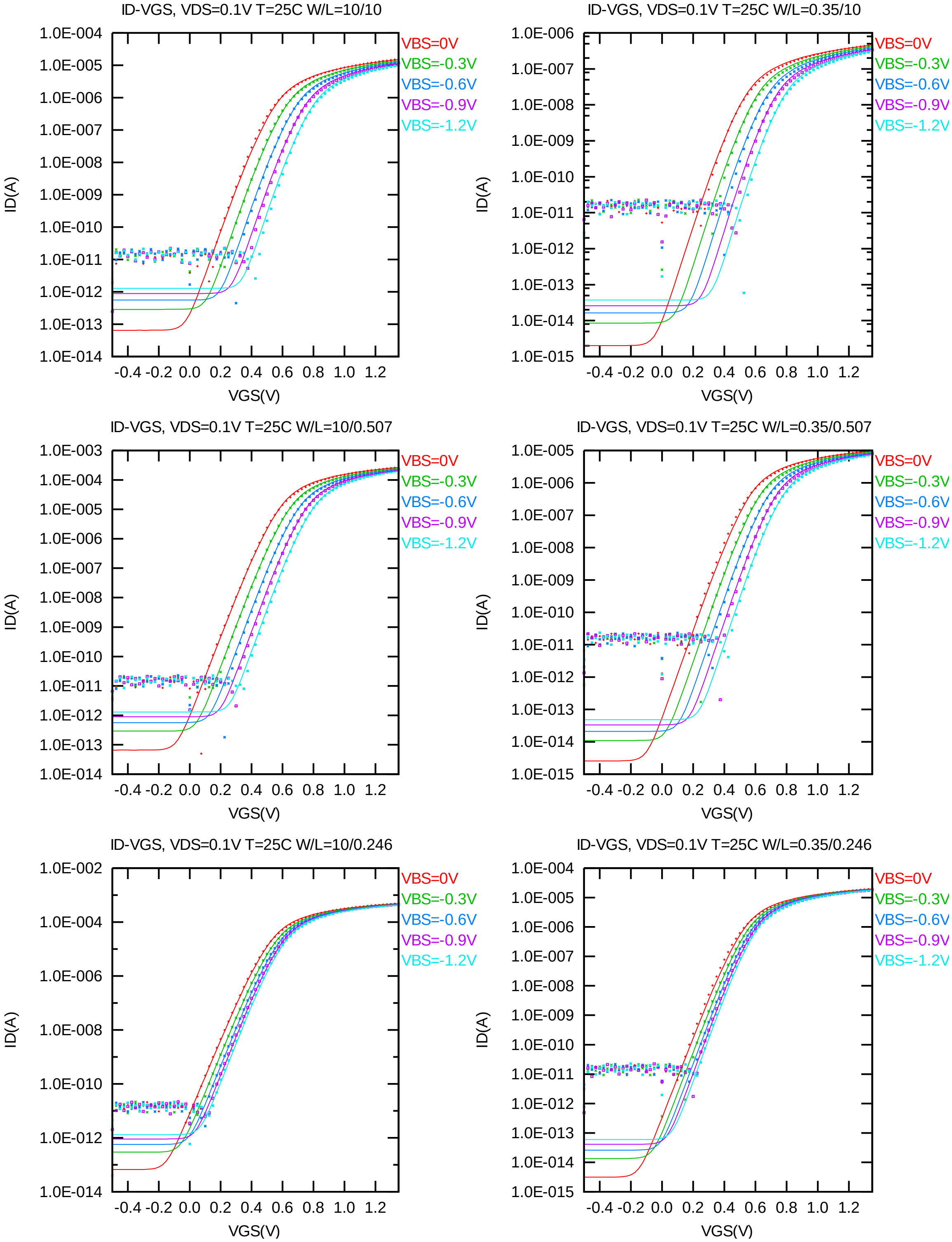

Figure 2-7 Log scaled I_D vs. V_{GS} at V_{DS} = 0.1 V for NMOS

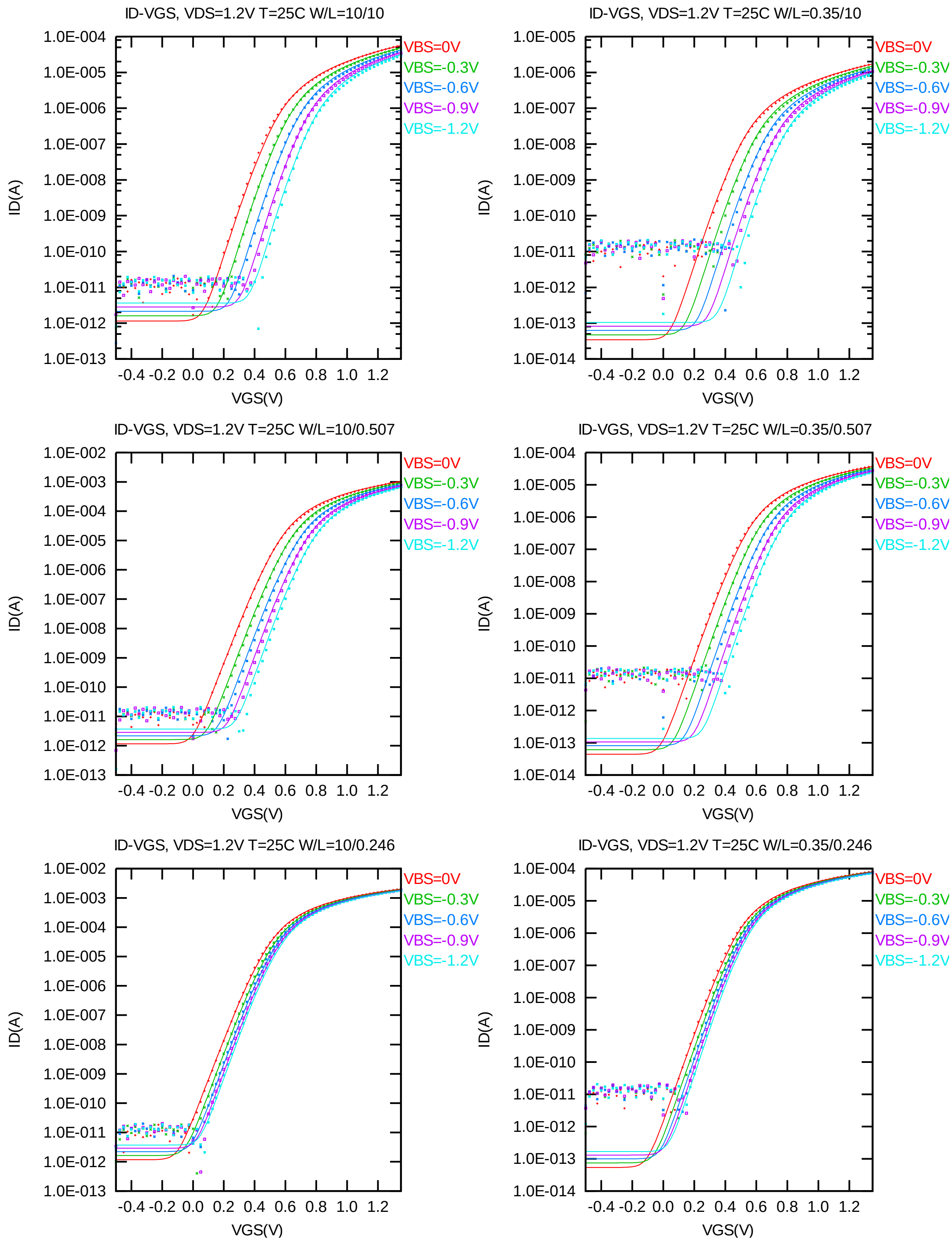

Figure 2-8 Log scaled I_D vs. V_{GS} at V_{DS} = 1.2V for NMOS

G_M vs. V_{GS} plots

The following plots show measured and simulated output I-V curves for selected devices at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

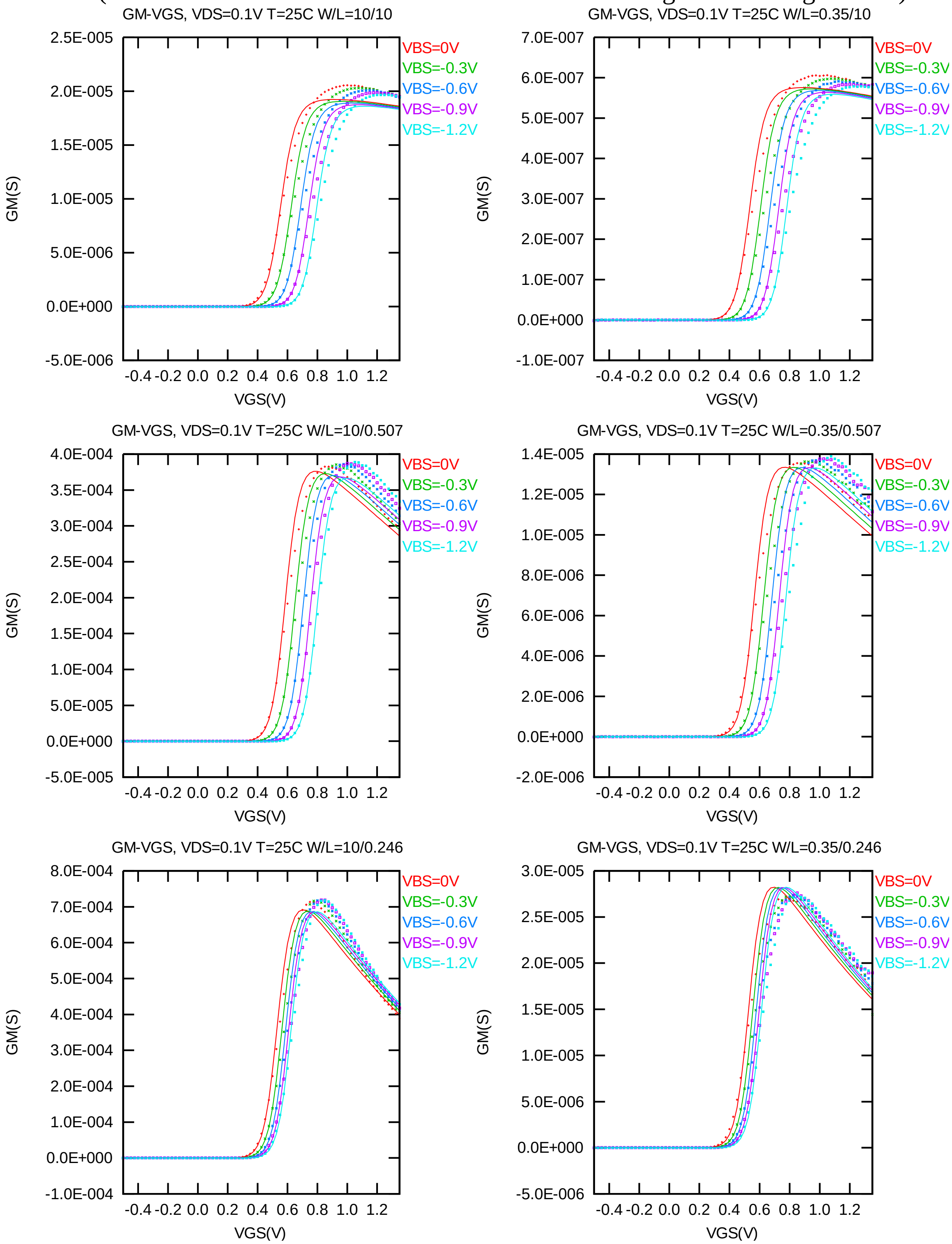

Figure 2-9 G_M vs. V_{GS} at V_{DS} = 0.1 V for NMOS

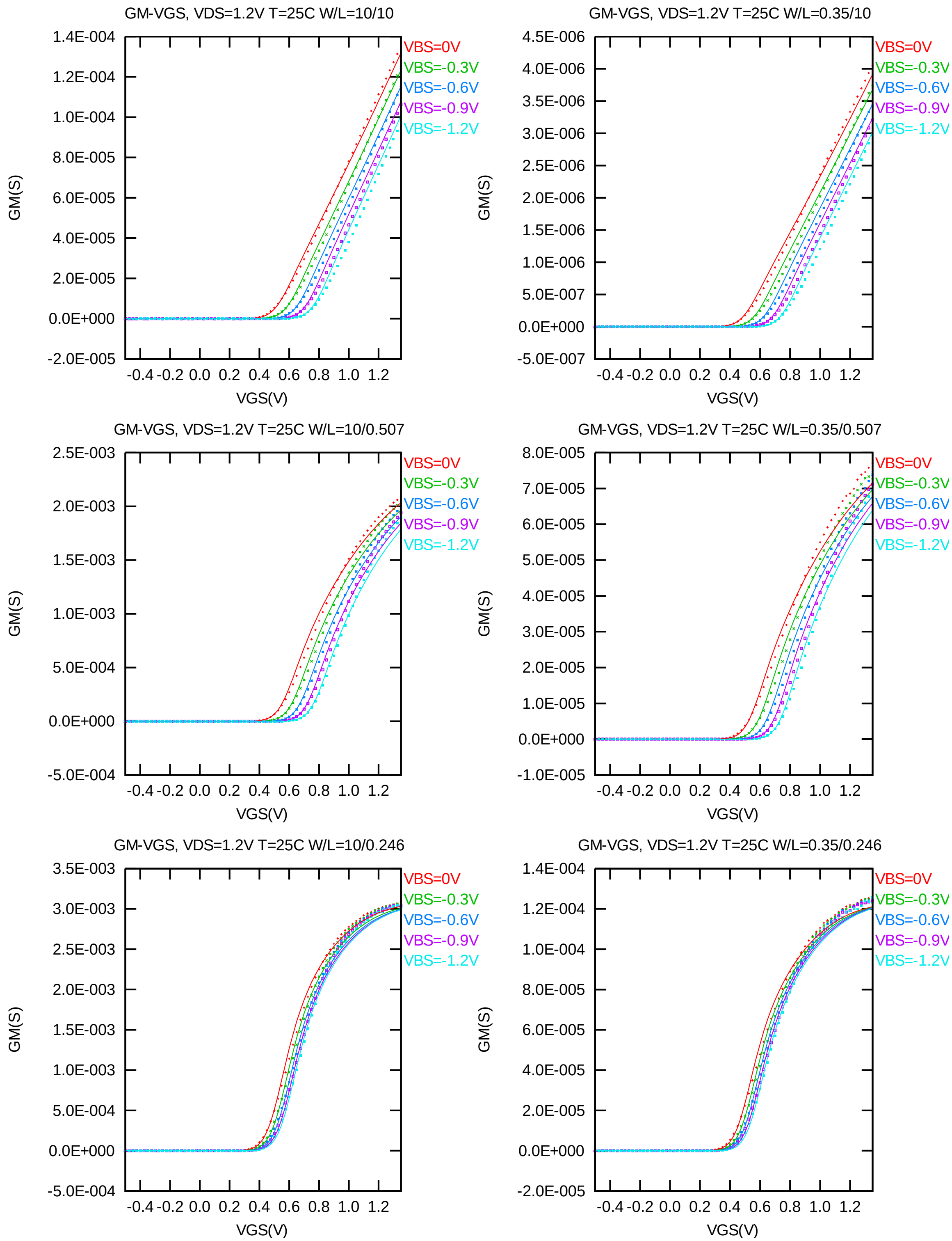

Figure 2-10 G_M vs. V_{GS} at V_{DS} = 1.2V for NMOS

The L and W in the plots refer to LDES and WDES respectively.

The following figures show the channel length and width dependence of linear and saturation threshold voltage extracted at ID =300nA*(WDES*0.9)/(LDES*0.9+0.012 μm) at 25 ℃. The plots show the physical dimension. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

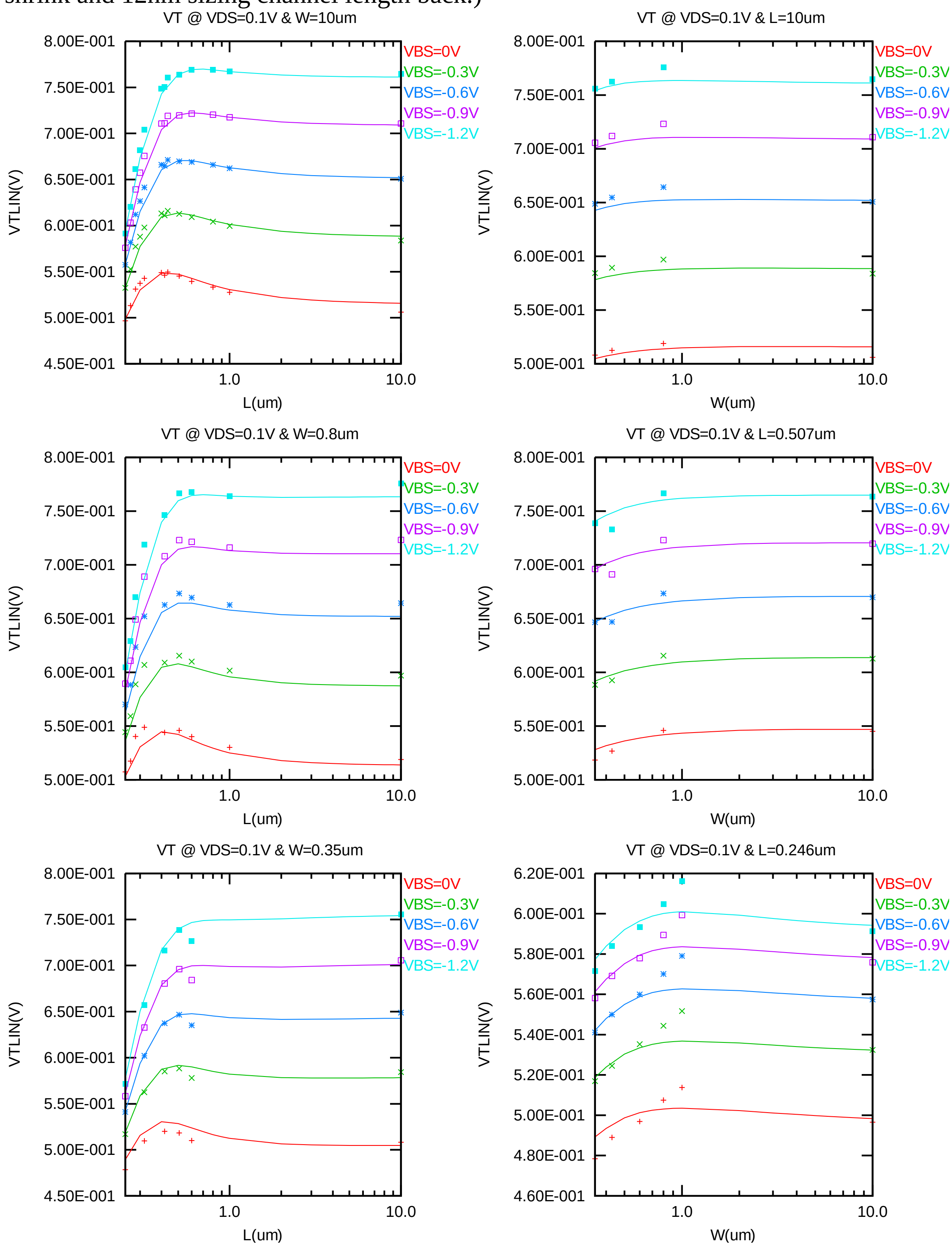

Figure 2-11 V_{TLIN} vs. channel length & channel width for low drain bias (V_{DS} = 0.1 V) for NMOS

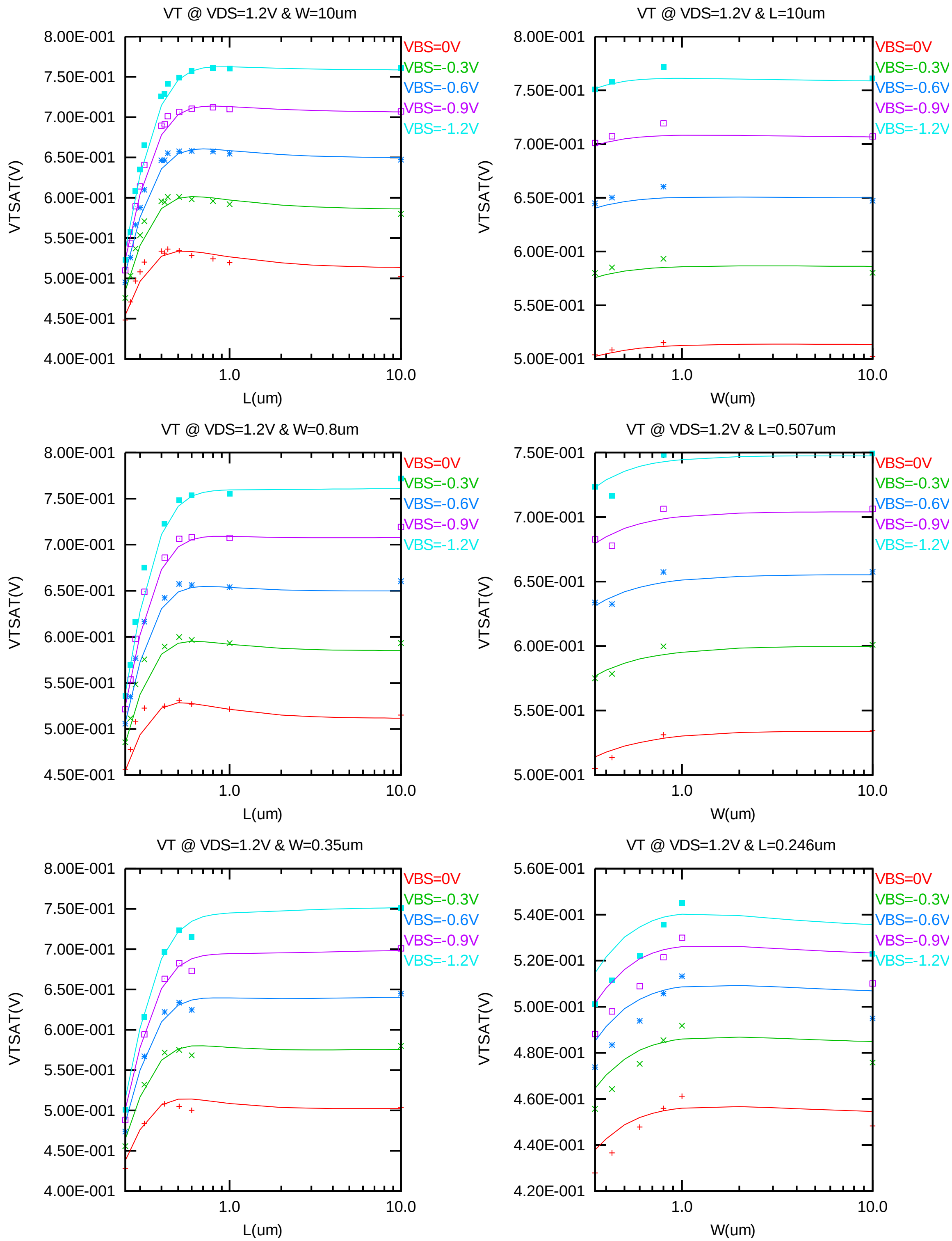

Figure 2-12 V_T vs. channel length & channel width for high drain bias (V_{DS} = 1.2V) for NMOS

The L and W in the plots refer to LDES and WDES respectively.

The following plots show the channel length and width dependence of I_{ON} extracted at $V_{DS} = V_{GS} = 1.2V$ for $V_{BS} = 0$ V and $V_{BS} = -1.2V$ at 25 C.(The dimensions are after 90% shrink and 12nm sizing channel length back.)

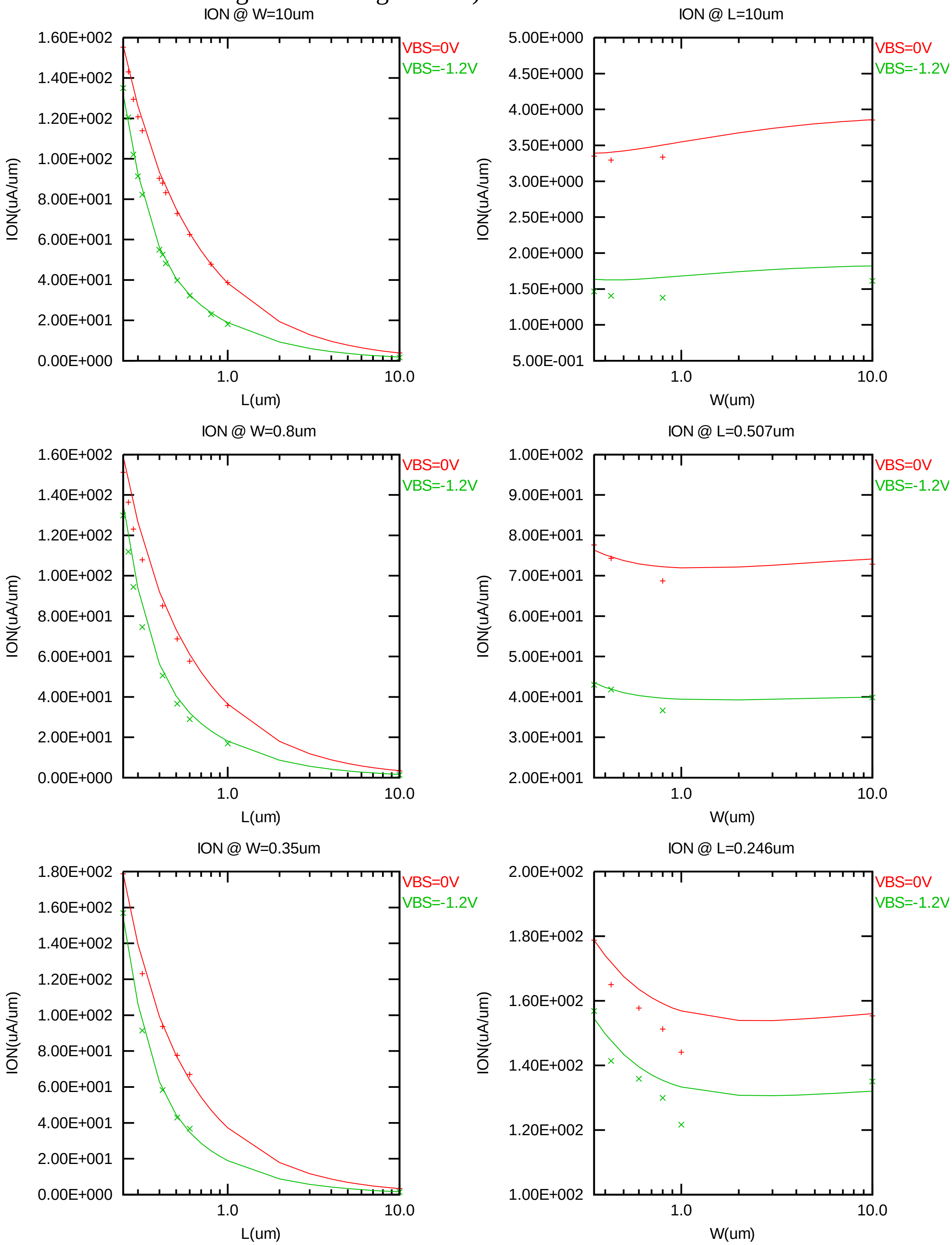

Figure 2-13 Saturation current I_{ON} vs. channel length and channel width for NMOS

The L and W in the plots refer to LDES and WDES respectively.

The following plots show the channel length and width dependence of I_{OFF} extracted at $V_{DS} = V_{CC}$ and $V_{GS} = 0V$ for $V_{BS} = 0$ V to -1.2V at 25 C. A large discrepancy of I_{OFF} can be seen in the graphs, which is due to the presence of junction leakage current.(The dimensions are after 90% shrink and 12nm sizing channel length back.)

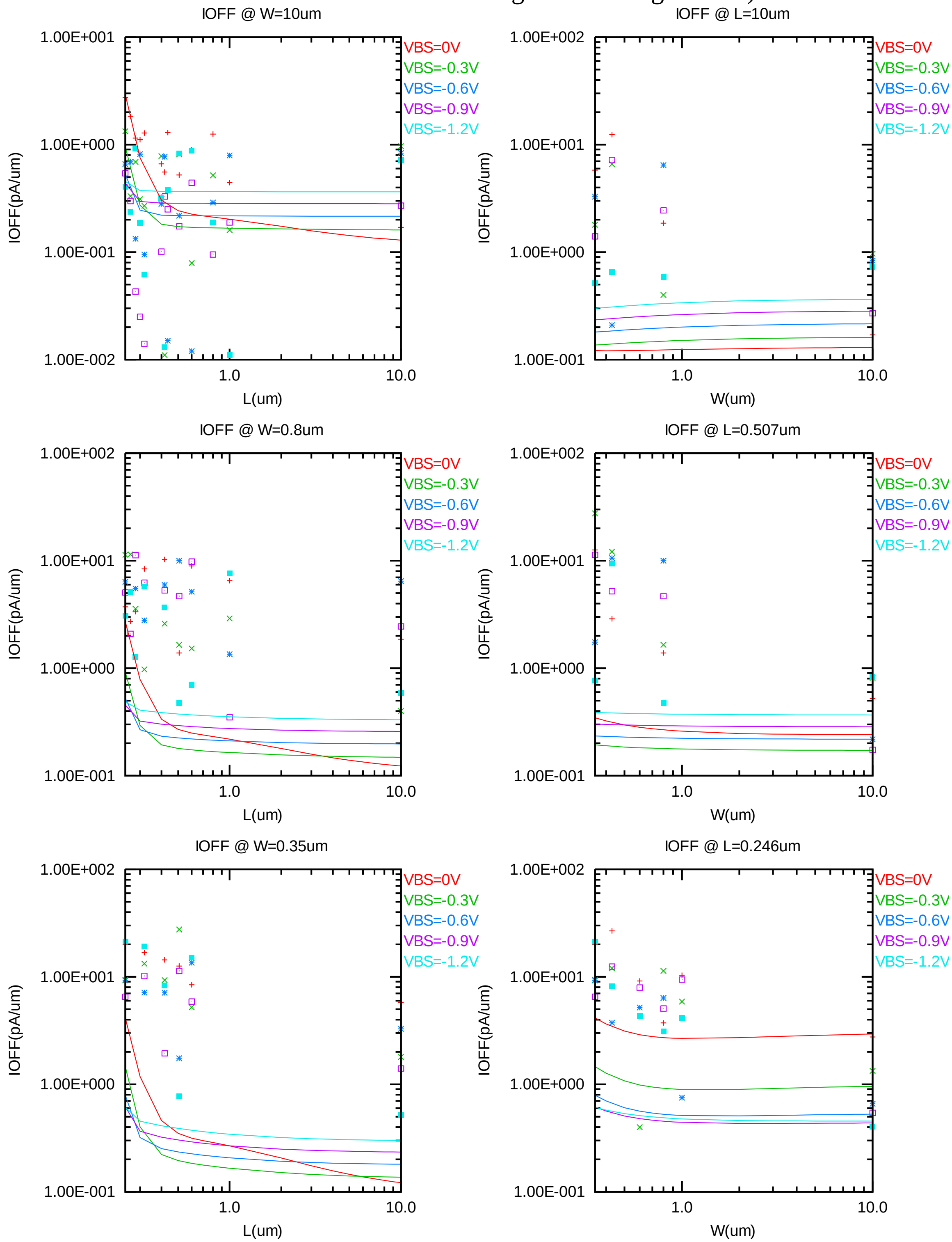

Figure 2-14 I_{OFF} vs. channel length and channel width for NMOS

V_{TLIN} vs. temperature

The following plots show the temperature dependence of the linear and saturation threshold voltage extracted at ID = 300nA*(WDES*0.9)/(LDES*0.9+0.012 µm). (The dimensions are after 90% shrink and 12nm sizing channel length back.)

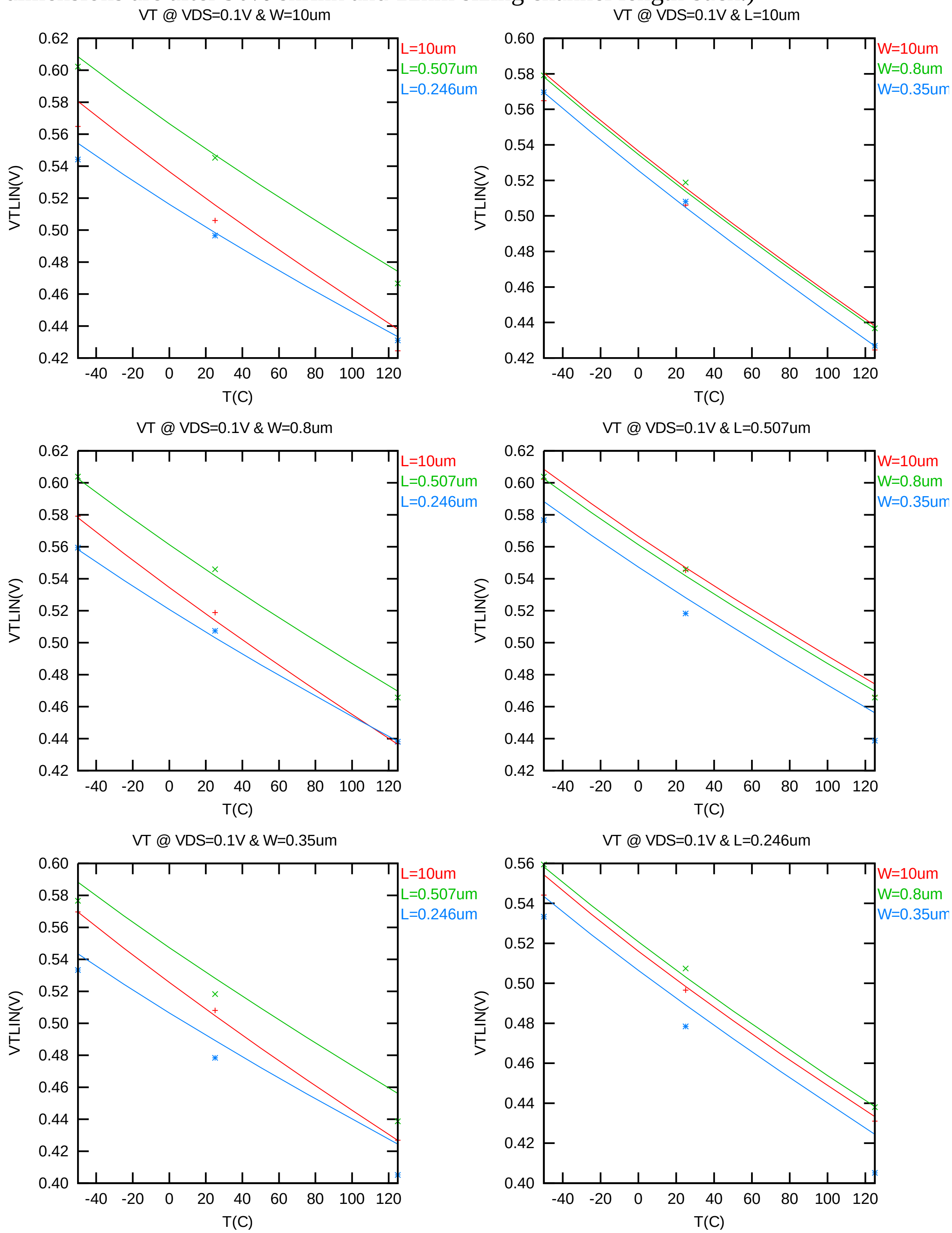

Figure 2-15 V_{TLIN} vs. temperature for NMOS

V_{TSAT} *vs. temperature*

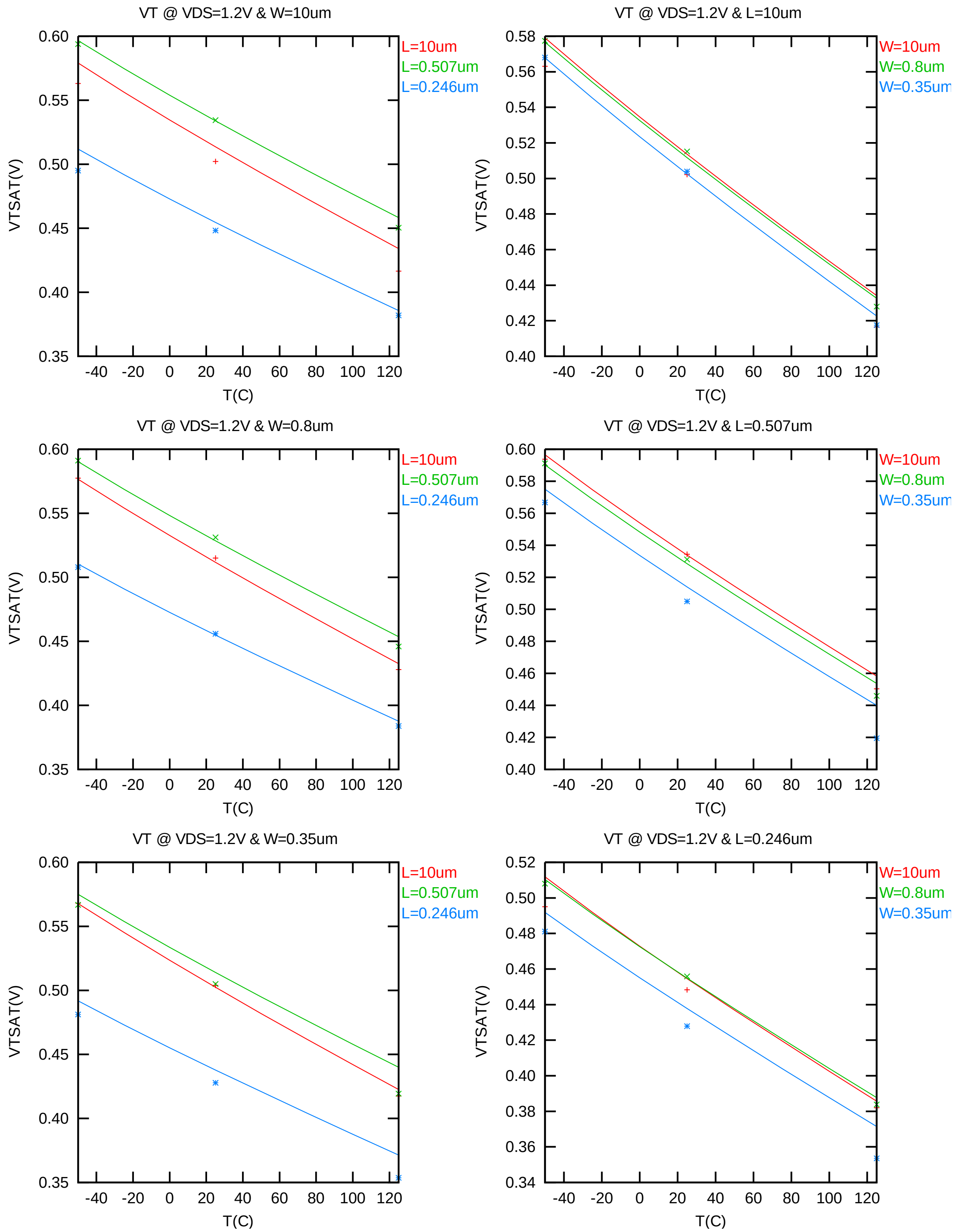

Figure 2-16 V_{TSAT} vs. temperature for NMOS

I_{ON} vs. temperature

The following plots show the temperature dependence of I_{ON} extracted at $V_{GS} = V_{DS} = 1.2V$, $V_{BS} = 0$ V. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

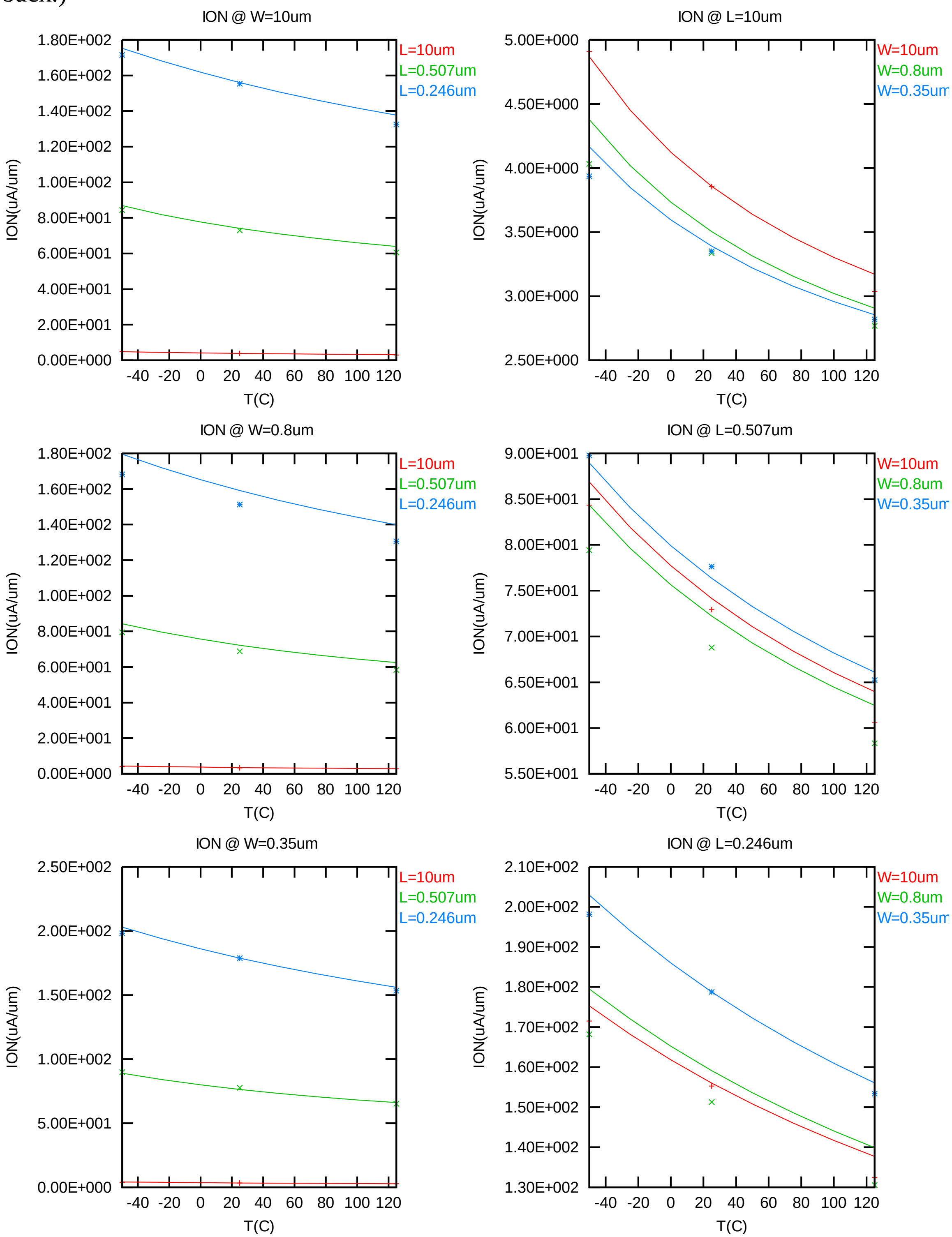

Figure 2-17 Saturation current vs. temperature for NMOS

I_{OFF} vs. temperature

The following plots show the temperature dependence of I_{OFF} extracted at V_{DS} = 1.2V, V_{GS} = V_{BS} = 0 V. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

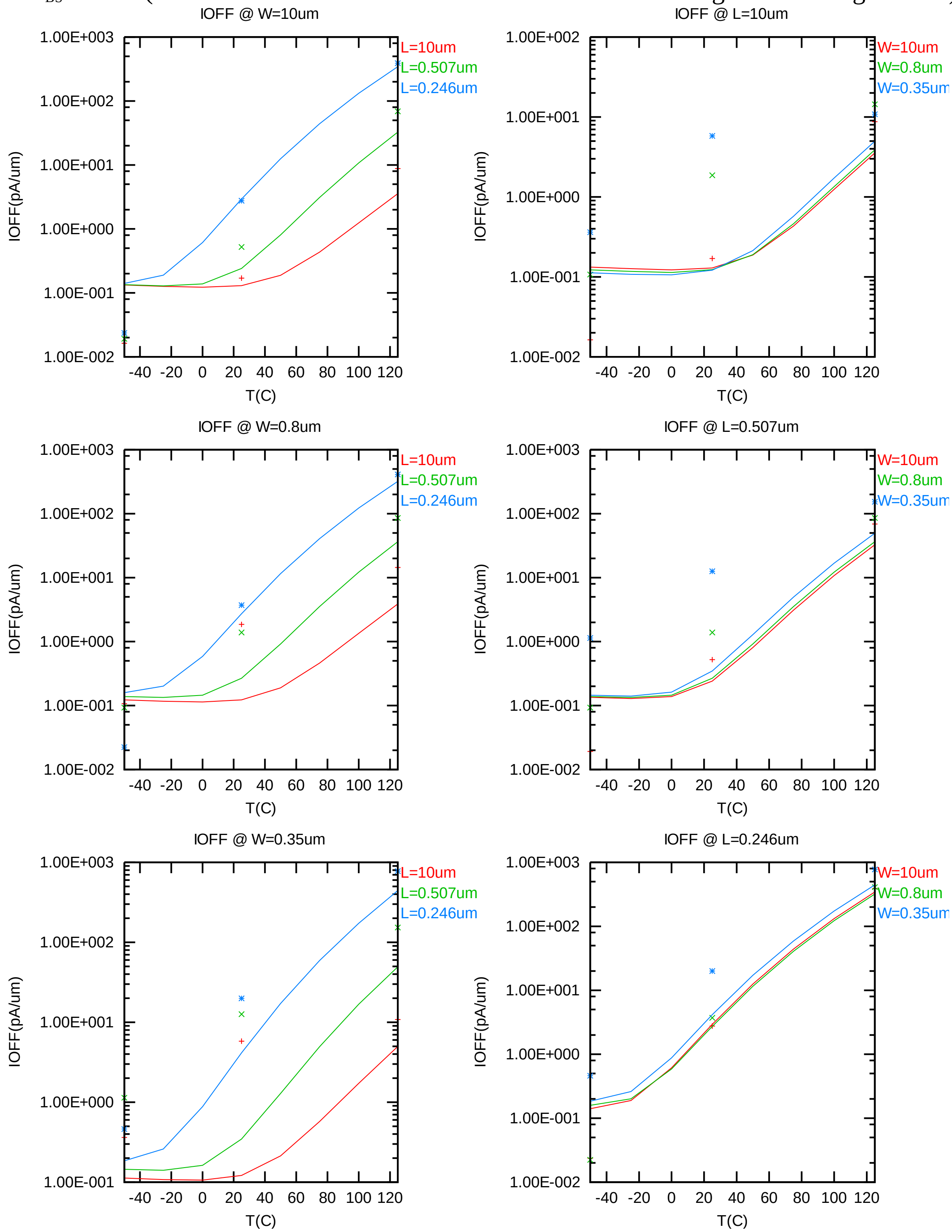

Figure 2-18 I_{OFF} vs. temperature for NMOS

2.3.4 Model fitting accuracy

The model fitting accuracy of parameters I_D, G_M and G_{DS} is quantified using the following equation:

error % = 100 * | measurement - simulation | / [(measurement + simulation) / 2]

Different error criteria are defined for different parameters (I_D, G_M and G_{DS}) as well as for different operation regions. These criteria are explained below.

1) I_D accuracy criteria

a. I_D in sub-threshold region, i.e. $V_{GS} < V_T$
b. I_D in moderate inversion region, i.e. $V_T \leqq V_{GS} < V_T + V_X$
c. I_D in strong inversion region, i.e. $V_T + V_X \leqq V_{GS} \leqq V_{CC}$

, where $V_X = (V_{CC} - V_T) / 2$

2) G_M and G_{DS} accuracy criteria

d. G_M and G_{DS} in sub-threshold region, i.e. $V_{GS} < V_T$
e. G_M and G_{DS} in inversion region, i.e. $V_T \leqq V_{GS} \leqq V_{CC}$

Table 2-5 Requirements for model fit quality

Target of Quality Check	$V_{GS} < V_T$ (sub-threshold)	$V_T \le V_{GS} < V_T+V_X$ (moderate inversion)	$V_T+V_X \le V_{GS} \le V_{CC}$ (strong inversion)
I_D	≤75%	≤25%	≤7%
G_M	≤50%	≤20%	≤20%
G_{DS}	≤100%	≤50%	≤50%

I_D accuracy

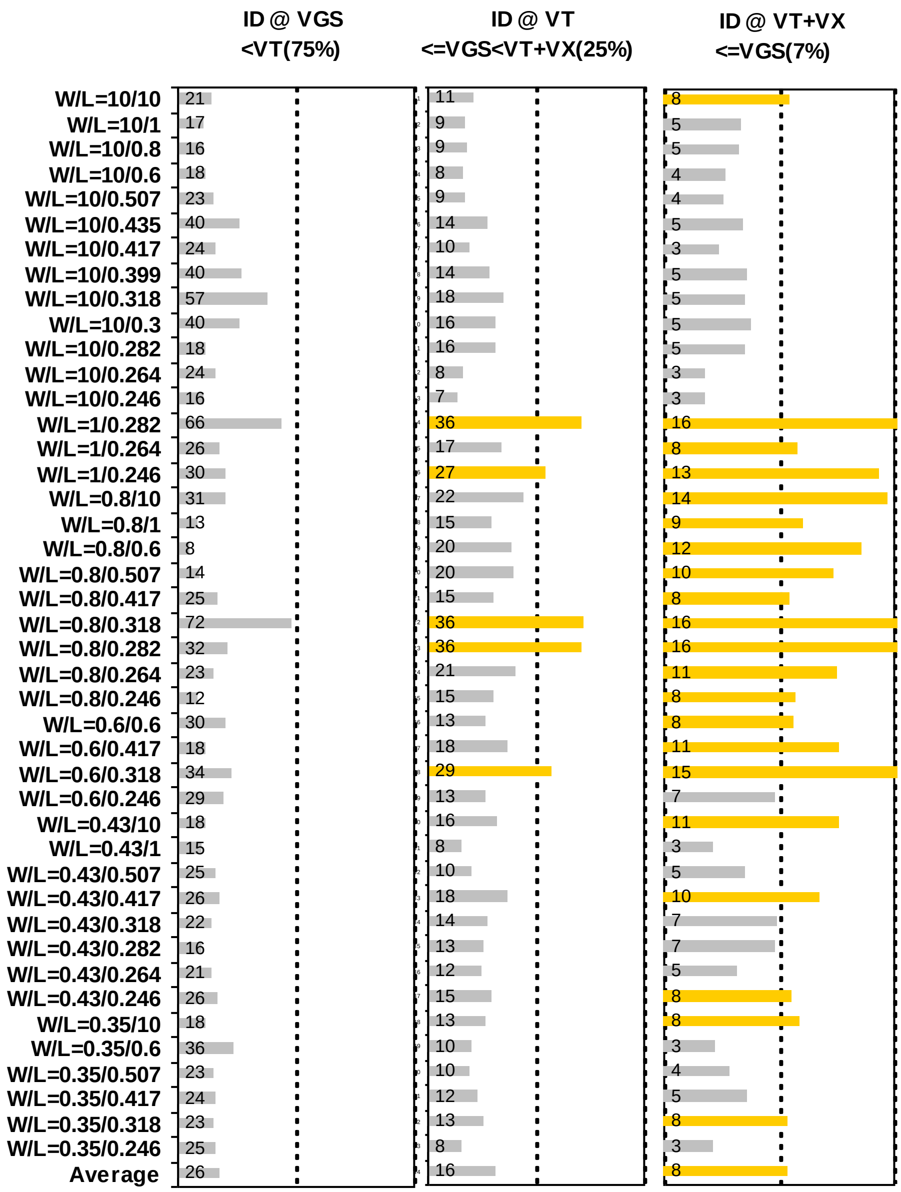

Figure 2-19 I_D accuracy at 25 C for NMOS

G_M and G_{DS} accuracy

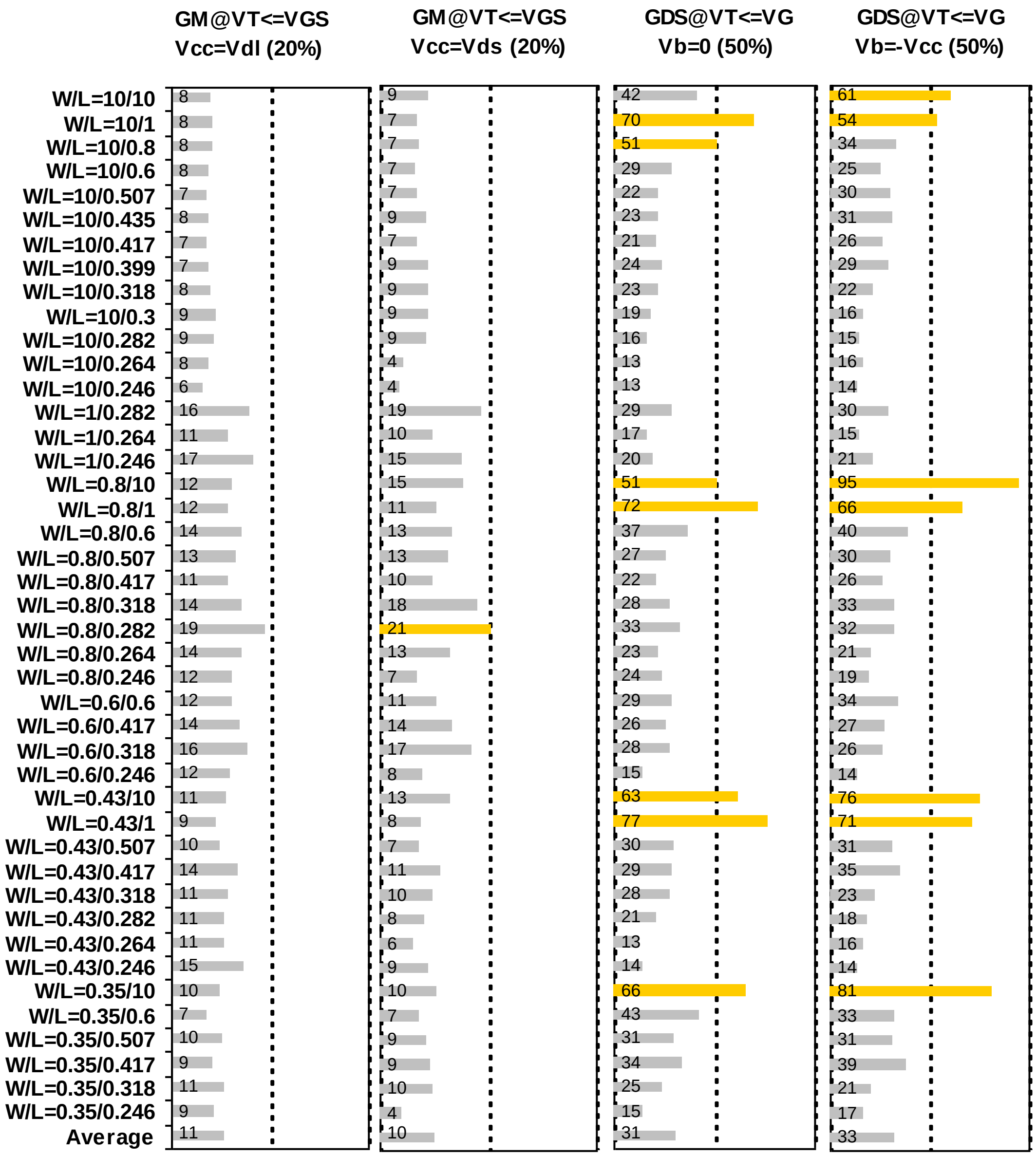

Figure 2-20 G_M and G_{DS} accuracy at 25 C for NMOS

2.4 MOSFET Capacitance Model

2.4.1 Test devices

The following test structures were used for MOSFET capacitance model parameter extraction.

Table 2-6 Measurement structure for NMOS Cgg capacitance model

Structure	L[μm]	W[μm]
Gate capacitance components, C_{GDS}	80	80

Table 2-7 Large area measurement structure for NMOS Cgc capacitance model

Structure	L[μm]	W[μm]
Gate capacitance components	10	10
	1	10
	.5	10
	.246	10
	10	1
	.417	10
	.417	1
	10	.5
	1	1
	.246	.5
	.5	.5
	.417	.25
	10	.16
	10	.25
	.417	.16
	.246	.16
	.5	.16

2.4.2 Measurement conditions

The measurement conditions for MOSFET capacitance are summarized in Table 2-8 below.

Table 2-8 Measurement conditions for NMOS capacitance model

V_{BIAS} [V]	From -3.63 to 3.63 V in 0.066 V step
Frequency [kHz]	100
V_{OSC} [mV]	50
Temp [°C]	25

2.4.3 T_{OX} extraction

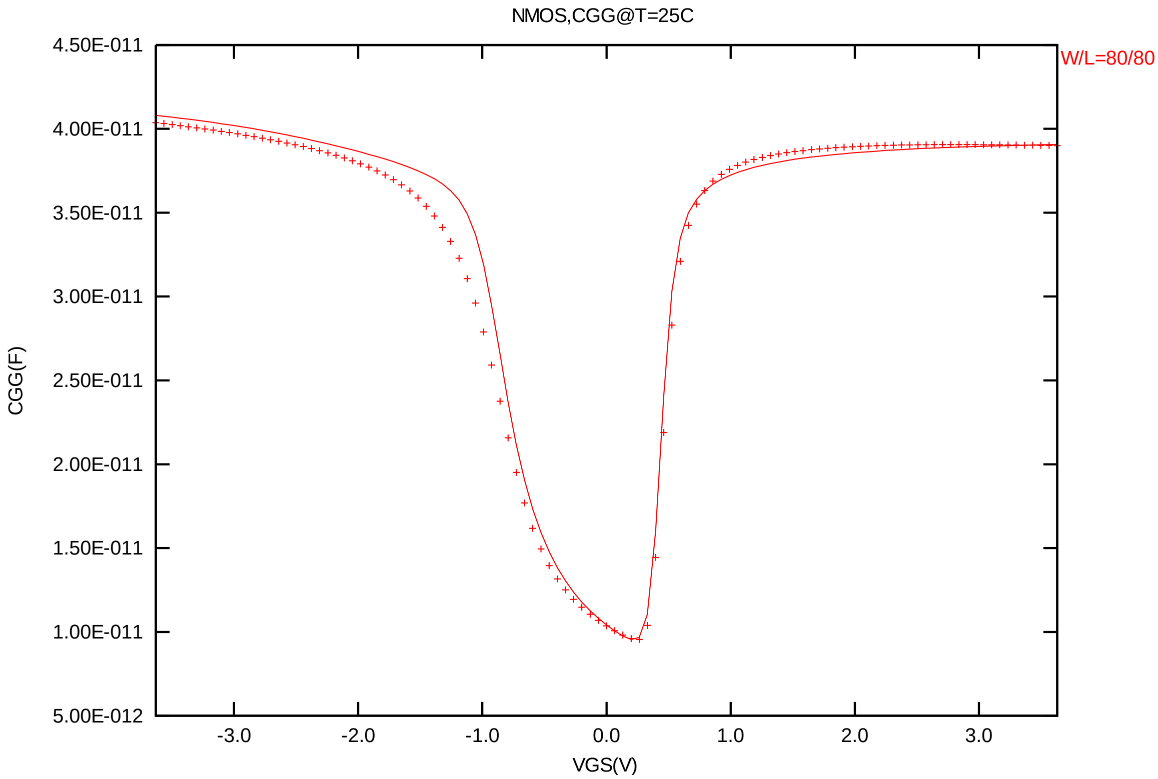

Figure 2-21 Measured and simulated C_{GG} curves

2.4.4 Width and length AC model accuracy

Width and length AC model accuracy

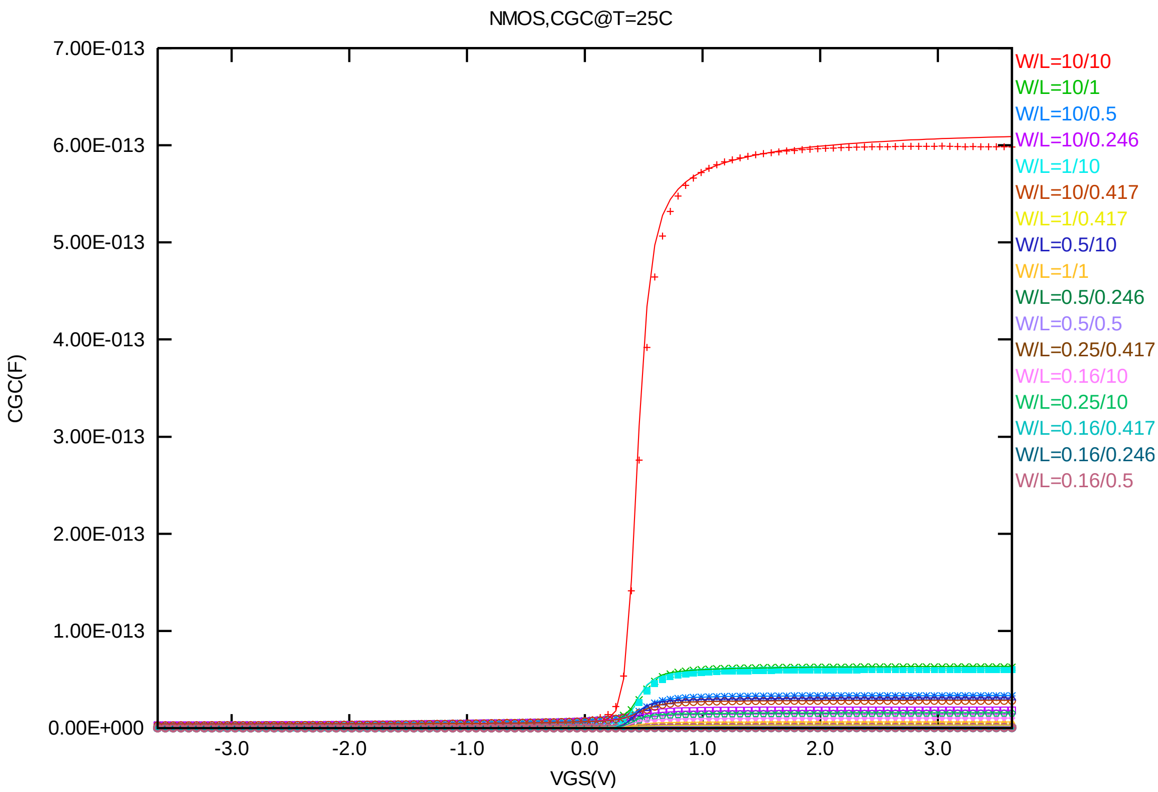

Figure 2-22 Measured and simulated C_{GC} curves for MOSFET measurement structure for NMOS

2.5 Parasitic Source / Drain Junction Capacitance Model

2.5.1 Test devices

The following test structures were used for parameter extraction.

Table 2-9 Junction capacitor structures for NMOS

Structures	Area[m^2]	Perimeter[m]	Description
Bottom area junction capacitance	5.00E-08	2.00E-03	Plate junction capacitor
STI bounded sidewall junction capacitance	3.60E-08	3.64E-02	Finger diode
Inner gate sidewall junction	2.75E-08	1.80E-02	Poly finger structure

2.5.2 Measurement conditions

All junction capacitance was measured under voltage sweep conditions summarized in the following table.

Table 2-10 Junction capacitor measurement conditions for NMOS

V_{BIAS} [V]	VH: pwell & VL: n+ From 0 to 3.63 V in 0.033 V step
Frequency [kHz]	100
V_{OSC} [mV]	50
Temp [°C]	-50,25,125

2.5.3 Extracted model parameters and verification plots

CJ, CJSW vs. Vbias plots

The following plots show the variation of bottom area junction and STI bounded sidewall at different bias voltages and temperatures.

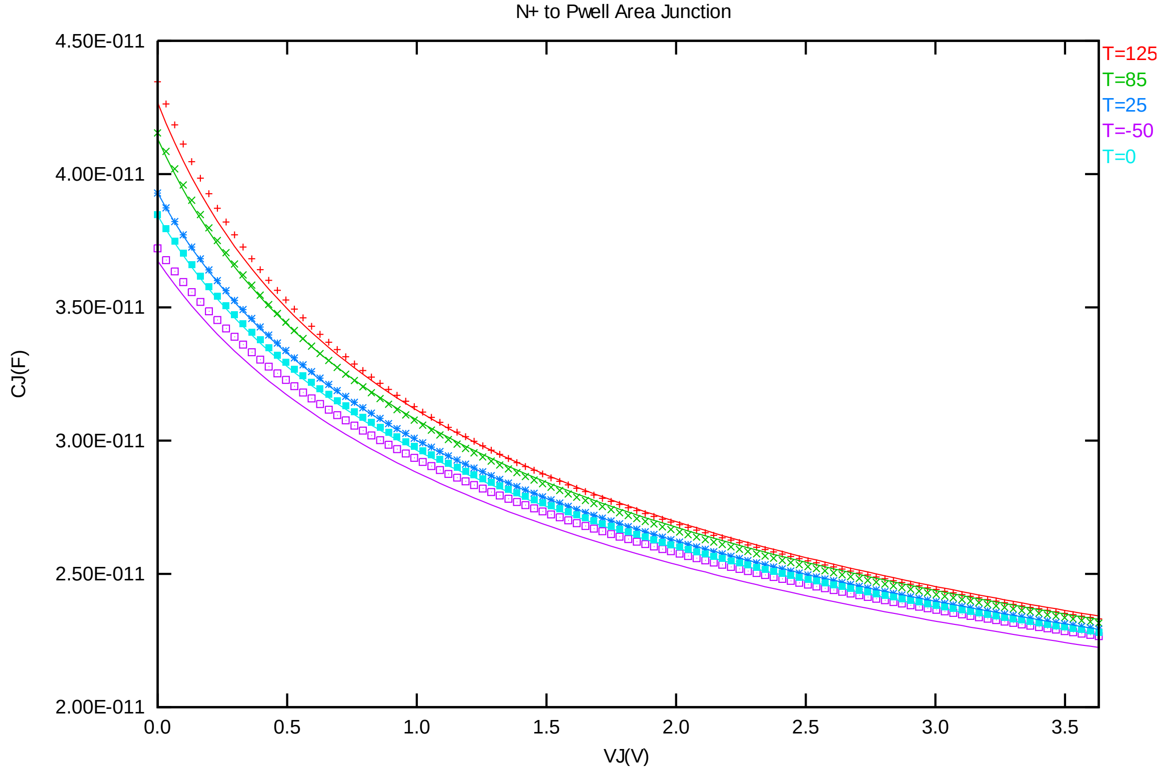

Figure 2-23 Bottom area junction capacitance at -50 C, 0 C, 25 C, 85 C and 125 C for NMOS

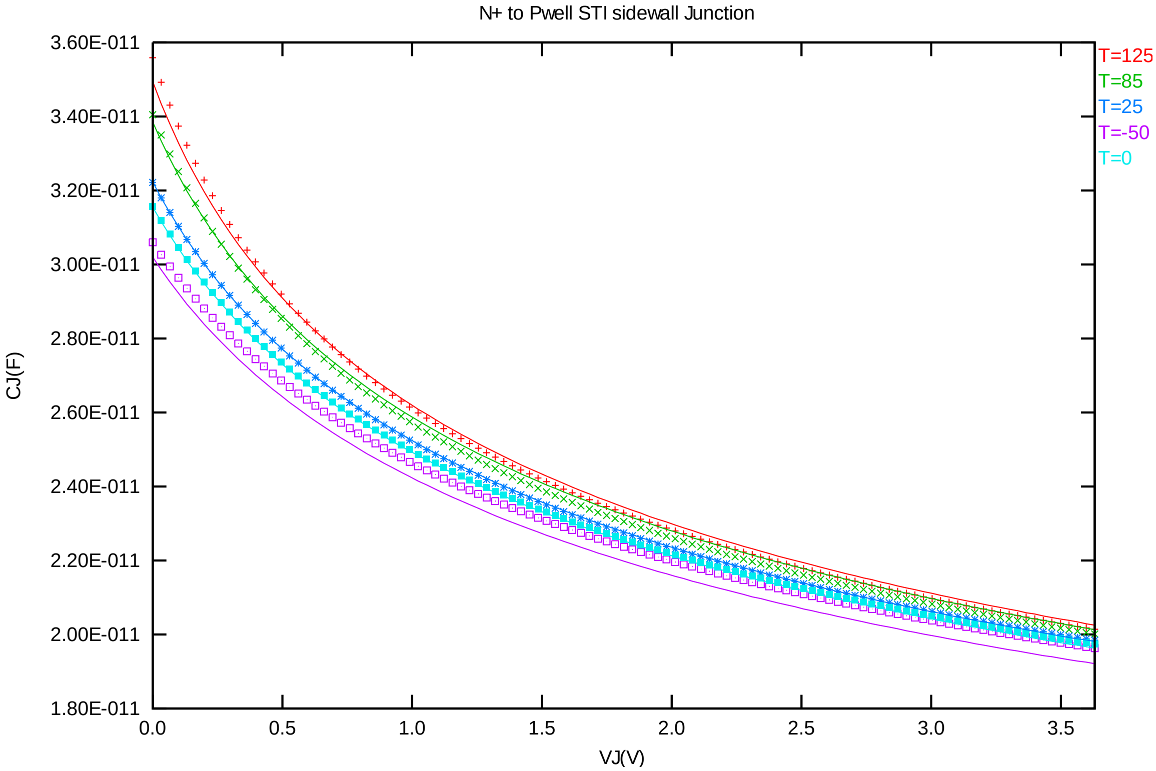

Figure 2-24 STI bounded sidewall junction capacitance at -50 C, 0 C, 25 C, 85 C and 125 C for NMOS

3 1.2V bpw N-ch MOSFET

3.1 Silicon Wafer for Modeling

The information about the silicon wafer used for characterization and model parameter extraction is summarized in the table below.

Table 3-1 Test wafer information for bpw NMOS

	MOSFET DC model	MOSFET AC Model	Junction DC Model	Junction AC Model
Mask ID	BSW0QA	BSW0QA	BSW0QA	BSW0QA
Lot ID	D5FHR.11	D5FHR.11	D5FHR.11	D5FHR.11
Wafer Number	11	11	11	11

3.2 Electrical Parameters

The important electrical parameters for the characterization of this device are summarized in the table below.

Linear and saturation threshold voltage is extracted at
ID =300nA*WDES*0.9/ (LDES*0.9+12nm) for NMOS.
The dimension is after 90% shrink and 12nm sizing channel length back.

Table 3-2 Electrical parameters for bpw NMOS

All parameters are given for T=25 ^{o}C , V_{BS}=0 V unless specified otherwise.

Target & Centered Value are given for SA=2.2u SB=2.2u

Measured & Extracted Value are given for SA=2.2u SB=2.2u

Parameter	Condition	Unit	Target	Measured Value	Extracted Value	Centered Value
Long channel device (W/L= 10 m m / 10 m m)						
V_{TLIN}	V_{DS} = 0.1 V	V	0.512	0.510	0.511	0.512
I_{ON}	V_{DS}=V_{GS}= 1.2 V	mA/ mm	3.70	3.70	3.72	3.70
Wide and short channel device (W/L= 10 m m / 0.246 m m)						
V_{TLIN}	V_{DS} = 0.1 V	V	0.503	0.501	0.505	0.503
V_{TSAT}	V_{DS} = 1.2 V	V	0.458	0.456	0.465	0.458
I_{ON}	V_{DS}=V_{GS}= 1.2 V	mA/mm	150.4	150.7	150.5	150.4
I_{OFF}	V_{DS}= 1.2 V, V_{GS}=0 V	A/mm	2.4E-12*	2.4E-12	1.9E-12	2.4E-12*
Narrow and long channel device (W/L= 0.35 m m / 10 m m)						
V_{TLIN}	V_{DS} = 0.1 V	V	0.510	0.506	0.507	0.510
V_{TSAT}	V_{DS}= 1.2 V	V	0.506	0.502	0.505	0.506
I_{ON}	V_{DS}=V_{GS}= 1.2 V	mA/mm	3.31	3.33	3.28	3.31
Narrow and short channel device (W/L= 0.35 mm / 0.246 mm)						
V_{TLIN}	V_{DS} = 0.1 V	V	0.492	0.491	0.493	0.492
V_{TSAT}	V_{DS}= 1.2 V	V	0.443	0.447	0.454	0.443
I_{ON}	V_{DS}=V_{GS}= 1.2 V	mA/mm	174.0	173.3	175.5	174.0
I_{OFF}	V_{DS}= 1.2 V, V_{GS}=0 V	A/mm	3.0E-12	3.2E-12	3.0E-12	3.0E-12*

Capacitance

CJ	V_J = 0 V	F/m^2	--	7.812E-04	7.812E-04	7.812E-04
CJSW	V_J = 0 V	F/m	--	1.125E-10	1.125E-10	1.125E-10
CJSWG	V_J = 0 V	F/m	--	2.250E-10	2.155E-10	2.155E-10
C_{OVL}	V_{GS} = -0.5 V	F/m	--	2.65E-10	2.55E-10	2.55E-10

*gmindc=1e-15

3.3 MOSFET DC Model

3.3.1 Test devices

The geometry of devices that were used for MOSFET DC model parameter extraction is summarized below.

Table 3-3 Geometry of bpw NMOS devices measured for DC parameter extraction

W\L	10	1	0.8	0.6	0.507	0.435	0.417	0.399	0.318	0.3	0.282	0.264	0.246
10	X	X	X	X	X	X	X	X	X	X	X	X	X
1											X	X	X
0.8	X	X		X	X		X		X		X	X	X
0.6				X			X		X				X
0.43	X	X			X		X		X		X	X	X
0.35	X			X	X		X		X				X

3.3.2 Measurement conditions

The drain current was measured at various bias conditions summarized in the table below:

Table 3-4 Test conditions for IV curves of bpw NMOS

I_D vs. V_{GS}

	Start	Stop	Step
V_{DS} (V)	0.1	1.2	1.1
V_{GS} (V)	-0.5	1.35	0.025
V_{BS} (V)	0	-1.2	-0.3

I_D vs. V_{DS}

	Start	Stop	Step
V_{DS} (V)	0	1.35	0.025
V_{GS} (V)	0.7	1.2	0.125
V_{BS} (V)	0	-1.2	-1.2

3.3.3 Model verification plots

The simulation results obtained from the extracted model were plotted together with the measurement results for comparison. The L and W in the plots refer to physical dimension (LDES *0.9+12nm and WDES *0.9) respectively.

I_D vs. V_{DS} plots

The following plots show measured and simulated output I-V curves for selected devices at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

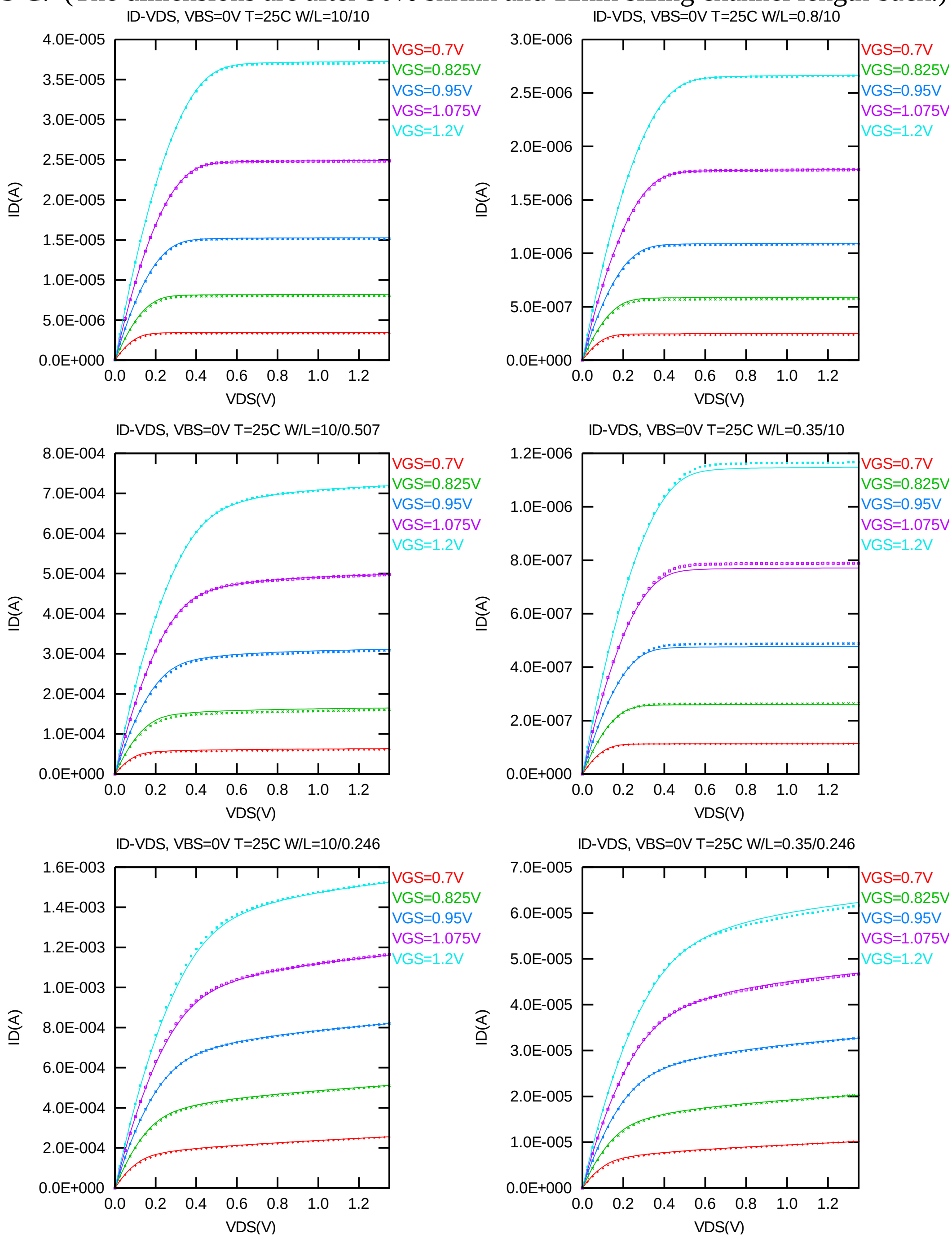

Figure 3-1 I_D vs. V_{DS} at V_{BS} = 0 V for bpw NMOS

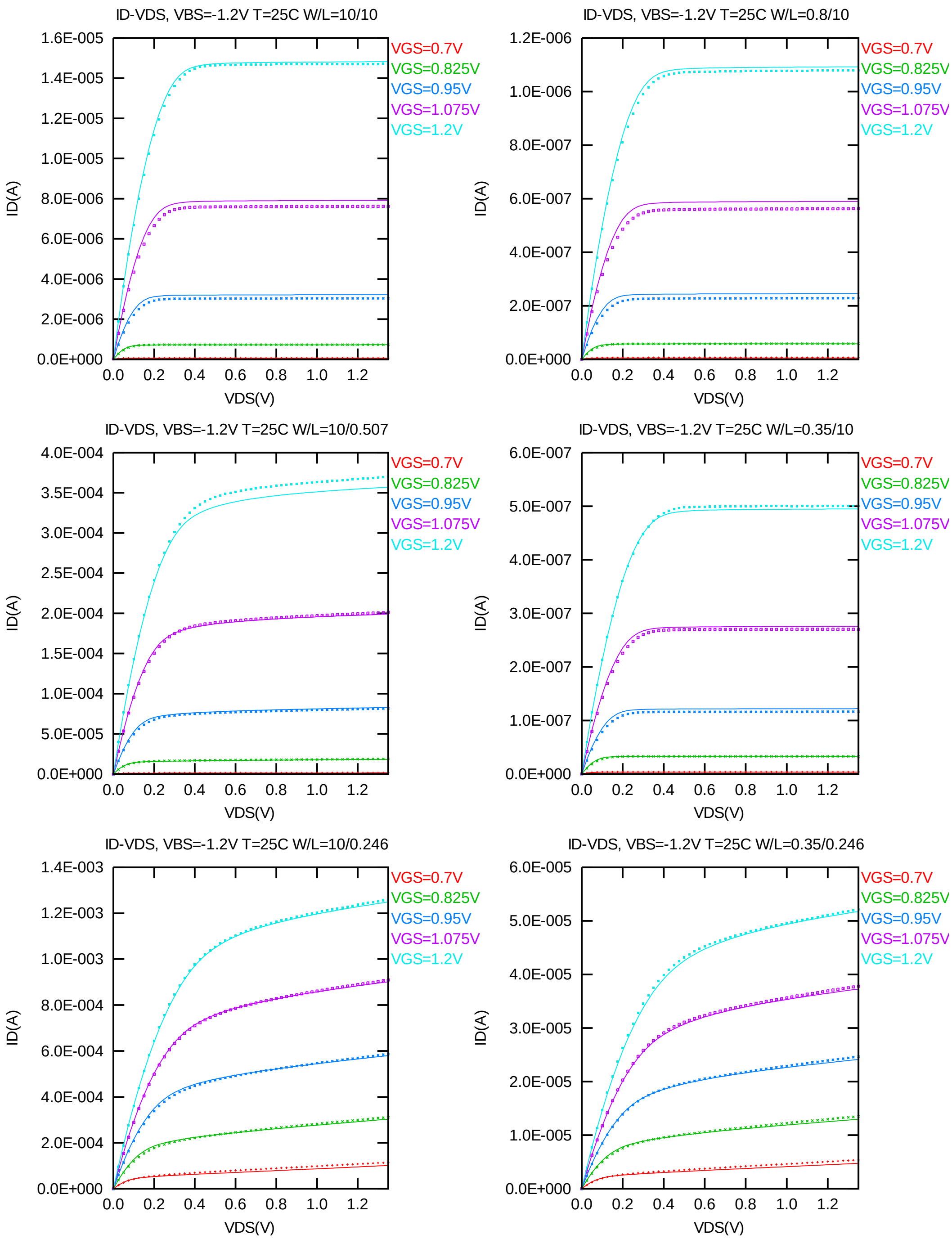

Figure 3-2 I_D vs. V_{DS} at V_{BS} = -1.2V for bpw NMOS

G_{DS} vs. V_{DS} plots

The following plots show measured and simulated output I-V curves for selected devices at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

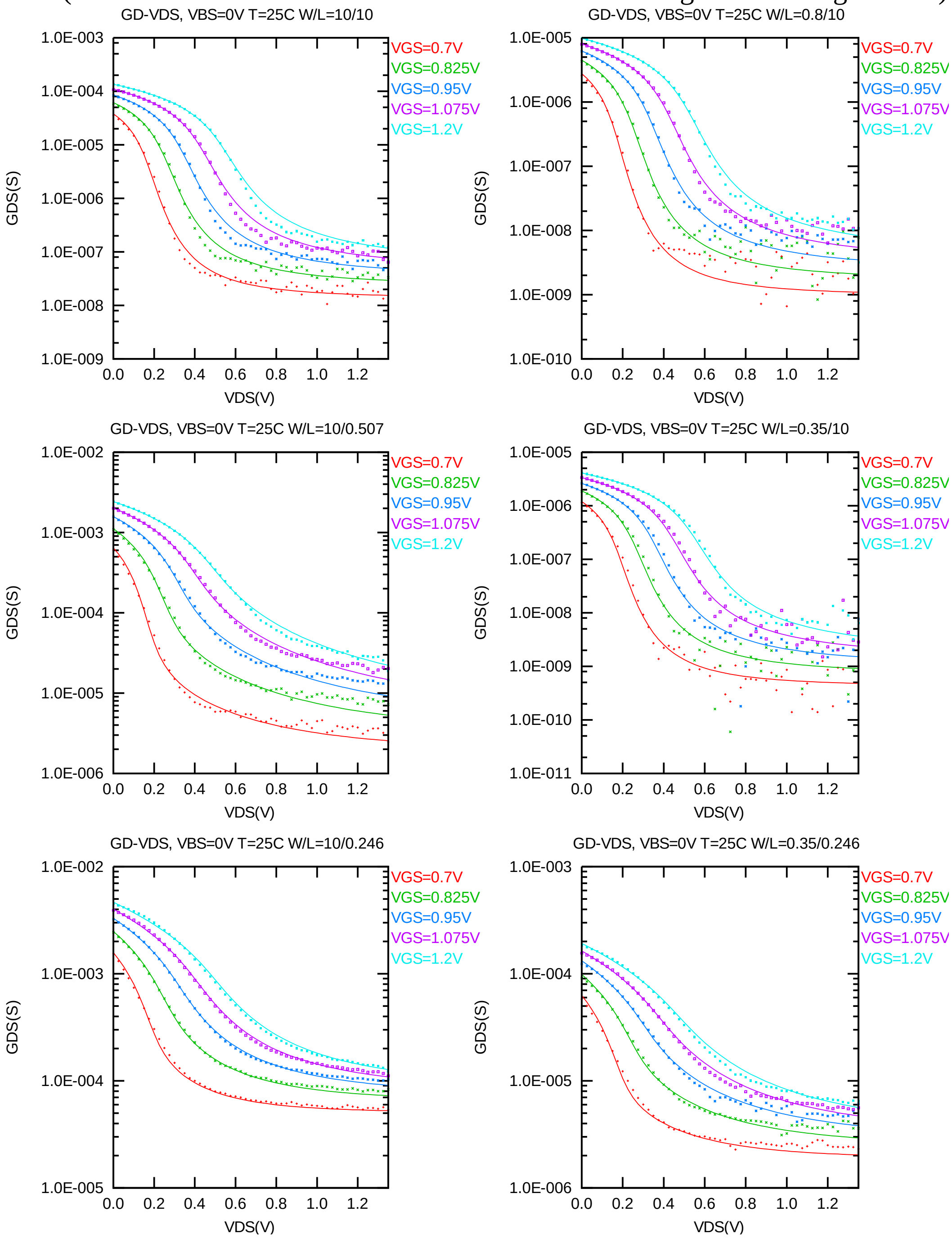

Figure 3-3 G_{DS} vs. V_{DS} at V_{BS} = 0 V for bpw NMOS

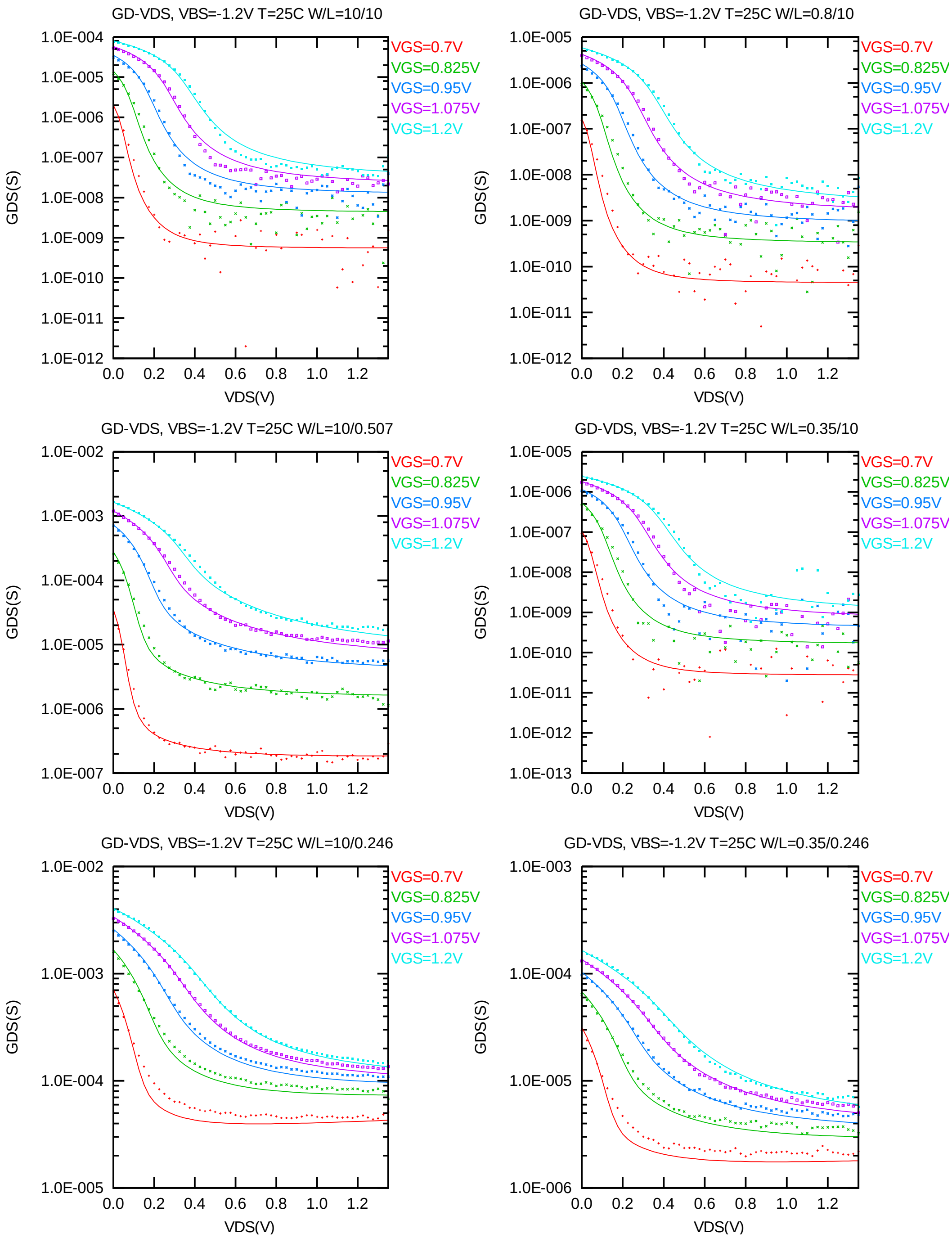

Figure 3-4 G_{DS} vs. V_{DS} at V_{BS} = -1.2V for bpw NMOS

I_D vs. V_{GS} plots (linear scale)

The following plots show measured and simulated output I-V curves for selected devices at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

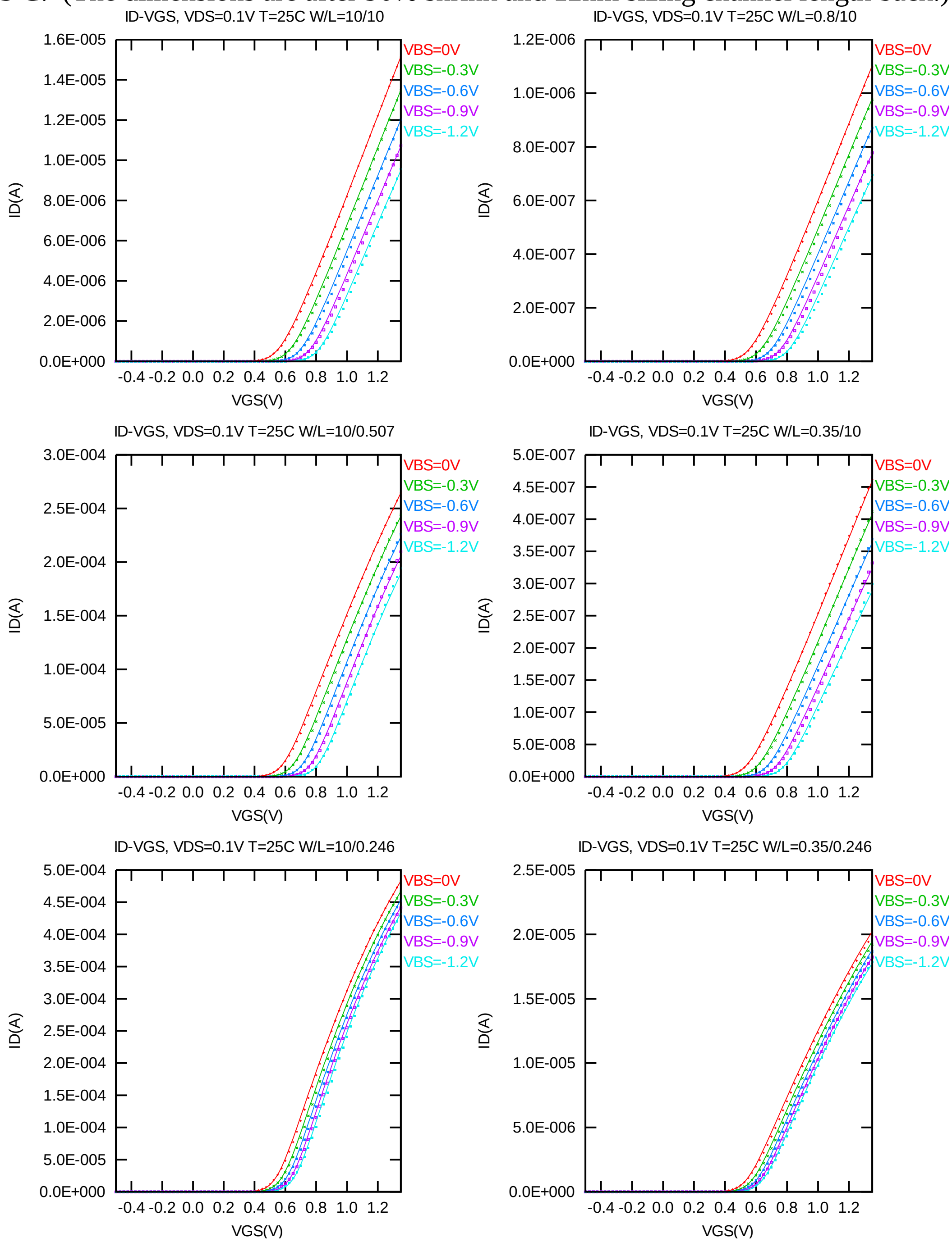

Figure 3-5 I_D vs. V_{GS} at V_{DS} = 0.1 V for bpw NMOS

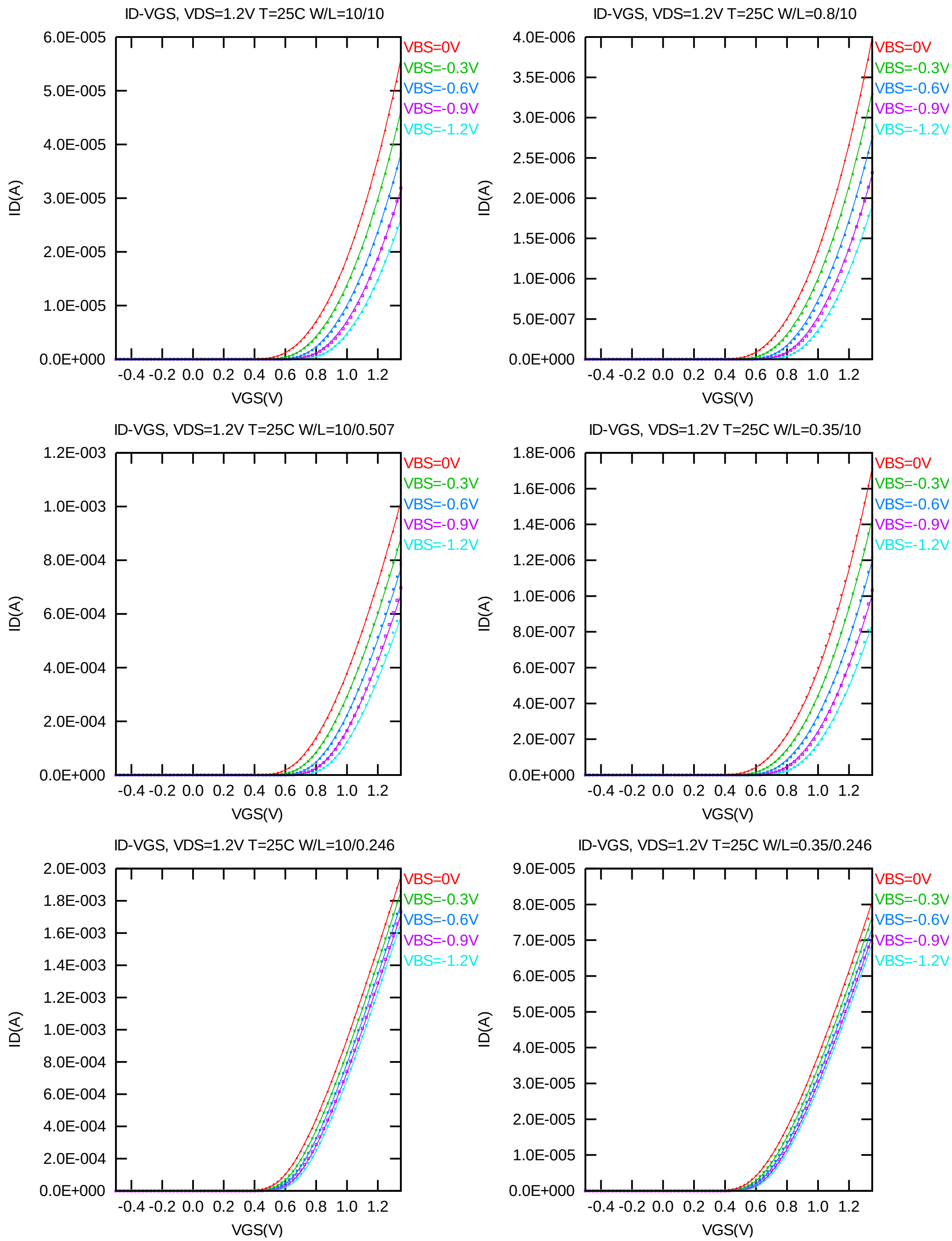

Figure 3-6 I_D vs. V_{GS} at V_{DS} = 1.2V for bpw NMOS

I_D vs. V_{GS} plots (logarithmic scale)

The following plots show measured and simulated output I-V curves for selected devices at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

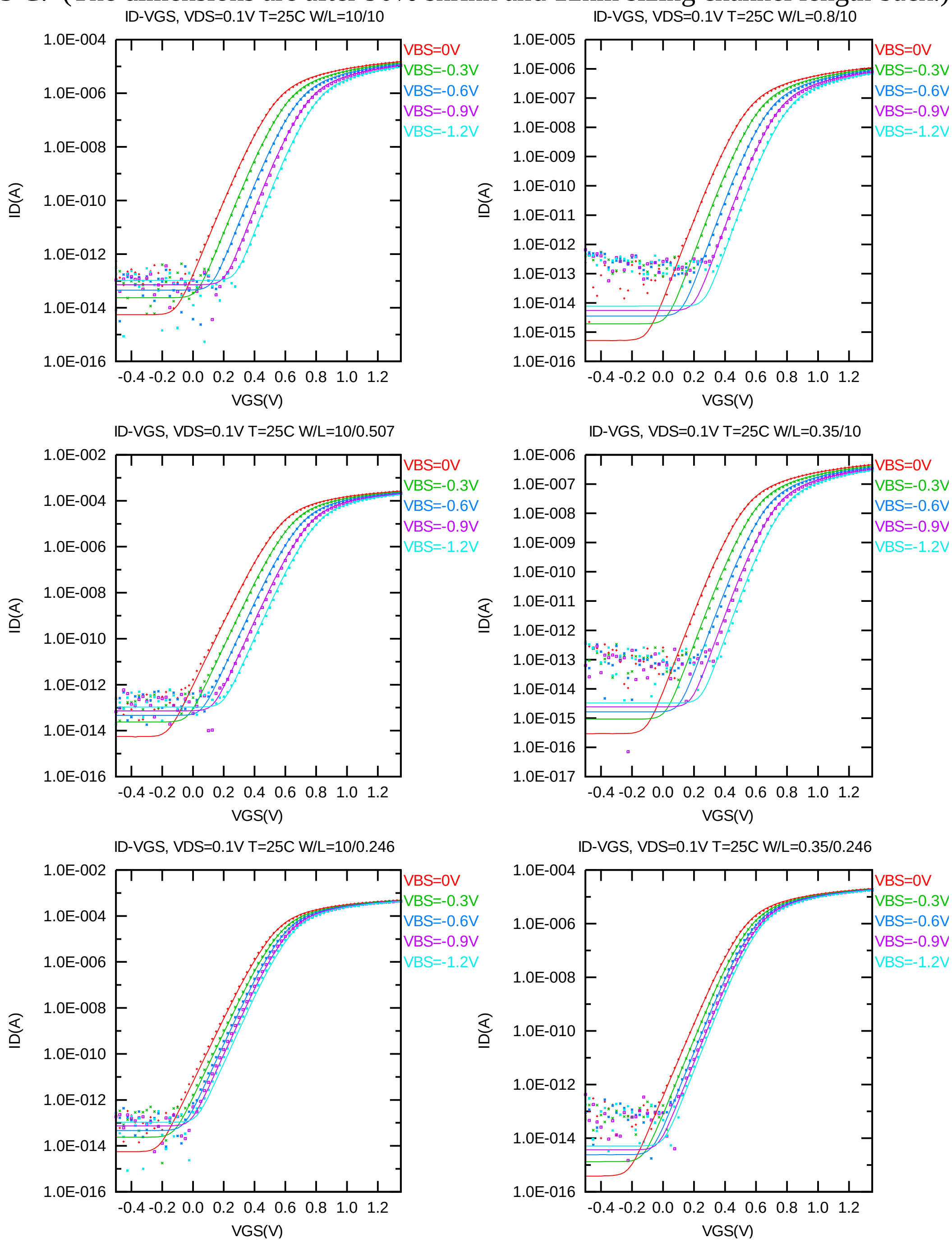

Figure 3-7 Log scaled I_D vs. V_{GS} at V_{DS} = 0.1 V for bpw NMOS

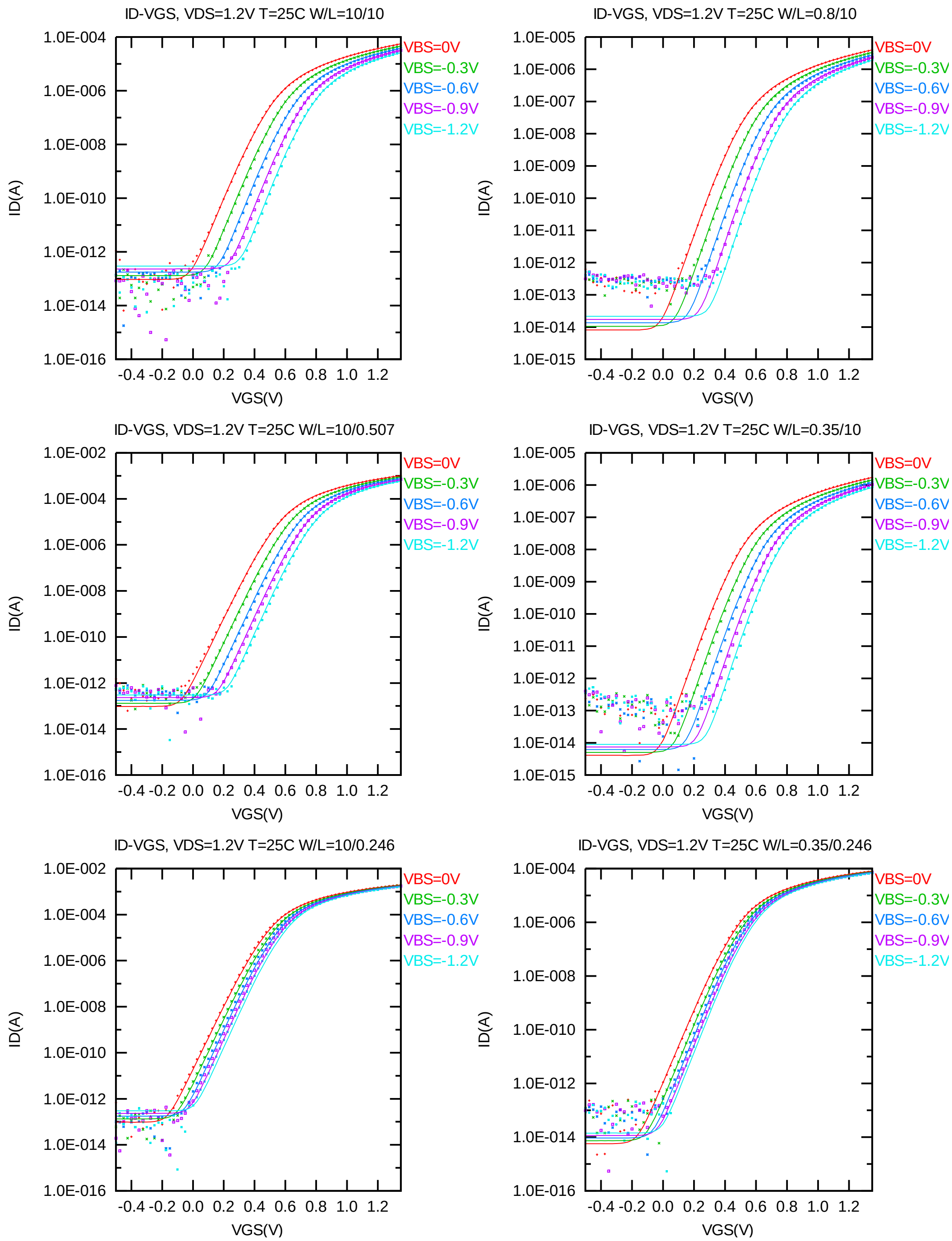

Figure 3-8 Log scaled I_D vs. V_{GS} at V_{DS} = 1.2V for bpw NMOS

G_M vs. V_{GS} plots

The following plots show measured and simulated output I-V curves for selected devices at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

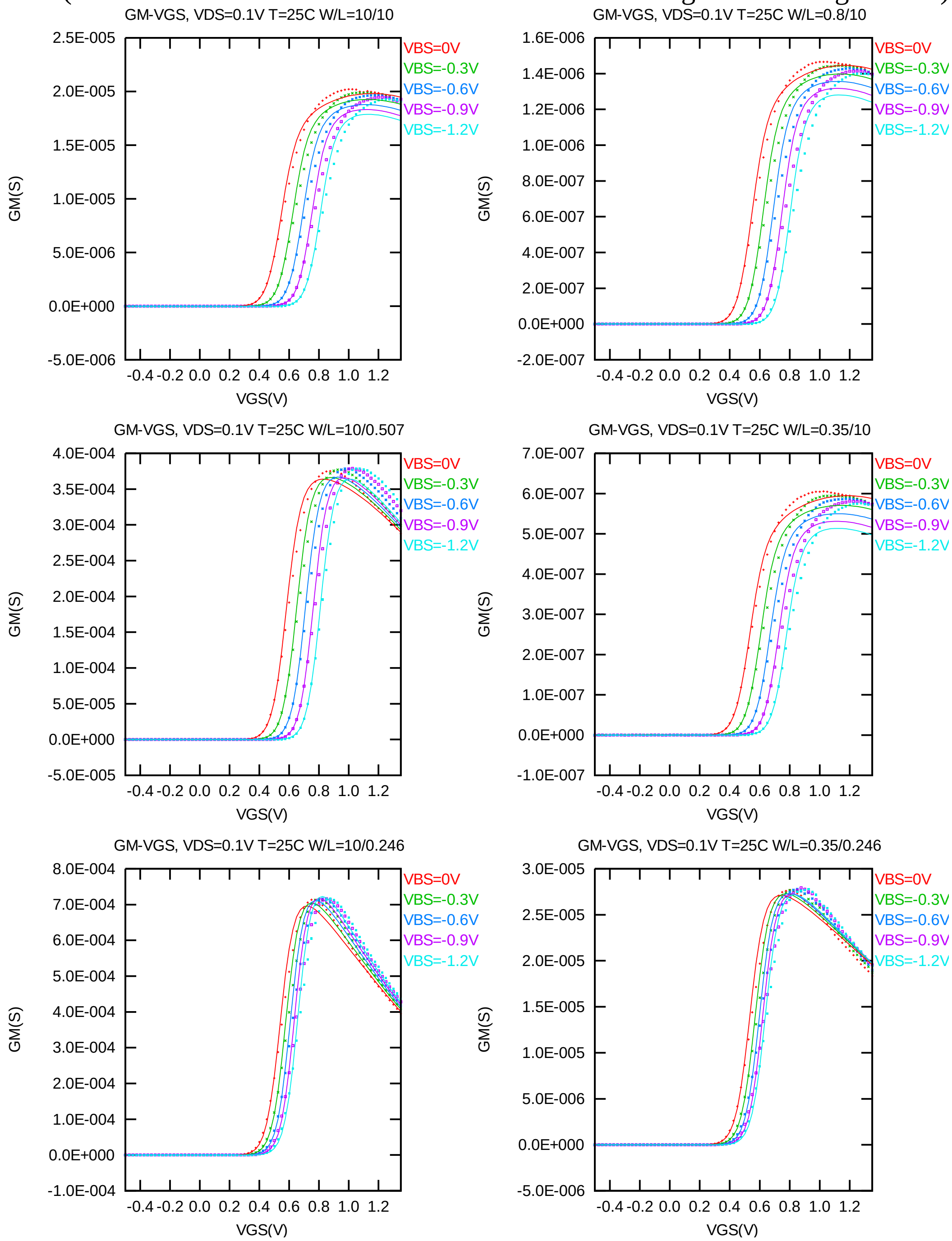

Figure 3-9 G_M vs. V_{GS} at V_{DS} = 0.1 V for bpw NMOS

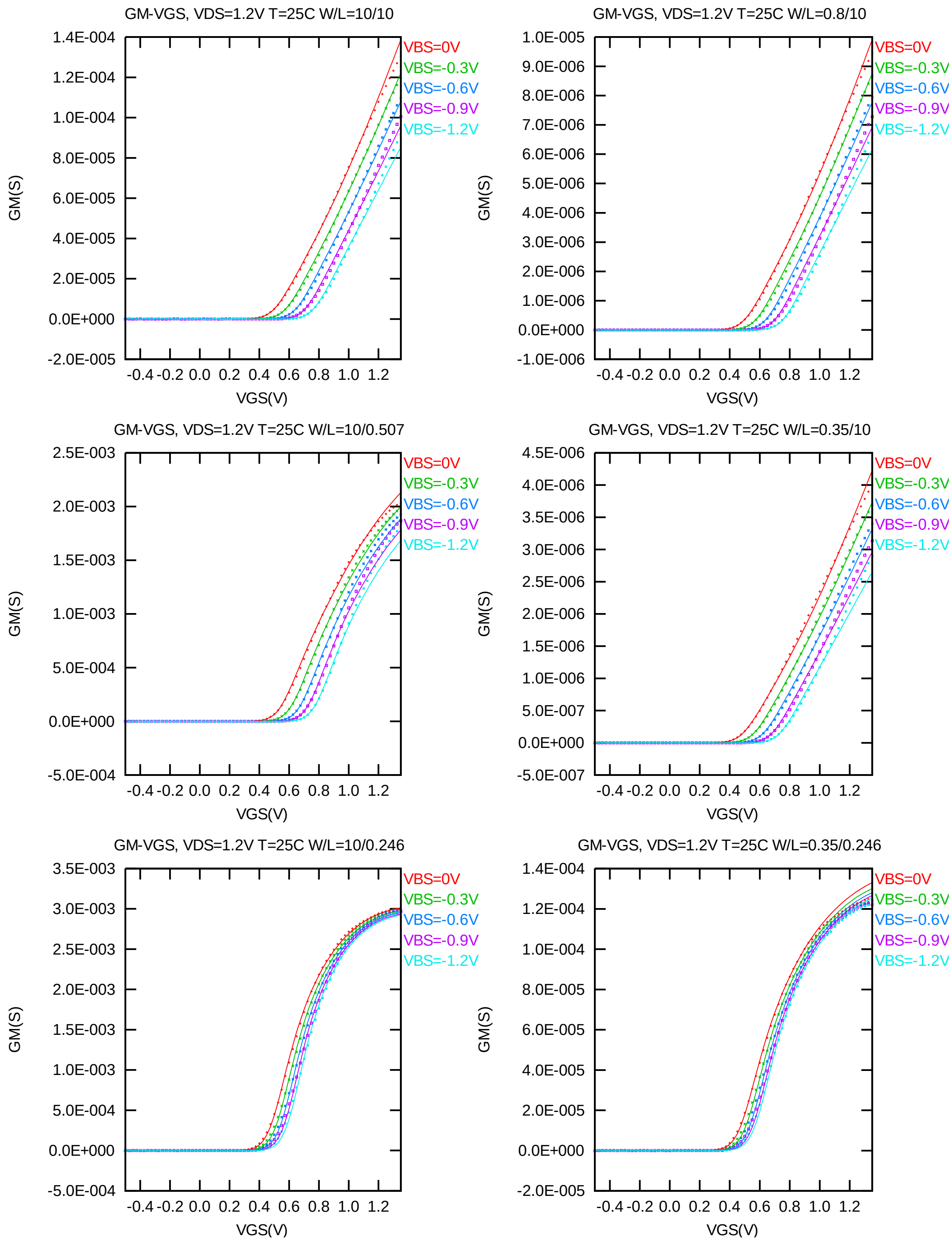

Figure 3-10 G_M vs. V_{GS} at V_{DS} = 1.2V for bpw NMOS

The L and W in the plots refer to LDES and WDES respectively.

The following figures show the channel length and width dependence of linear and saturation threshold voltage extracted at ID =300nA*(WDES*0.9)/(LDES*0.9+0.012 μm) at 25 ℃. The plots show the physical dimension. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

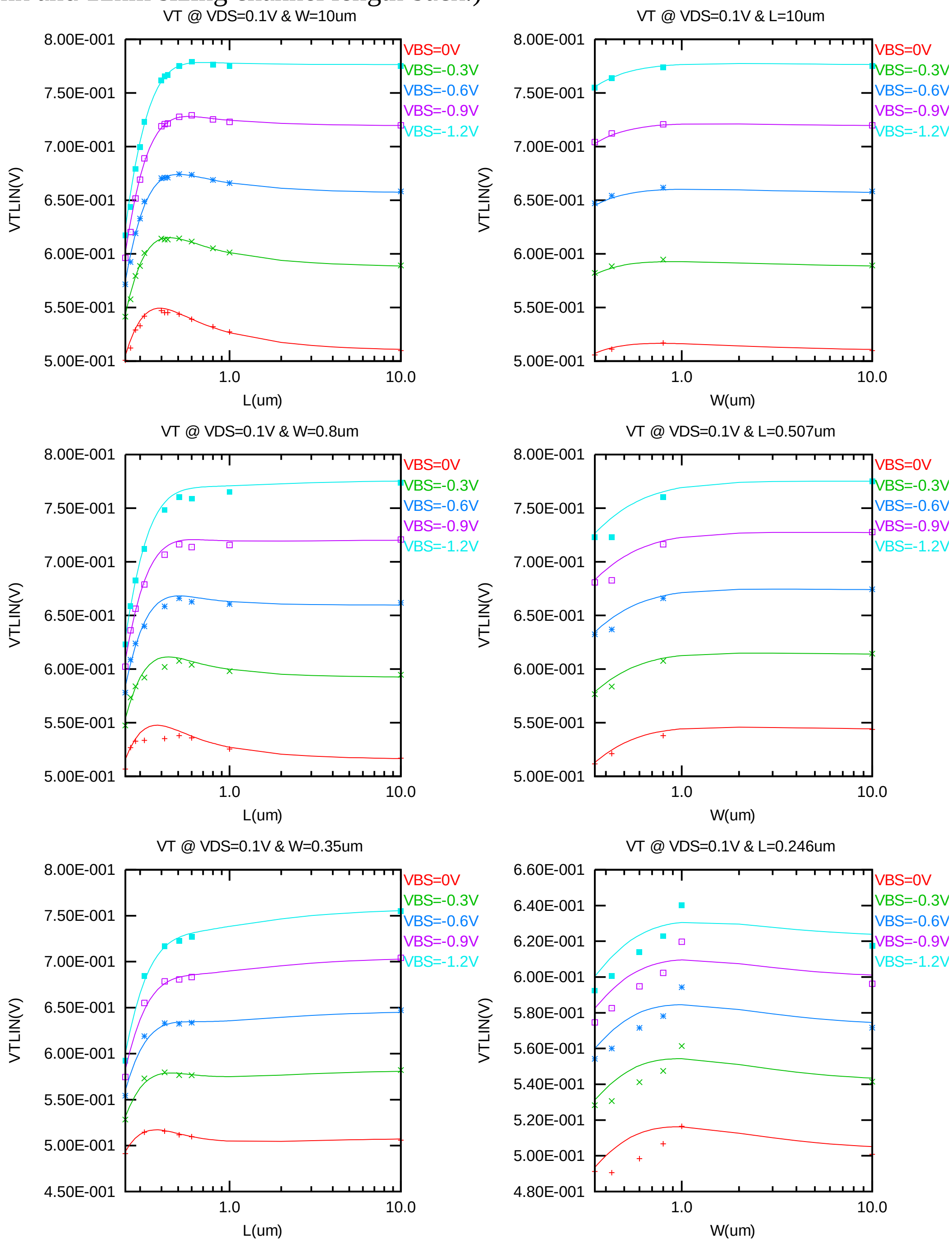

Figure 3-11 V_{TLIN} vs. channel length & channel width for low drain bias (V_{DS} = 0.1 V) for bpw NMOS

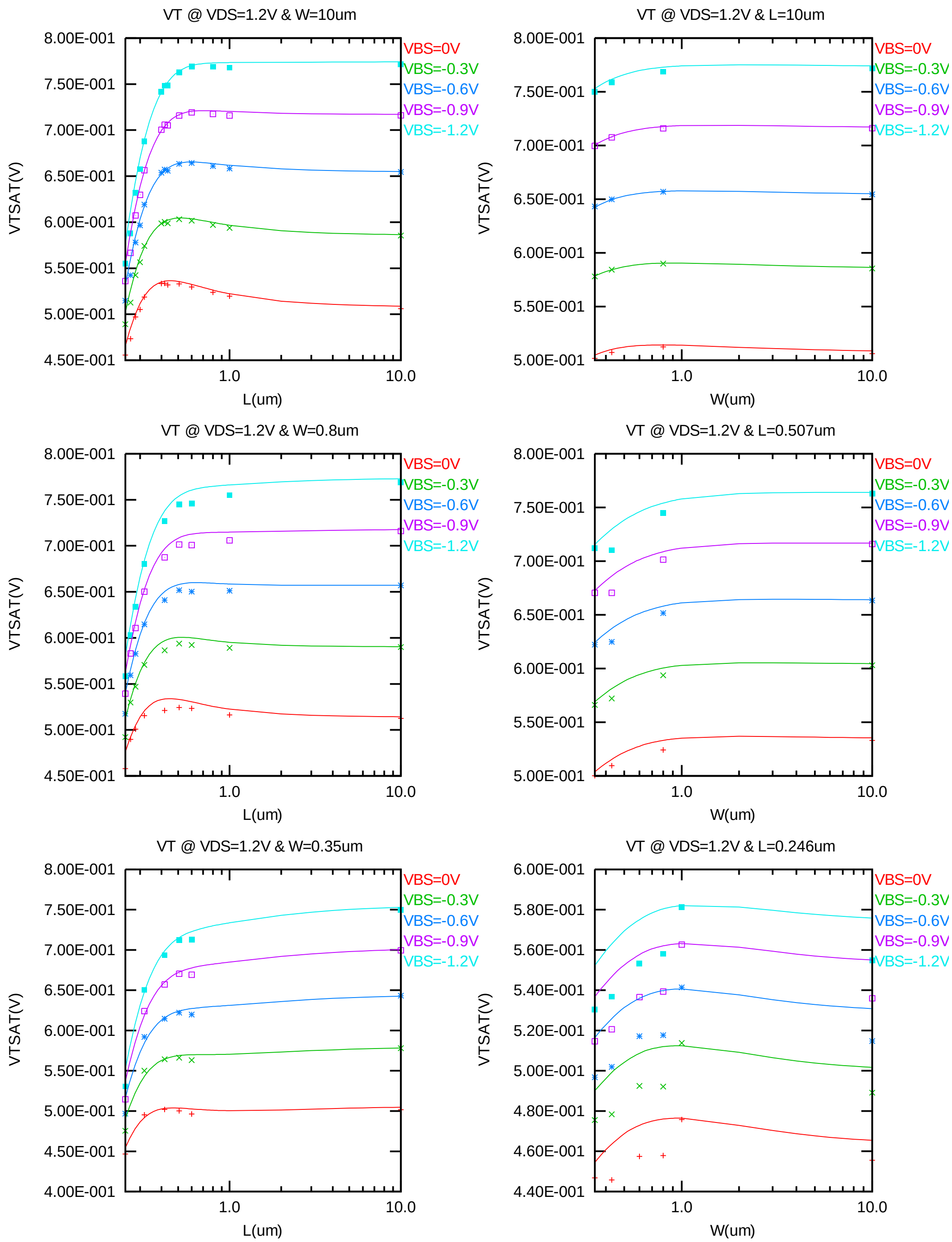

Figure 3-12 V_T vs. channel length & channel width for high drain bias (V_{DS} = 1.2V) for bpw NMOS

The L and W in the plots refer to LDES and WDES respectively.

The following plots show the channel length and width dependence of I_{ON} extracted at $V_{DS} = V_{GS} = 1.2V$ for $V_{BS} = 0$ V to -1.2V at 25 C.(The dimensions are after 90% shrink and 12nm sizing channel length back.)

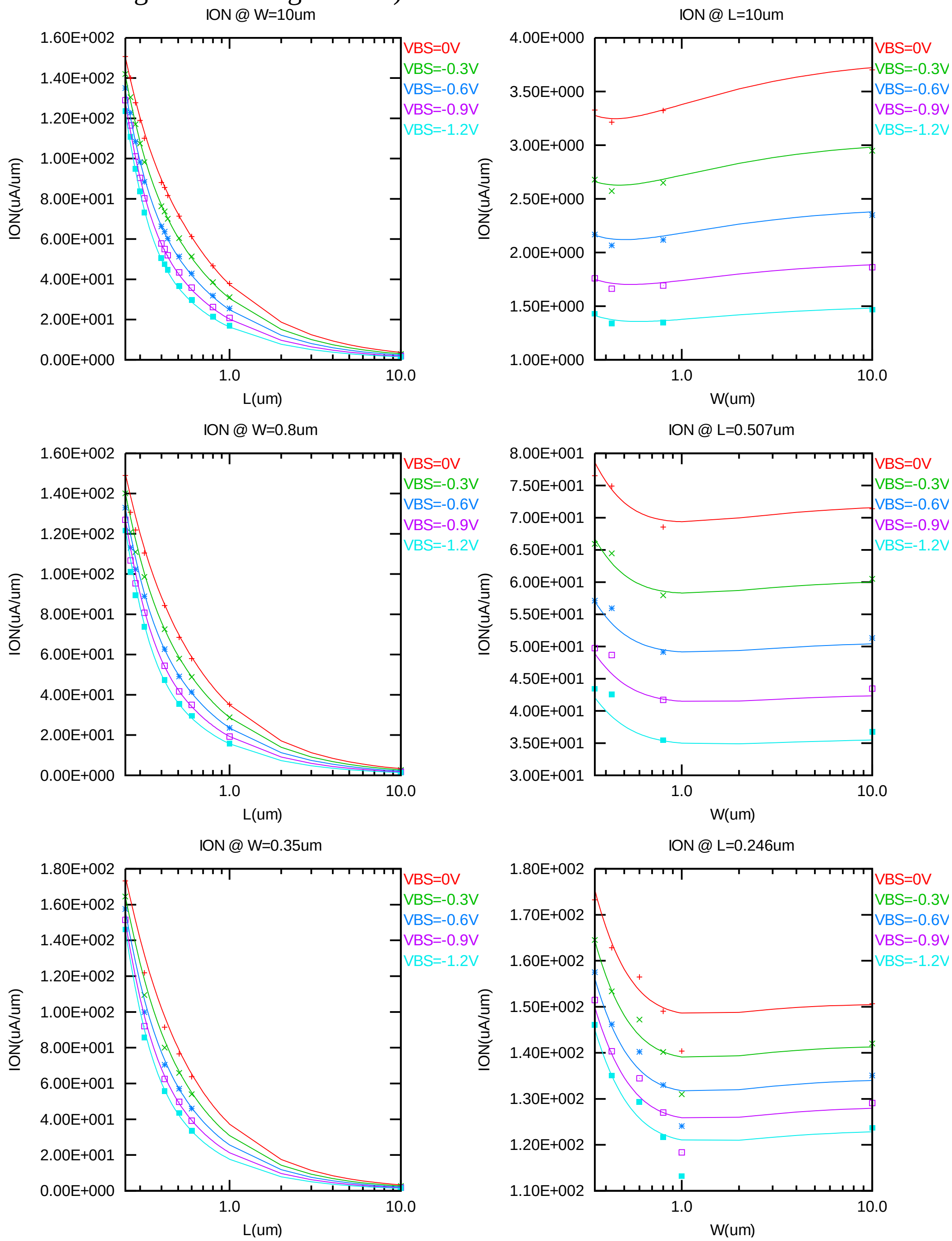

Figure 3-13 Saturation current I_{ON} vs. channel length and channel width for bpw NMOS

The L and W in the plots refer to LDES and WDES respectively.

The following plots show the channel length and width dependence of I_{OFF} extracted at $V_{DS} = V_{CC}$ and $V_{GS} = 0V$ for $V_{BS} = 0$ V to -1.2V at 25 C. A large discrepancy of I_{OFF} can be seen in the graphs, which is due to the presence of junction leakage current.(The dimensions are after 90% shrink and 12nm sizing channel length back.)

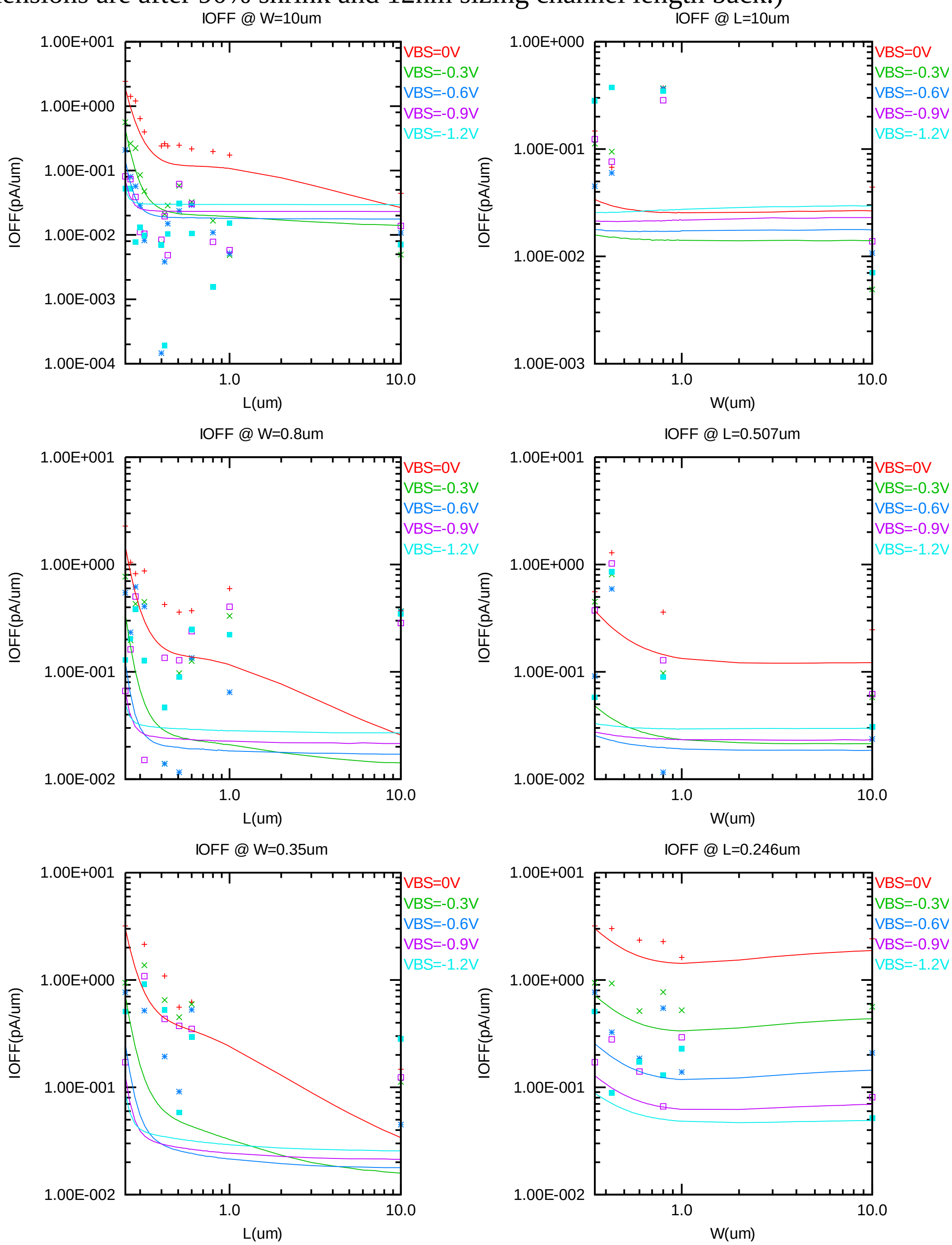

Figure 3-14 I_{OFF} vs. channel length and channel width for bpw NMOS

V_{TLIN} vs. temperature

The following plots show the temperature dependence of the linear and saturation threshold voltage extracted at ID = 300nA*(WDES*0.9)/(LDES*0.9+0.012 μm). (The dimensions are after 90% shrink and 12nm sizing channel length back.)

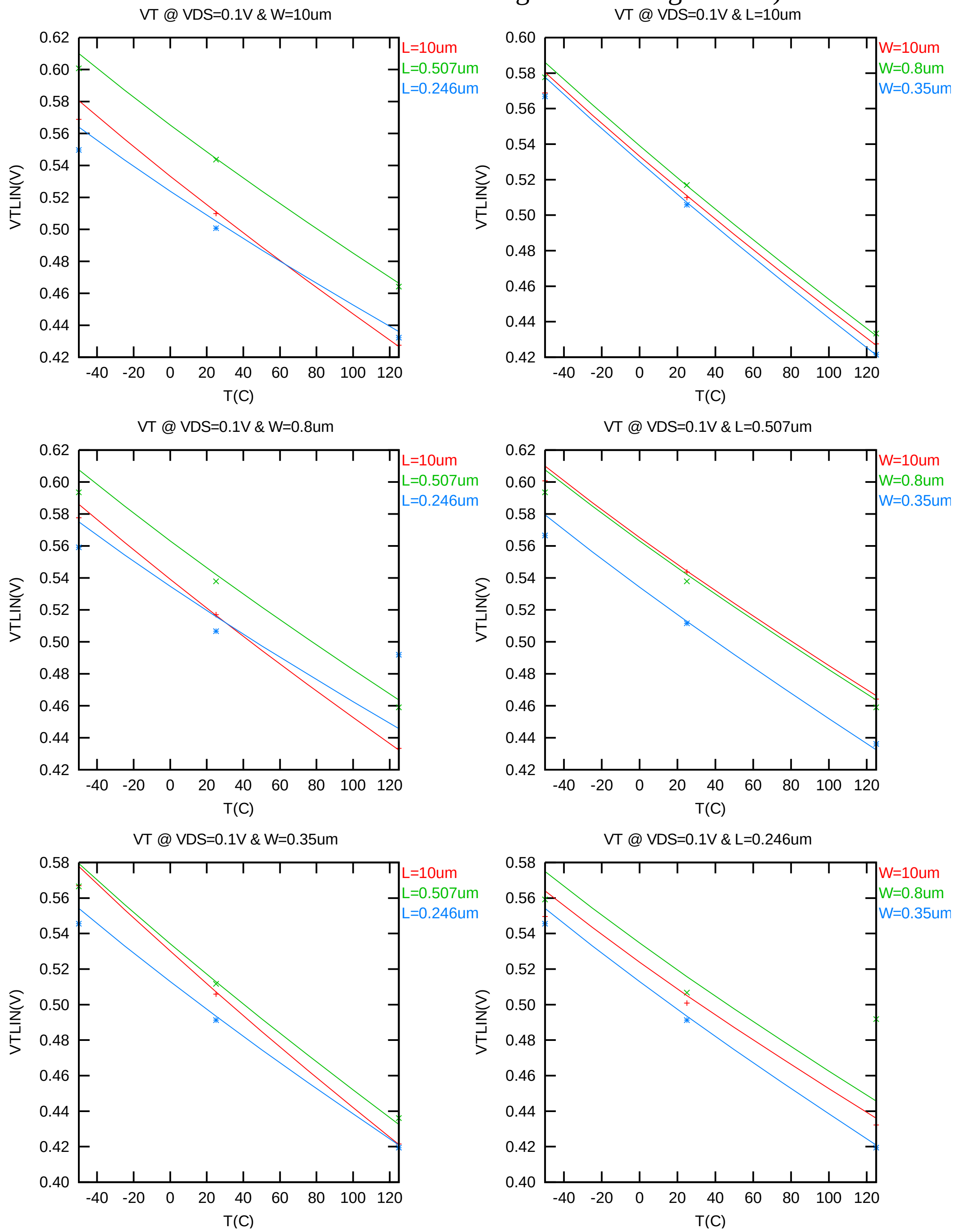

Figure 3-15 V_{TLIN} vs. temperature for bpw NMOS

V_{TSAT} vs. temperature

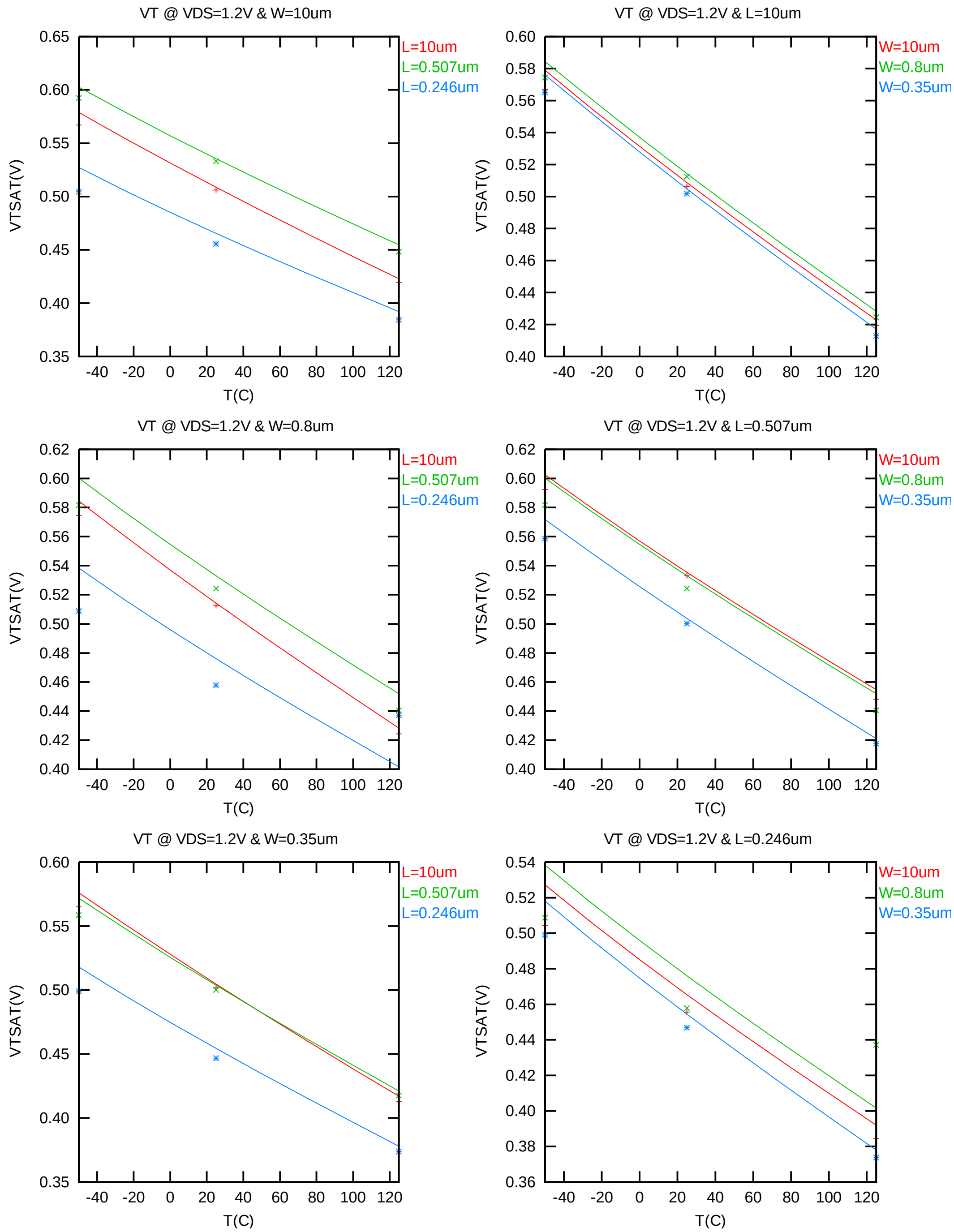

Figure 3-16 V_{TSAT} vs. temperature for bpw NMOS

I_{ON} vs. temperature

The following plots show the temperature dependence of I_{ON} extracted at $V_{GS} = V_{DS} = 1.2V$, $V_{BS} = 0$ V. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

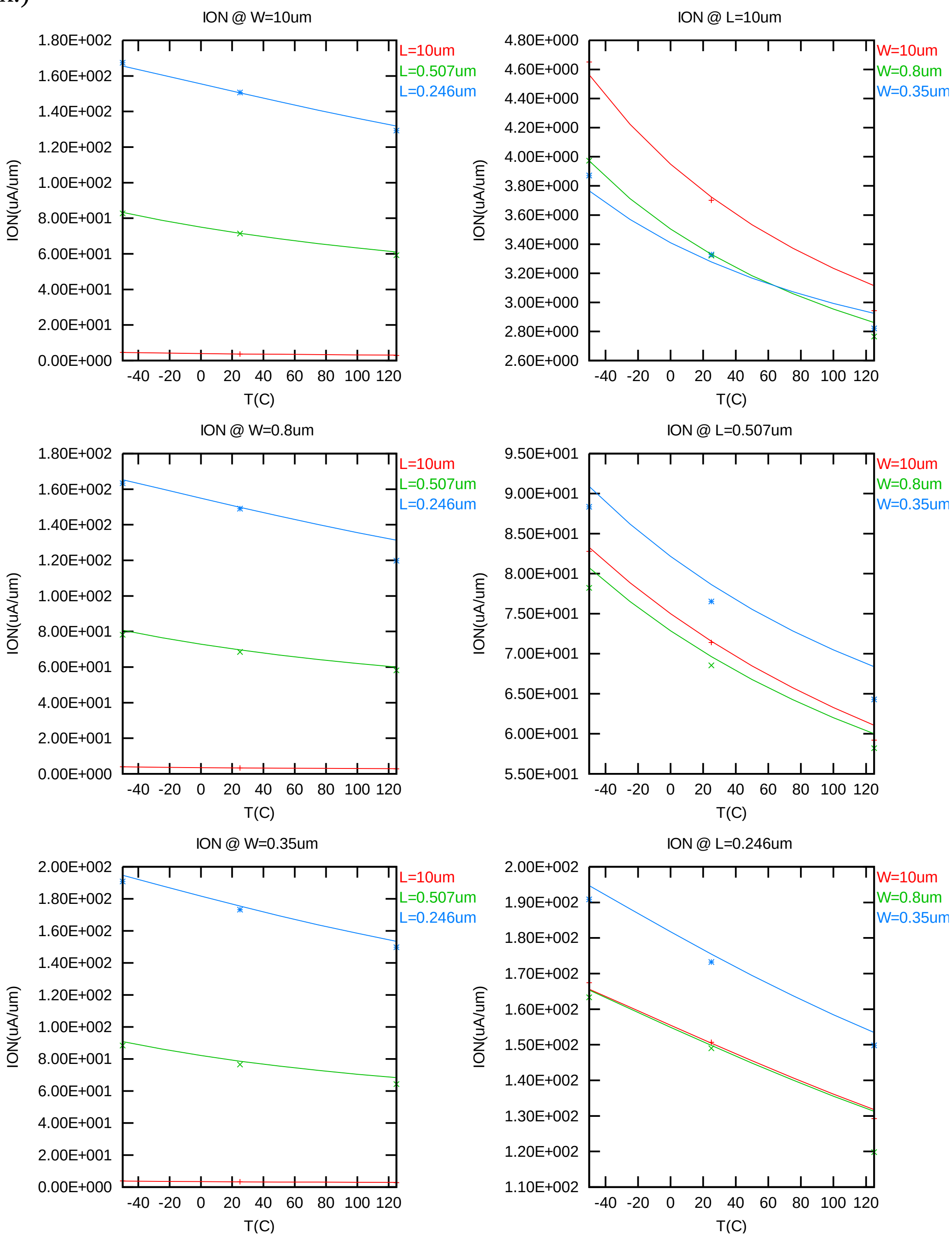

Figure 3-17 Saturation current vs. temperature for bpw NMOS

I_{OFF} vs. temperature

The following plots show the temperature dependence of I_{OFF} extracted at V_{DS} = 1.2V, V_{GS} = V_{BS} = 0 V. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

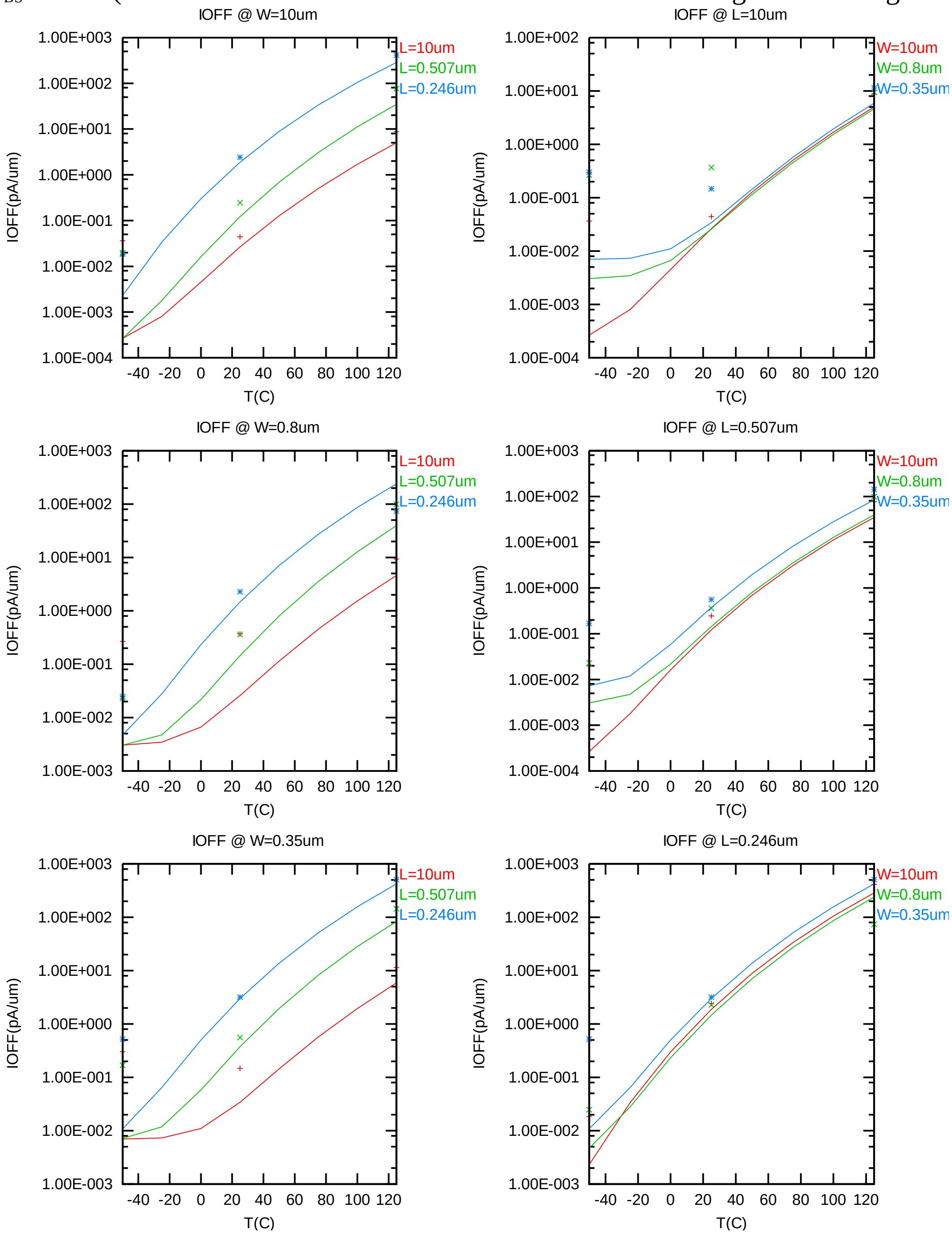

Figure 3-18 I_{OFF} vs. temperature for bpw NMOS

3.3.4 Model fitting accuracy

The model fitting accuracy of parameters I_D, G_M and G_{DS} is quantified using the following equation:

error % = 100 * | measurement - simulation | / [(measurement + simulation) / 2]

Different error criteria are defined for different parameters (I_D, G_M and G_{DS}) as well as for different operation regions. These criteria are explained below.

1) I_D accuracy criteria

f. I_D in sub-threshold region, i.e. $V_{GS} < V_T$
g. I_D in moderate inversion region, i.e. $V_T \leqq V_{GS} < V_T + V_X$
h. I_D in strong inversion region, i.e. $V_T + V_X \leqq V_{GS} \leqq V_{CC}$

, where $V_X = (V_{CC} - V_T) / 2$

2) G_M and G_{DS} accuracy criteria

i. G_M and G_{DS} in sub-threshold region, i.e. $V_{GS} < V_T$
j. G_M and G_{DS} in inversion region, i.e. $V_T \leqq V_{GS} \leqq V_{CC}$

Table 3-5 Requirements for model fit quality

Target of Quality Check	$V_{GS}< V_T$ (sub-threshold)	$V_T \le V_{GS} < V_T+V_X$ (moderate inversion)	$V_T+V_X \le V_{GS} \le V_{CC}$ (strong inversion)
I_D	$\le$75%	$\le$25%	$\le$7%
G_M	$\le$50%	$\le$20%	$\le$20%
G_{DS}	$\le$100%	$\le$50%	$\le$50%

I_D accuracy

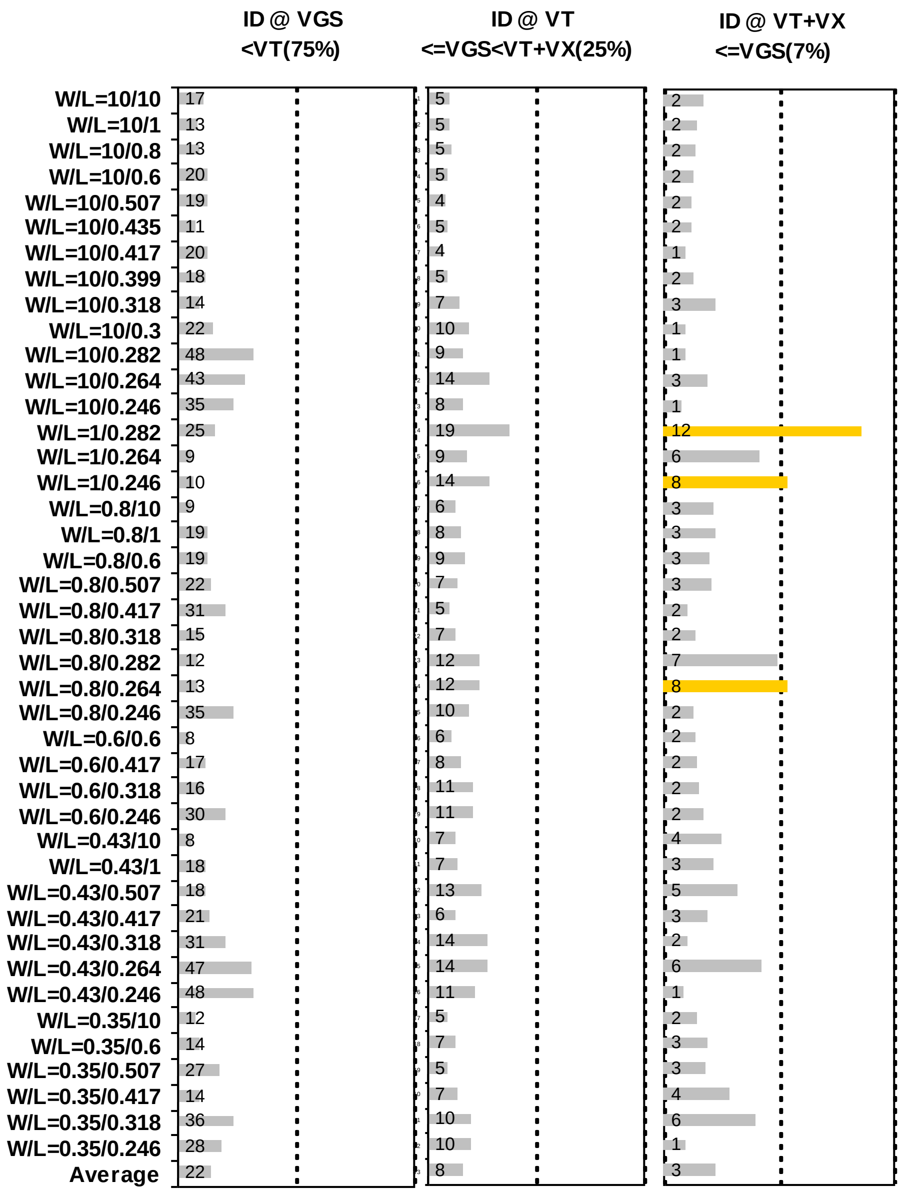

Figure 3-19 I_D accuracy at 25 C for bpw NMOS

G_M and G_{DS} accuracy

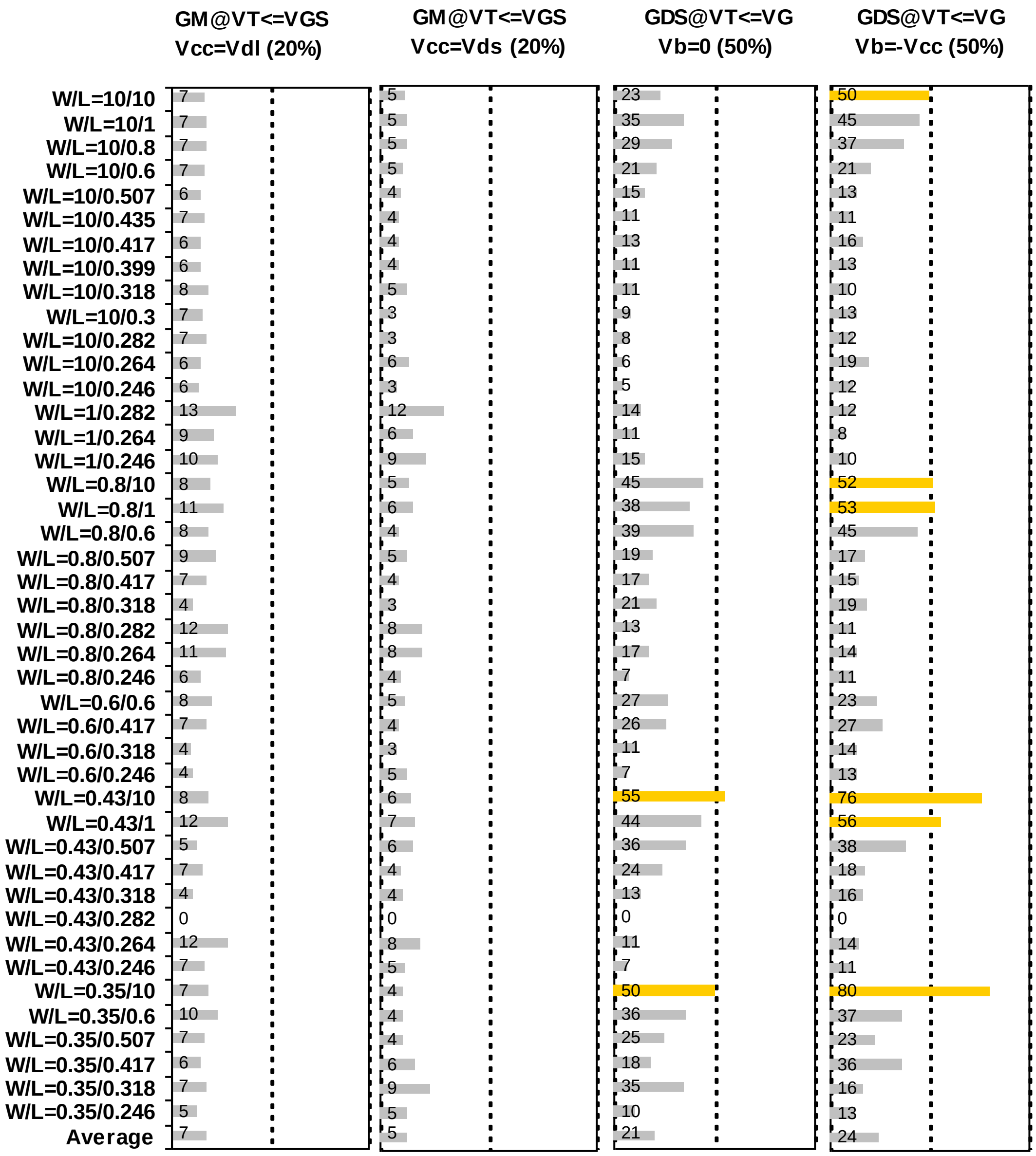

Figure 3-20 G_M and G_{DS} accuracy at 25 C for bpw NMOS

4 1.2V P-ch MOSFET

4.1 Silicon Wafer for Modeling

The information about the silicon wafer used for characterization and model parameter extraction is summarized in the table below.

Table 4-1 Test wafer information for PMOS

	MOSFET DC model	MOSFET AC Model	Junction DC Model	Junction AC Model
Mask ID	BSW0QA	BSW0QA	BSW0QA	BSW0QA
Lot ID	D5FHR.11	D5FHR.11	D5FHR.11	D5FHR.11
Wafer Number	11	11	11	11

4.2 Electrical Parameters

The important electrical parameters for the characterization of this device are summarized in the table below.

Linear and saturation threshold voltage is extracted at
ID =300nA*WDES*0.9/(LDES*0.9+12nm) for NMOS and
ID =-70nA*WDES*0.9/(LDES*0.9+12nm) for PMOS, respectively.
The dimension is after 90% shrink and 12nm sizing channel length back.

All parameters are given for T=25 C, V_{BS}=0 V unless specified otherwise.

Table 4-2 Electrical parameters for PMOS

All parameters are given for T=25 ^oC , V_{BS}=0 V unless specified otherwise.

Target & Centered Value are given for SA=2.2u SB=2.2u

Measured & Extracted Value are given for SA=2.2u SB=2.2u

Parameter	Condition	Unit	Target	Measured Value	Extracted Value	Centered Value

Long channel device (W/L= 10 μ m / 10 μ m)

V_{TLIN}	V_{DS} = -0.1 V	V	-0.498	-0.496	-0.498	-0.498
I_{ON}	V_{DS}=V_{GS}= -1.2 V	μA/ μm	-0.91	-0.91	-0.91	-0.91

Wide and short channel device (W/L= 10 μ m / 0.246 μ m)

V_{TLIN}	V_{DS} = -0.1 V	V	-0.438	-0.438	-0.438	-0.438
V_{TSAT}	V_{DS} = -1.2 V	V	-0.418	-0.416	-0.417	-0.418
I_{ON}	V_{DS}=V_{GS}= -1.2 V	μA/μm	-51.25	-51.40	-51.25	-51.25
I_{OFF}	V_{DS}= -1.2 V, V_{GS}=0 V	A/μm	--	-1.6E-12	-5.4E-12	-5.2E-12 *

Narrow and long channel device (W/L= 0.35 μ m / 10 μ m)

V_{TLIN}	V_{DS} = -0.1 V	V	-0.494	-0.488	-0.491	-0.494
V_{TSAT}	V_{DS}= -1.2 V	V	-0.491	-0.484	-0.491	-0.493
I_{ON}	V_{DS}=V_{GS}= -1.2 V	μA/μm	-0.87	-0.86	-0.89	-0.87

Narrow and short channel device (W/L= 0.35 μm / 0.246 μm)

V_{TLIN}	V_{DS} = -0.1 V	V	-0.440	-0.439	-0.439	-0.440
V_{TSAT}	V_{DS}= -1.2 V	V	-0.421	-0.421	-0.419	-0.420
I_{ON}	V_{DS}=V_{GS}= -1.2 V	μA/μm	-45.03	-45.33	-44.55	-45.02
I_{OFF}	V_{DS}= -1.2 V, V_{GS}=0 V	A/μm	--	-3.7E-11	-5.4E-12	-5.2E-12*

Capacitance

CJS	V_J = 0 V	F/m^2	- -	9.63E-04	9.63E-04	9.63E-04
CJSWS	V_J = 0 V	F/m	- -	1.05E-10	1.05E-10	1.05E-10
CJSWGS	V_J = 0 V	F/m	- -	3.37E-10	3.37E-10	3.37E-10
C_{OVL}	V_{GS} = 0.5 V	F/m	- -	3.21E-10	2.93E-10	2.93E-10

*gmindc=1e-15

4.3 MOSFET DC Model

4.3.1 Test devices

The geometry of devices that were used for MOSFET DC model parameter extraction is summarized below.

Table 4-3 Geometry of PMOS devices measured for DC parameter extraction

W\L	10	1	0.8	0.6	0.507	0.435	0.417	0.399	0.318	0.3	0.282	0.264	0.246
10	X	X	X	X	X	X	X	X	X	X	X	X	X
1											X	X	X
0.8	X	X		X	X		X		X		X	X	X
0.6				X			X		X				X
0.43	X	X			X		X		X		X	X	X
0.35	X			X	X		X		X				X

4.3.2 Measurement conditions

The drain current was measured at various bias conditions summarized in the table below:

Table 4-4 Test conditions for IV curves of PMOS

I_D vs. V_{GS}

	Start	Stop	Step
V_{DS} (V)	-0.1	-1.2	-1.1
V_{GS} (V)	0.5	-1.35	-0.025
V_{BS} (V)	0	1.2	0.3

I_D vs. V_{DS}

	Start	Stop	Step
V_{DS} (V)	0	-1.35	-0.025
V_{GS} (V)	-0.7	-1.2	-0.125
V_{BS} (V)	0	1.2	1.2

4.3.3 Model verification plots

The simulation results obtained from the extracted model were plotted together with the measurement results for comparison. The L and W in the plots refer to physical dimension (LDES *0.9+12nm and WDES *0.9) respectively.

I_D vs. V_{DS} plots

The following plots show measured and simulated output I-V curves for selected devices at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

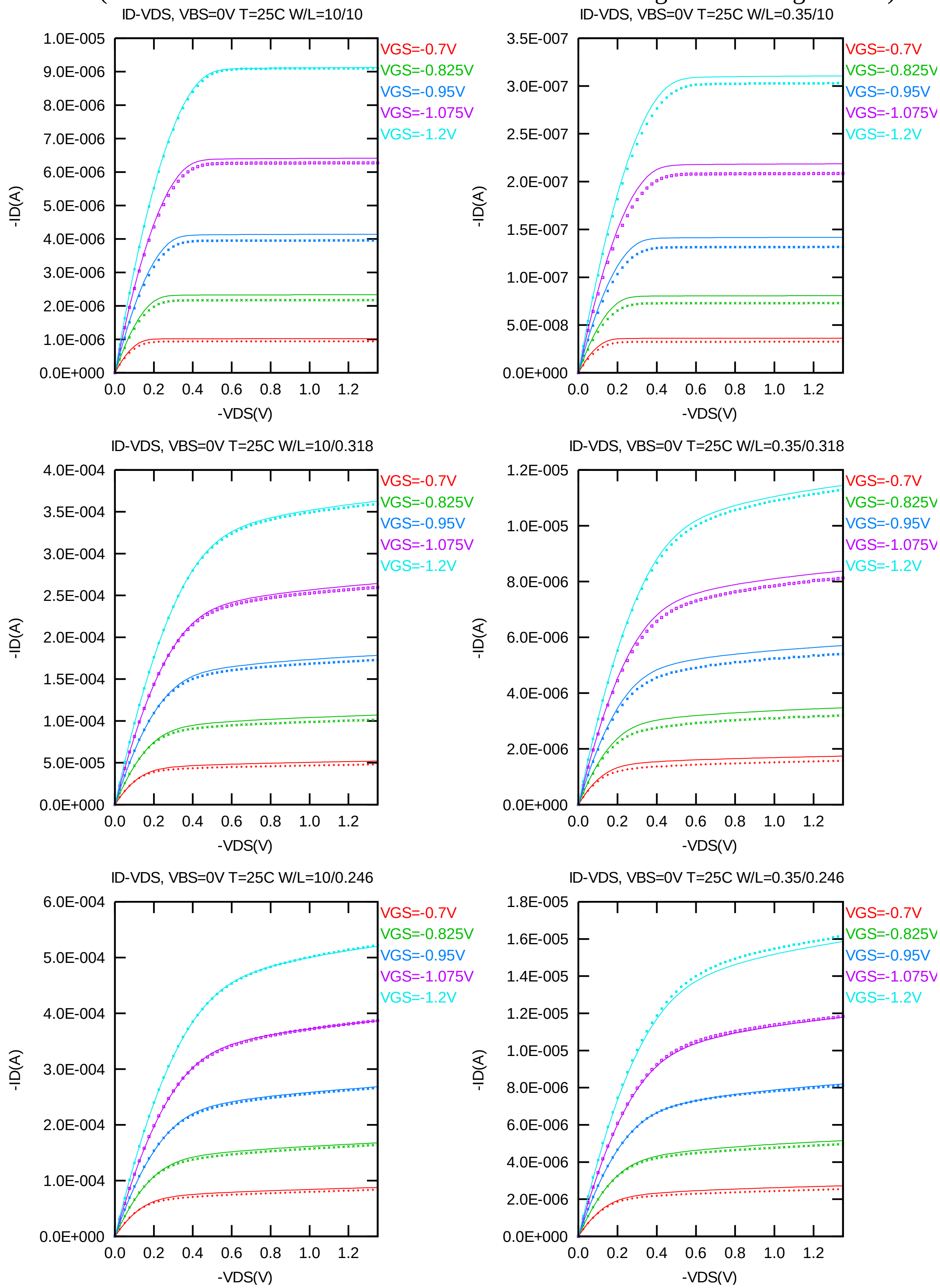

Figure 4-1 I_D vs. V_{DS} at V_{BS} = 0 V for PMOS

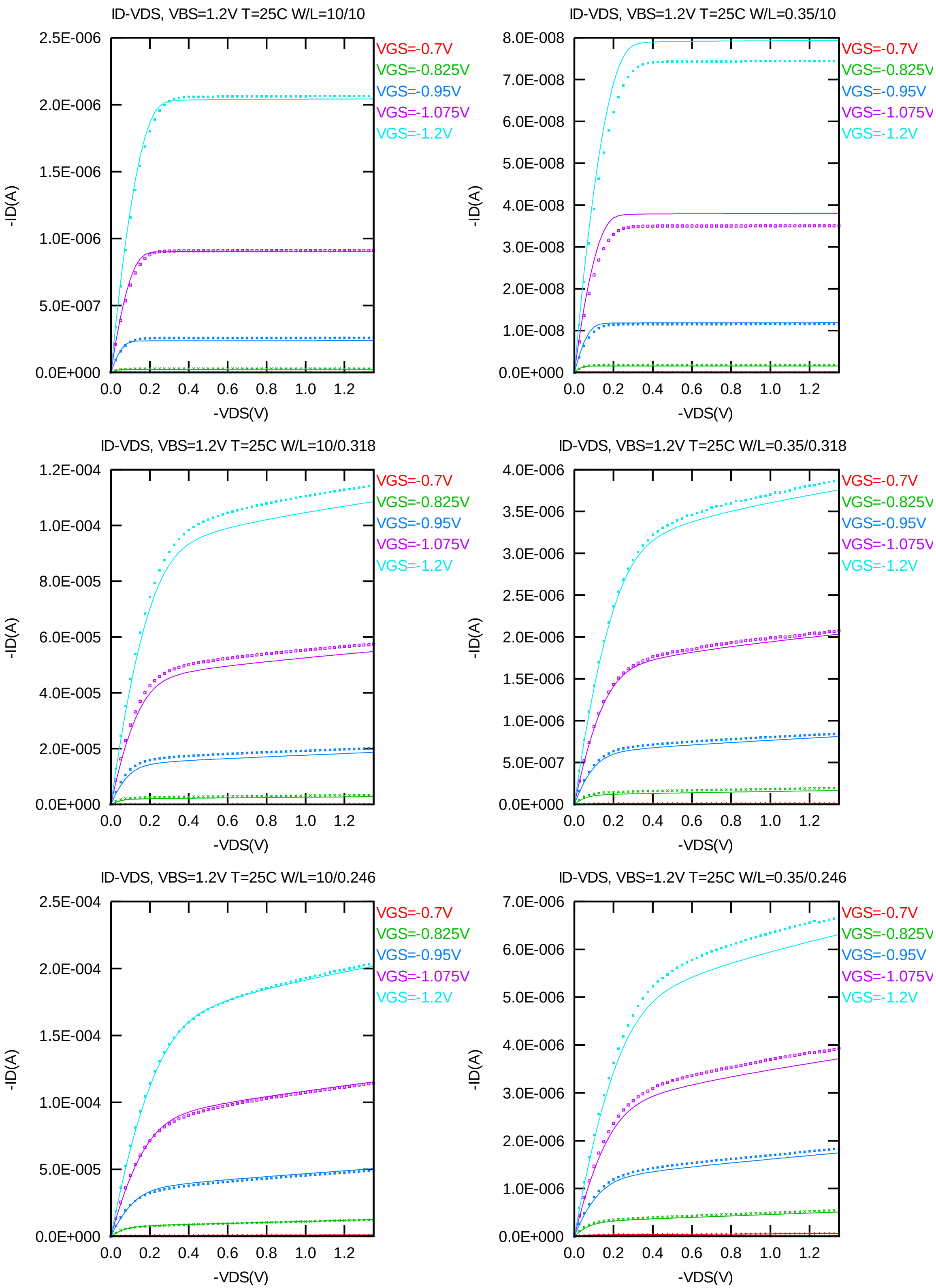

Figure 4-2 I_D vs. V_{DS} at V_{BS}= 1.2V for PMOS

G_{DS} vs. V_{DS} plots

The following plots show measured and simulated output I-V curves for selected devices at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

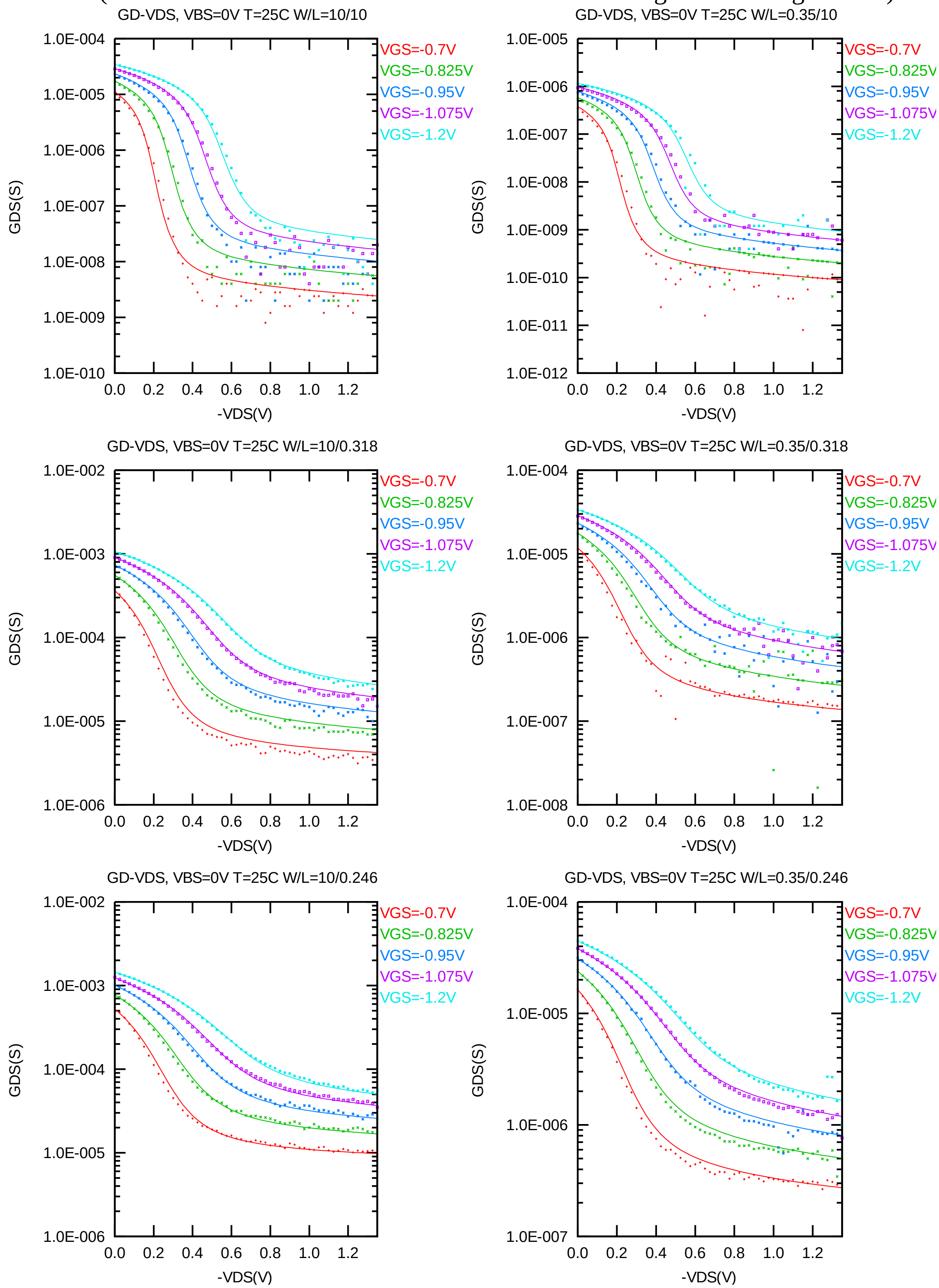

Figure 4-3 G_{DS} vs. V_{DS} at V_{BS} = 0 V for PMOS

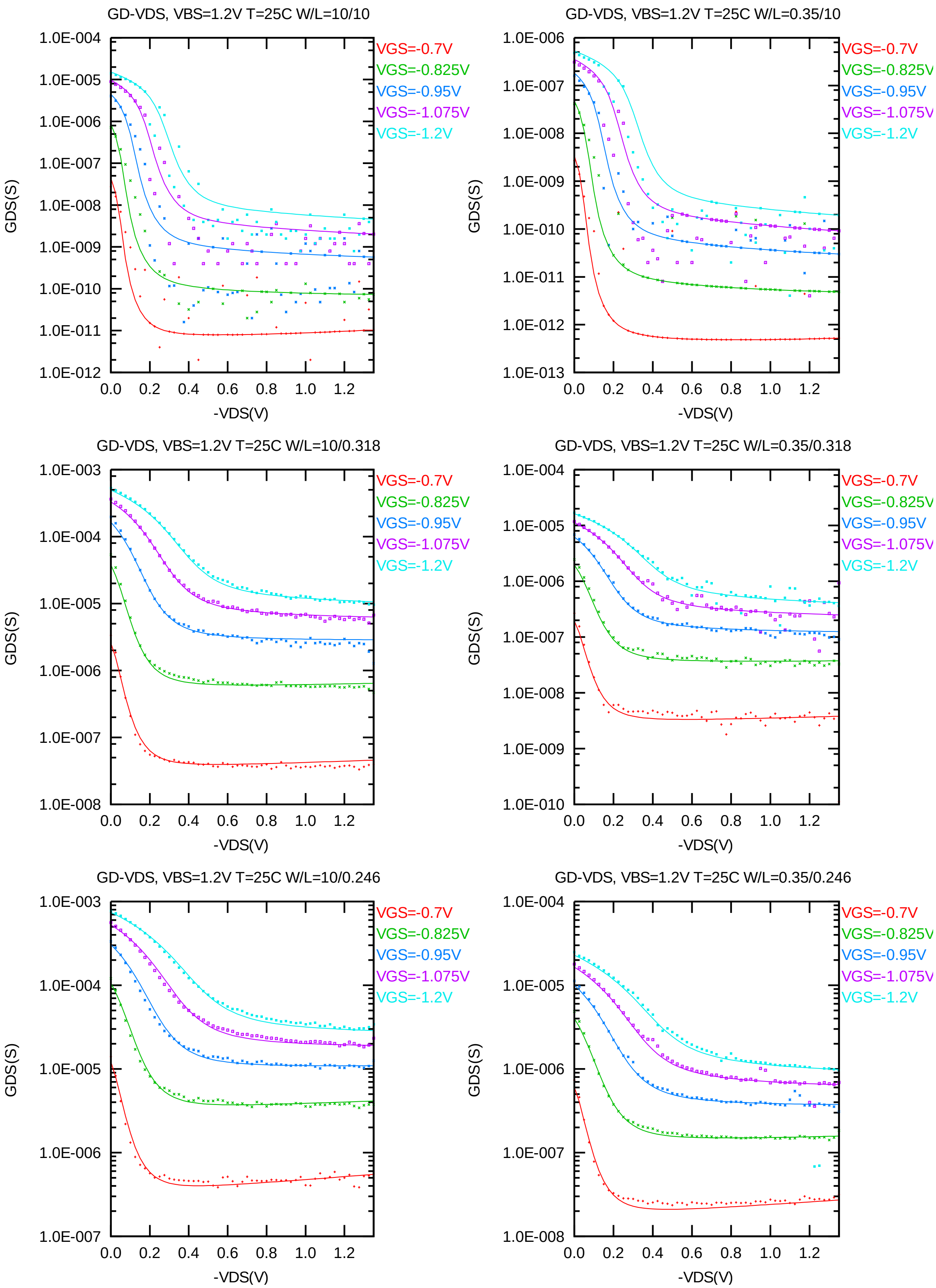

Figure 4-4 G_{DS} vs. V_{DS} at V_{BS} = 1.2V for PMOS

I_D vs. V_{GS} plots (linear scale)

The following plots show measured and simulated output I-V curves for selected devices at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

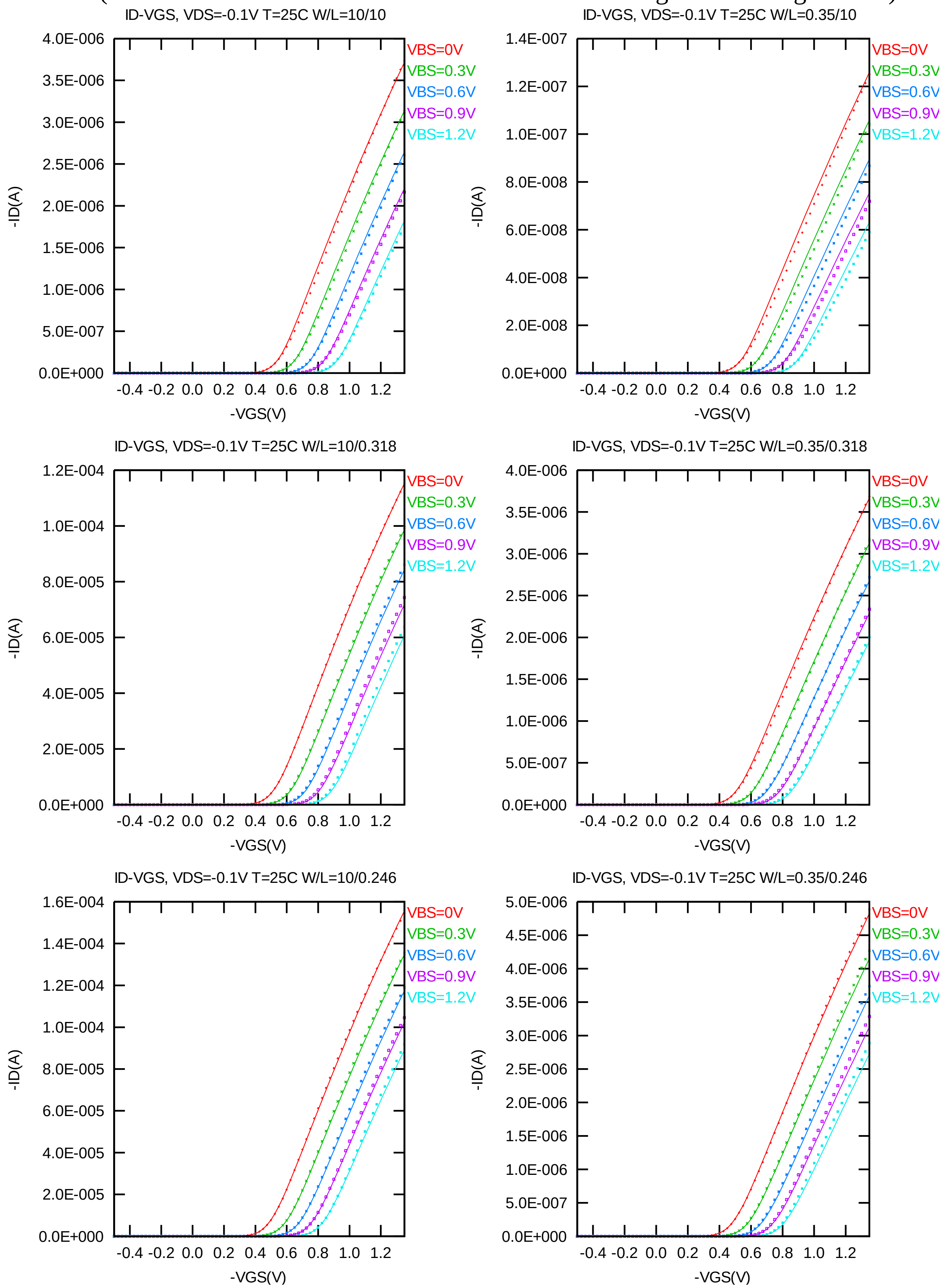

Figure 4-5 I_D vs. V_{GS} at V_{DS} = -0.1 V for PMOS

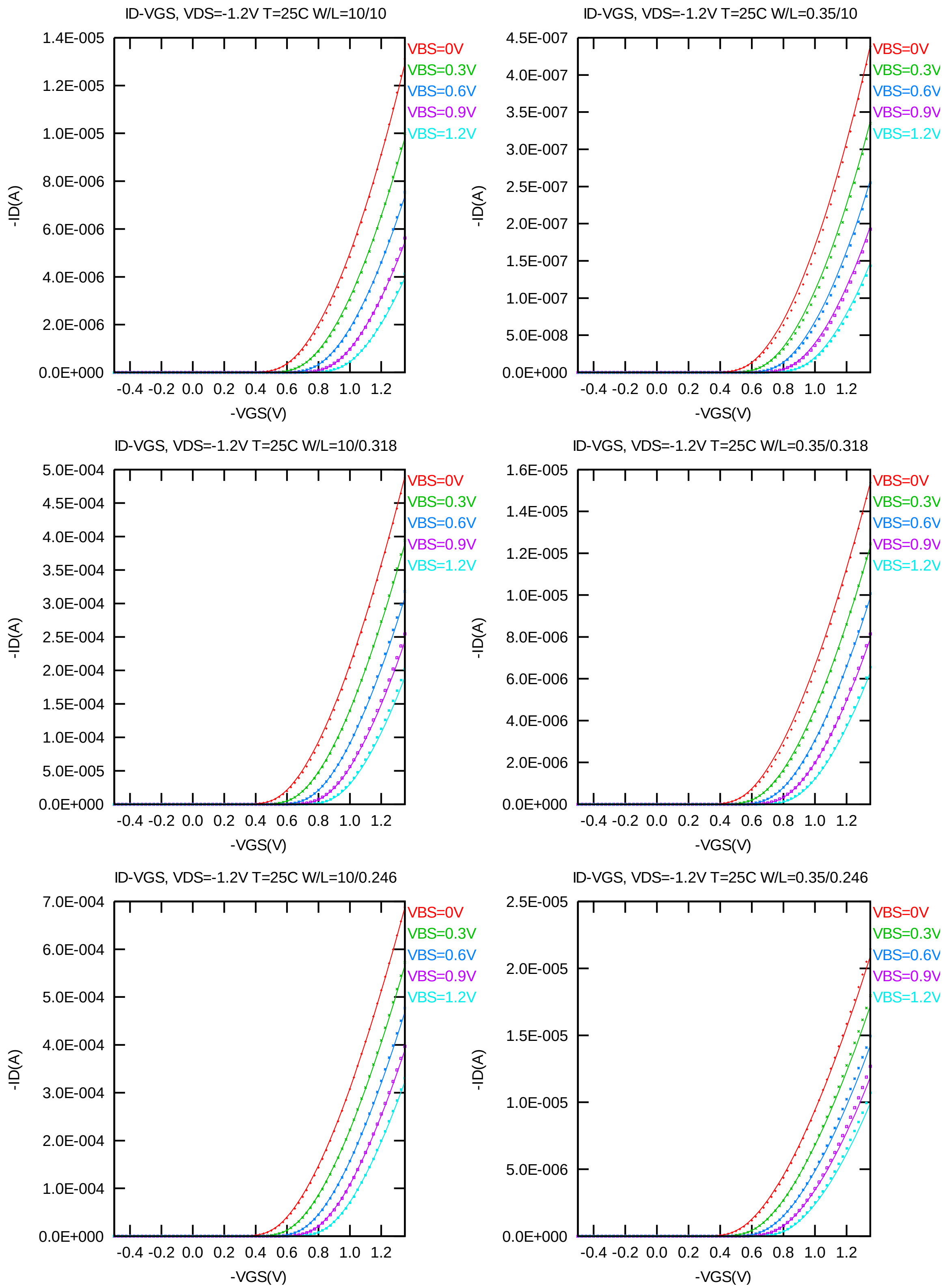

Figure 4-6 I_D vs. V_{GS} at V_{DS} = -1.2V for PMOS

I_D vs. V_{GS} plots (logarithmic scale)

The following plots show measured and simulated output I-V curves for selected devices at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

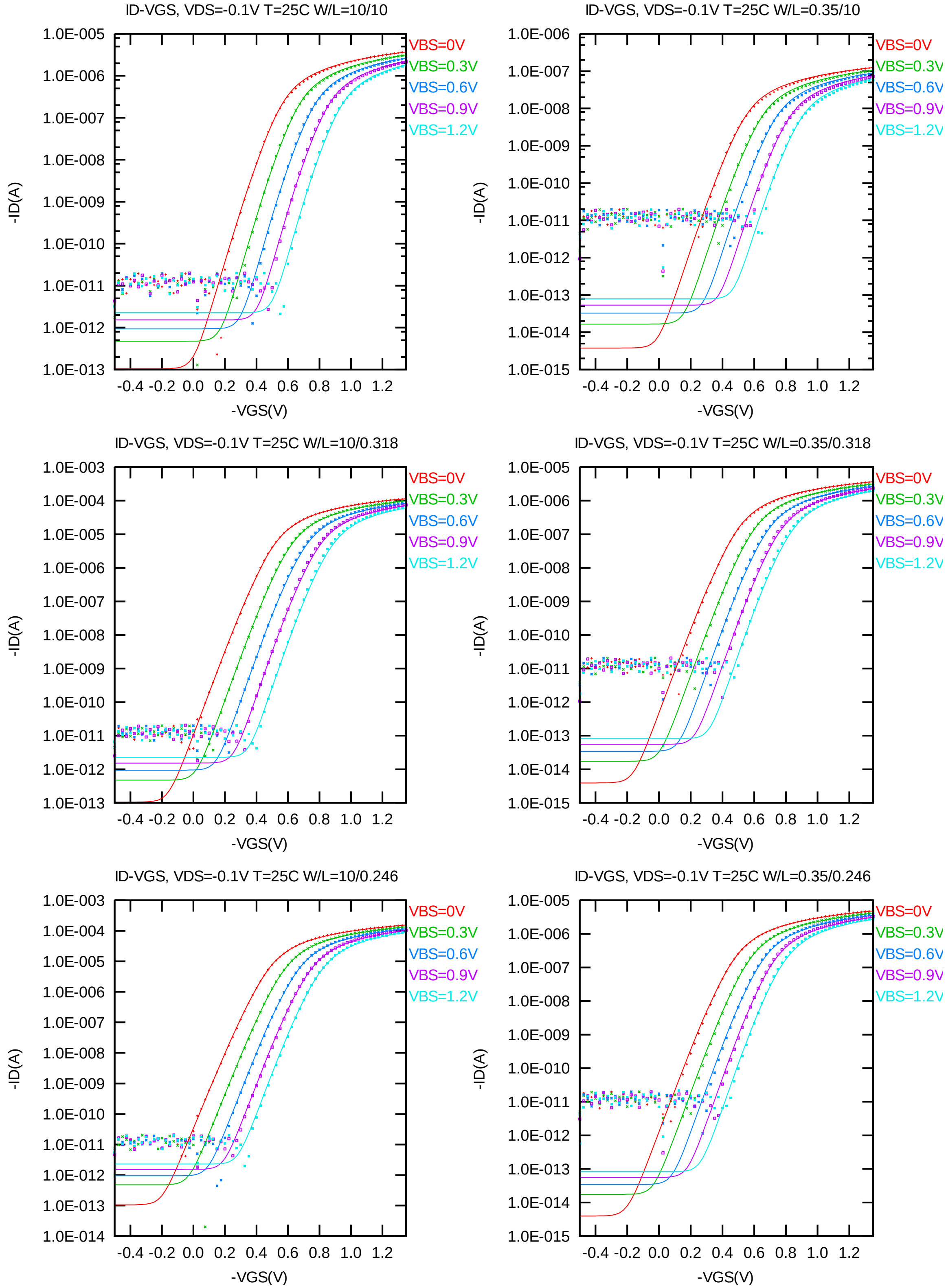

Figure 4-7 Log scaled I_D vs. V_{GS} at V_{DS} = -0.1 V for PMOS

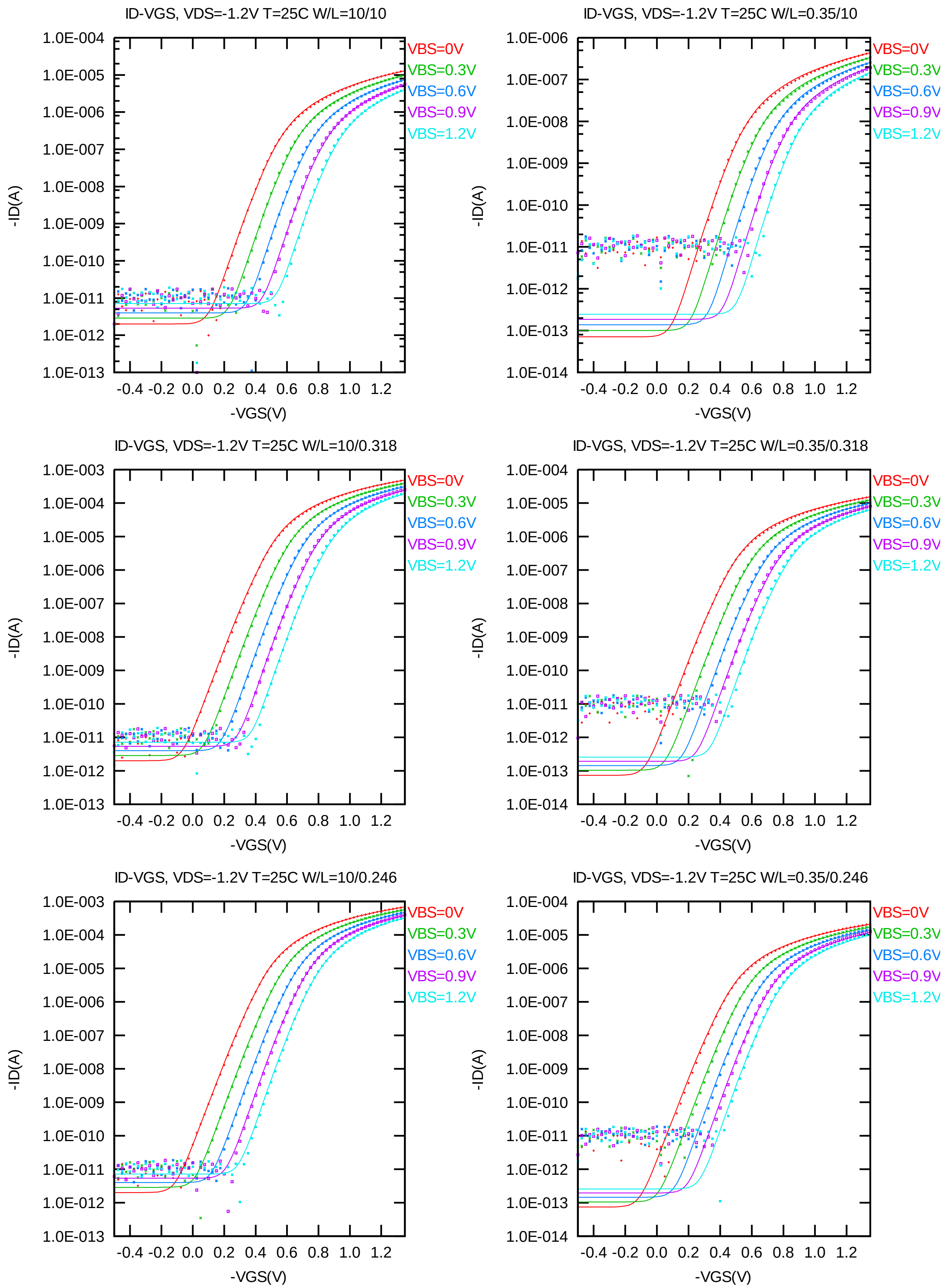

Figure 4-8 Log scaled I_D vs. V_{GS} at V_{DS} = -1.2V for PMOS

G_M vs. V_{GS} plots

The following plots show measured and simulated output I-V curves for selected devices at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

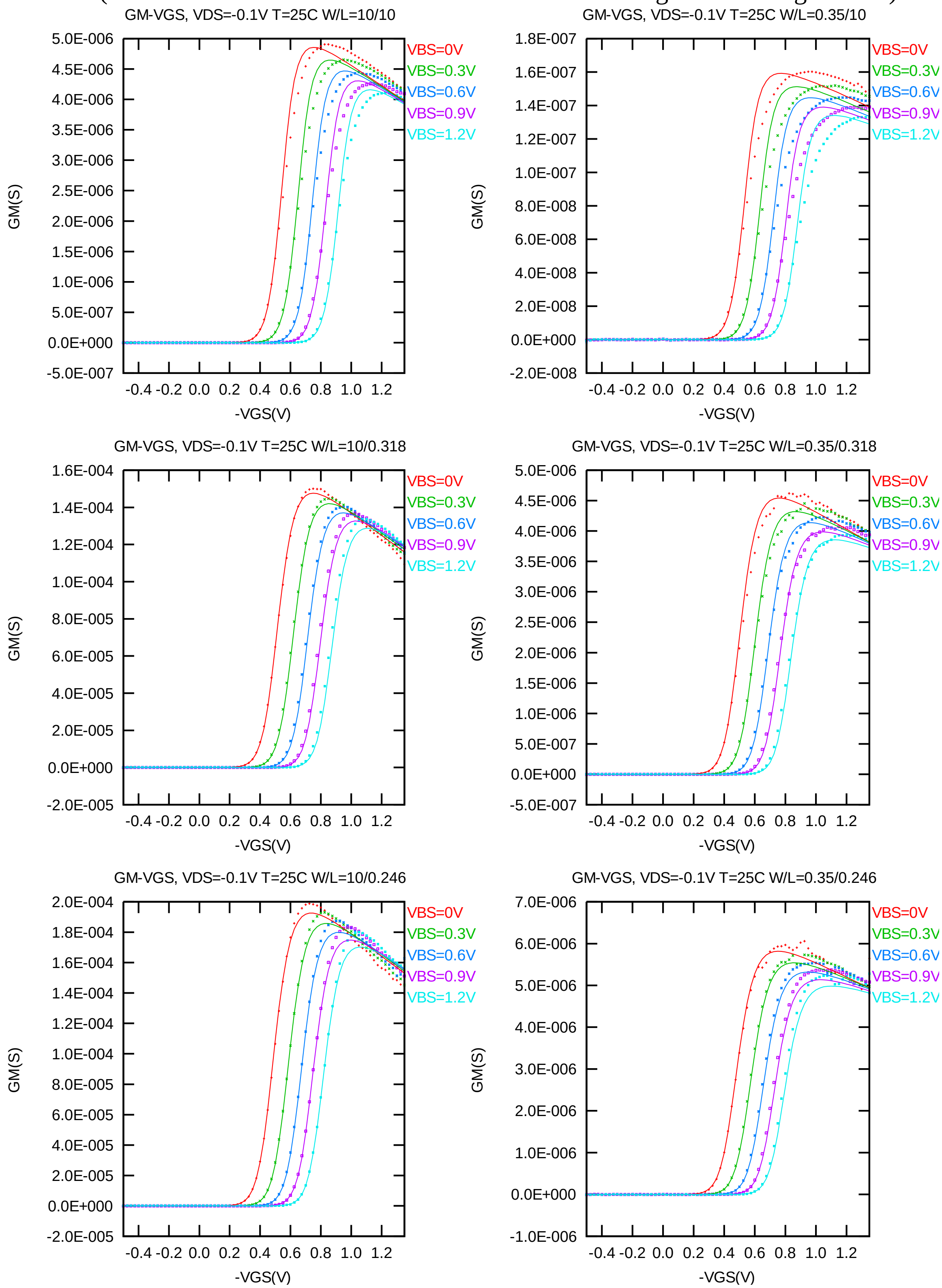

Figure 4-9 G_M vs. V_{GS} at V_{DS} = -0.1 V for PMOS

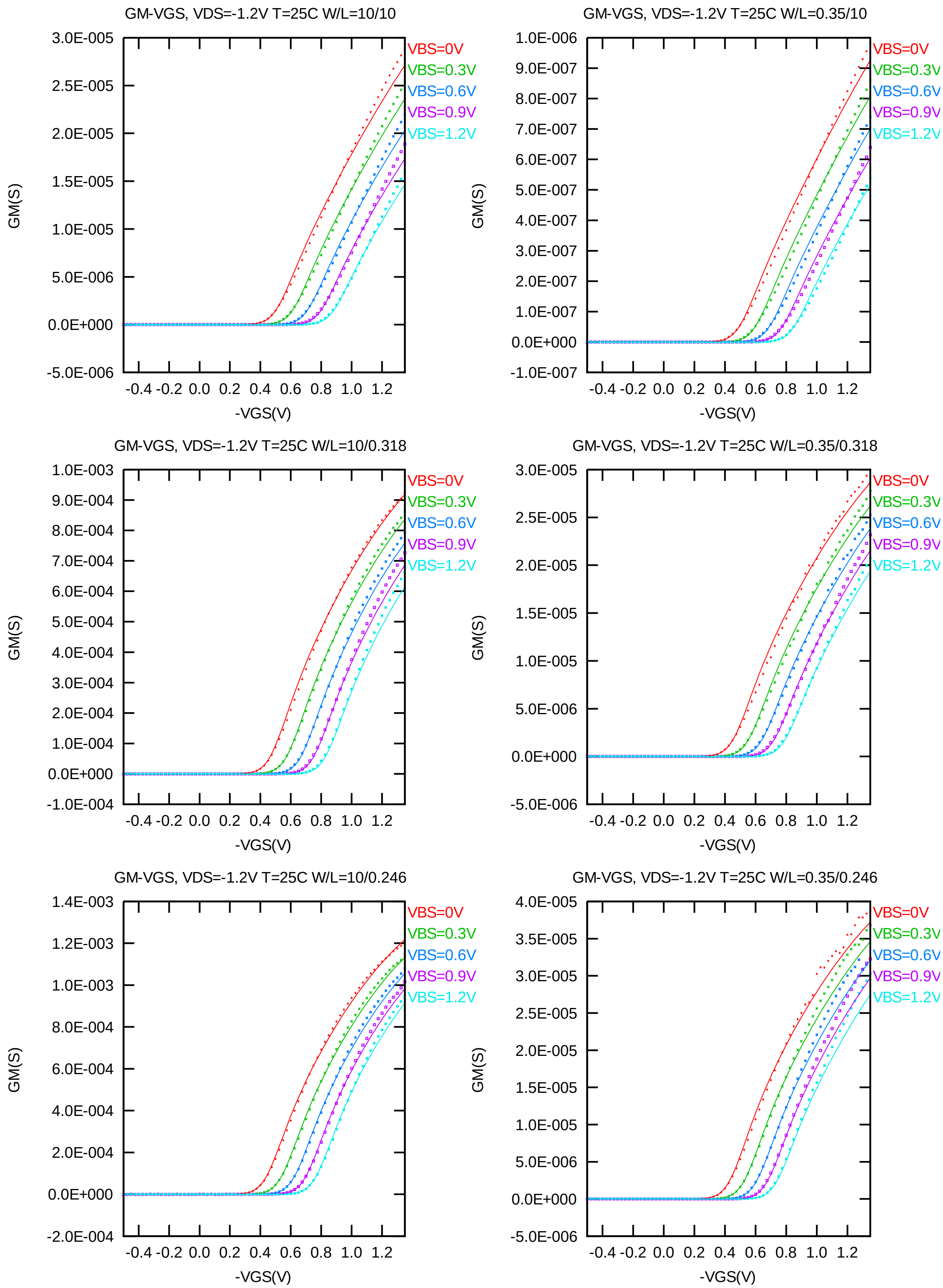

Figure 4-10 G_M vs. V_{GS} at V_{DS} = -1.2V for PMOS

The L and W in the plots refer to LDES and WDES respectively.

The following figures show the channel length and width dependence of linear and saturation threshold voltage extracted at ID =-70nA*(WDES*0.9)/(LDES*0.9+0.012 μm) at 25 ℃. The plots show the physical dimension. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

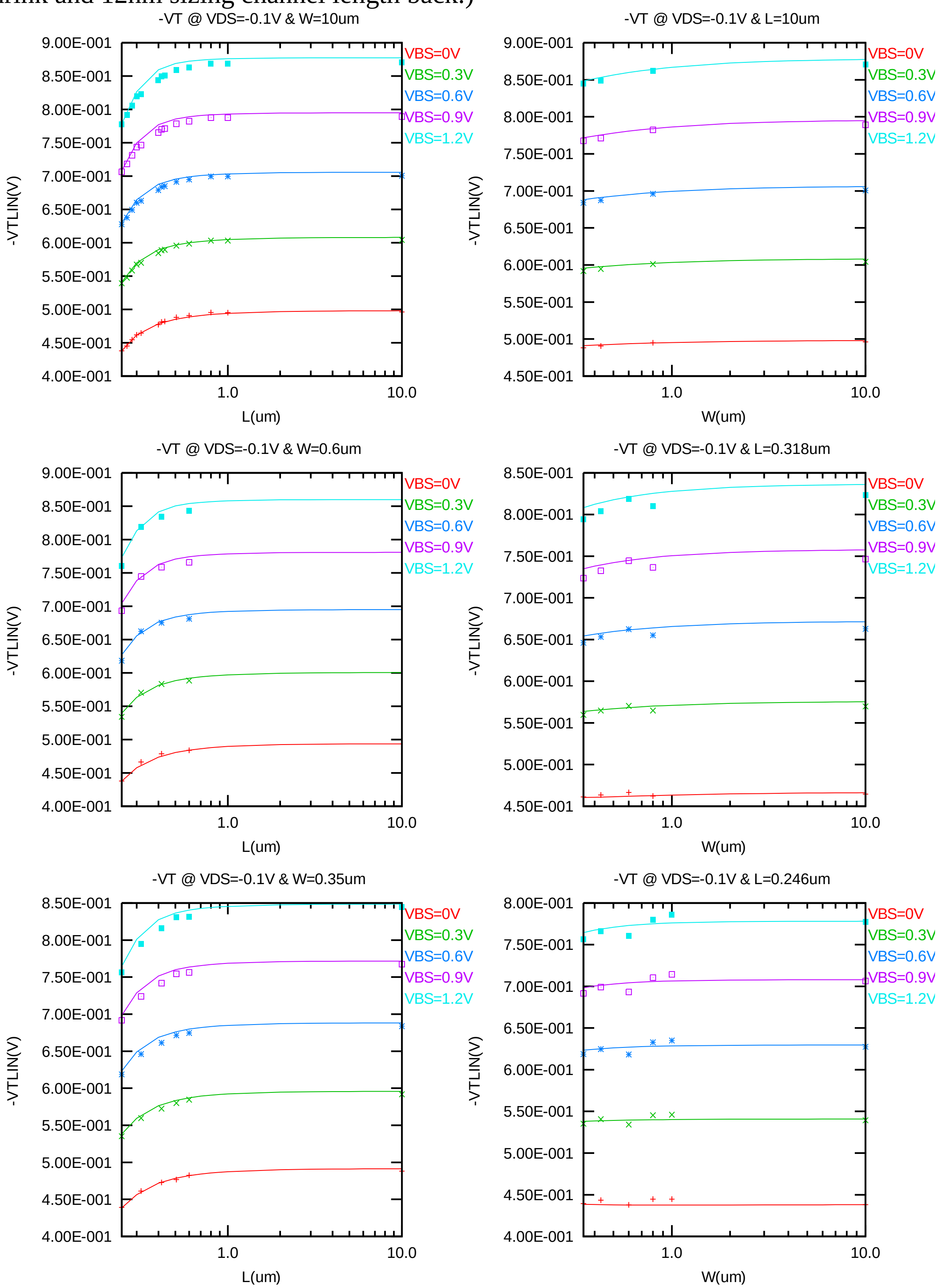

Figure 4-11 V_{TLIN} vs. channel length & channel width for low drain bias (V_{DS} = -0.1 V) for PMOS

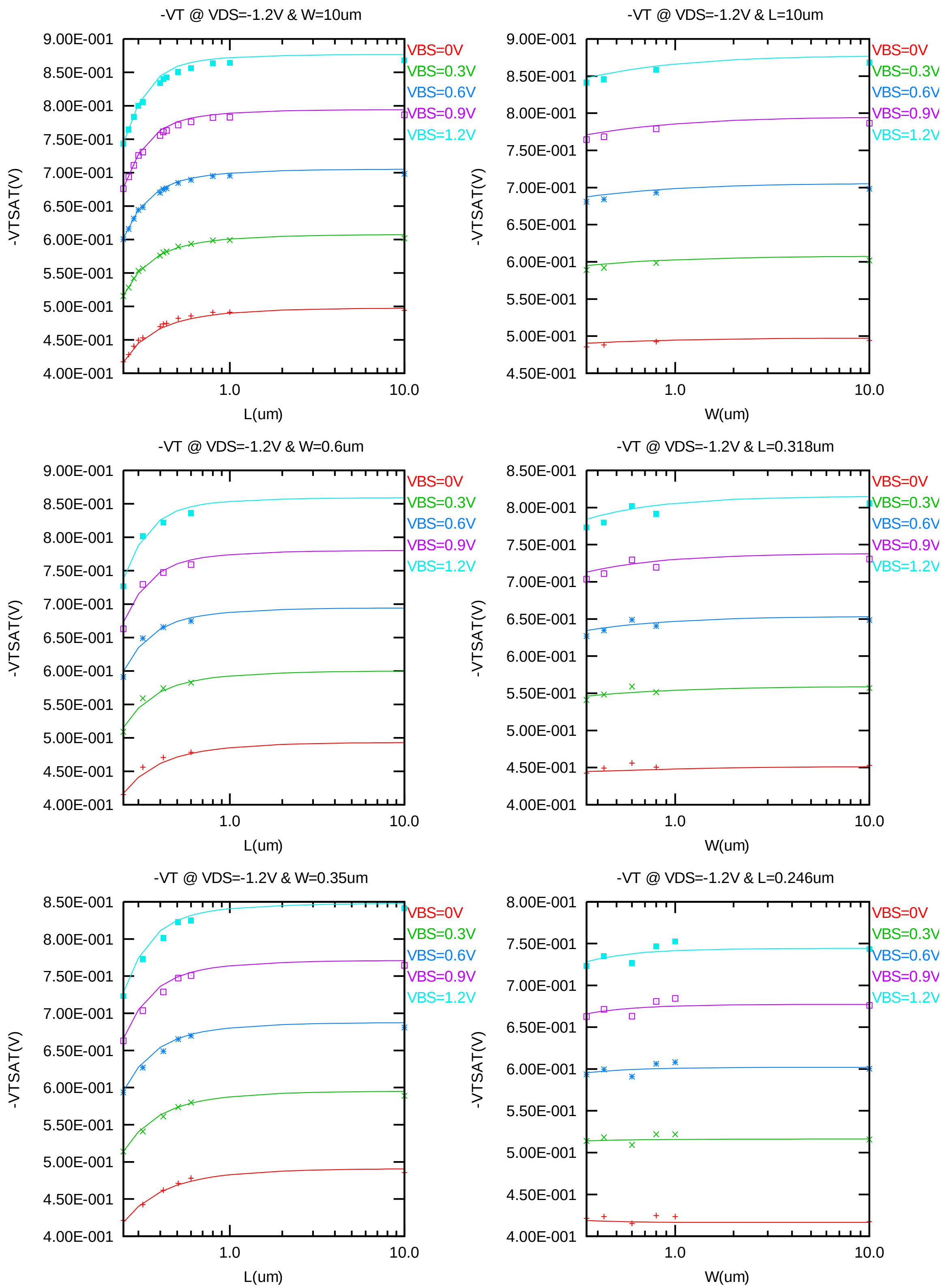

Figure 4-12 V_T vs. channel length & channel width for high drain bias (V_{DS} = -1.2V) for PMOS

The L and W in the plots refer to LDES and WDES respectively.

The following plots show the channel length and width dependence of I_{ON} extracted at $V_{DS} = V_{GS} = -1.2V$ for $V_{BS} = 0$ V and $V_{BS} = 1.2V$ at 25 C.(The dimensions are after 90% shrink and 12nm sizing channel length back.)

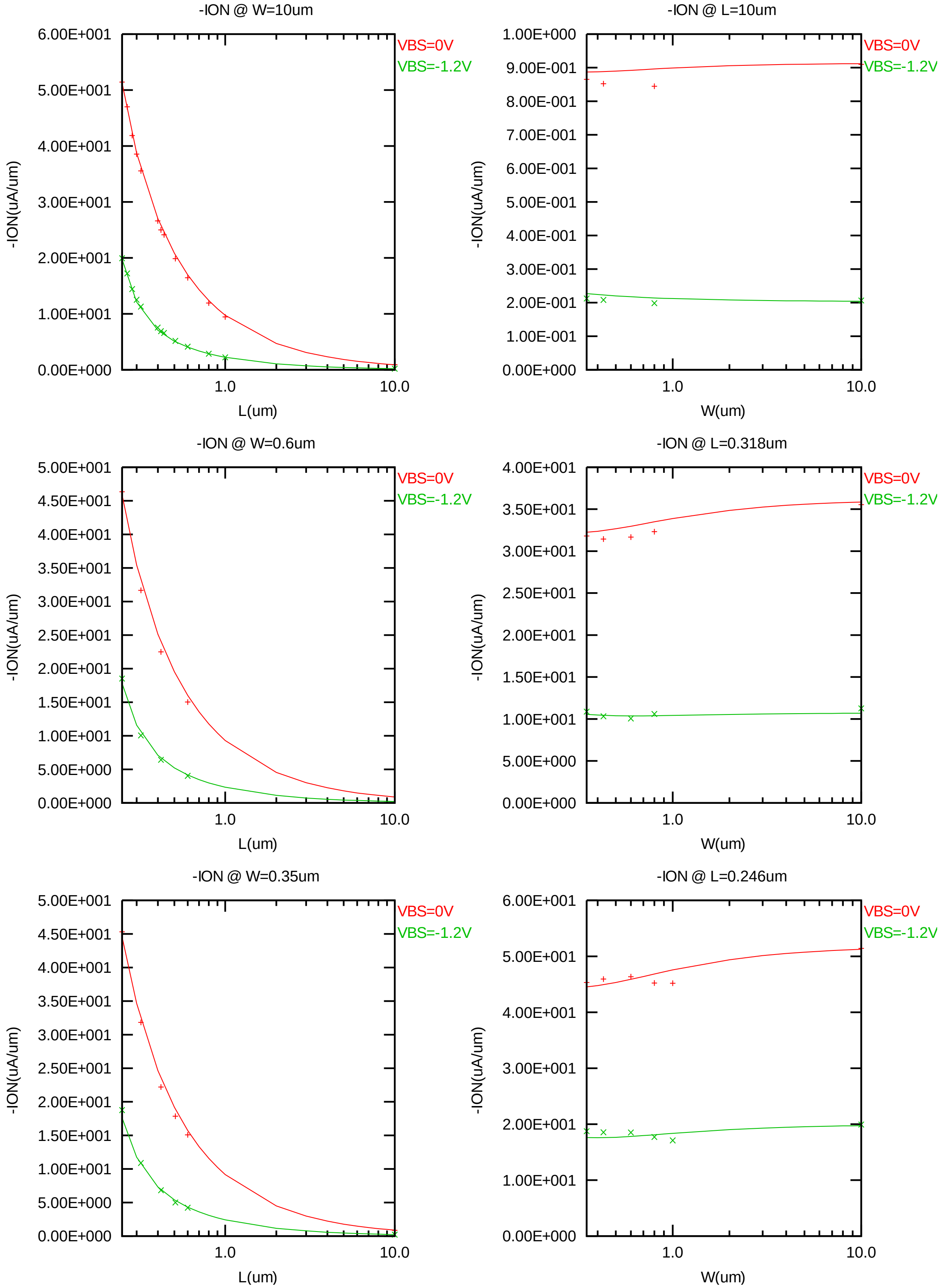

Figure 4-13 Saturation current I_{ON} vs. channel length and channel width for PMOS

The L and W in the plots refer to LDES and WDES respectively.

The following plots show the channel length and width dependence of I_{OFF} extracted at V_{DS} = -V_{CC} and V_{GS} = 0V for V_{BS} = 0 V to 1.2V at 25 C. A large discrepancy of I_{OFF} can be seen in the graphs, which is due to the presence of junction leakage current.(The dimensions are after 90% shrink and 12nm sizing channel length back.)

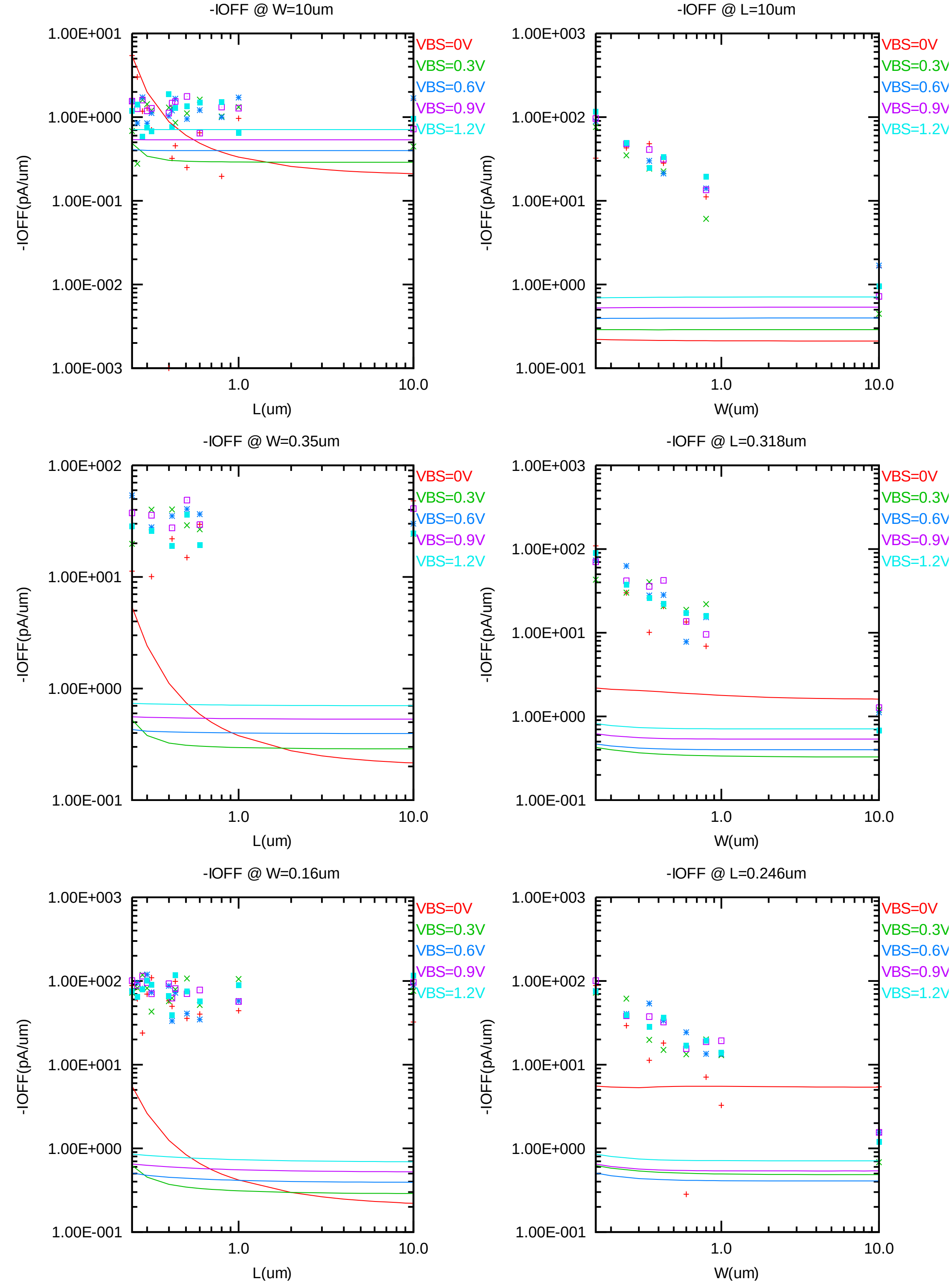

Figure 4-14 I_{OFF} vs. channel length and channel width for PMOS

V_{TLIN} vs. temperature

The following plots show the temperature dependence of the linear and saturation threshold voltage extracted at ID = -70nA*(WDES*0.9)/(LDES*0.9+0.012 µm). (The dimensions are after 90% shrink and 12nm sizing channel length back.)

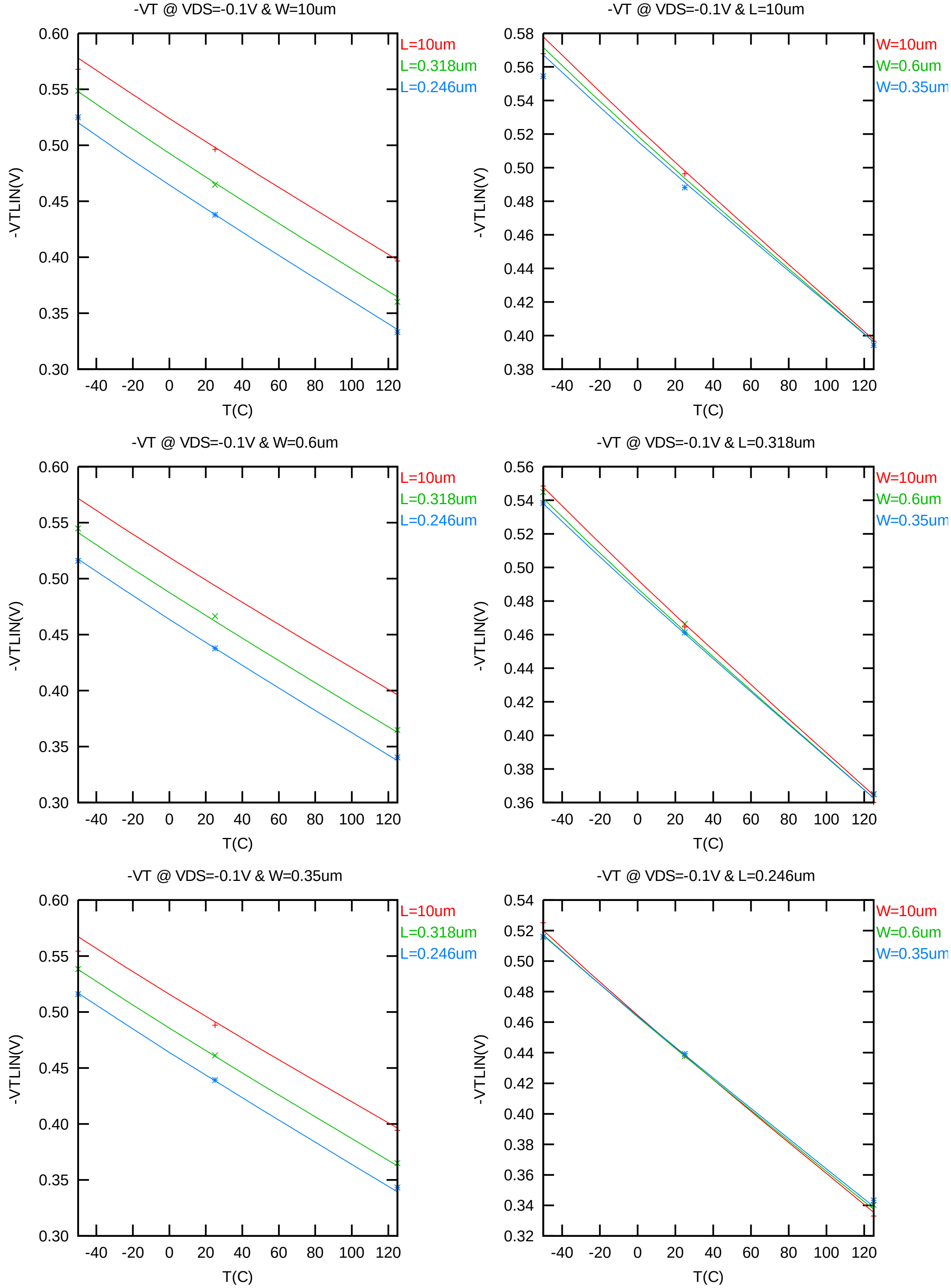

Figure 4-15 V_{TLIN} vs. temperature for PMOS

V_{TSAT} vs. temperature

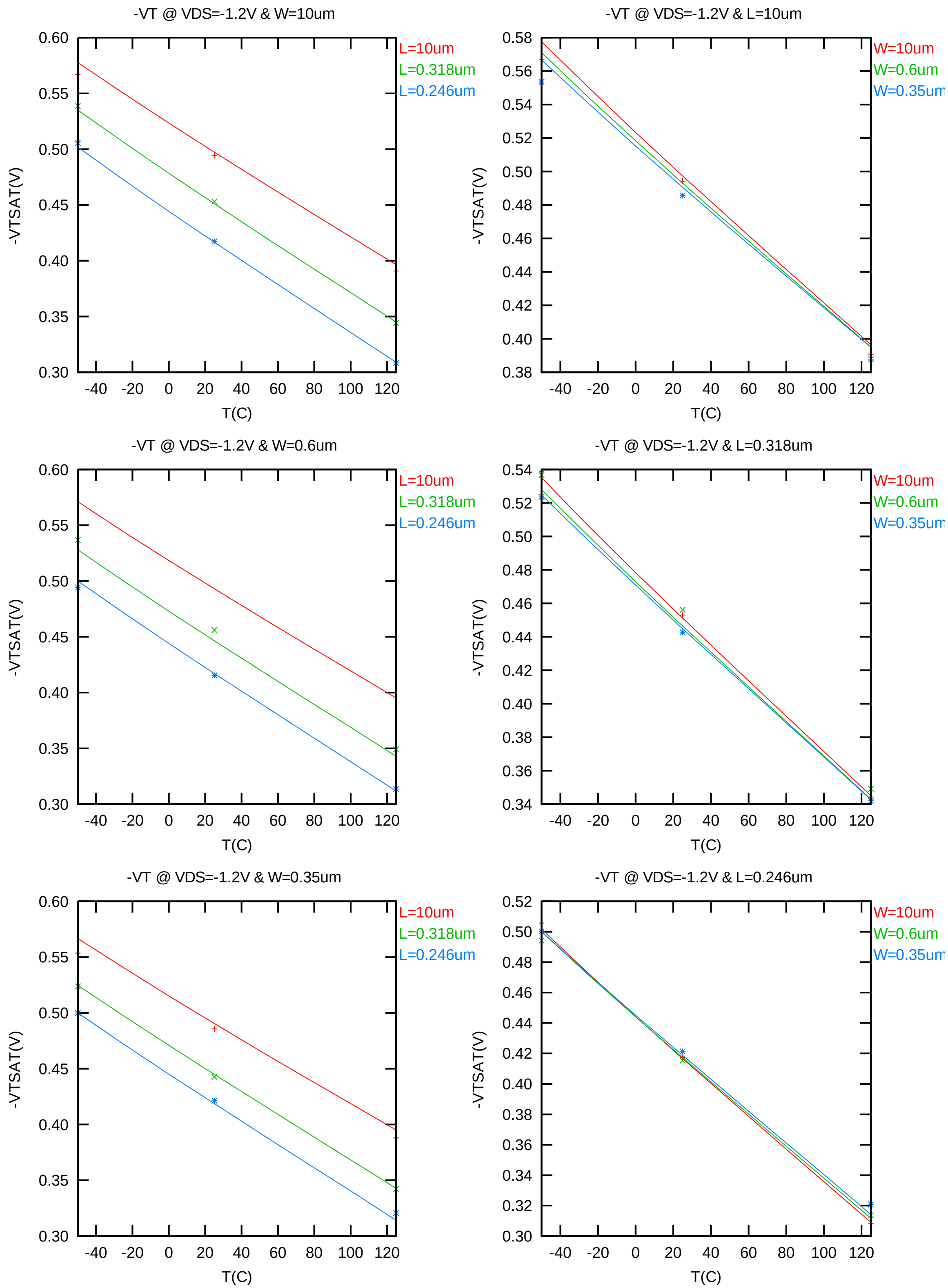

Figure 4-16 V_{TSAT} vs. temperature for PMOS

I_{ON} vs. temperature

The following plots show the temperature dependence of I_{ON} extracted at $V_{GS} = V_{DS}$ = -1.2V, V_{BS} = 0 V. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

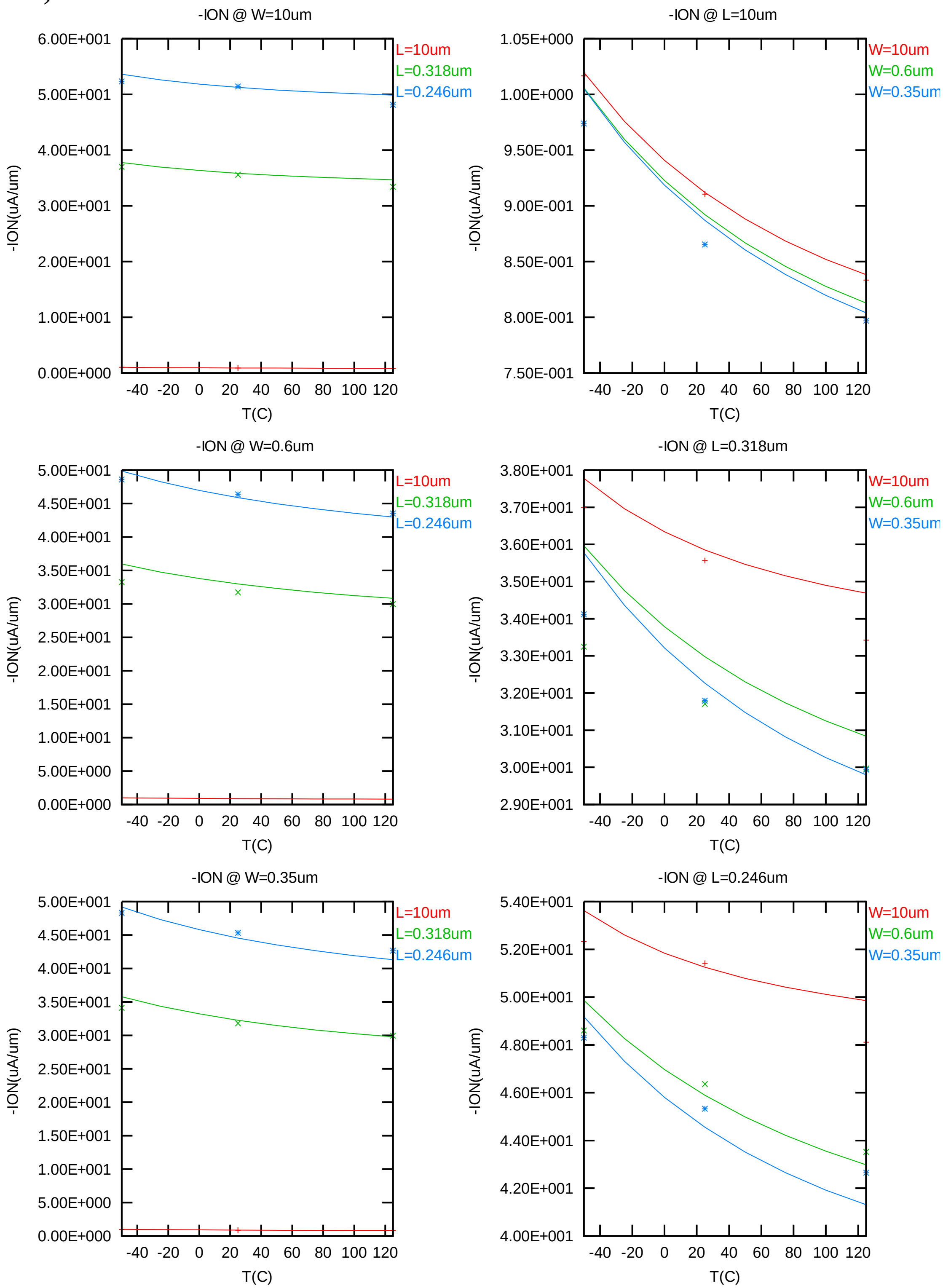

Figure 4-17 Saturation current vs. temperature for PMOS

I_{OFF} vs. temperature

The following plots show the temperature dependence of I_{OFF} extracted at V_{DS} = -1.2V, $V_{GS} = V_{BS}$ = 0 V. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

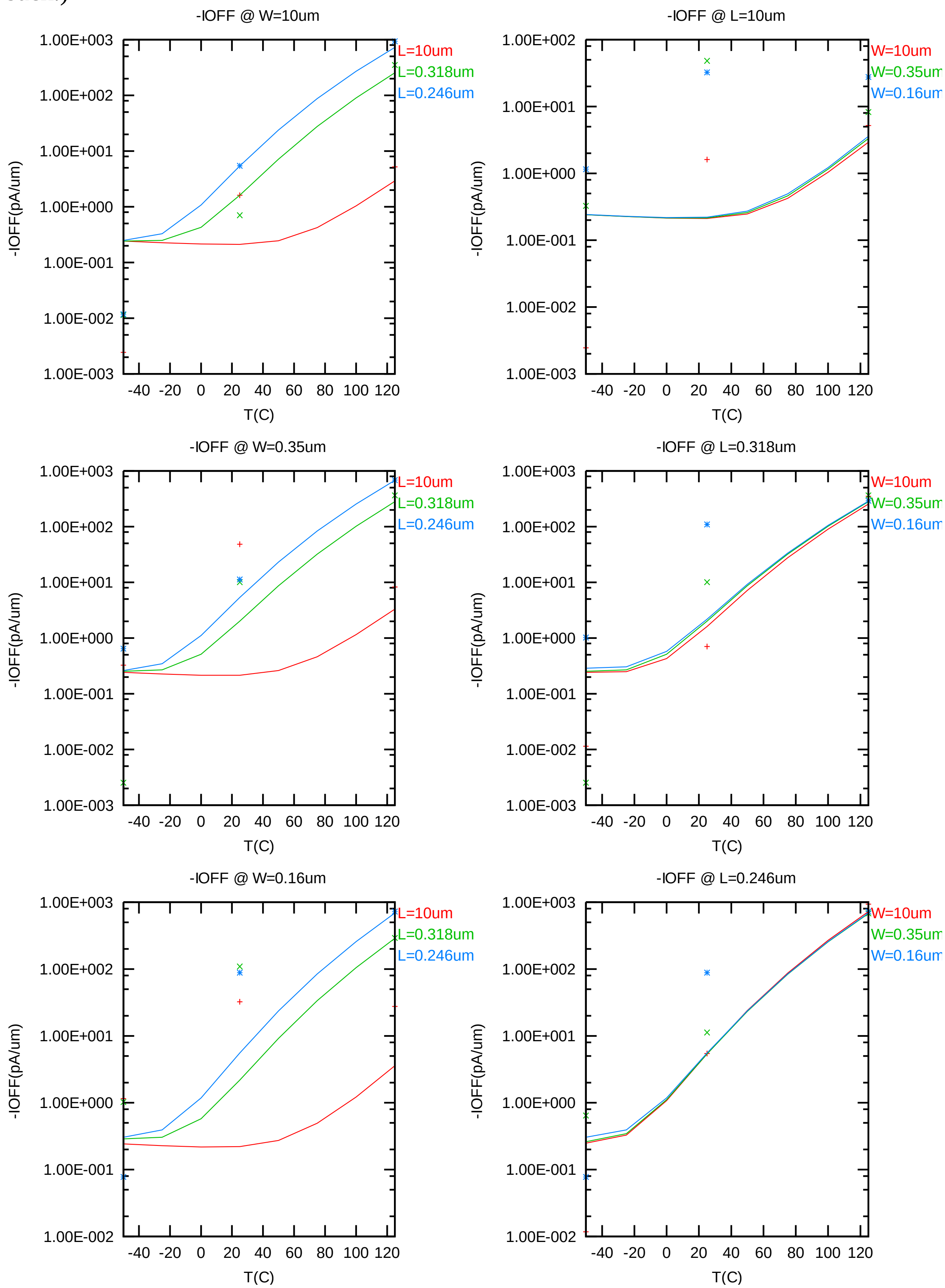

Figure 4-18 I_{OFF} vs. temperature for PMOS

4.3.4 Model fitting accuracy

The model fitting accuracy of parameters I_D, G_M and G_{DS} is quantified using the following equation:

error % = 100 * | measurement - simulation | / [(measurement + simulation) / 2]

Different error criteria are defined for different parameters (I_D, G_M and G_{DS}) as well as for different operation regions. These criteria are explained below.

1) I_D accuracy criteria

k. I_D in sub-threshold region, i.e. $V_{GS} < V_T$
l. I_D in moderate inversion region, i.e. $V_T \leqq V_{GS} < V_T + V_X$
m. I_D in strong inversion region, i.e. $V_T + V_X \leqq V_{GS} \leqq V_{CC}$

, where $V_X = (V_{CC} - V_T) / 2$

2) G_M and G_{DS} accuracy criteria

n. G_M and G_{DS} in sub-threshold region, i.e. $V_{GS} < V_T$
o. G_M and G_{DS} in inversion region, i.e. $V_T \leqq V_{GS} \leqq V_{CC}$

Table 4-5 Requirements for model fit quality

Target of Quality Check	$V_{GS} < V_T$ (sub-threshold)	$V_T \leq V_{GS} < V_T+V_X$ (moderate inversion)	$V_T+V_X \leq V_{GS} \leq V_{CC}$ (strong inversion)
I_D	$\leq$75%	$\leq$25%	$\leq$7%
G_M	$\leq$50%	$\leq$20%	$\leq$20%
G_{DS}	$\leq$100%	$\leq$50%	$\leq$50%

I_D accuracy

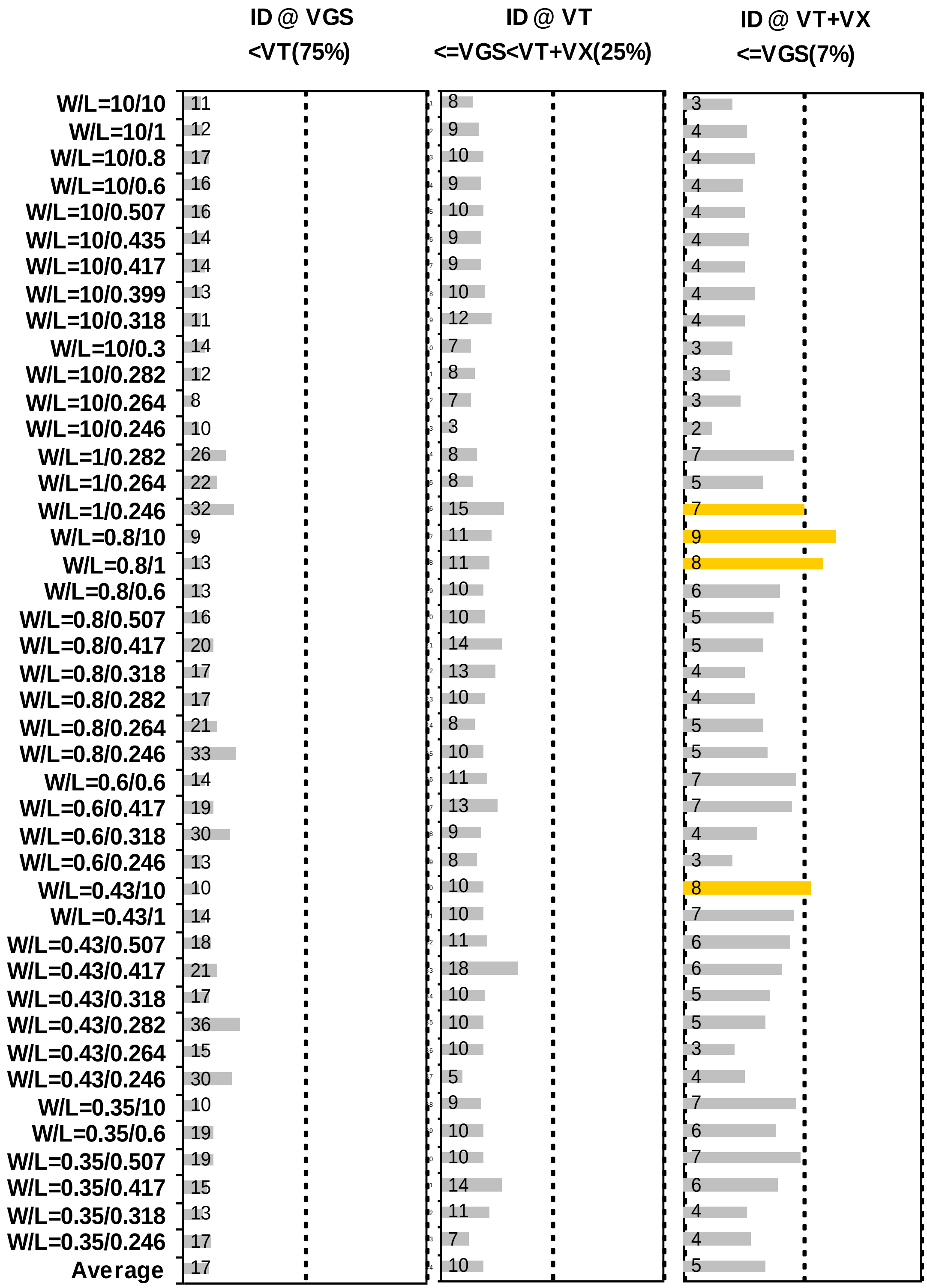

Figure 4-19 I_D accuracy at 25 C for PMOS

G_M and G_{DS} accuracy

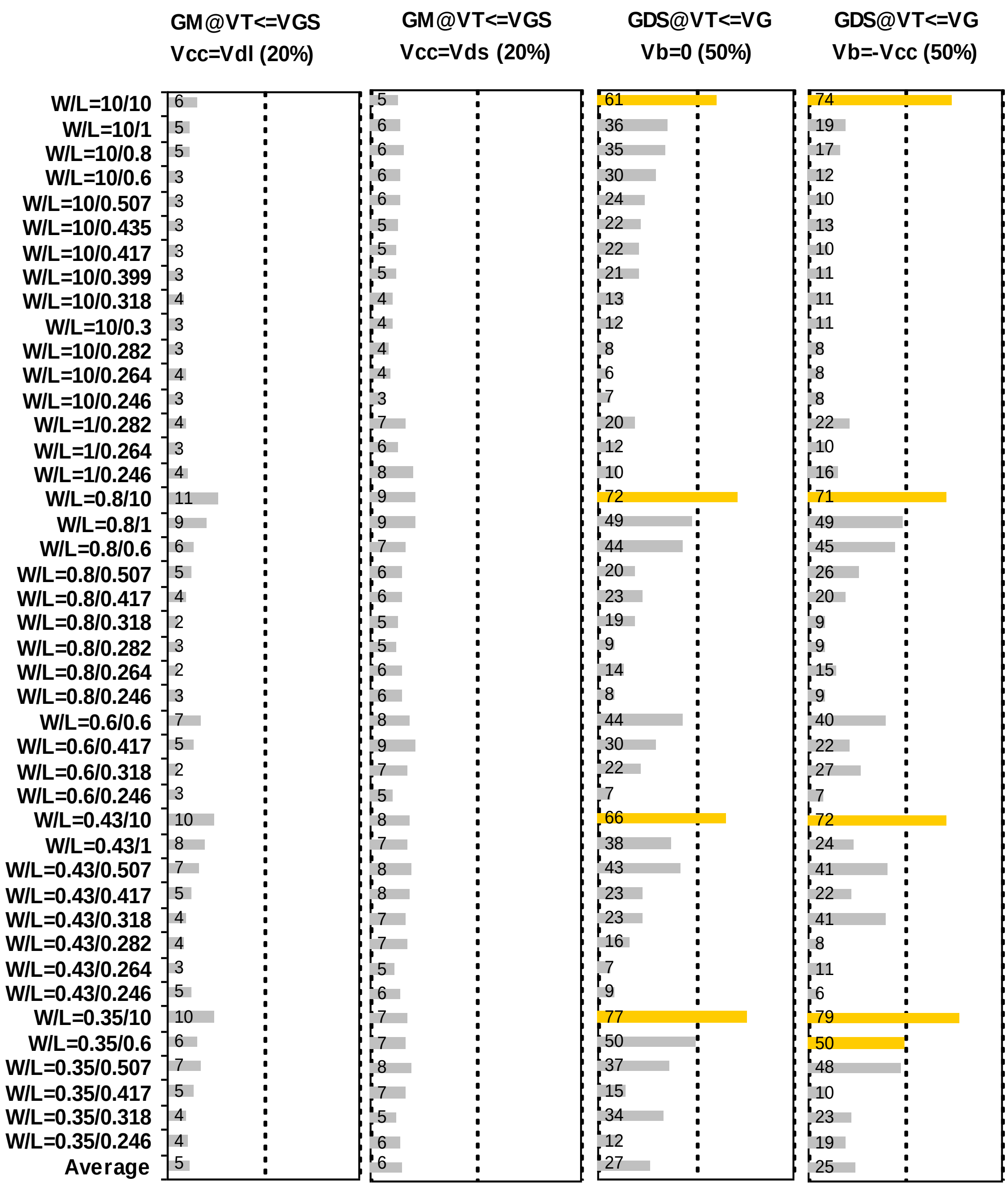

Figure 4-20 G_M and G_{DS} accuracy at 25 C for PMOS

4.4 MOSFET Capacitance Model

4.4.1 Test devices

The following test structures were used for MOSFET capacitance model parameter extraction.

Table 4-6 Measurement structure for PMOS Cgg capacitance model

Structure	L[μm]	W[μm]
Gate capacitance components, C_{GDS}	80	80

Table 4-7 Large area measurement structure for PMOS Cgc capacitance model

Structure	L[μm]	W[μm]
Gate capacitance components	10	10
	1	10
	.5	10
	.246	10
	10	1
	.417	10
	.417	1
	10	.5
	1	1
	.246	.5
	.5	.5
	.417	.25
	10	.16
	10	.25
	.417	.16
	.246	.16
	.5	.16

4.4.2 Measurement conditions

The measurement conditions for MOSFET capacitance are summarized in Table 2-8 below.

Table 4-8 Measurement conditions for PMOS capacitance model

V_{BIAS} [V]	From -3.63 to 3.63 V in 0.066 V step
Frequency [kHz]	100
V_{OSC} [mV]	50
Temp [°C]	25

4.4.3 T_{OX} extraction

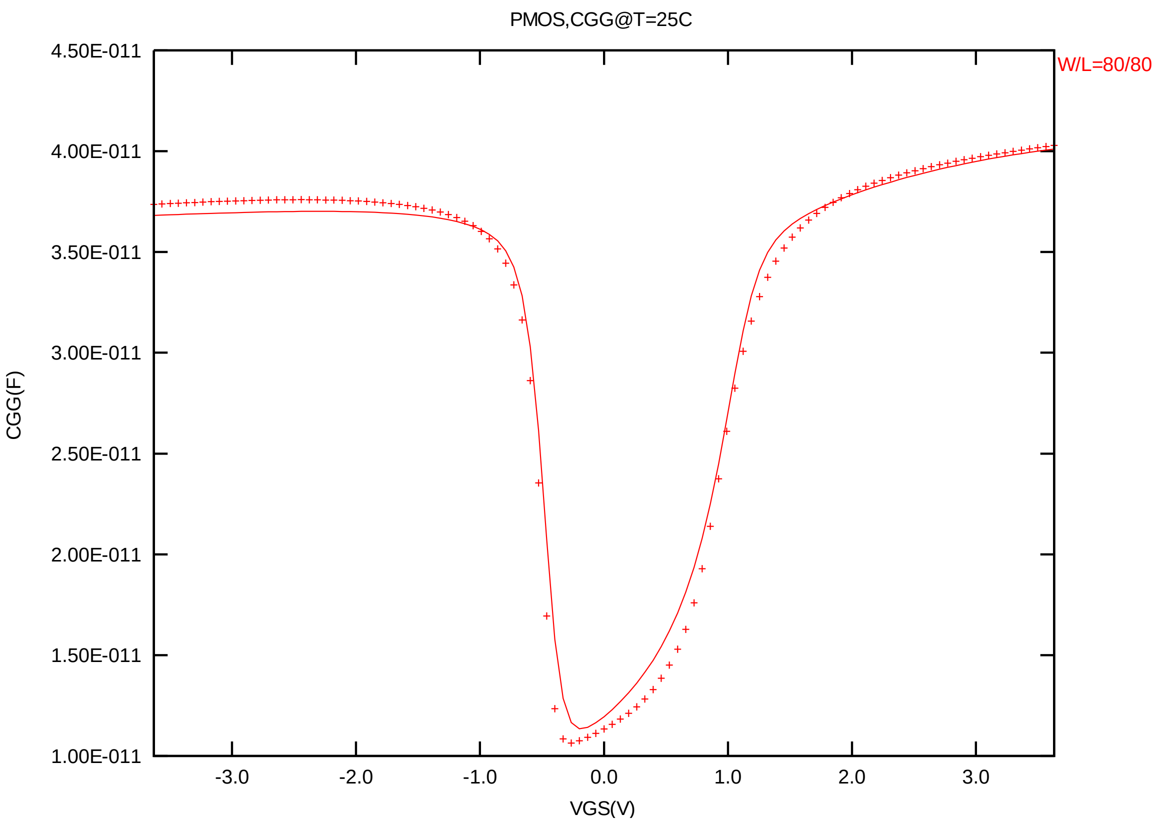

Figure 4-21 Measured and simulated C_{GG} curves

4.4.4 Width and length AC model accuracy

Width and length AC model accuracy

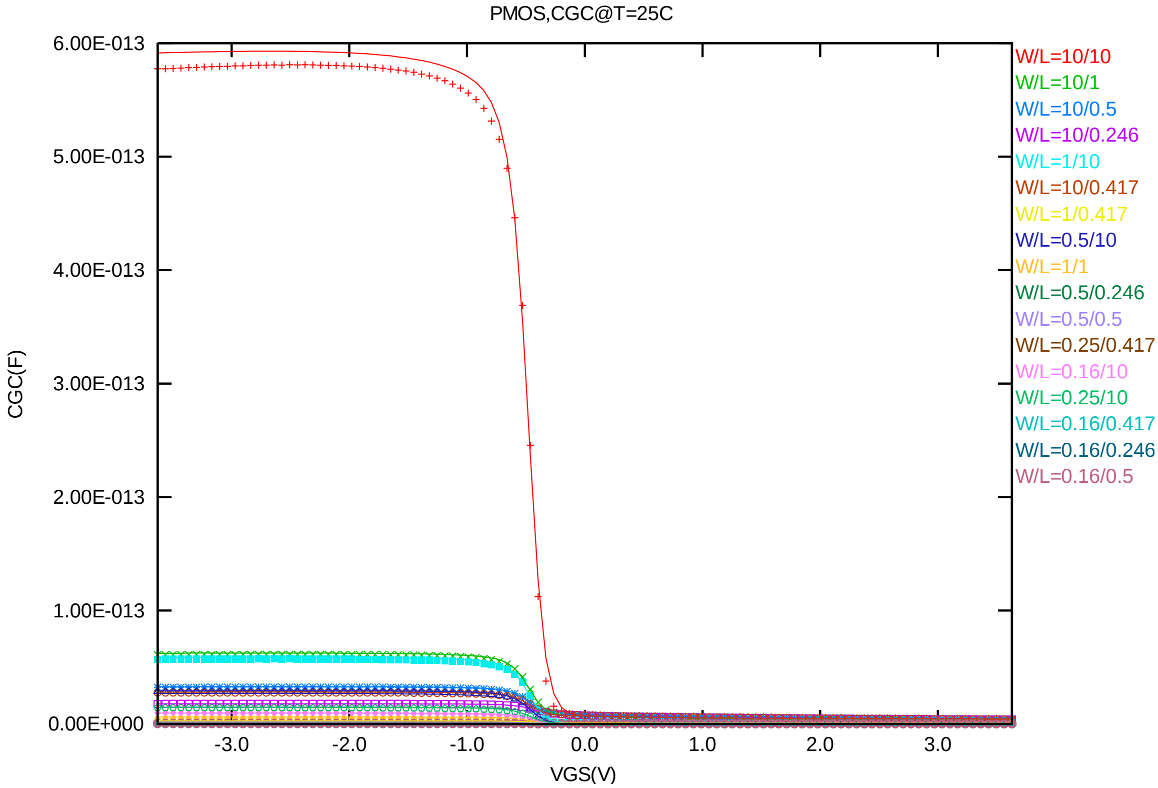

Figure 4-22 Measured and simulated C_{GC} curves for MOSFET measurement structure for PMOS

4.5 Parasitic Source / Drain Junction Capacitance Model

4.5.1 Test devices

The following test structures were used for parameter extraction.

Table 4-9 Junction capacitor structures for PMOS

Structures	Area[m^2]	Perimeter[m]	Description
Bottom area junction capacitance	5.00E-08	2.00E-03	Plate junction capacitor
STI bounded sidewall junction capacitance	3.60E-08	3.64E-02	Finger diode
Inner gate sidewall junction	2.70E-08	1.830E-02	Poly finger structure

4.5.2 Measurement conditions

All junction capacitance was measured under voltage sweep conditions summarized in the following table.

Table 4-10 Junction capacitor measurement conditions for PMOS

V_{BIAS} [V]	VH: nwell & VL: p+ From 0 to 3.63 V in 0.033 V step
Frequency [kHz]	100
V_{OSC} [mV]	50
Temp [°C]	-50,25,125

4.5.3 Extracted model parameters and verification plots

CJ, CJSW vs. Vbias plots

The following plots show the variation of bottom area junction and STI bounded sidewall at different bias voltages and temperatures.

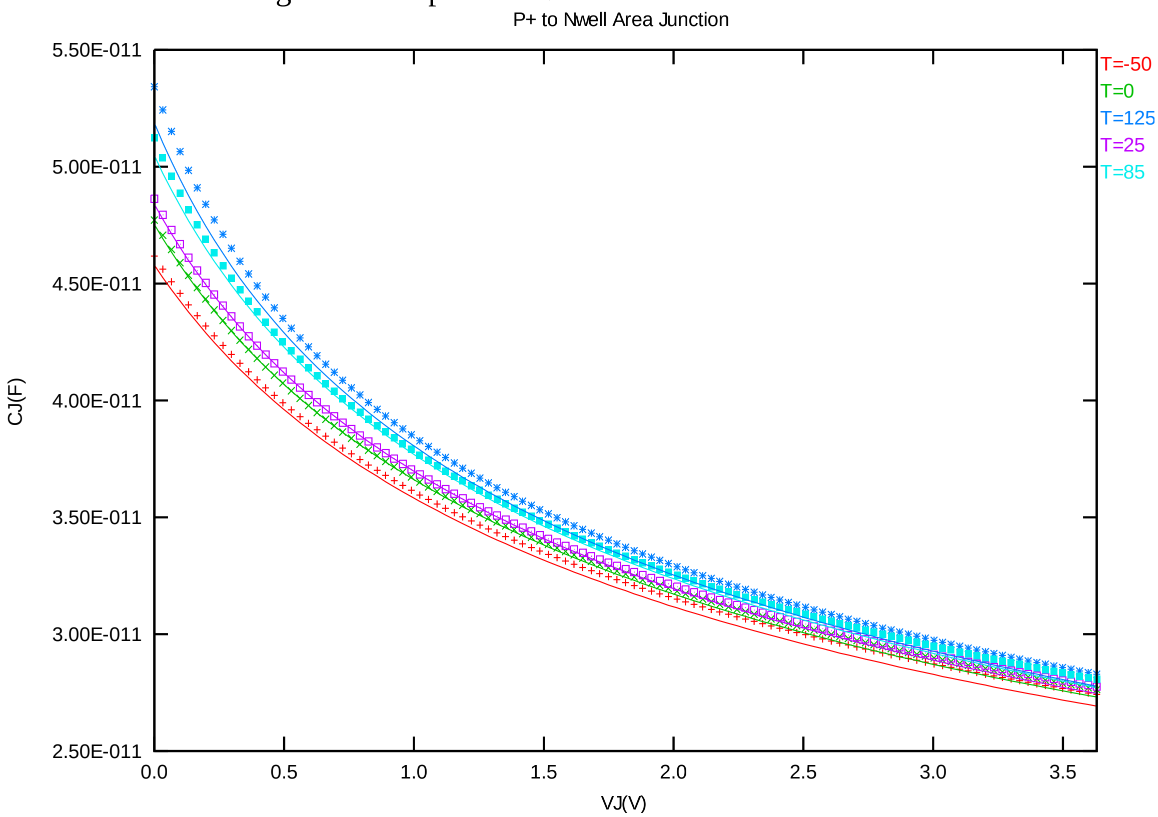

Figure 4-23 Bottom area junction capacitance at -50 C, 0 C, 25 C, 85 C and 125 C for PMOS

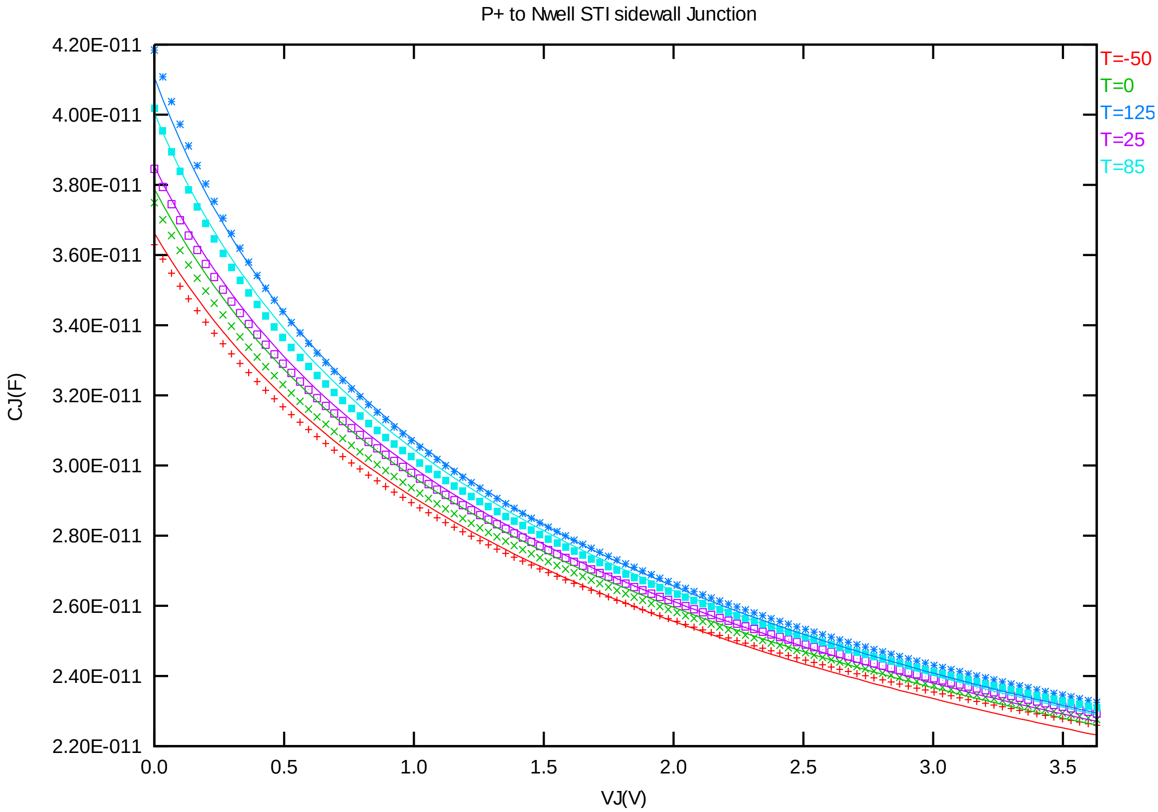

Figure 4-24 STI bounded sidewall junction capacitance at -50 C, 0 C, 25 C, 85 C and 125 C for PMOS

5 Worst Case Model

5.1 Wafer Data vs. Corner Plots

Threshold voltage in linear region V_{TLIN}

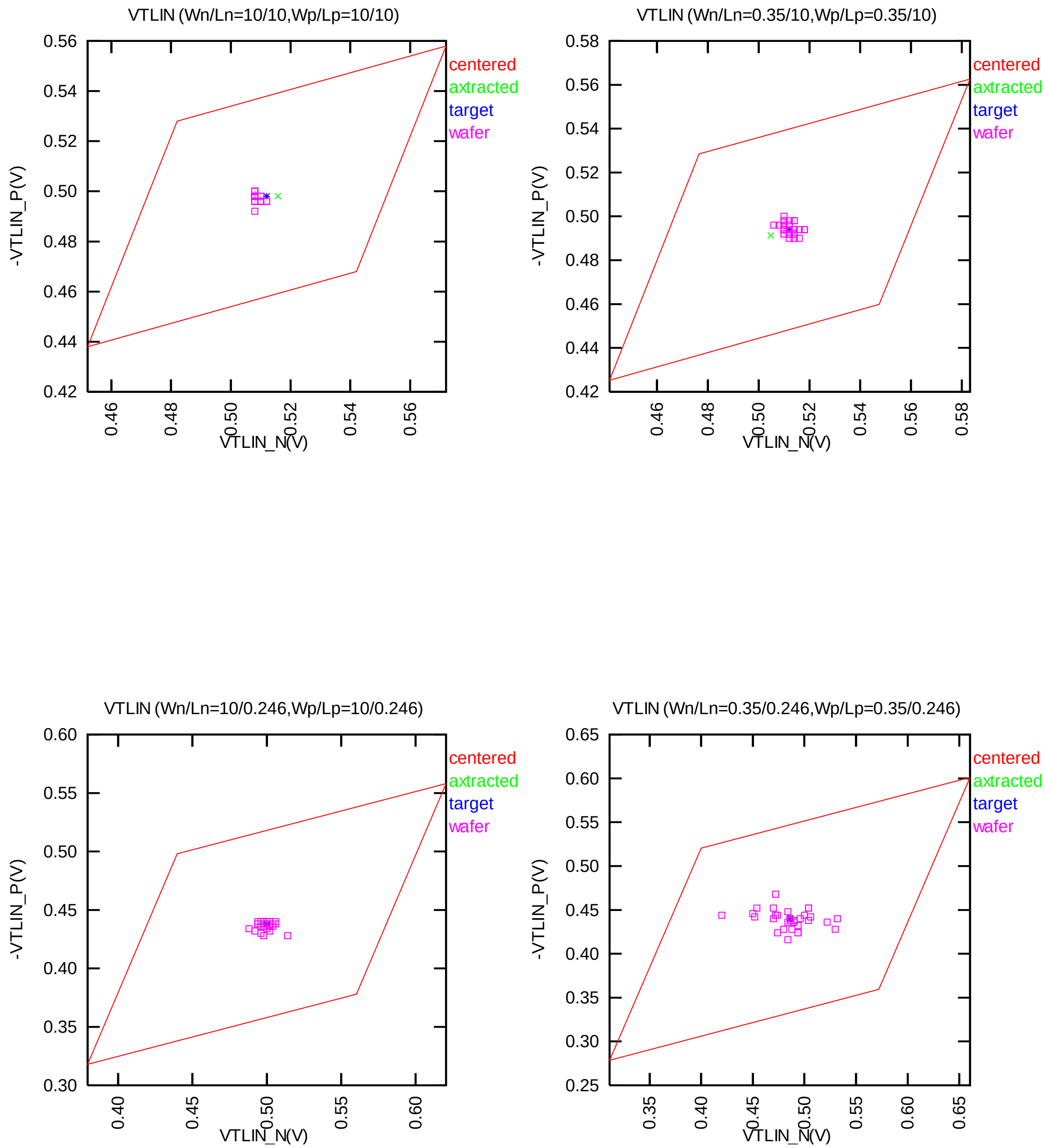

Figure 5-1 Corner plots of threshold voltage at low drain bias

Threshold voltage in saturation region V_{TSAT}

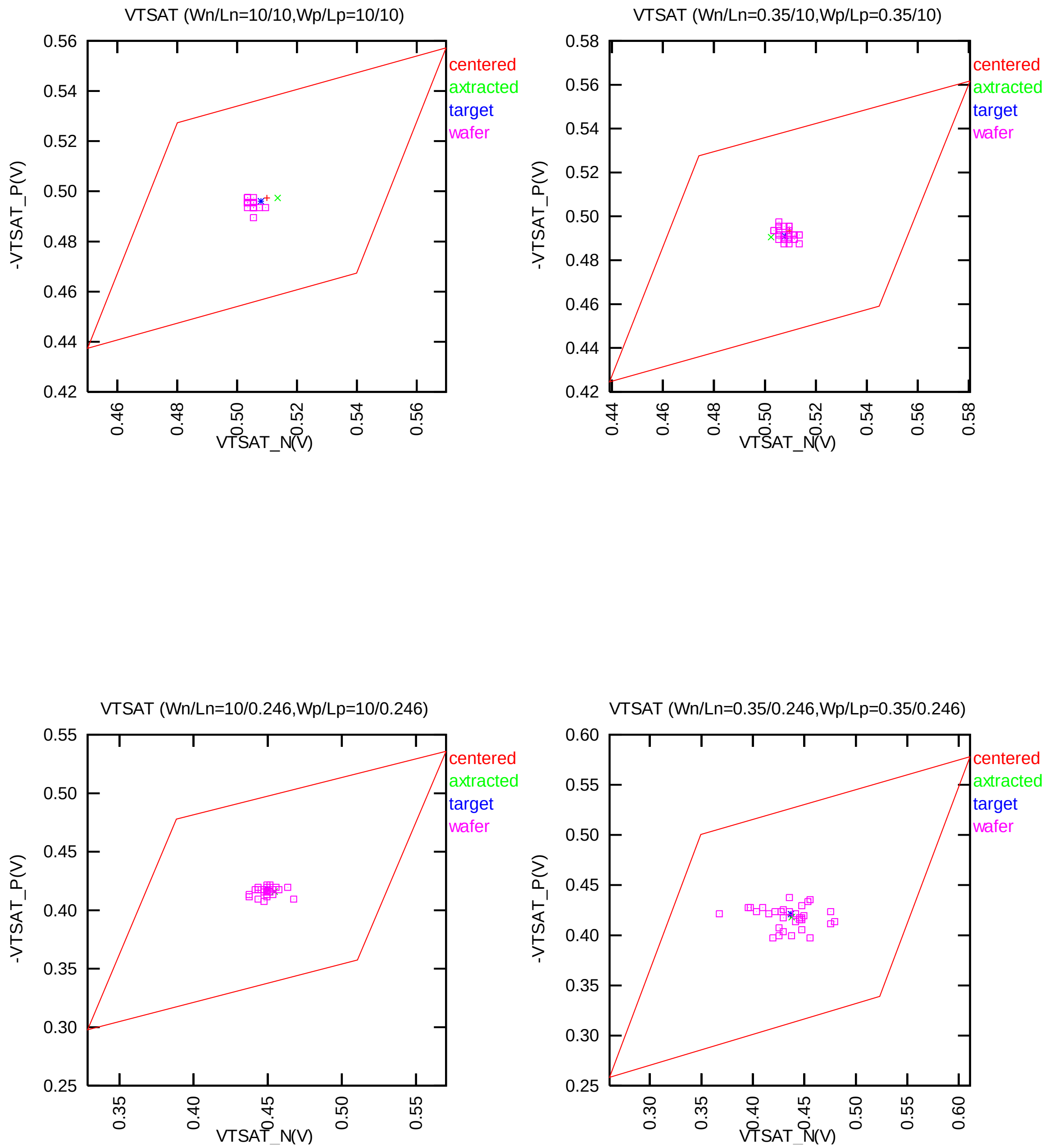

Figure 5-2 Corner plots of threshold voltage at high drain bias

Saturation current I_{ON}

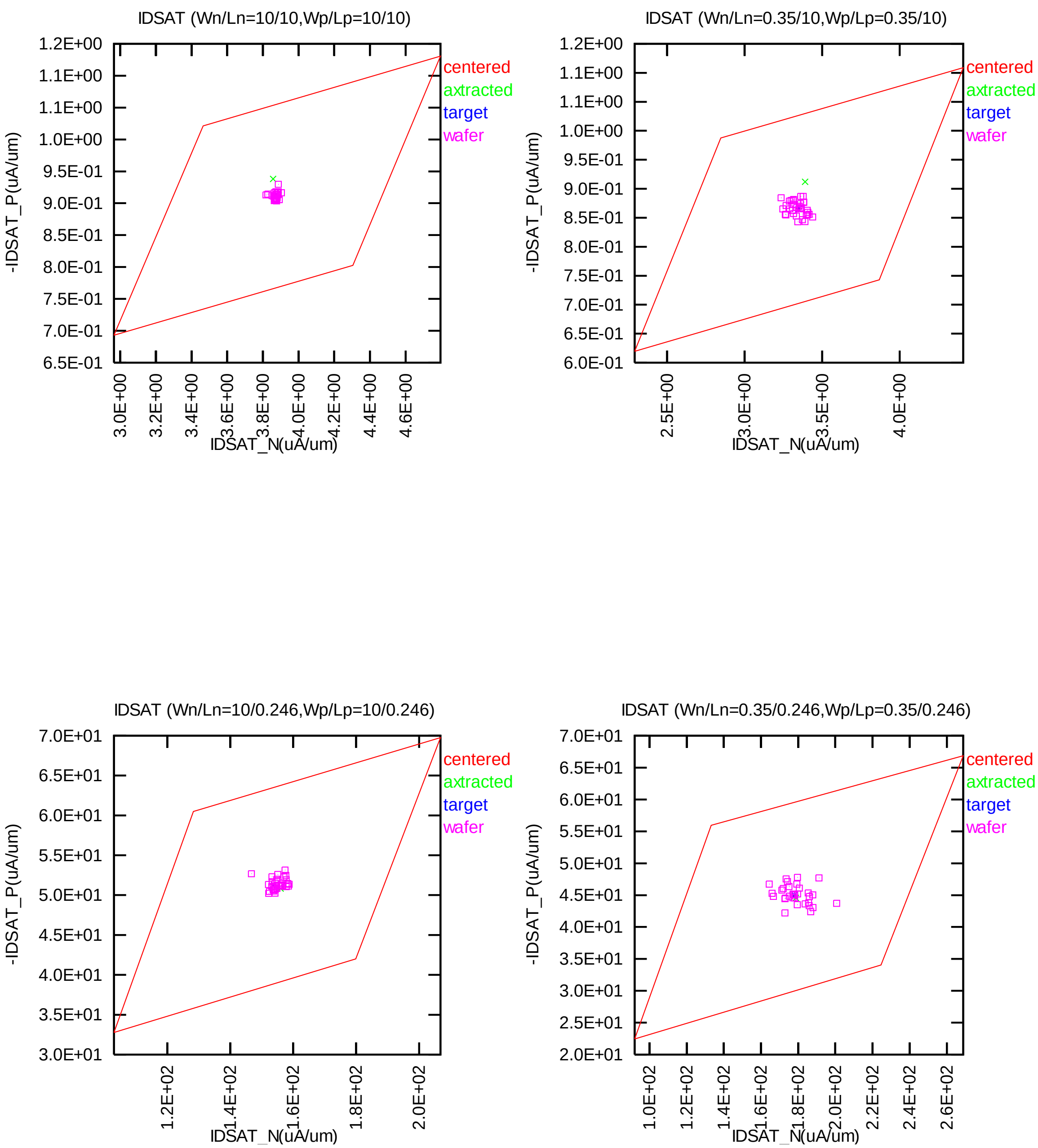

Figure 5-3 Corner plots of on-current

Off state current I_{OFF}

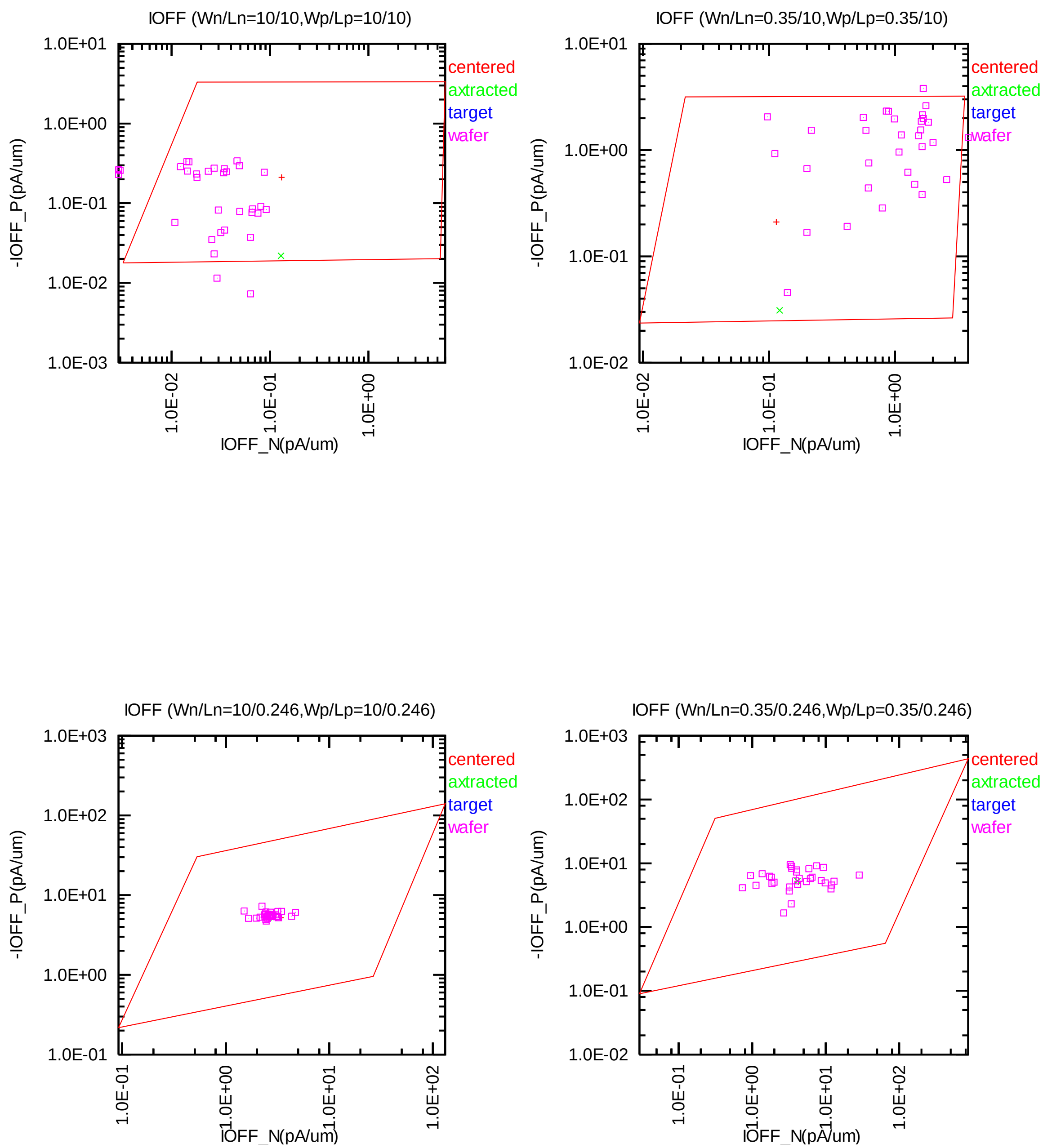

Figure 5-4 Corner plots of off-current

Saturation current I_{ON} v.s. Threshold voltage in linear region V_{TLIN} For bpw NMOS

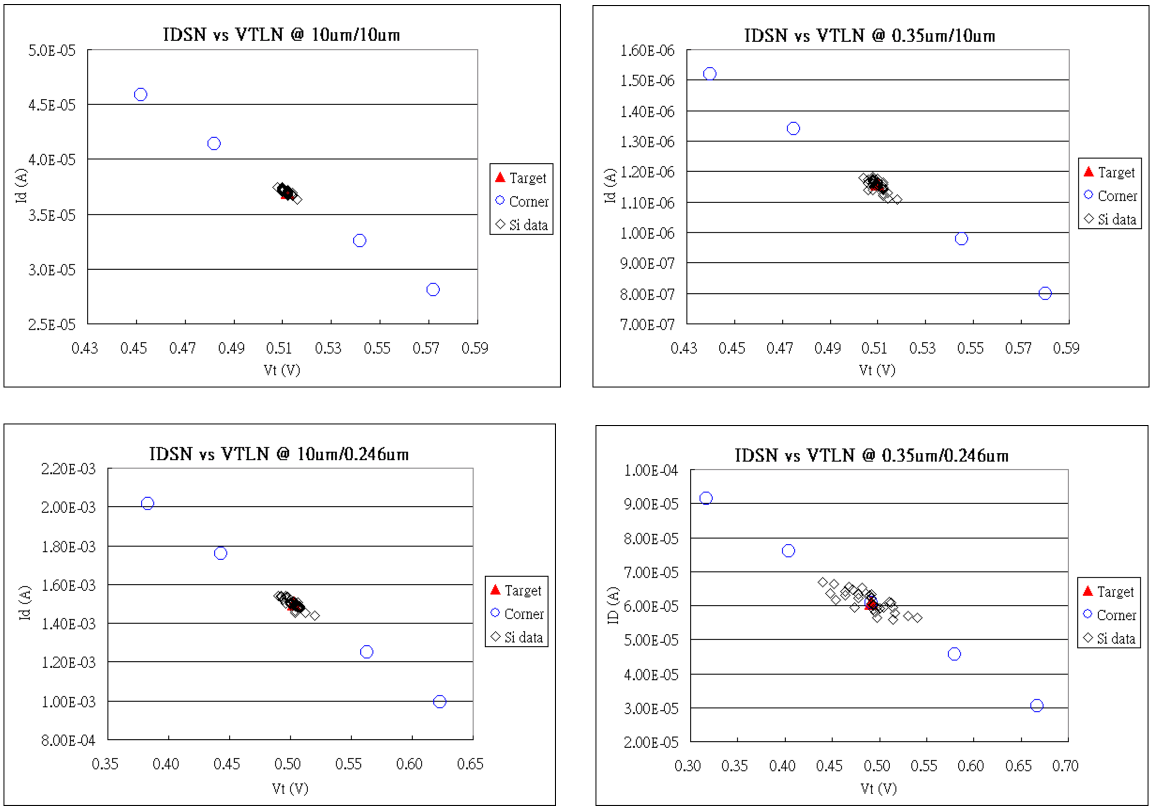

Figure 5-5 Corner plots of on-current v.s. threshold voltage at low drain bias

5.2 Device Performance Reference Tables

** The model simulation results in the following tables have already reflected silicon data that is sizing channel length back for 12nm. It means: The EDR value of size W/L: 10um/0.26um would be the same as the model card simulation result in size of W/L: 10um/0.234um (After 90% shrink without 12nm sizing channel length back).

Table 5-1 NMOS ID current at V_{GS}= 1.2V, V_{DS}= 0.6V and V_{GS}= 1.2V, V_{DS}= 1.2V.

NMOS	Temp. (℃)	-50 ℃		25 ℃		125 ℃	
IDSAT (μA)	VDS	0.6V	1.2V	0.6V	1.2V	0.6V	1.2V
TT	9/9	0.51	48.94	1.30	38.63	2.39	31.57
SS	9/9	0.13	36.71	0.51	29.55	1.26	24.64
FF	9/9	1.57	61.76	2.74	47.87	4.03	38.36
SNFP	9/9	0.26	43.33	0.84	34.55	1.78	28.53
FNSP	9/9	0.93	54.93	1.92	42.96	3.13	34.76
TT	9/0.234	45.79	1549.50	83.96	1384.80	127.16	1224.20
SS	9/0.234	2.89	1023.30	12.04	925.91	32.85	833.98
FF	9/0.234	229.56	2084.40	268.59	1860.70	305.17	1632.00
SNFP	9/0.234	12.93	1288.40	35.40	1153.00	69.31	1024.80
FNSP	9/0.234	117.86	1808.40	162.99	1618.40	207.35	1426.70
TT	0.35/9	0.02	1.60	0.05	1.30	0.09	1.10
SS	0.35/9	0.00	1.07	0.02	0.89	0.04	0.77
FF	0.35/9	0.07	2.16	0.11	1.72	0.16	1.41
SNFP	0.35/9	0.01	1.34	0.03	1.11	0.06	0.94
FNSP	0.35/9	0.04	1.87	0.08	1.51	0.12	1.25
TT	0.35/0.234	2.46	70.62	4.04	62.32	5.73	54.43
SS	0.35/0.234	0.04	35.85	0.20	32.21	0.68	28.90
FF	0.35/0.234	16.92	106.68	17.50	94.09	18.07	81.29
SNFP	0.35/0.234	0.40	52.65	1.15	46.63	2.33	41.10
FNSP	0.35/0.234	8.00	89.02	9.59	78.60	11.11	68.31

Table 5-2 NMOS ID current density at V_{GS}= 1.2V, V_{DS}= 0.6V and V_{GS}= 1.2V, V_{DS}= 1.2V.

NMOS	Temp. (℃)	-50 ℃		25 ℃		125 ℃	
IDSAT (μA/um)	VDS	0.6V	1.2V	0.6V	1.2V	0.6V	1.2V
TT	9/9	0.06	5.44	0.14	4.29	0.27	3.51
SS	9/9	0.01	4.08	0.06	3.28	0.14	2.74
FF	9/9	0.17	6.86	0.30	5.32	0.45	4.26
SNFP	9/9	0.03	4.81	0.09	3.84	0.20	3.17
FNSP	9/9	0.10	6.10	0.21	4.77	0.35	3.86
TT	9/0.234	5.09	172.17	9.33	153.87	14.13	136.02
SS	9/0.234	0.32	113.70	1.34	102.88	3.65	92.66
FF	9/0.234	25.51	231.60	29.84	206.74	33.91	181.33
SNFP	9/0.234	1.44	143.16	3.93	128.11	7.70	113.87
FNSP	9/0.234	13.10	200.93	18.11	179.82	23.04	158.52
TT	0.35/9	0.06	4.57	0.14	3.72	0.25	3.13
SS	0.35/9	0.01	3.04	0.05	2.54	0.12	2.19
FF	0.35/9	0.20	6.17	0.32	4.91	0.46	4.03
SNFP	0.35/9	0.03	3.84	0.08	3.16	0.18	2.69
FNSP	0.35/9	0.11	5.35	0.22	4.30	0.35	3.58
TT	0.35/0.234	7.03	201.77	11.53	178.05	16.37	155.50
SS	0.35/0.234	0.10	102.44	0.57	92.02	1.93	82.58
FF	0.35/0.234	48.35	304.80	49.99	268.82	51.62	232.26
SNFP	0.35/0.234	1.15	150.42	3.27	133.22	6.65	117.41
FNSP	0.35/0.234	22.85	254.33	27.41	224.58	31.73	195.17

Table 5-3 NMOS V_{TLIN}.

NMOS	Temp. (℃)	-50 ℃		25 ℃		125 ℃	
VTH (V)	VDS	0.1V	1.2V	0.1V	1.2V	0.1V	1.2V
TT	9/9	0.577	0.576	0.512	0.510	0.434	0.430
SS	9/9	0.636	0.635	0.572	0.570	0.496	0.492
FF	9/9	0.518	0.516	0.452	0.450	0.373	0.370
SNFP	9/9	0.607	0.605	0.542	0.540	0.465	0.461
FNSP	9/9	0.547	0.546	0.482	0.480	0.404	0.400
TT	9/0.234	0.556	0.507	0.500	0.450	0.435	0.380
SS	9/0.234	0.676	0.627	0.621	0.571	0.557	0.502
FF	9/0.234	0.436	0.386	0.379	0.328	0.313	0.259
SNFP	9/0.234	0.616	0.568	0.560	0.511	0.495	0.442
FNSP	9/0.234	0.496	0.446	0.440	0.388	0.374	0.319
TT	0.35/9	0.577	0.575	0.512	0.510	0.433	0.429
SS	0.35/9	0.647	0.645	0.584	0.581	0.508	0.502
FF	0.35/9	0.507	0.505	0.441	0.439	0.361	0.357
SNFP	0.35/9	0.612	0.610	0.548	0.545	0.470	0.465
FNSP	0.35/9	0.542	0.540	0.476	0.474	0.397	0.393
TT	0.35/0.234	0.541	0.491	0.486	0.436	0.421	0.370
SS	0.35/0.234	0.713	0.663	0.661	0.611	0.599	0.547
FF	0.35/0.234	0.367	0.316	0.311	0.261	0.245	0.193
SNFP	0.35/0.234	0.626	0.577	0.572	0.523	0.509	0.457
FNSP	0.35/0.234	0.455	0.404	0.400	0.349	0.334	0.282

Table 5-4 NMOS ID current at V_{GS}=0 and V_{DS}=0.1V or V_{DS}= 1.2V (off current).

NMOS	Temp. (℃)	-50 ℃		25 ℃		125 ℃	
IOFF (A)	VDS	0.1V	1.2V	0.1V	1.2V	0.1V	1.2V
TT	9/9	5.92E-14	1.19E-12	2.16E-13	1.19E-12	3.60E-11	3.80E-11
SS	9/9	2.22E-16	1.78E-15	2.86E-14	3.20E-14	1.04E-11	1.07E-11
FF	9/9	3.04E-12	6.15E-11	3.93E-12	5.39E-11	1.36E-10	1.82E-10
SNFP	9/9	6.11E-15	1.22E-13	6.87E-14	1.71E-13	1.84E-11	1.91E-11
FNSP	9/9	2.74E-12	5.53E-11	3.11E-12	4.81E-11	7.33E-11	1.13E-10
TT	9/0.234	6.79E-14	1.27E-12	6.89E-12	3.05E-11	1.13E-09	3.49E-09
SS	9/0.234	2.78E-16	3.55E-15	1.94E-13	8.20E-13	7.80E-11	2.35E-10
FF	9/0.234	4.05E-12	6.98E-11	2.61E-10	1.19E-09	1.70E-08	5.28E-08
SNFP	9/0.234	6.77E-15	1.28E-13	1.10E-12	4.72E-12	2.87E-10	8.71E-10
FNSP	9/0.234	2.91E-12	5.76E-11	4.57E-11	2.39E-10	4.47E-09	1.40E-08
TT	0.35/9	2.01E-15	3.88E-14	8.86E-15	4.08E-14	1.56E-12	1.63E-12
SS	0.35/9	2.01E-16	2.39E-15	1.14E-15	3.36E-15	3.90E-13	4.07E-13
FF	0.35/9	6.93E-14	1.40E-12	1.23E-13	1.26E-12	7.23E-12	8.40E-12
SNFP	0.35/9	3.71E-16	5.83E-15	2.74E-15	7.83E-15	7.32E-13	7.63E-13
FNSP	0.35/9	5.69E-14	1.15E-12	7.64E-14	1.01E-12	3.45E-12	4.33E-12
TT	0.35/0.234	3.54E-15	6.41E-14	3.54E-13	1.49E-12	5.48E-11	1.61E-10
SS	0.35/0.234	2.01E-16	2.39E-15	2.18E-15	1.02E-14	1.14E-12	3.16E-12
FF	0.35/0.234	6.32E-13	6.19E-12	7.13E-11	3.03E-10	2.85E-09	8.40E-09
SNFP	0.35/0.234	4.86E-16	7.97E-15	2.60E-14	1.10E-13	7.65E-12	2.20E-11
FNSP	0.35/0.234	1.38E-13	2.61E-12	5.00E-12	2.27E-11	3.92E-10	1.17E-09

*gmindc=1e-15

Table 5-5 bpw NMOS ID current at V_{GS}= 1.2V, V_{DS}= 0.6V and V_{GS}= 1.2V, V_{DS}= 1.2V.

NMOS	Temp. (℃)	-50 ℃		25 ℃		125 ℃	
IDSAT (μA)	VDS	0.6V	1.2V	0.6V	1.2V	0.6V	1.2V
TT	9/9	0.46	45.22	1.21	36.90	2.27	30.86
SS	9/9	0.12	33.34	0.49	28.00	1.21	24.21
FF	9/9	1.36	57.61	2.50	45.82	3.78	37.19
SNFP	9/9	0.24	39.18	0.79	32.45	1.68	27.59
FNSP	9/9	0.81	51.38	1.77	41.36	2.96	34.05
TT	9/0.234	39.95	1483.60	77.59	1353.30	118.41	1189.60
SS	9/0.234	2.58	1015.50	11.26	892.03	29.82	762.71
FF	9/0.234	209.60	1946.80	255.43	1814.60	294.46	1625.20
SNFP	9/0.234	11.31	1251.80	32.76	1122.60	63.69	973.65
FNSP	9/0.234	104.22	1713.60	151.58	1583.90	195.63	1408.50
TT	0.35/9	0.02	1.48	0.05	1.29	0.09	1.15
SS	0.35/9	0.00	0.98	0.02	0.89	0.04	0.83
FF	0.35/9	0.06	2.00	0.10	1.69	0.16	1.46
SNFP	0.35/9	0.01	1.23	0.03	1.09	0.06	0.99
FNSP	0.35/9	0.03	1.74	0.07	1.49	0.12	1.31
TT	0.35/0.234	1.91	67.26	3.58	60.90	5.47	53.43
SS	0.35/0.234	0.03	36.00	0.19	30.45	0.66	25.60
FF	0.35/0.234	14.57	97.95	16.31	91.34	17.90	82.03
SNFP	0.35/0.234	0.30	51.95	1.01	45.67	2.19	39.23
FNSP	0.35/0.234	6.58	82.43	8.70	76.12	10.78	67.88

Table 5-6 bpw NMOS ID current density at V_{GS}= 1.2V, V_{DS}= 0.6V and V_{GS}= 1.2V, V_{DS}= 1.2V.

NMOS	Temp. (℃)	-50 ℃		25 ℃		125 ℃	
IDSAT (μA/um)	VDS	0.6V	1.2V	0.6V	1.2V	0.6V	1.2V
TT	9/9	0.05	5.02	0.13	4.10	0.25	3.43
SS	9/9	0.01	3.70	0.05	3.11	0.13	2.69
FF	9/9	0.15	6.40	0.28	5.09	0.42	4.13
SNFP	9/9	0.03	4.35	0.09	3.61	0.19	3.07
FNSP	9/9	0.09	5.71	0.20	4.60	0.33	3.78
TT	9/0.234	4.44	164.84	8.62	150.37	13.16	132.18
SS	9/0.234	0.29	112.83	1.25	99.11	3.31	84.75
FF	9/0.234	23.29	216.31	28.38	201.62	32.72	180.58
SNFP	9/0.234	1.26	139.09	3.64	124.73	7.08	108.18
FNSP	9/0.234	11.58	190.40	16.84	175.99	21.74	156.50
TT	0.35/9	0.05	4.22	0.13	3.68	0.25	3.30
SS	0.35/9	0.01	2.81	0.05	2.54	0.12	2.36
FF	0.35/9	0.17	5.71	0.30	4.83	0.45	4.18
SNFP	0.35/9	0.02	3.50	0.08	3.11	0.18	2.84
FNSP	0.35/9	0.10	4.96	0.20	4.26	0.34	3.74
TT	0.35/0.234	5.45	192.17	10.24	173.99	15.63	152.66
SS	0.35/0.234	0.09	102.85	0.55	86.99	1.88	73.14
FF	0.35/0.234	41.63	279.86	46.59	260.98	51.14	234.38
SNFP	0.35/0.234	0.86	148.42	2.89	130.49	6.27	112.09
FNSP	0.35/0.234	18.79	235.51	24.86	217.48	30.79	193.94

Table 5-7 bpw NMOS V_{TLIN} and V_{TSAT}.

NMOS	Temp. (℃)	-50 ℃		25 ℃		125 ℃	
VTH (V)	VDS	0.1V	1.2V	0.1V	1.2V	0.1V	1.2V
TT	9/9	0.582	0.580	0.512	0.510	0.428	0.424
SS	9/9	0.640	0.639	0.572	0.570	0.491	0.486
FF	9/9	0.523	0.521	0.452	0.450	0.366	0.362
SNFP	9/9	0.611	0.610	0.542	0.540	0.460	0.455
FNSP	9/9	0.552	0.551	0.482	0.480	0.397	0.393
TT	9/0.234	0.562	0.520	0.503	0.458	0.434	0.384
SS	9/0.234	0.680	0.637	0.624	0.578	0.559	0.508
FF	9/0.234	0.444	0.401	0.383	0.337	0.311	0.262
SNFP	9/0.234	0.621	0.579	0.563	0.518	0.496	0.446
FNSP	9/0.234	0.503	0.461	0.443	0.398	0.372	0.322
TT	0.35/9	0.580	0.579	0.510	0.507	0.424	0.420
SS	0.35/9	0.648	0.646	0.580	0.577	0.498	0.492
FF	0.35/9	0.511	0.510	0.440	0.437	0.352	0.348
SNFP	0.35/9	0.615	0.613	0.545	0.542	0.461	0.456
FNSP	0.35/9	0.546	0.545	0.475	0.472	0.388	0.383
TT	0.35/0.234	0.554	0.507	0.492	0.442	0.418	0.364
SS	0.35/0.234	0.722	0.676	0.667	0.617	0.601	0.547
FF	0.35/0.234	0.382	0.335	0.317	0.267	0.239	0.186
SNFP	0.35/0.234	0.639	0.592	0.579	0.530	0.509	0.455
FNSP	0.35/0.234	0.468	0.422	0.404	0.355	0.328	0.275

Table 5-8 bpw NMOS ID current at V_{GS}=0 and V_{DS}=0.1V or V_{DS}= 1.2V (off current).

NMOS	Temp. (℃)	-50 ℃		25 ℃		125 ℃	
IOFF (A)	VDS	0.1V	1.2V	0.1V	1.2V	0.1V	1.2V
TT	9/9	3.33E-16	3.55E-15	1.67E-13	2.50E-13	4.30E-11	4.84E-11
SS	9/9	2.22E-16	2.66E-15	3.04E-14	3.29E-14	1.23E-11	1.27E-11
FF	9/9	1.28E-14	2.39E-13	3.89E-12	5.30E-11	3.32E-10	2.90E-09
SNFP	9/9	2.22E-16	2.66E-15	6.68E-14	7.82E-14	2.20E-11	2.30E-11
FNSP	9/9	9.10E-15	1.77E-13	2.58E-12	3.87E-11	2.11E-10	2.10E-09
TT	9/0.234	5.27E-15	3.02E-14	5.80E-12	2.15E-11	1.11E-09	3.09E-09
SS	9/0.234	2.22E-16	3.55E-15	1.78E-13	6.31E-13	7.64E-11	2.04E-10
FF	9/0.234	6.17E-13	3.55E-12	2.20E-10	8.69E-10	1.73E-08	5.06E-08
SNFP	9/0.234	6.11E-16	4.44E-15	9.44E-13	3.45E-12	2.77E-10	7.63E-10
FNSP	9/0.234	6.60E-14	4.86E-13	3.86E-11	1.74E-10	4.60E-09	1.45E-08
TT	0.35/9	2.08E-16	2.44E-15	8.01E-15	1.20E-14	2.00E-12	2.14E-12
SS	0.35/9	2.01E-16	2.41E-15	1.34E-15	3.55E-15	5.05E-13	5.23E-13
FF	0.35/9	5.52E-16	8.13E-15	1.31E-13	1.32E-12	1.33E-11	7.54E-11
SNFP	0.35/9	1.98E-16	2.39E-15	2.97E-15	5.38E-15	9.43E-13	9.81E-13
FNSP	0.35/9	3.89E-16	5.83E-15	6.53E-14	7.85E-13	6.83E-12	4.44E-11
TT	0.35/0.234	4.02E-16	3.66E-15	2.50E-13	1.07E-12	5.19E-11	1.61E-10
SS	0.35/0.234	2.01E-16	2.41E-15	1.65E-15	8.22E-15	1.02E-12	2.87E-12
FF	0.35/0.234	2.63E-13	1.73E-12	5.72E-11	2.47E-10	3.05E-09	9.40E-09
SNFP	0.35/0.234	2.08E-16	2.41E-15	1.74E-14	7.43E-14	6.66E-12	2.03E-11
FNSP	0.35/0.234	7.83E-15	5.62E-14	3.91E-12	1.79E-11	4.15E-10	1.35E-09

*gmindc=1e-15

Table 5-9 PMOS ID current at V_{GS}= -1.2V, V_{DS}= -0.6V and V_{GS}= -1.2V, V_{DS}= -1.2V.

PMOS	Temp. (℃)	-50 ℃		25 ℃		125 ℃	
-IDSAT (μA)	-VDS	0.6V	1.2V	0.6V	1.2V	0.6V	1.2V
TT	9/9	0.11	10.19	0.37	9.11	0.80	8.38
SS	9/9	0.03	7.52	0.15	6.92	0.42	6.57
FF	9/9	0.35	12.96	0.76	11.30	1.34	10.09
SNFP	9/9	0.21	11.57	0.54	10.21	1.04	9.24
FNSP	9/9	0.06	8.83	0.24	8.02	0.59	7.49
TT	9/0.234	14.72	481.68	32.74	460.76	58.96	448.93
SS	9/0.234	1.41	292.31	6.99	294.42	20.49	303.27
FF	9/0.234	61.72	685.59	89.07	627.29	121.84	582.34
SNFP	9/0.234	33.28	582.51	57.21	544.00	87.96	516.35
FNSP	9/0.234	5.11	384.44	16.33	377.61	36.36	378.70
TT	0.35/9	0.01	0.38	0.01	0.34	0.03	0.31
SS	0.35/9	0.00	0.26	0.01	0.24	0.01	0.23
FF	0.35/9	0.02	0.50	0.03	0.43	0.05	0.38
SNFP	0.35/9	0.01	0.44	0.02	0.38	0.04	0.34
FNSP	0.35/9	0.00	0.32	0.01	0.29	0.02	0.27
TT	0.35/0.234	0.57	17.38	1.15	15.76	1.97	14.64
SS	0.35/0.234	0.02	8.18	0.13	7.86	0.44	7.71
FF	0.35/0.234	3.14	27.07	3.94	23.40	4.94	20.67
SNFP	0.35/0.234	1.55	22.16	2.33	19.58	3.30	17.70
FNSP	0.35/0.234	0.14	12.74	0.45	11.91	1.02	11.43

Table 5-10 PMOS ID current density at V_{GS}= -1.2V, V_{DS}= -0.6V and V_{GS}= -1.2V, V_{DS}= -1.2V.

PMOS	Temp. (℃)	-50 ℃		25 ℃		125 ℃	
-IDSAT (μA/um)	-VDS	0.6V	1.2V	0.6V	1.2V	0.6V	1.2V
TT	9/9	0.01	1.13	0.04	1.01	0.09	0.93
SS	9/9	0.00	0.84	0.02	0.77	0.05	0.73
FF	9/9	0.04	1.44	0.08	1.26	0.15	1.12
SNFP	9/9	0.02	1.29	0.06	1.13	0.12	1.03
FNSP	9/9	0.01	0.98	0.03	0.89	0.07	0.83
TT	9/0.234	1.64	53.52	3.64	51.20	6.55	49.88
SS	9/0.234	0.16	32.48	0.78	32.71	2.28	33.70
FF	9/0.234	6.86	76.18	9.90	69.70	13.54	64.70
SNFP	9/0.234	3.70	64.72	6.36	60.44	9.77	57.37
FNSP	9/0.234	0.57	42.72	1.81	41.96	4.04	42.08
TT	0.35/9	0.01	1.09	0.04	0.96	0.08	0.87
SS	0.35/9	0.00	0.75	0.01	0.69	0.04	0.64
FF	0.35/9	0.05	1.43	0.09	1.23	0.15	1.08
SNFP	0.35/9	0.03	1.26	0.06	1.10	0.11	0.98
FNSP	0.35/9	0.01	0.92	0.03	0.82	0.06	0.76
TT	0.35/0.234	1.63	49.66	3.30	45.02	5.62	41.84
SS	0.35/0.234	0.06	23.38	0.38	22.45	1.26	22.03
FF	0.35/0.234	8.98	77.33	11.27	66.87	14.10	59.04
SNFP	0.35/0.234	4.43	63.32	6.65	55.94	9.42	50.56
FNSP	0.35/0.234	0.39	36.41	1.29	34.04	2.91	32.65

Table 5-11 PMOS V_{TLIN}.

PMOS	Temp. (℃)	-50 ℃		25 ℃		125 ℃	
-VTH (V)	-VDS	0.1V	1.2V	0.1V	1.2V	0.1V	1.2V
TT	9/9	0.578	0.577	0.498	0.497	0.398	0.396
SS	9/9	0.636	0.636	0.558	0.557	0.460	0.459
FF	9/9	0.519	0.519	0.438	0.437	0.335	0.334
SNFP	9/9	0.549	0.548	0.468	0.467	0.366	0.365
FNSP	9/9	0.607	0.607	0.528	0.527	0.429	0.428
TT	9/0.234	0.520	0.503	0.438	0.418	0.336	0.310
SS	9/0.234	0.638	0.619	0.558	0.536	0.459	0.432
FF	9/0.234	0.401	0.384	0.318	0.298	0.213	0.187
SNFP	9/0.234	0.461	0.443	0.378	0.357	0.274	0.248
FNSP	9/0.234	0.579	0.562	0.498	0.478	0.397	0.372
TT	0.35/9	0.570	0.569	0.494	0.493	0.400	0.398
SS	0.35/9	0.637	0.636	0.563	0.562	0.471	0.470
FF	0.35/9	0.502	0.502	0.425	0.424	0.328	0.327
SNFP	0.35/9	0.536	0.536	0.460	0.459	0.364	0.362
FNSP	0.35/9	0.603	0.603	0.529	0.528	0.435	0.434
TT	0.35/0.234	0.518	0.501	0.440	0.420	0.341	0.315
SS	0.35/0.234	0.676	0.656	0.601	0.578	0.508	0.479
FF	0.35/0.234	0.360	0.342	0.278	0.258	0.173	0.147
SNFP	0.35/0.234	0.439	0.422	0.359	0.339	0.257	0.231
FNSP	0.35/0.234	0.597	0.580	0.521	0.500	0.424	0.399

Table 5-12 PMOS ID current at V_{GS}=0 and V_{DS}=-0.1V or V_{DS}= -1.2V (off current).

PMOS	Temp. (℃)	-50 ℃		25 ℃		125 ℃	
-IOFF (A)	-VDS	0.1V	1.2V	0.1V	1.2V	0.1V	1.2V
TT	9/9	9.59E-14	2.17E-12	1.98E-13	1.91E-12	2.70E-11	2.88E-11
SS	9/9	7.55E-15	1.69E-13	2.84E-14	1.63E-13	8.93E-12	9.25E-12
FF	9/9	1.57E-12	3.55E-11	2.08E-12	3.01E-11	9.30E-11	1.17E-10
SNFP	9/9	1.57E-12	3.55E-11	1.79E-12	2.98E-11	5.16E-11	7.49E-11
FNSP	9/9	7.55E-15	1.69E-13	5.21E-14	1.87E-13	1.49E-11	1.53E-11
TT	9/0.234	1.37E-13	2.23E-12	3.04E-11	4.68E-11	4.31E-09	6.34E-09
SS	9/0.234	8.22E-15	1.70E-13	1.20E-12	1.95E-12	3.68E-10	5.43E-10
FF	9/0.234	4.80E-12	4.06E-11	8.20E-10	1.27E-09	5.00E-08	7.50E-08
SNFP	9/0.234	1.94E-12	3.61E-11	1.62E-10	2.74E-10	1.50E-08	2.25E-08
FNSP	9/0.234	1.22E-14	1.76E-13	5.75E-12	8.59E-12	1.23E-09	1.78E-09
TT	0.35/9	3.75E-15	8.28E-14	8.47E-15	7.42E-14	1.13E-12	1.20E-12
SS	0.35/9	4.75E-16	8.60E-15	1.25E-15	8.38E-15	3.78E-13	3.97E-13
FF	0.35/9	5.84E-14	1.32E-12	8.93E-14	1.13E-12	4.41E-12	5.31E-12
SNFP	0.35/9	5.84E-14	1.32E-12	7.02E-14	1.11E-12	2.26E-12	3.14E-12
FNSP	0.35/9	4.75E-16	8.60E-15	2.30E-15	9.38E-15	6.10E-13	6.32E-13
TT	0.35/0.234	5.81E-15	8.97E-14	1.22E-12	1.81E-12	1.61E-10	2.30E-10
SS	0.35/0.234	4.96E-16	8.94E-15	1.66E-14	3.14E-14	5.96E-12	8.49E-12
FF	0.35/0.234	7.16E-13	2.38E-12	1.02E-10	1.52E-10	4.43E-09	6.56E-09
SNFP	0.35/0.234	9.69E-14	1.44E-12	1.14E-11	1.77E-11	8.70E-10	1.27E-09
FNSP	0.35/0.234	5.90E-16	9.05E-15	1.32E-13	1.94E-13	2.97E-11	4.16E-11

*gmindc=1e-15

6 Shallow Trench Isolation (STI) or LOD Stress Effect

6.1 Simulation Results for NMOS at SA=SB=0.318 um (SAREF=2.2um)

Note: The LOD stress effect model was based on very limit and early silicon data. SB is equal to SA, SD=0 in all following plots and tables. The following table show the channel length and width dependence of linear and saturation threshold voltage extracted at ID =300nA*(W_{DES}*0.9)/ (L_{DES}*0.9+0.012μm). The dimension is after 90% shrink and 12nm sizing channel length back.

The simulation results for the NMOS characterization are summarized in the table below.

Table 6-1 Simulation results for NMOS

All parameters are given for T=25 ^{o}C ,V_{BS}=0 V unless specified otherwise.

Parameter	Condition	Unit	Simulation Value
Long channel device (W/L= 10 μ m / 10 μ m)			
V_{TLIN}	V_{DS} = 0.1 V	V	0.513
I_{ON}	V_{DS}=V_{GS}= 1.2 V	μA/ μm	3.84
Wide and short channel device (W/L= 10 μ m / 0.246 μ m)			
V_{TLIN}	V_{DS} = 0.1 V	V	0.521
V_{TSAT}	V_{DS} = 1.2 V	V	0.473
I_{ON}	V_{DS}=V_{GS}= 1.2 V	μA/μm	146.53
I_{OFF}	V_{DS}= 1.2 V, V_{GS}=0 V	A/μm	1.7E-12*
Narrow and long channel device (W/L= 0.35 μ m / 10 μ m)			
V_{TLIN}	V_{DS} = 0.1 V	V	0.513
V_{TSAT}	V_{DS}= 1.2 V	V	0.510
I_{ON}	V_{DS}=V_{GS}= 1.2 V	μA/μm	3.32
Narrow and short channel device (W/L= 0.35 μm / 0.246 μm)			
V_{TLIN}	V_{DS} = 0.1 V	V	0.507
V_{TSAT}	V_{DS}= 1.2 V	V	0.460
I_{ON}	V_{DS}=V_{GS}= 1.2 V	μA/μm	169.97
I_{OFF}	V_{DS}= 1.2 V, V_{GS}=0 V	A/μm	2.1E-12*

*gmindc=1e-15

6.1.1 Model verification plots for NMOS

The L and W in the plots refer to LDES and WDES respectively.

V_T vs. SA

The following figures show the SA dependence of linear and saturation threshold voltage extracted at ID=300nA*WDES*0.9/(LDES*0.9'+12nm) at 25℃. The dimension is after 90% shrink and 12nm sizing channel length back.

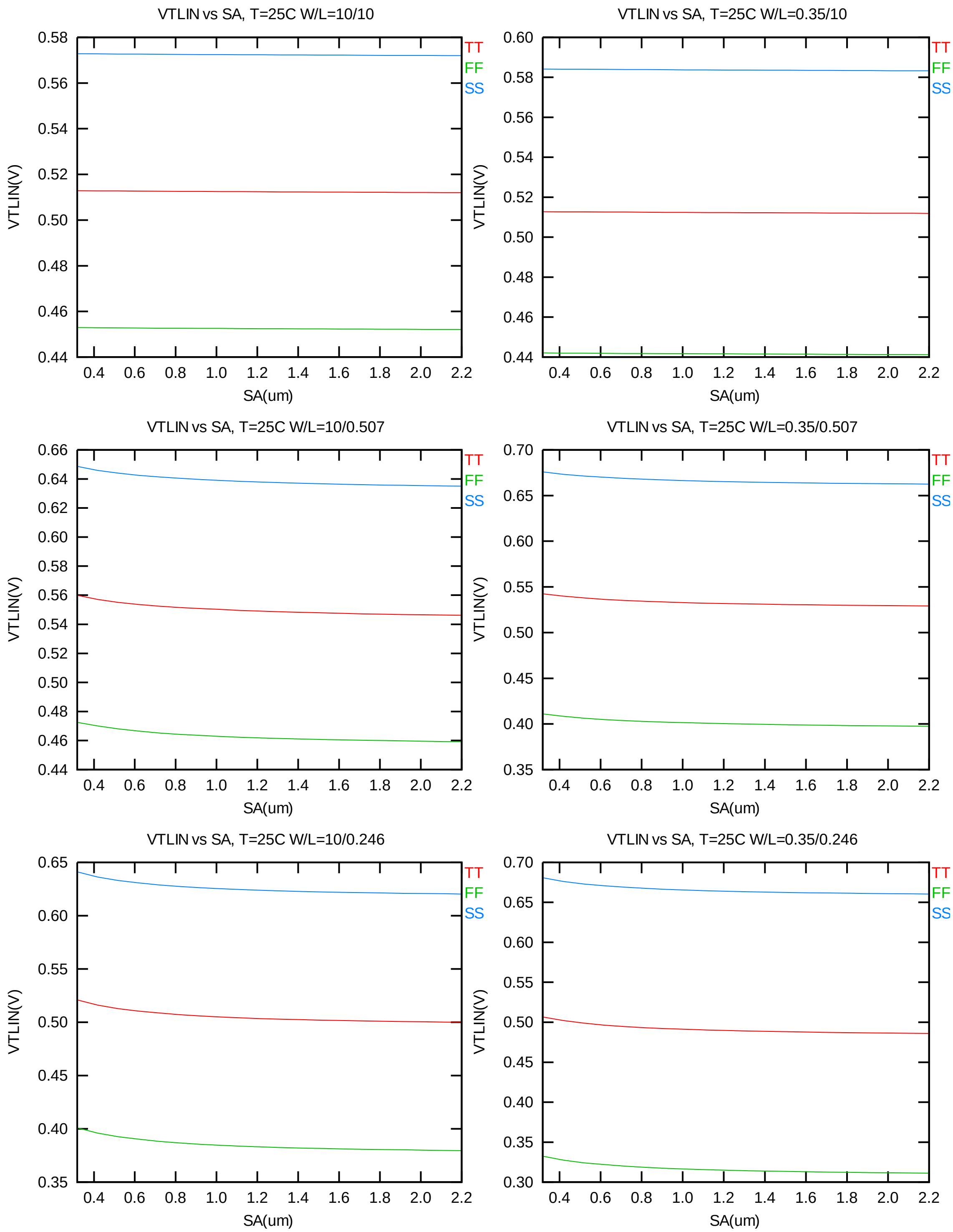

Figure 6-1 V_T vs. SA for low drain bias (V_{DS} = 0.1 V) for NMOS

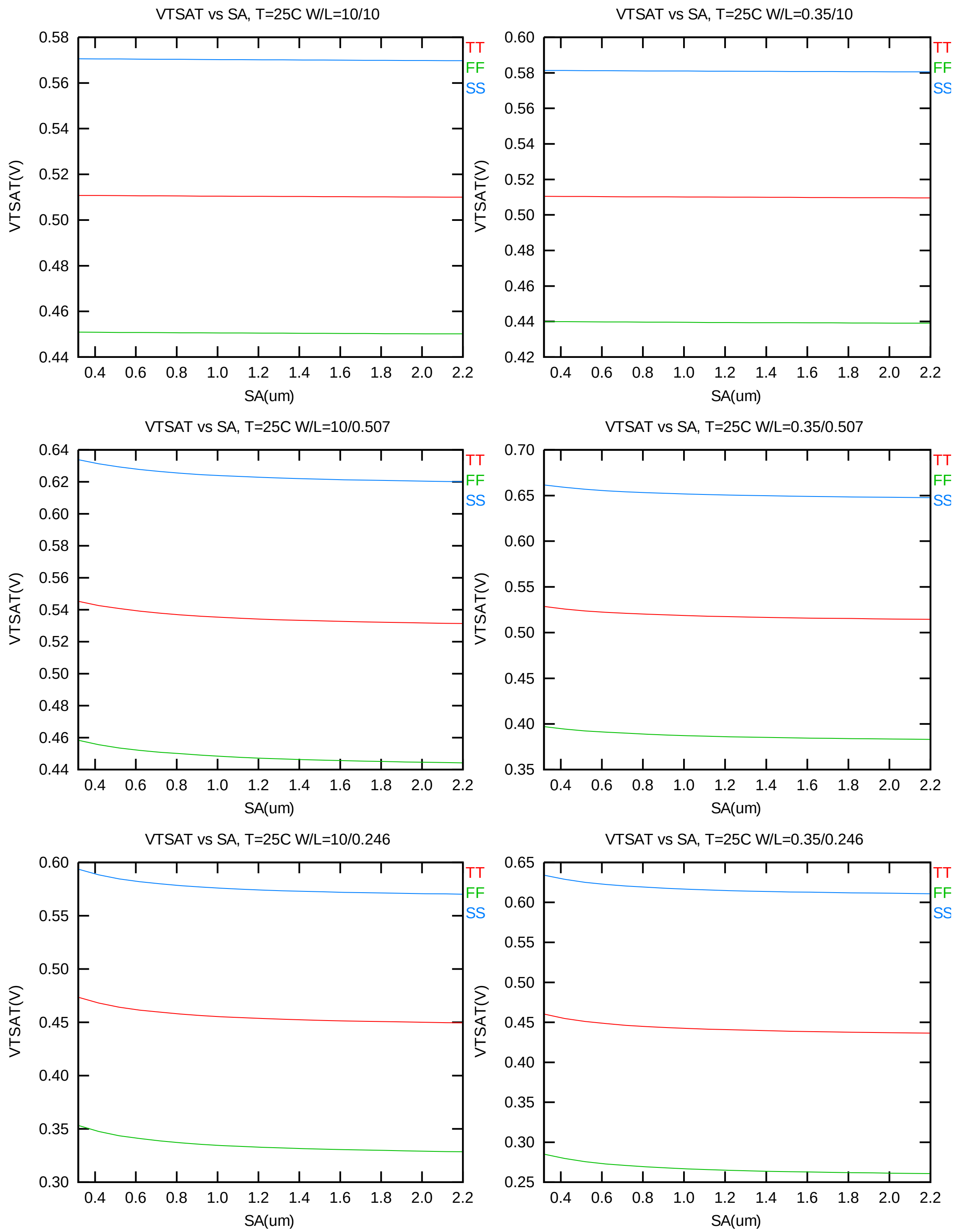

Figure 6-2 V_T vs. SA for high drain bias (V_{DS} = 1.2V) for NMOS

I_{ON} vs. SA

The following plots show the SA dependence of I_{ON} extracted at V_{DS}=V_{GS}=1.2V for V_{BS}=0 at 25C.

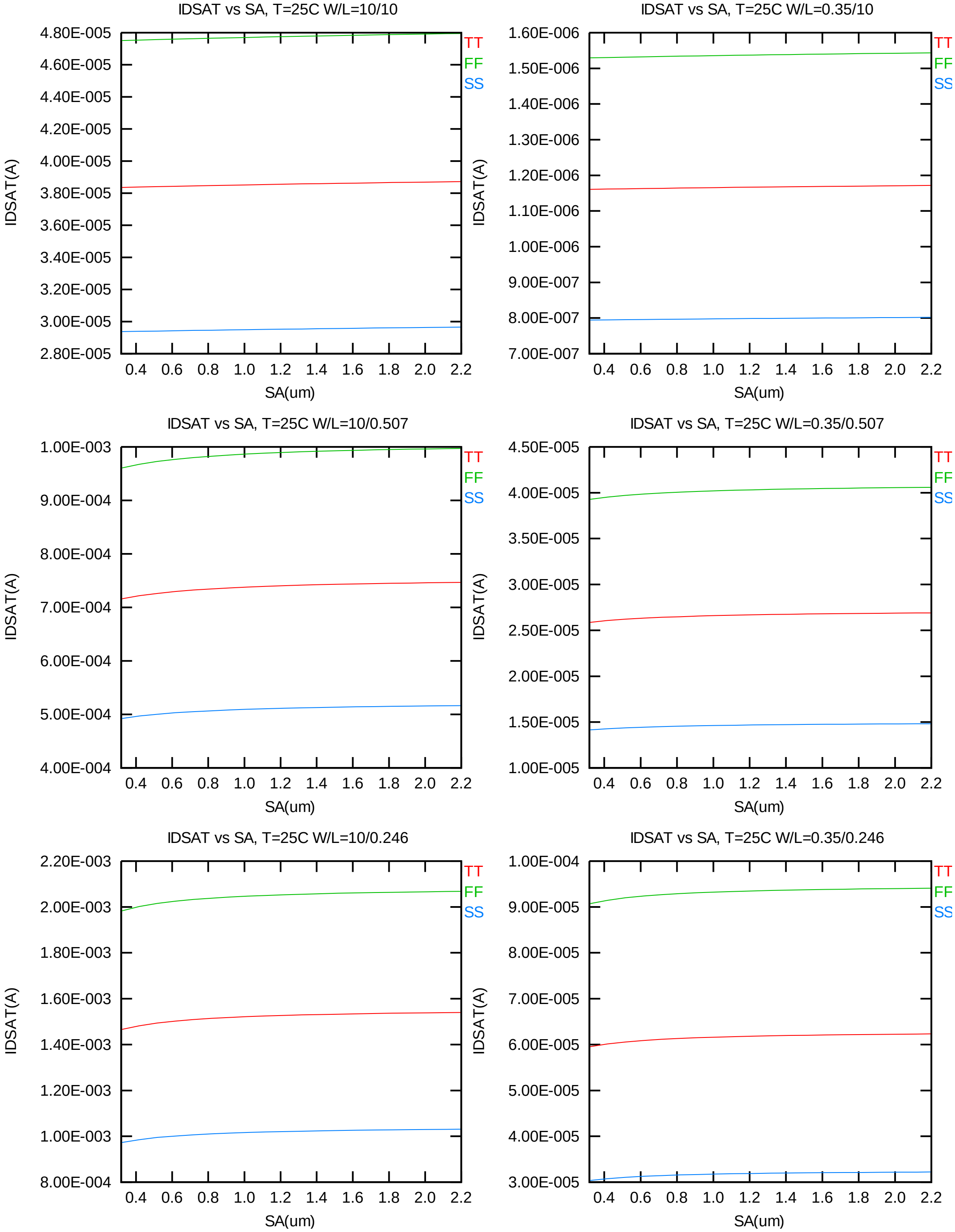

Figure 6-3 Saturation current I_{ON} vs. SA for NMOS

6.2 Simulation Results for PMOS at SA=0.318 um (SAREF=2.2um)

Note: The LOD stress effect model was based on very limit and early silicon data. SB is equal to SA, SD=0 in all following plots and tables. The following table show the channel length and width dependence of linear and saturation threshold voltage extracted at ID =-70nA*(W_{DES}*0.9)/ (L_{DES}*0.9+0.012μm). The dimension is after 90% shrink and 12nm sizing channel length back.

The simulation results for the PMOS characterization are summarized in the table below.

Table 6-2 Simulation results for PMOS

All parameters are given for T=25 °C , V_{BS}=0 V unless specified otherwise.

Parameter	Condition	Unit	Simulation Value
Long channel device (W/L= 10 μ m / 10 μ m)			
V_{TLIN}	V_{DS} = -0.1 V	V	-0.498
I_{ON}	V_{DS}=V_{GS}= -1.2 V	μA/ μm	-0.92
Wide and short channel device (W/L= 10 μ m / 0.246 μ m)			
V_{TLIN}	V_{DS} = -0.1 V	V	-0.439
V_{TSAT}	V_{DS} = -1.2 V	V	-0.419
I_{ON}	V_{DS}=V_{GS}= -1.2 V	μA/μm	-54.60
I_{OFF}	V_{DS}= -1.2 V, V_{GS}=0 V	A/μm	-4.9E-12*
Narrow and long channel device (W/L= 0.35 μ m / 10 μ m)			
V_{TLIN}	V_{DS} = -0.1 V	V	-0.494
V_{TSAT}	V_{DS}= -1.2 V	V	-0.493
I_{ON}	V_{DS}=V_{GS}= -1.2 V	μA/μm	-0.87
Narrow and short channel device (W/L= 0.35 μm / 0.246 μm)			
V_{TLIN}	V_{DS} = -0.1 V	V	-0.441
V_{TSAT}	V_{DS}= -1.2 V	V	-0.421
I_{ON}	V_{DS}=V_{GS}= -1.2 V	μA/μm	-47.79
I_{OFF}	V_{DS}= -1.2 V, V_{GS}=0 V	A/μm	-4.9E-12 *

*gmindc=1e-15

6.2.1 Model verification plots for PMOS

The L and W in the plots refer to LDES and WDES respectively

V_T vs. SA

The following figures show the SA dependence of linear and saturation threshold voltage extracted at ID=-70nA*WDES*0.9/(LDES*0.9'+12nm) at 25℃. The dimension is after 90% shrink and 12nm sizing channel length back.

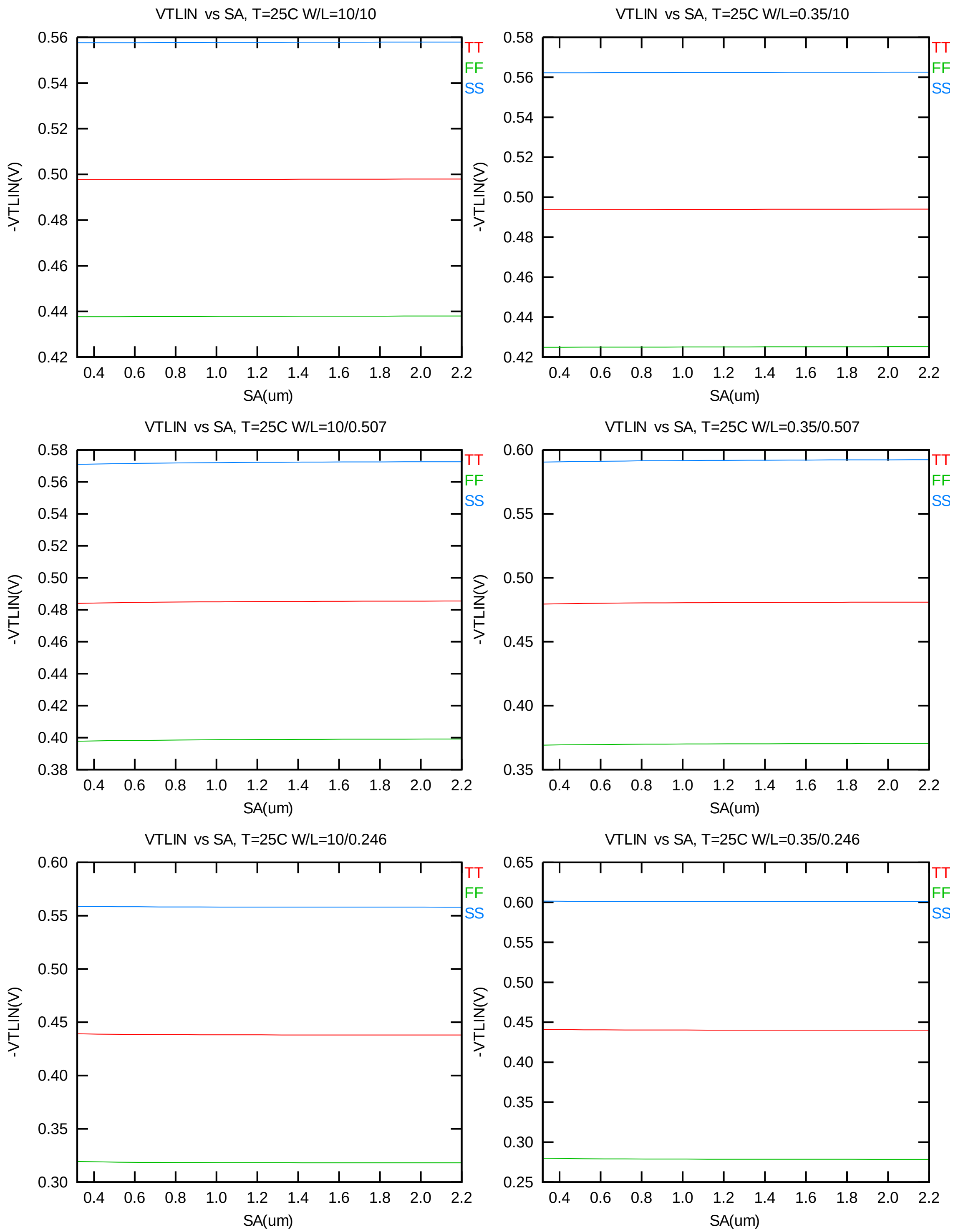

Figure 6-4 V_T vs. SA for low drain bias (V_{DS}= -0.1V) for PMOS

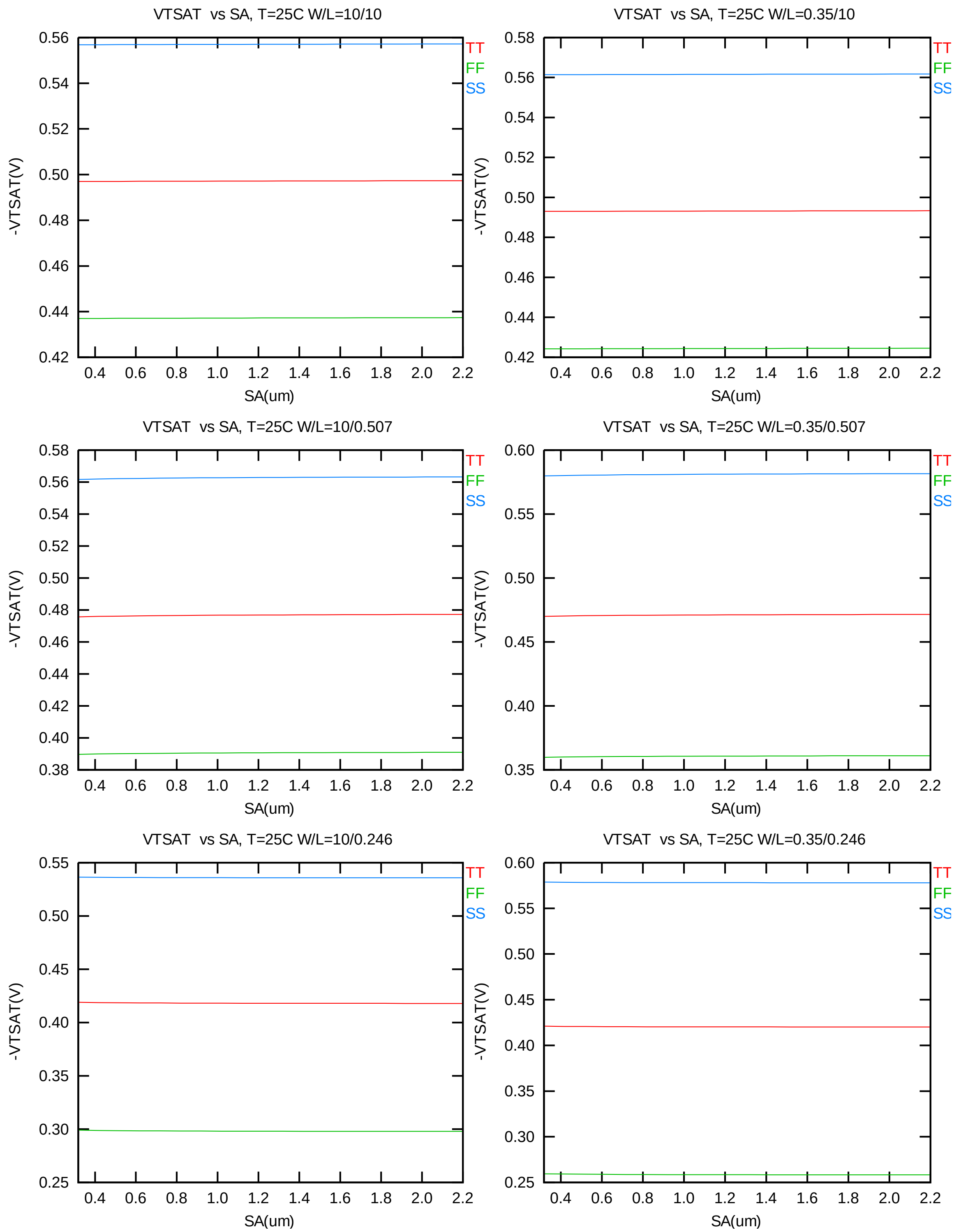

Figure 6-5 V_T vs. SA for high drain bias (V_{DS}= -1.2V) for PMOS

I_{ON} vs. SA

The following plots show the SA dependence of I_{ON} extracted at V_{DS}=V_{GS}= -1.2V for V_{BS}=0 at 25C.

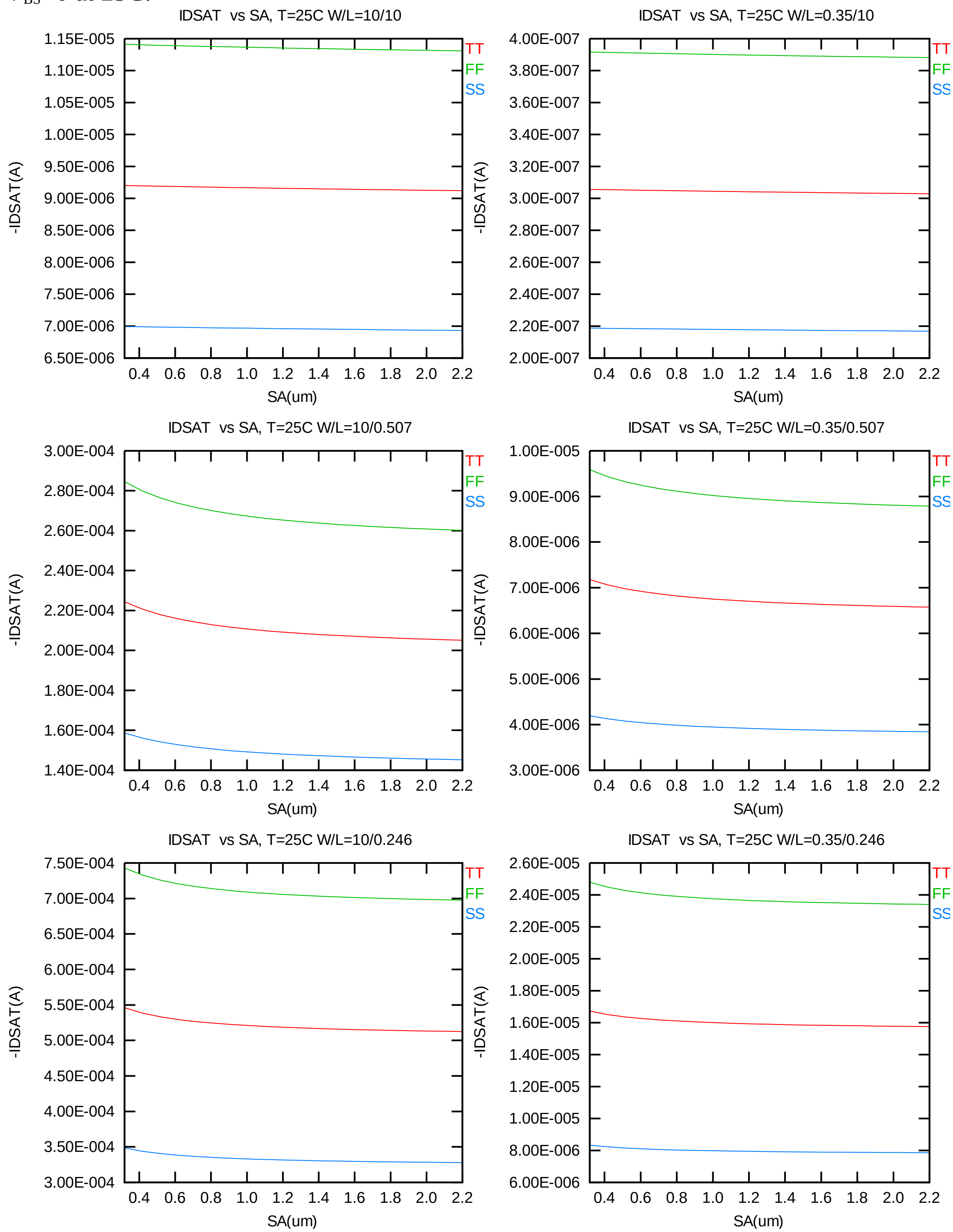

Figure 6-6 Saturation current I_{ON} vs. SA for PMOS

6.3 V_T and I_{ON} Variation v. SA for both N/PMOS

Delta V_T is defined as V_T (SA)-V_T (2.2um) for NMOS and -[V_T (SA)-V_T (2.2um)] for PMOS. Delta I_{ON} is defined as [I_{ON} (SA)-I_{ON} (2.2um)]/I_{ON} (2.2um)*100%.

Delta V_T and I_{ON} vs. SA

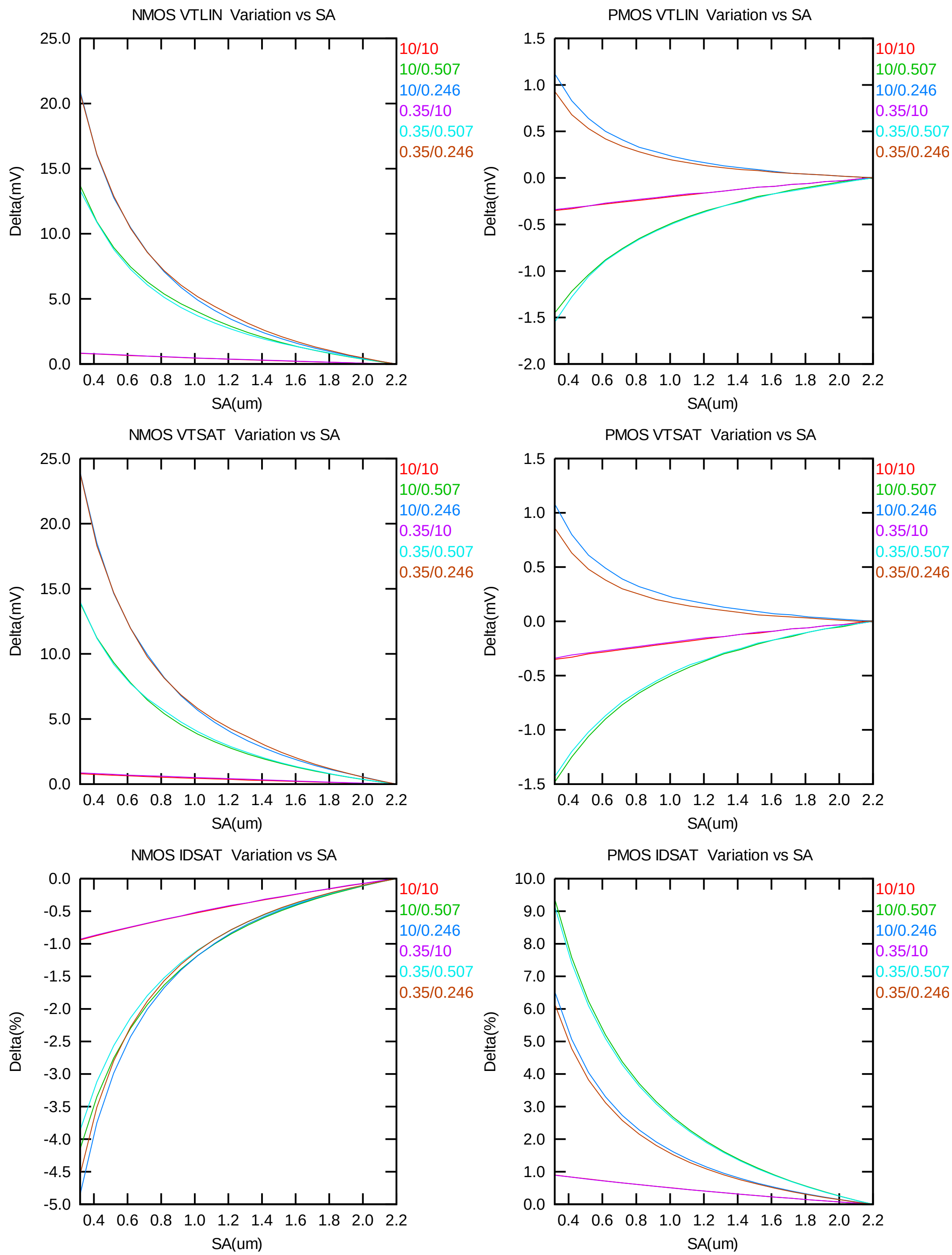

Figure 6-7 Delta V_T and I_{ON} vs. SA for both N/PMOS

7 Dynamic Performance

7.1 Test Circuit

To assess the dynamic performance of these devices, a ring oscillator circuit with the following configurations is adopted.

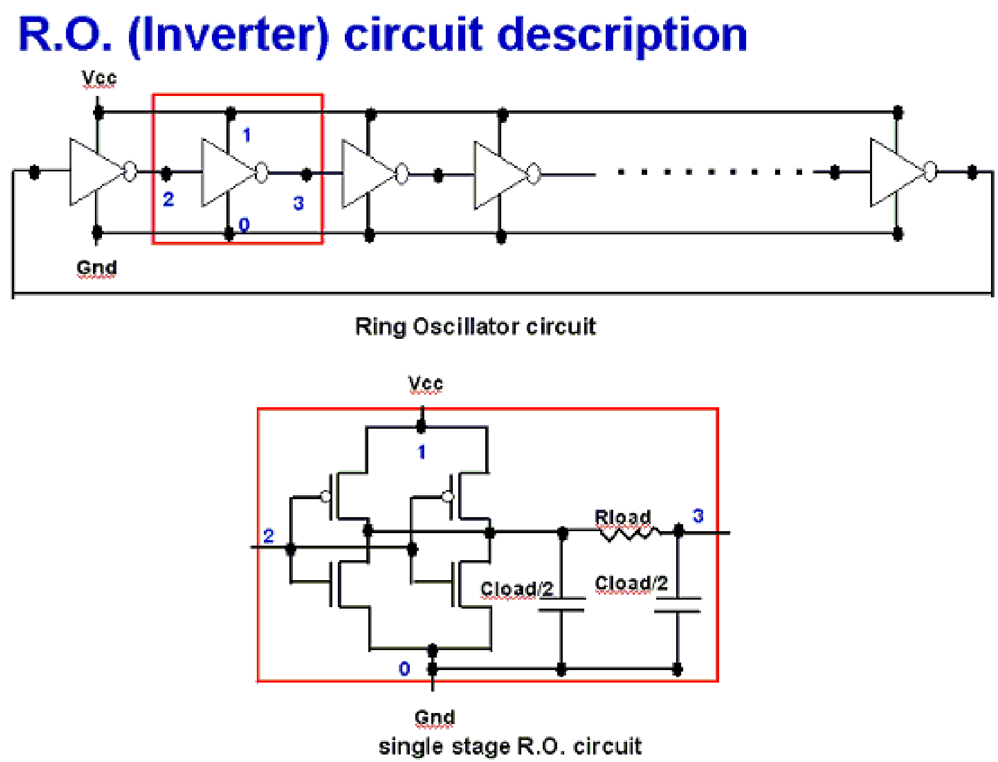

Figure 7-1 Ring oscillator netlist description

7.2 Dynamic Performance for SA= SB=0.318um

Table 7-1 Ring oscillator (inverter) structure description for SA= SB=0.318um

R.O. (Inverter) Structure description		
Contact size	0.144 um X 0.144 um	
Polysilicon gate to contact spacing	0.12 um	
Contact to diffusion edge spacing	0.054 um	
Without RC loading	R = 0 ohm	C = 0 fF
Inverter		
Logic gate type	Inverter	
Channel width for n-ch MOSFET	5 um X 1	
Channel length for n-ch MOSFET	0.246 um	
Channel width for p-ch MOSFET	6.5 um X 1	
Channel Length for p-ch MOSFET	0.246 um	

Table 7-2 Electrical parameters for ring oscillator (ps/stage) for SA=SB=0.318um

Parameter	Condition	Unit	Target	Centered Value
T_{pd}	V_{cc} = 1.2V	ps/stage	--	80.2

The simulation results at different temperatures and using different worst-case models are listed below.

Table 7-3 Ring oscillator simulation results (ps/stage) for SA= SB=0.318um

Inverter, R=0, C=0f, Fanout=1, Vcc=1.2V			
Temp(C)	TT	FF	SS
-50	73.383	50.406	122.697
0	78.188	54.723	127.064
25	80.234	56.636	128.770
85	84.411	60.747	131.948
125	86.715	63.171	133.541

Inverter, R=0, C=0f, Fanout=1, T=25C			
Vcc(V)	TT	FF	SS
0.90	153.679	90.936	310.431
1.00	117.660	75.152	213.439
1.10	95.274	64.385	160.930
1.20	80.234	56.636	128.770
1.32	67.738	49.853	103.891

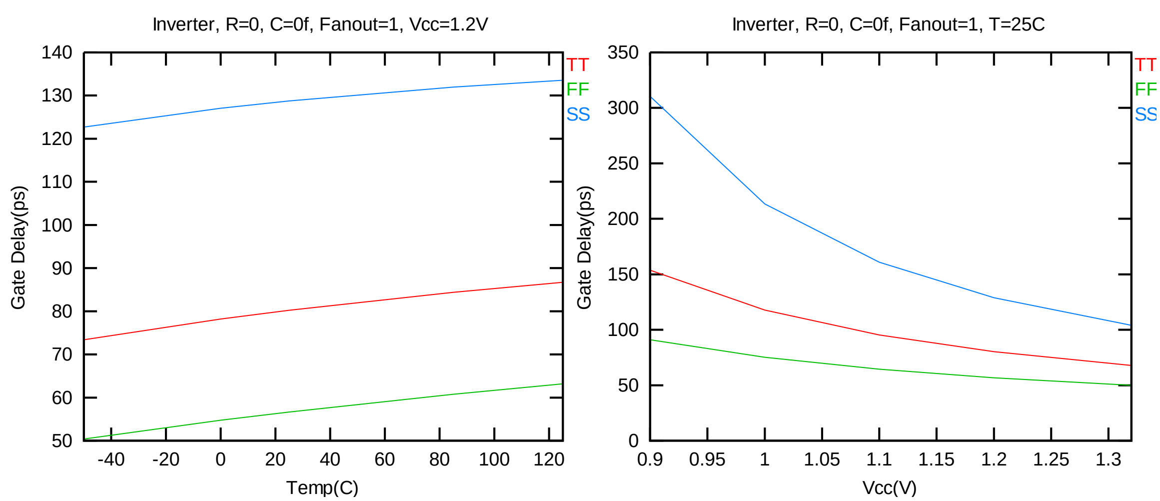

Figure 7-2 Ring oscillator (inverter) simulation for SA=SB=0.318um

The (Inverter) ring oscillator gate delay simulation vs. width for SA= SB=0.318um

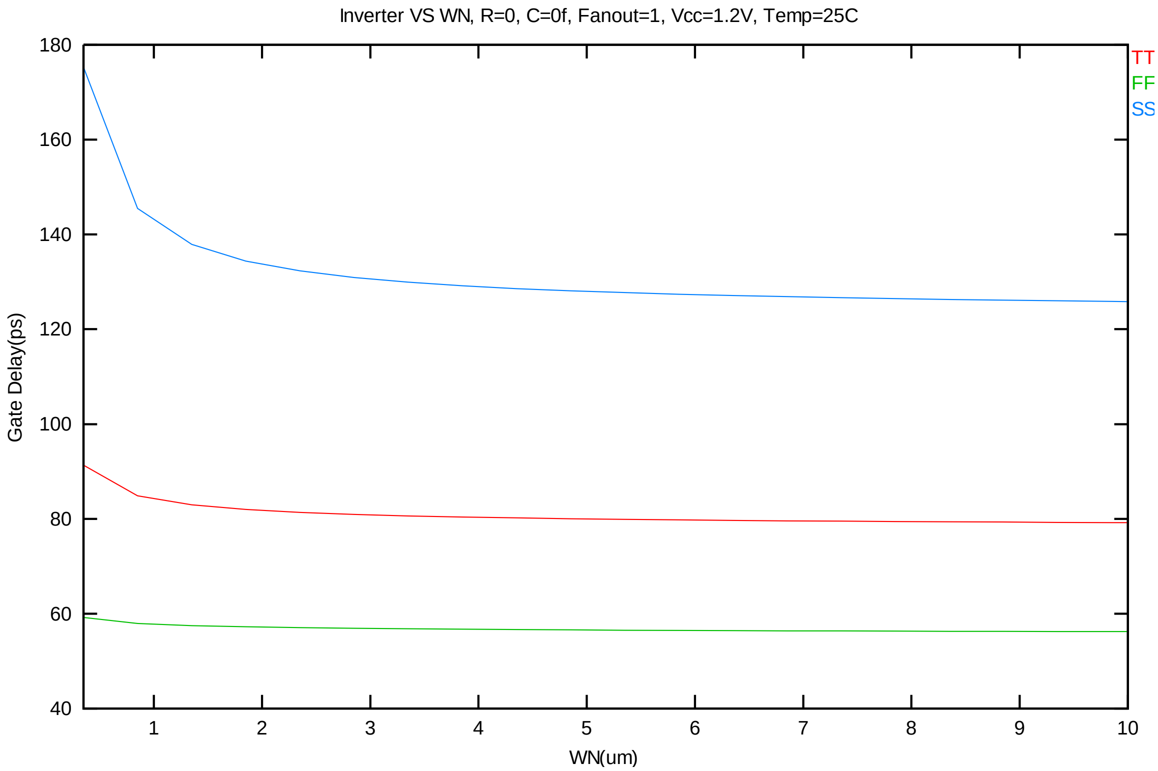

Figure 7-3 Ring oscillator (inverter) gate delay simulation vs. width for SA= SB=0.318um

8 Noise Characteristics

8.1 Verification Plots

NMOS, W=10um L=10um

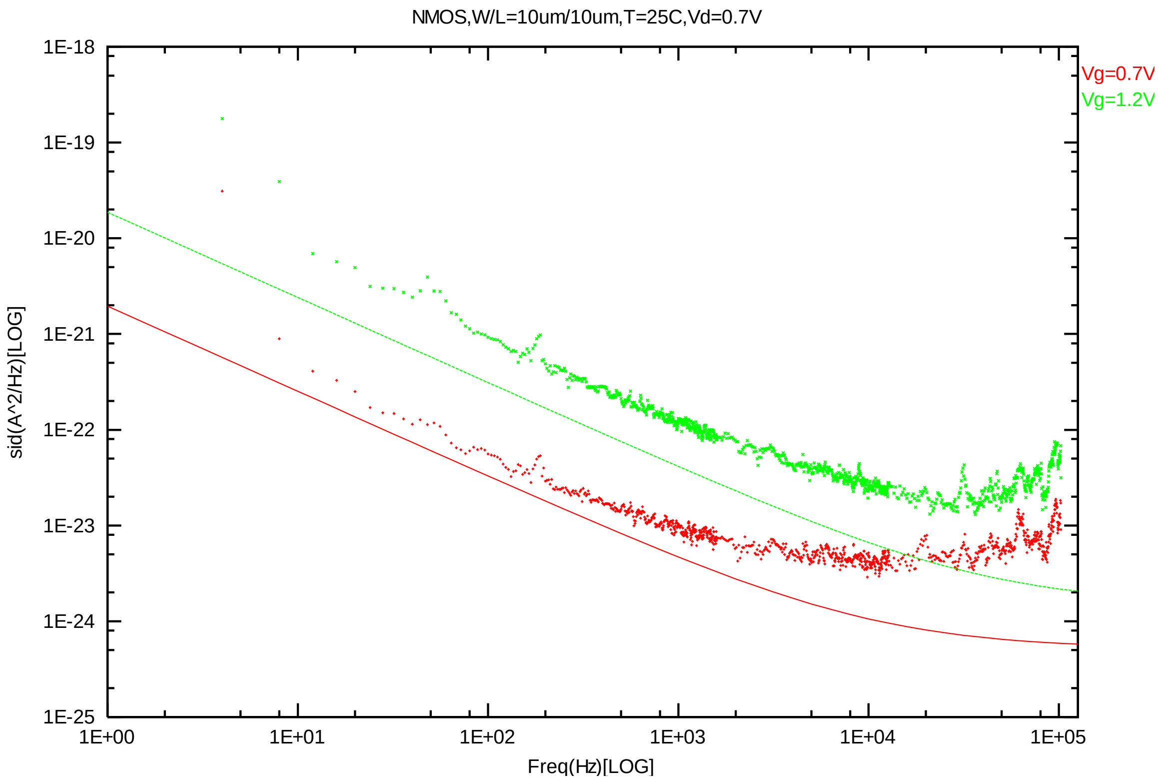

Figure 8-1 NMOS W/L=10um/10um @ Vd=0.7V

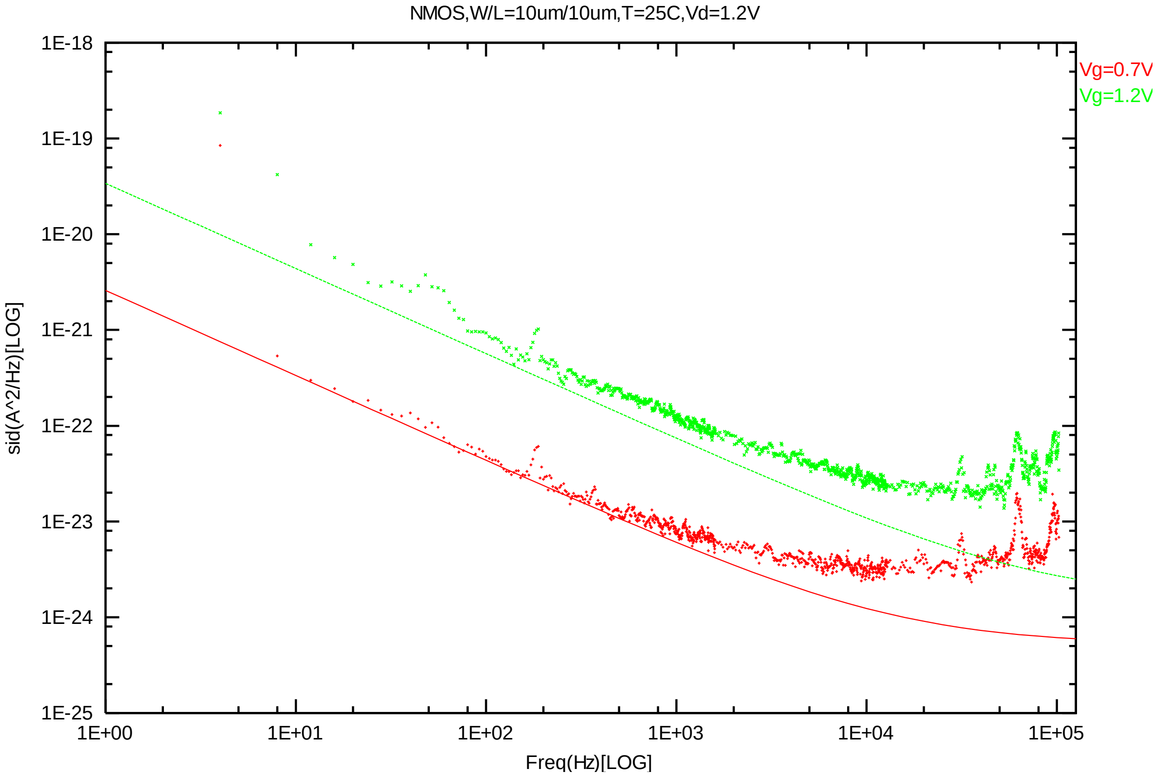

Figure 8-2 NMOS W/L=10um/10um @ Vd=1.2V

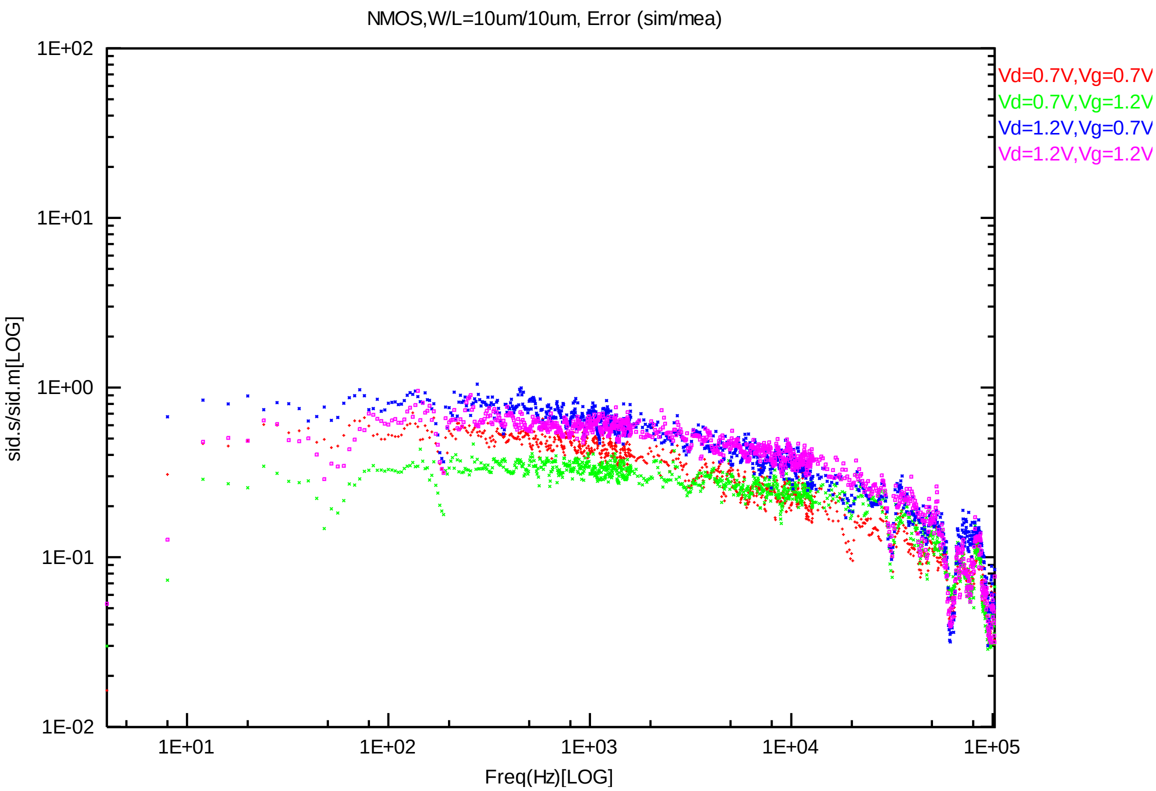

Figure 8-3 NMOS W/L=10um/10um error distribution plot

NOMS, W=10um L=1um

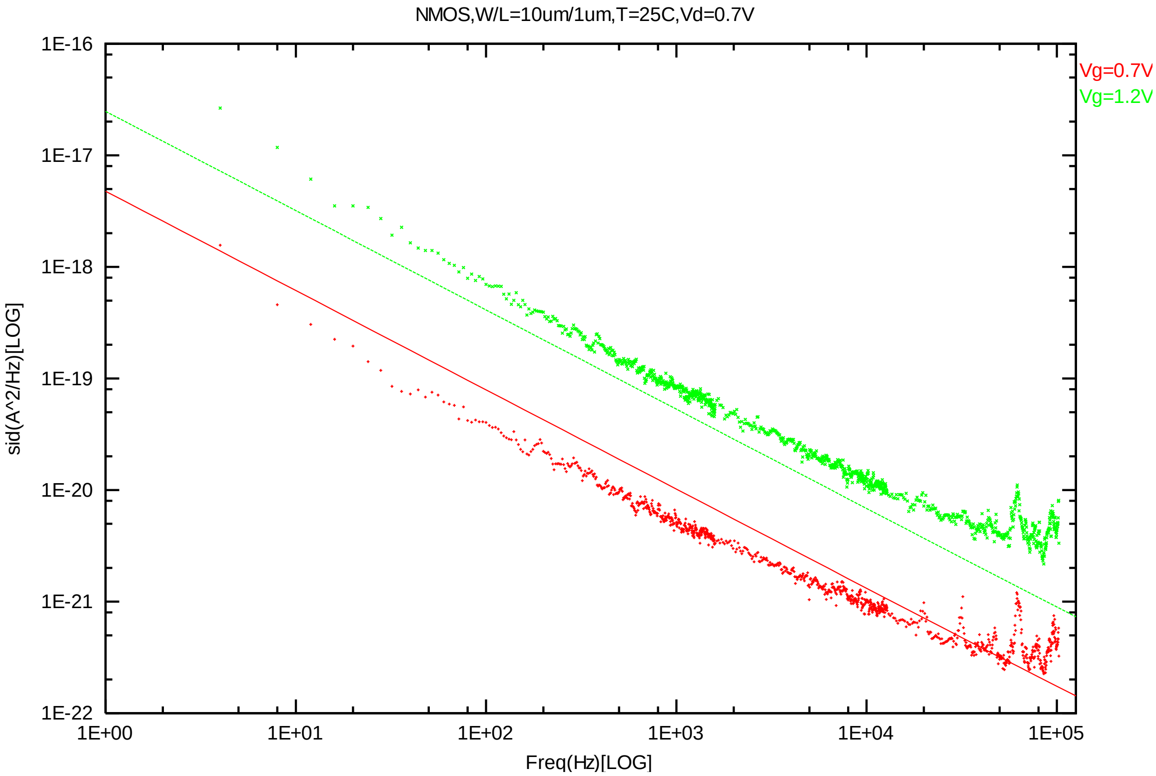

Figure 8-4 NMOS W/L=10um/1um @ Vd=0.7V

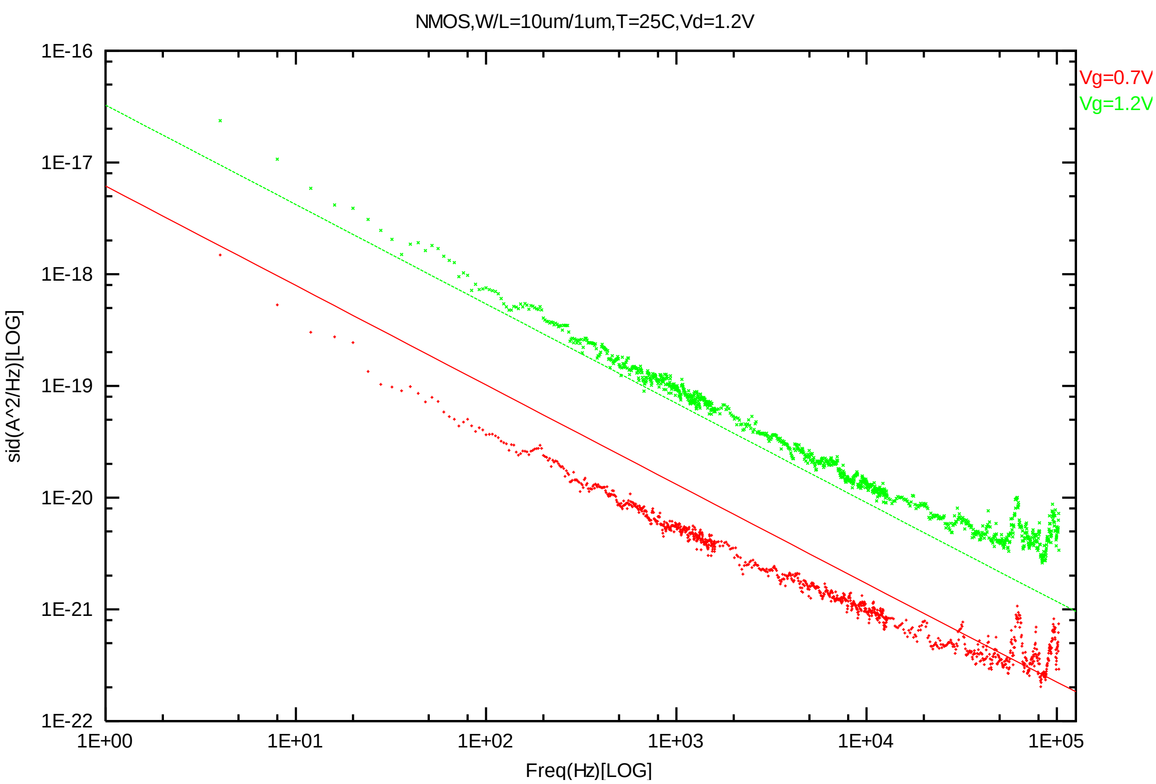

Figure 8-5 NMOS W/L=10um/1um @ Vd=1.2V

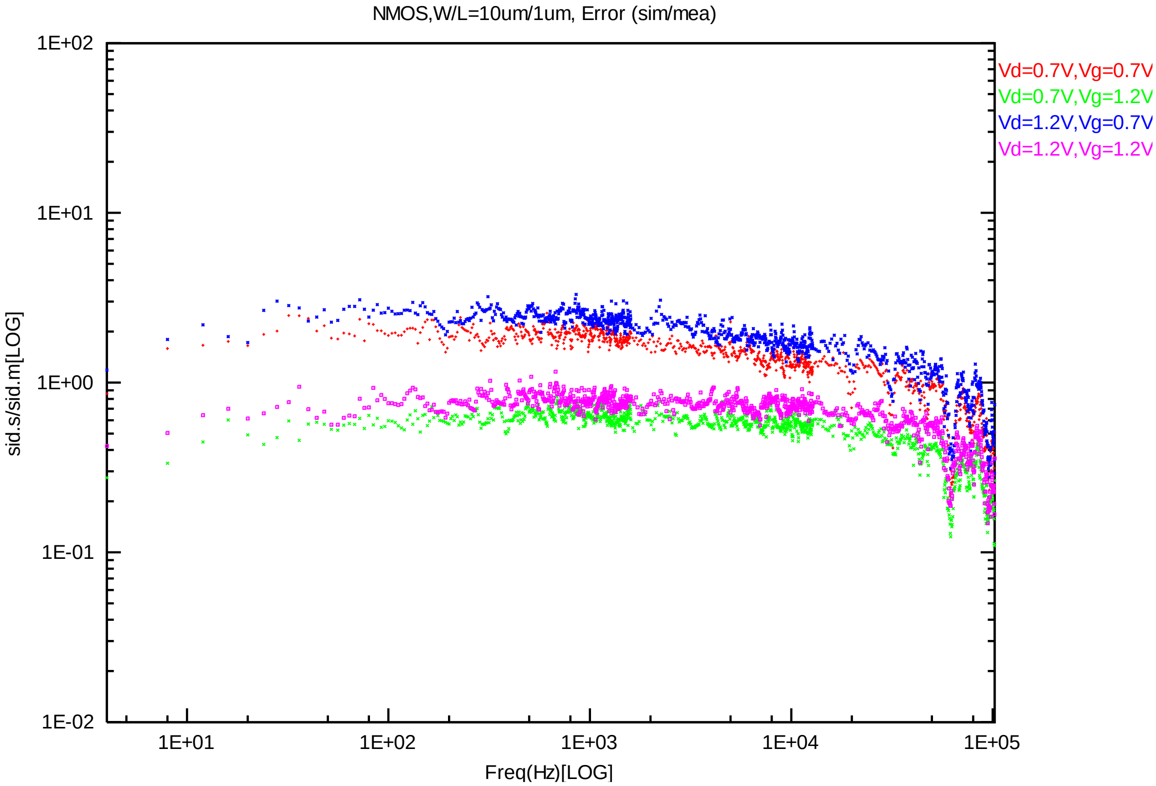

Figure 8-6 NMOS W/L=10um/1um error distribution plot

NMOS, W=10um L=0.507um

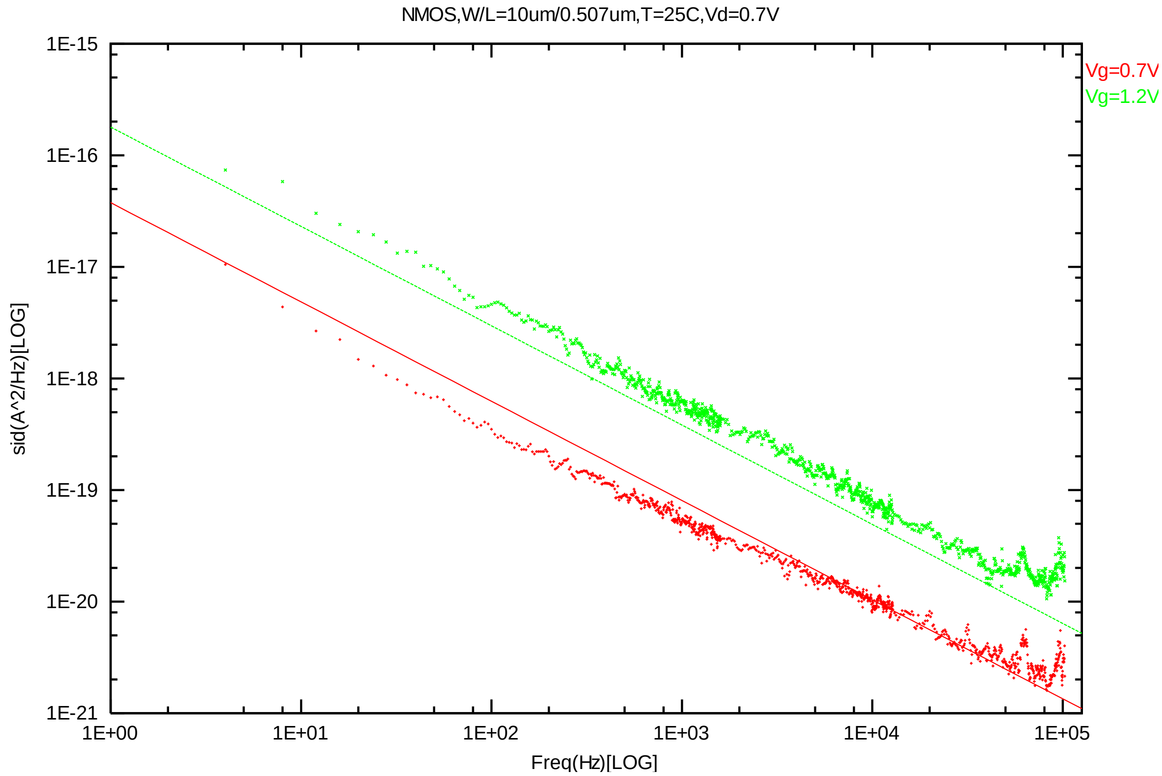

Figure 8-7 NMOS W/L=10um/0.507um @ Vd=0.7V

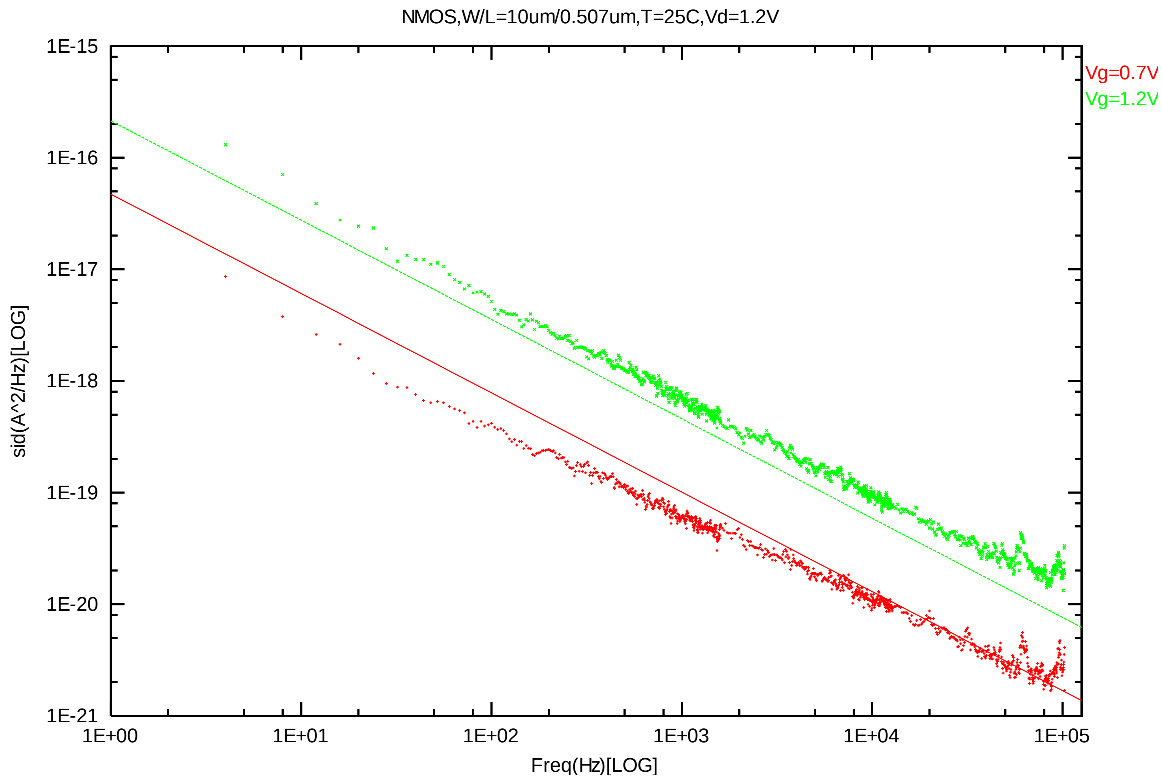

Figure 8-8 NMOS W/L=10um/0.507um @ Vd=1.2V

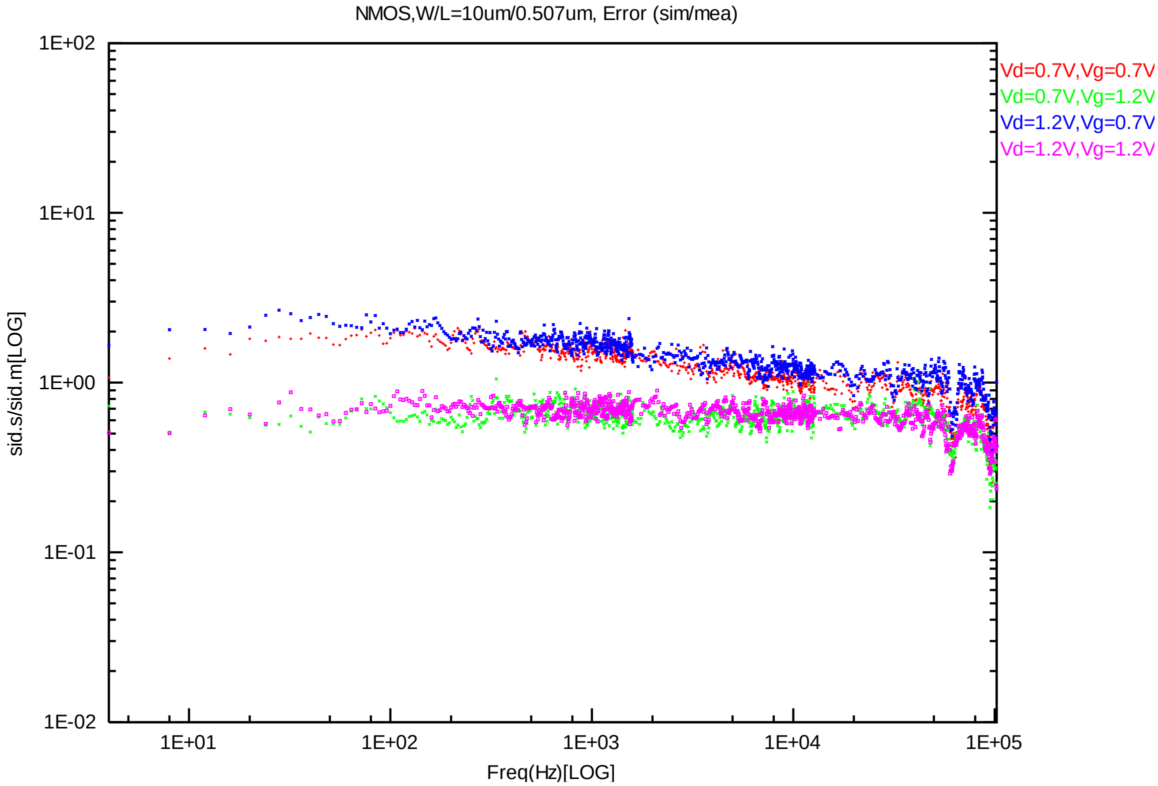

Figure 8-9 NMOS W/L=10um/0.507um error distribution plot

NMOS, W=10um L=0.417um

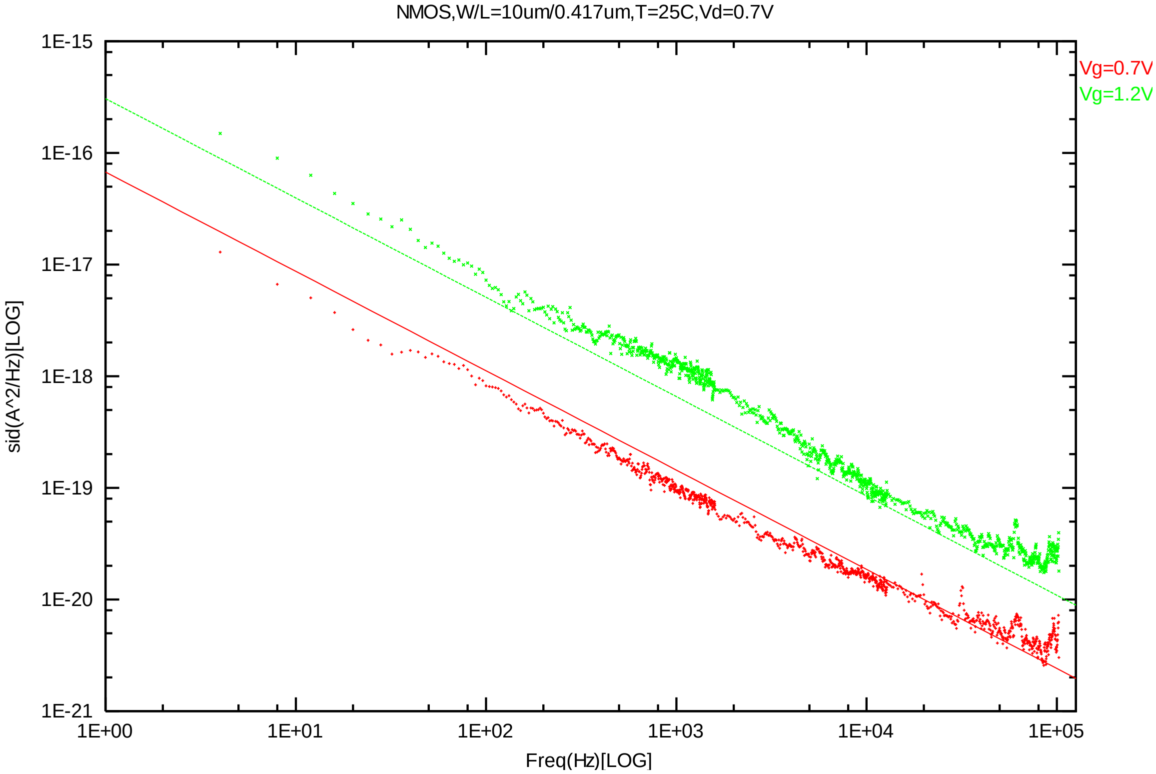

Figure 8-10 NMOS W/L=10um/0.417um @ Vd=0.7V

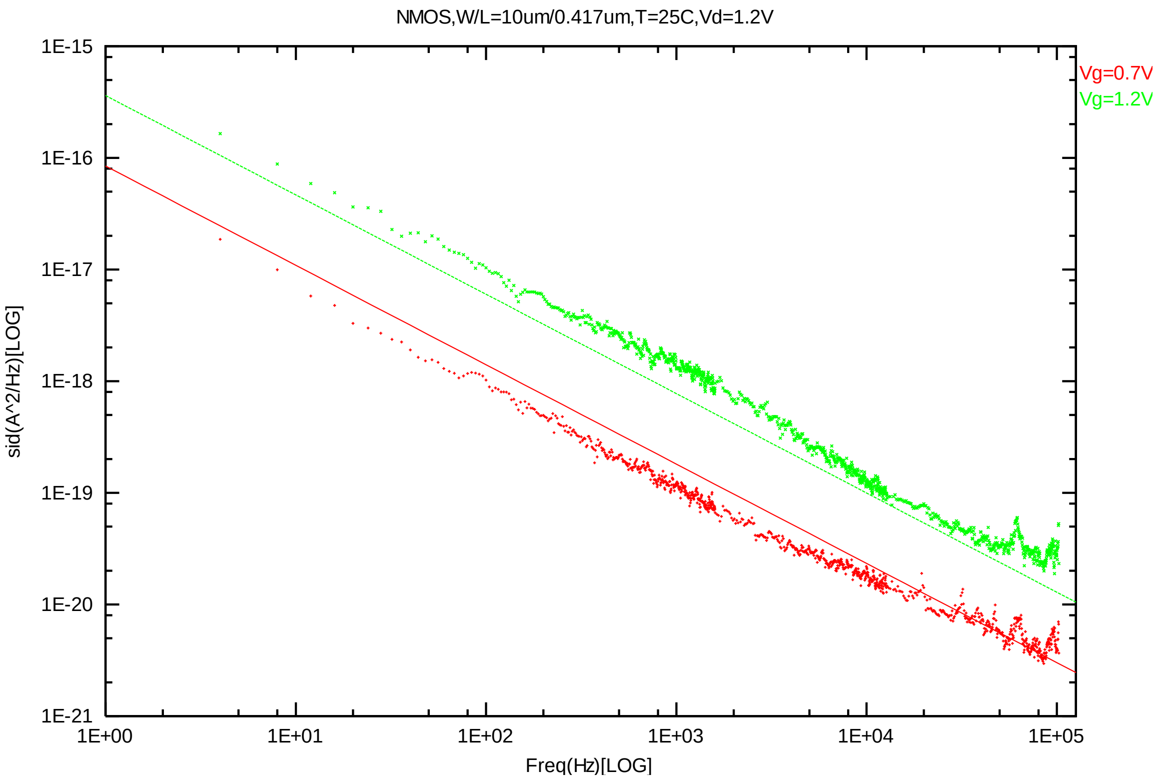

Figure 8-11 NMOS W/L=10um/0.417um @ Vd=1.2V

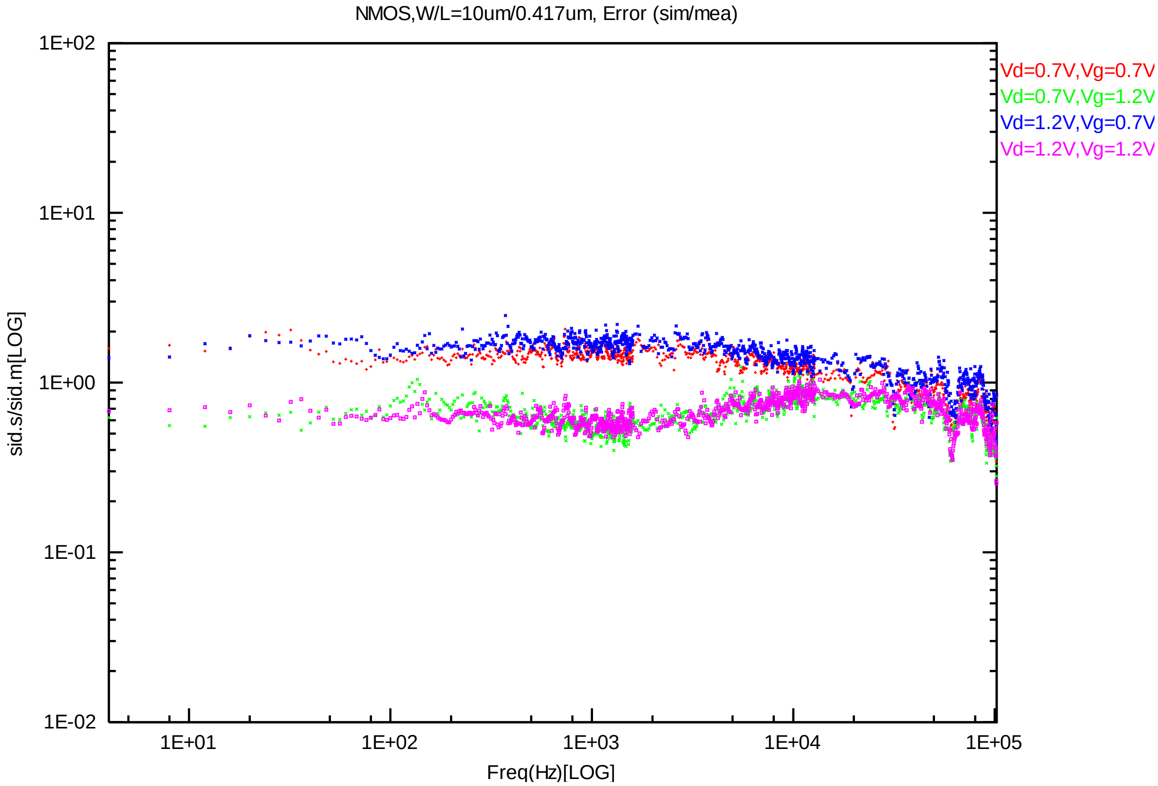

Figure 8-12 NMOS W/L=10um/0.417um error distribution plot

NMOS, W=10um L=0.246um

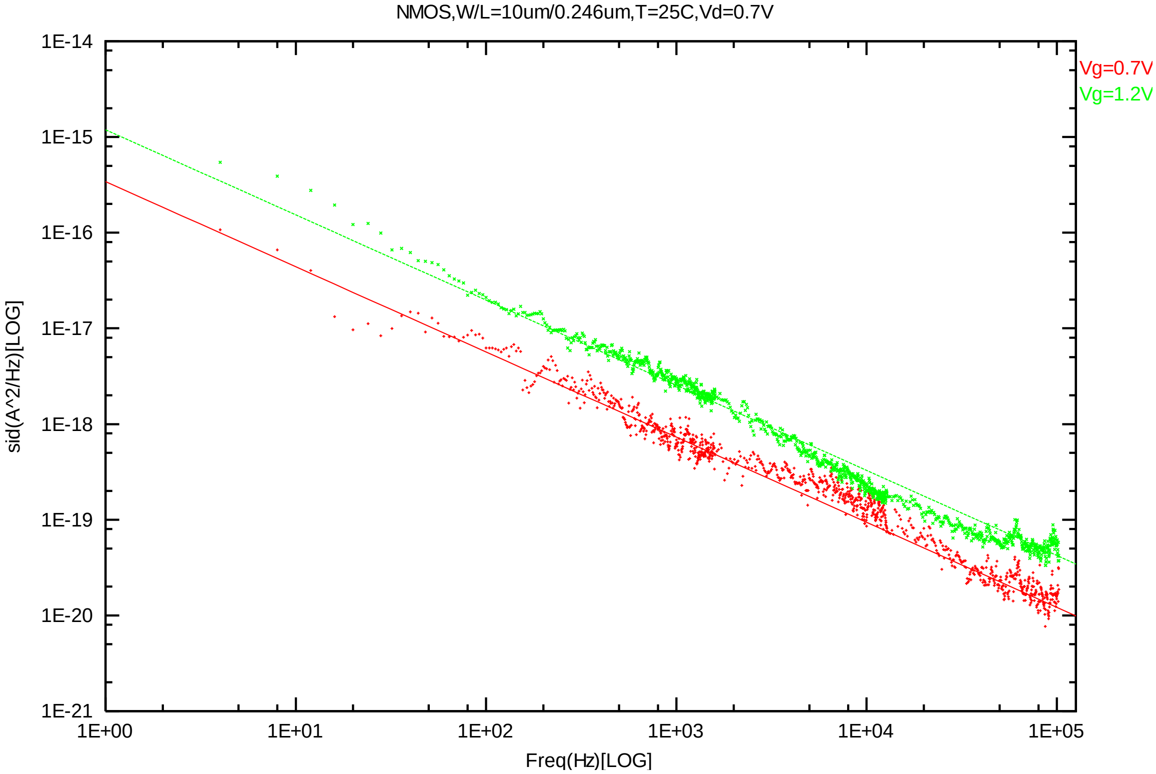

Figure 8-13 NMOS W/L=10um/0.246um @ Vd=0.7V

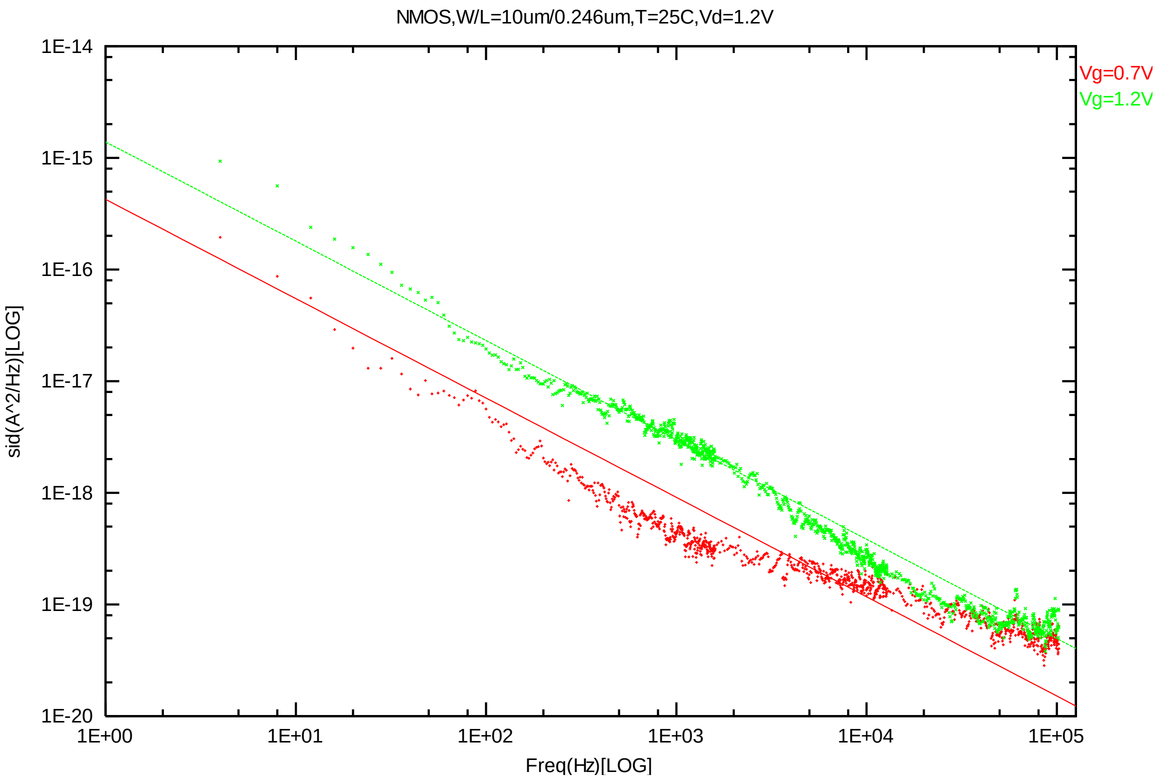

Figure 8-14 NMOS W/L=10um/0.246um @ Vd=1.2V

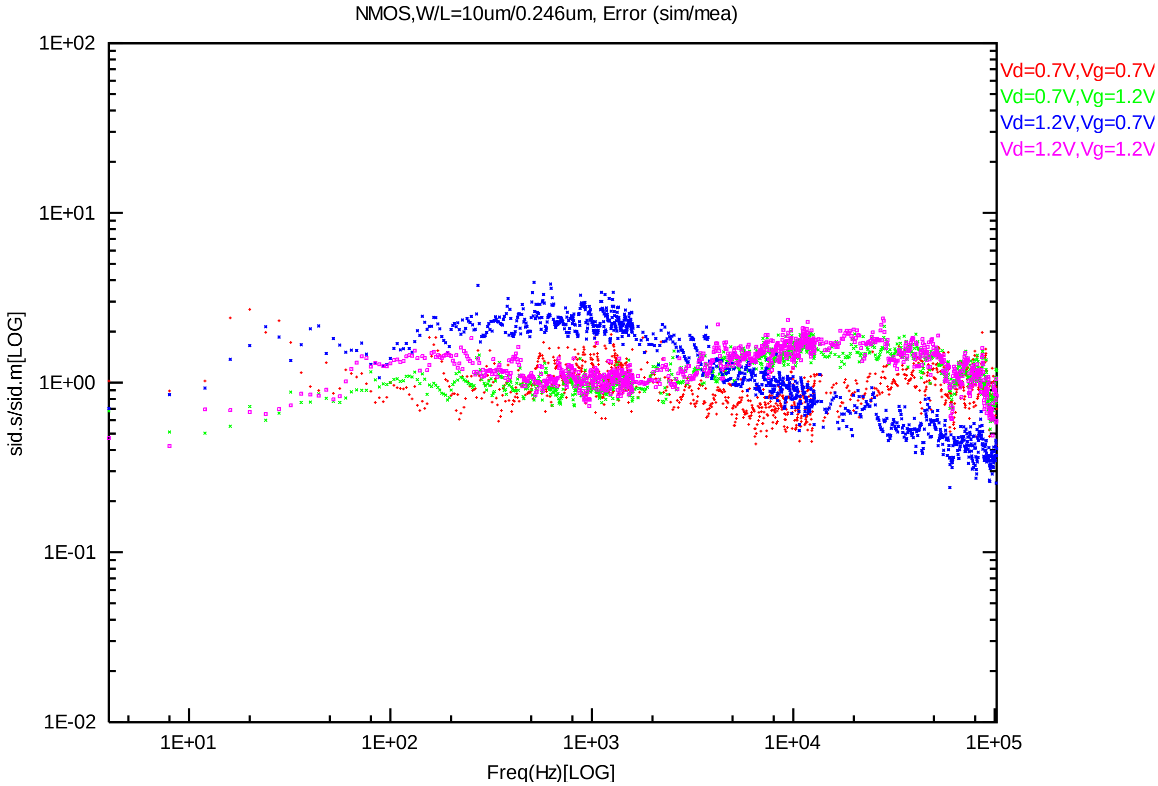

Figure 8-15 NMOS W/L=10um/0.246um error distribution plot

PMOS, W=10um L=10um

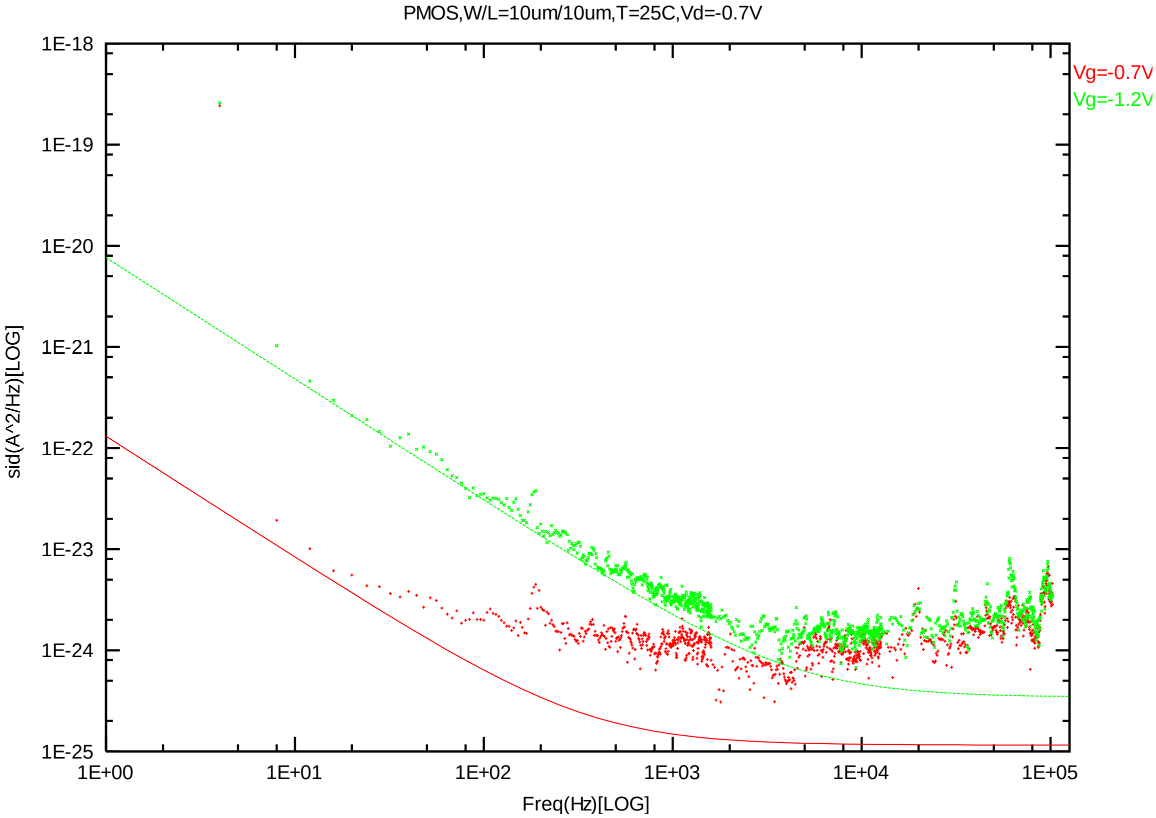

Figure 8-16 PMOS W/L=10um/10um @ Vd=-0.7V

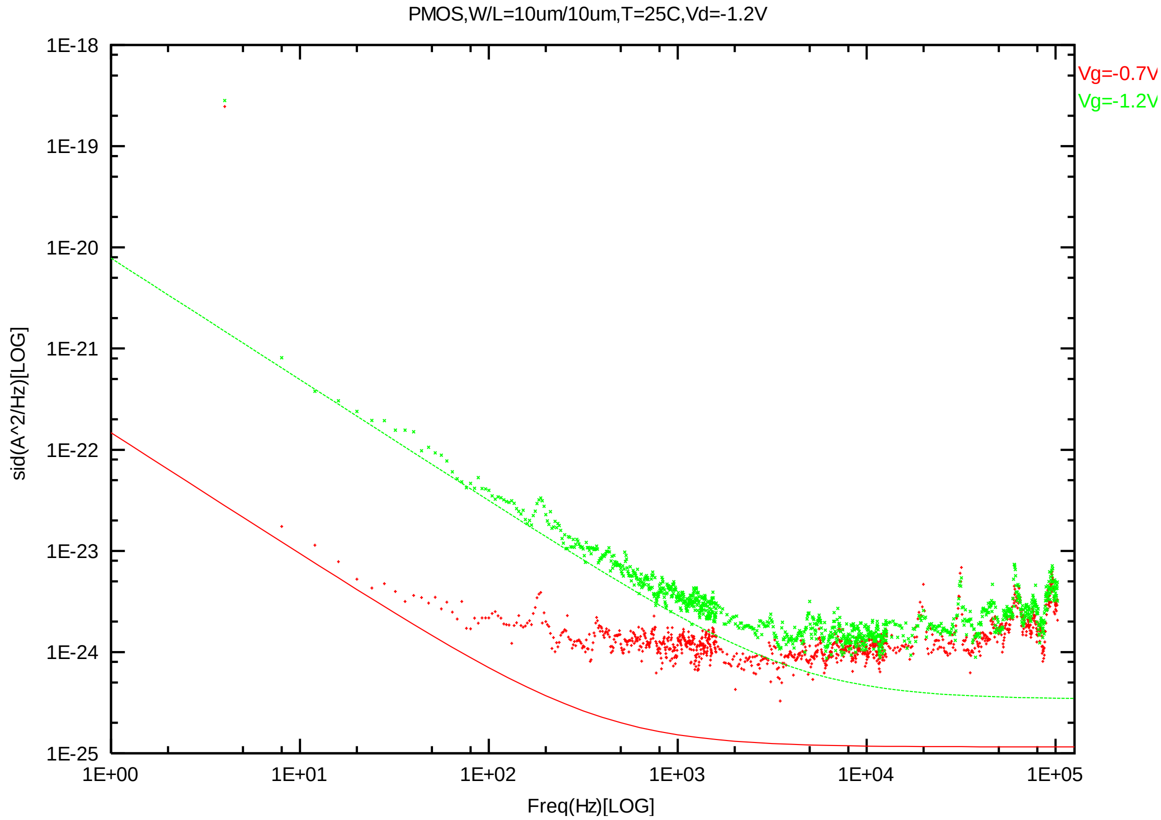

Figure 8-17 PMOS W/L=10um/10um @ Vd=-1.2V

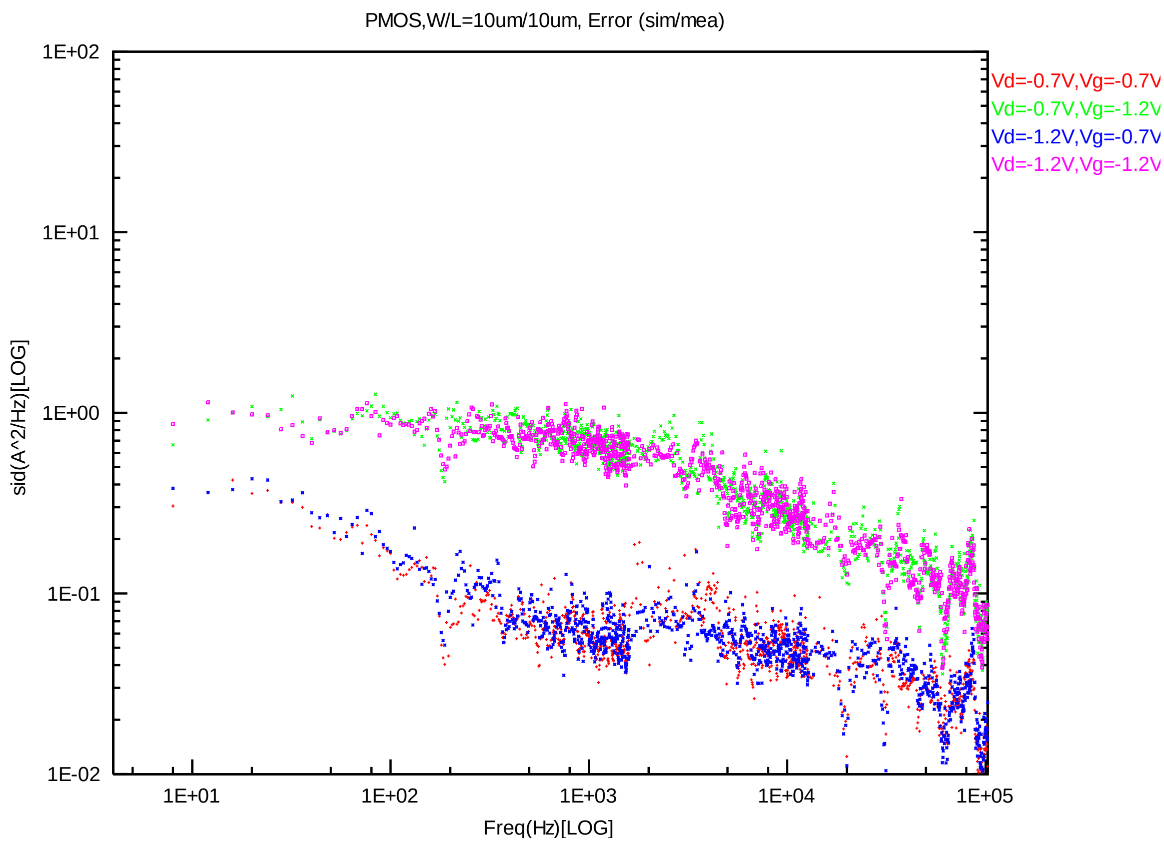

Figure 8-18 PMOS W/L=10um/10um error distribution plot

PMOS, W=10um L=1um

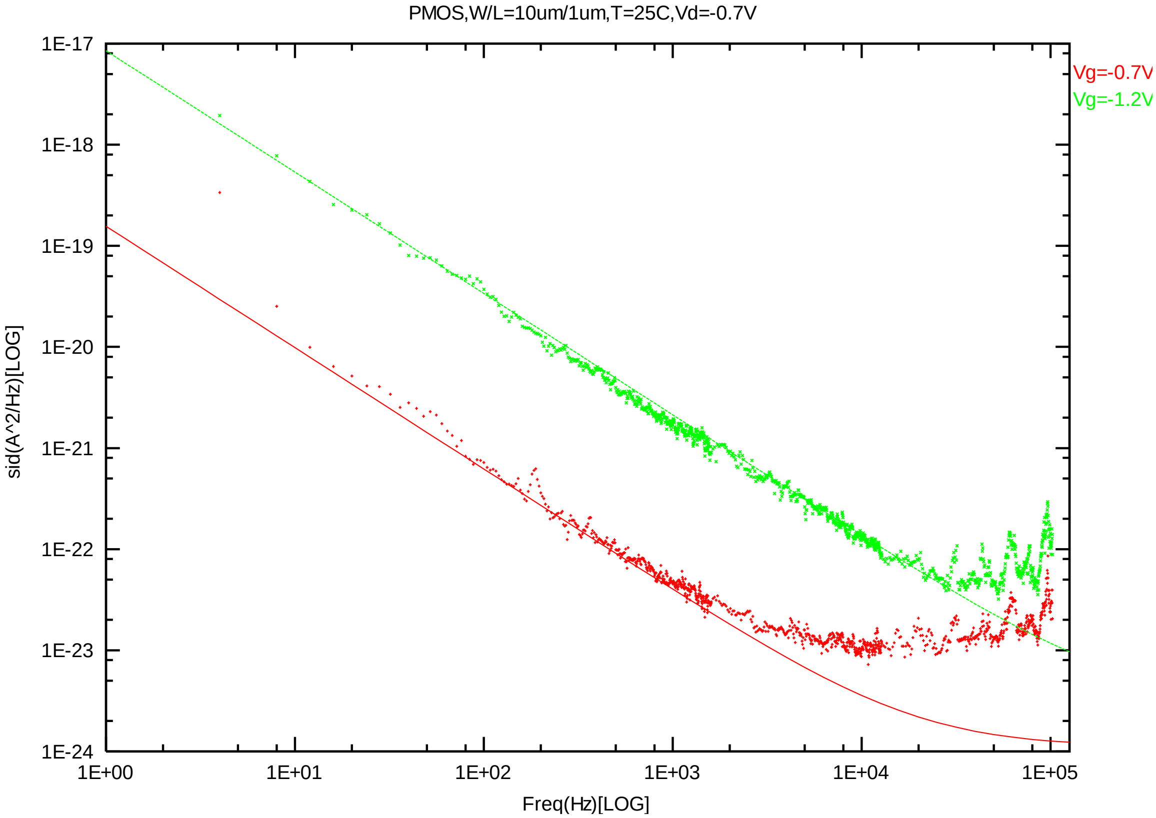

Figure 8-19 PMOS W/L=10um/1um @ Vd=-0.7V

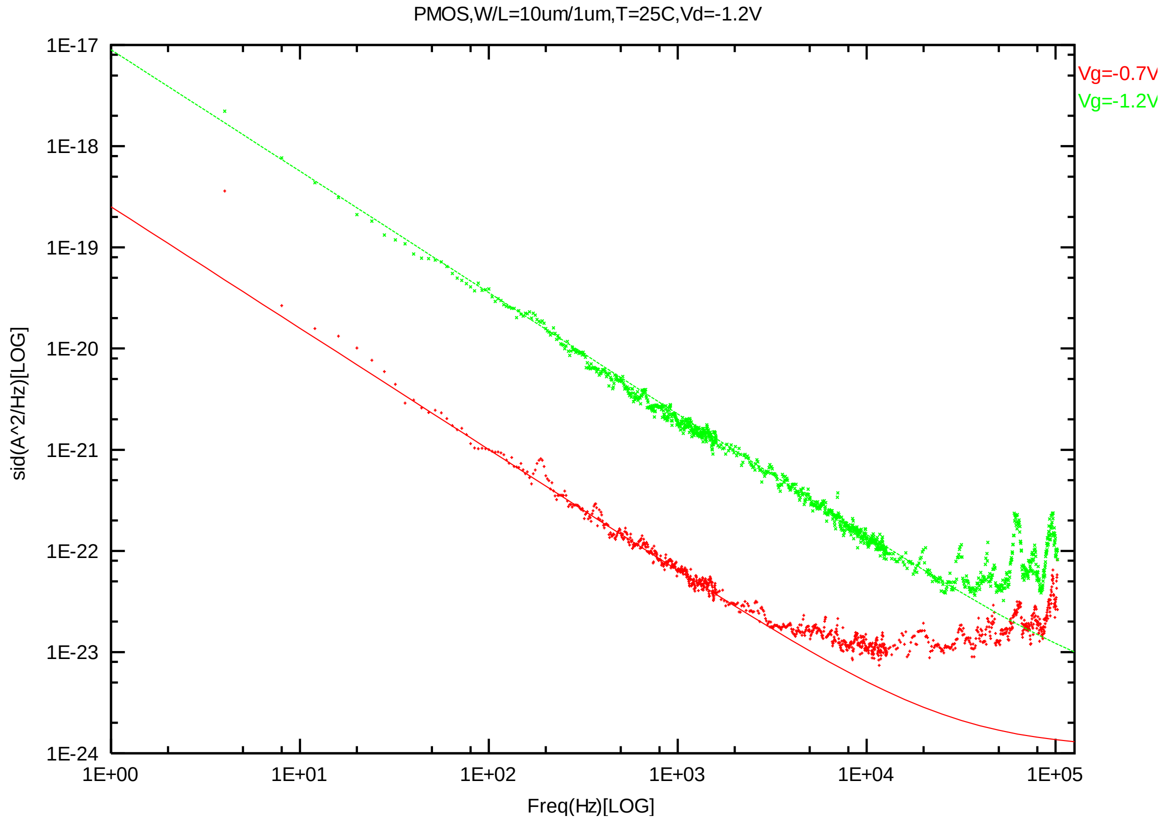

Figure 8-20 PMOS W/L=10um/1um @ Vd=-1.2V

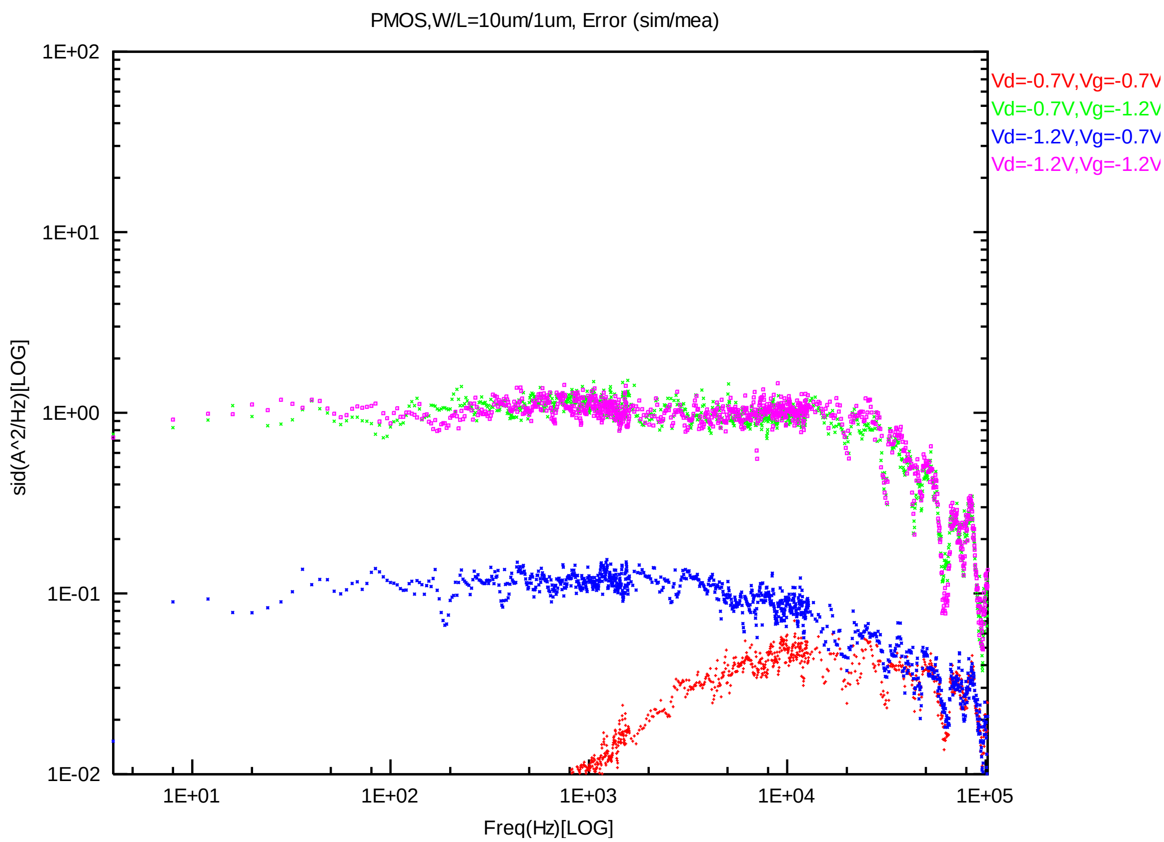

Figure 8-21 PMOS W/L=10um/1um error distribution plot

PMOS, W=10um L=0.507um

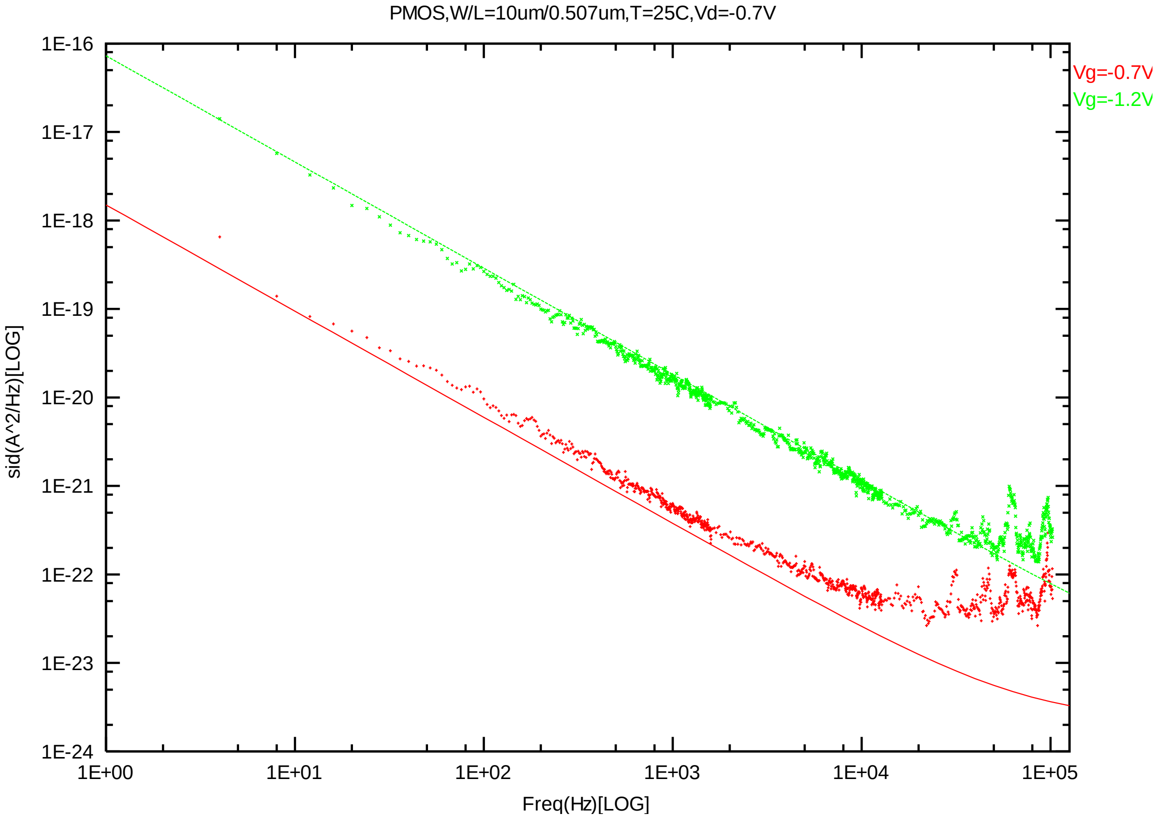

Figure 8-22 PMOS W/L=10um/0.507um @ Vd=-0.7V

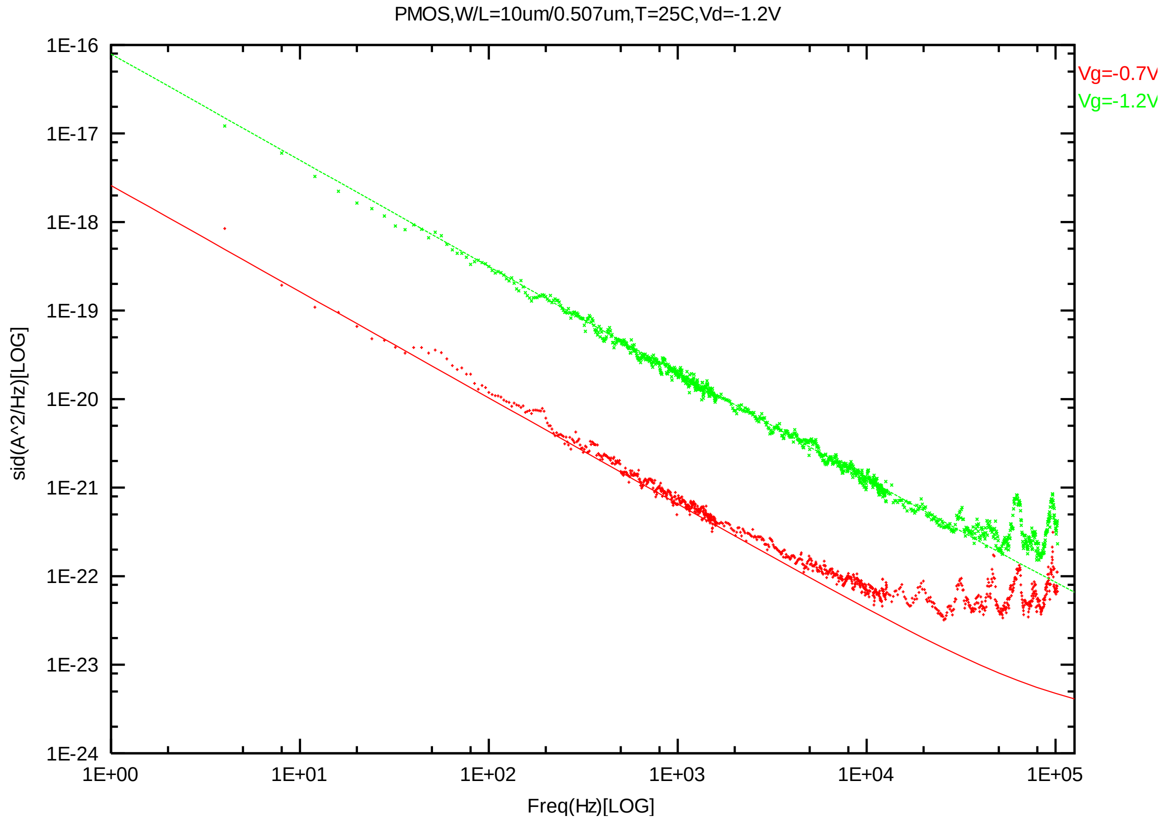

Figure 8-23 PMOS W/L=10um/0.507um @ Vd=-1.2V

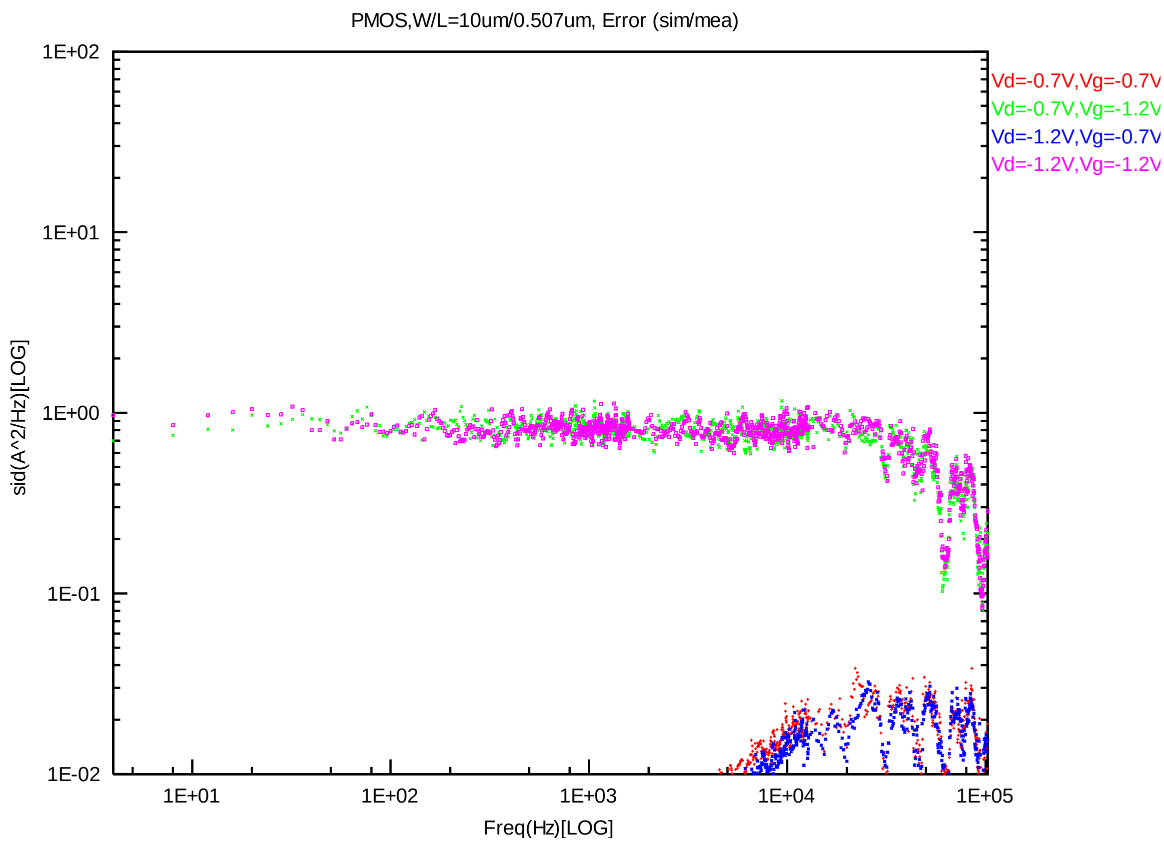

Figure 8-24 PMOS W/L=10um/0.507um error distribution plot

PMOS, W=10um L=0.417um

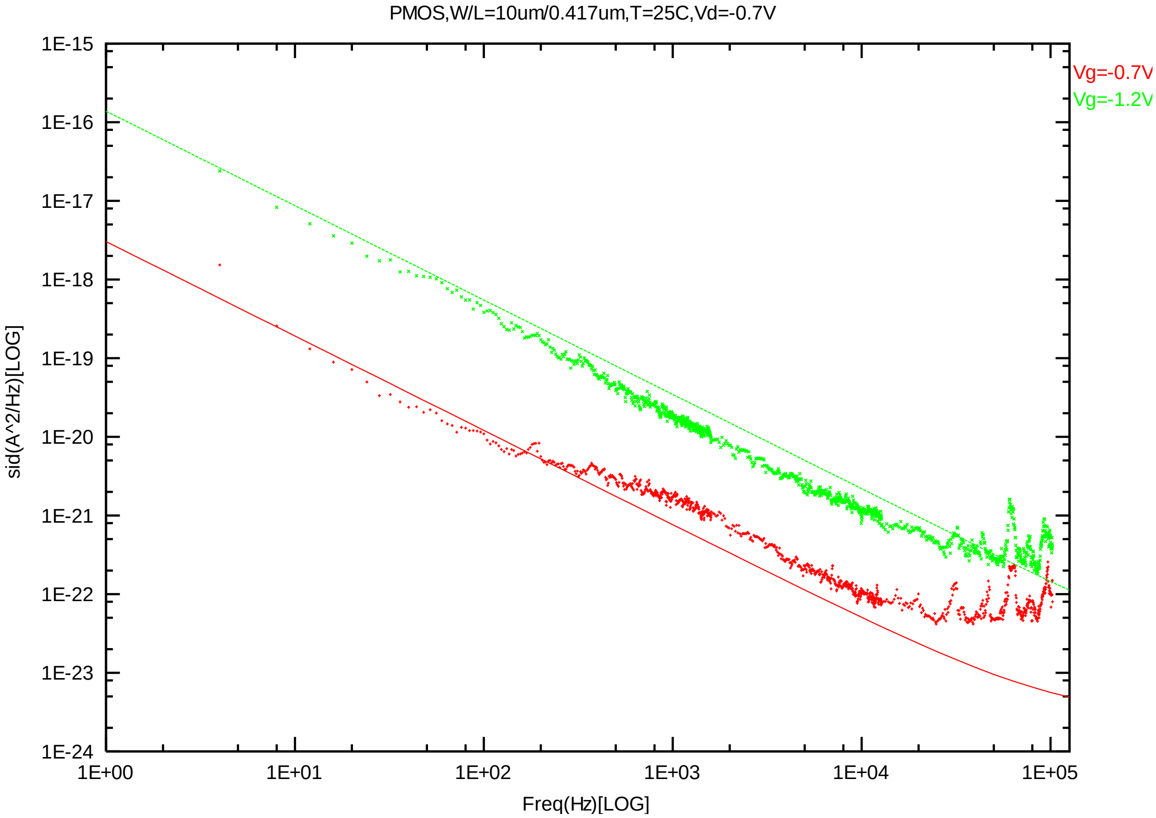

Figure 8-25 PMOS W/L=10um/0.417um @ Vd=-0.7V

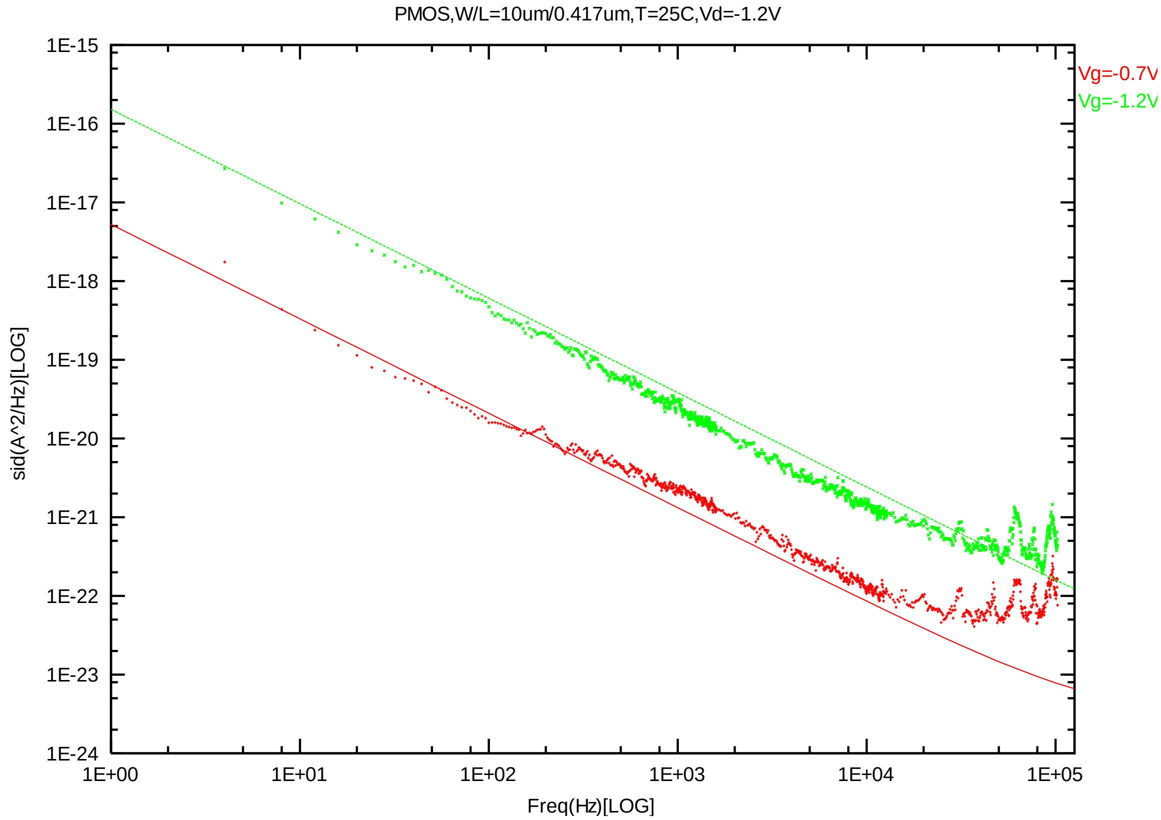

Figure 8-26 PMOS W/L=10um/0.417um @ Vd=-1.2V

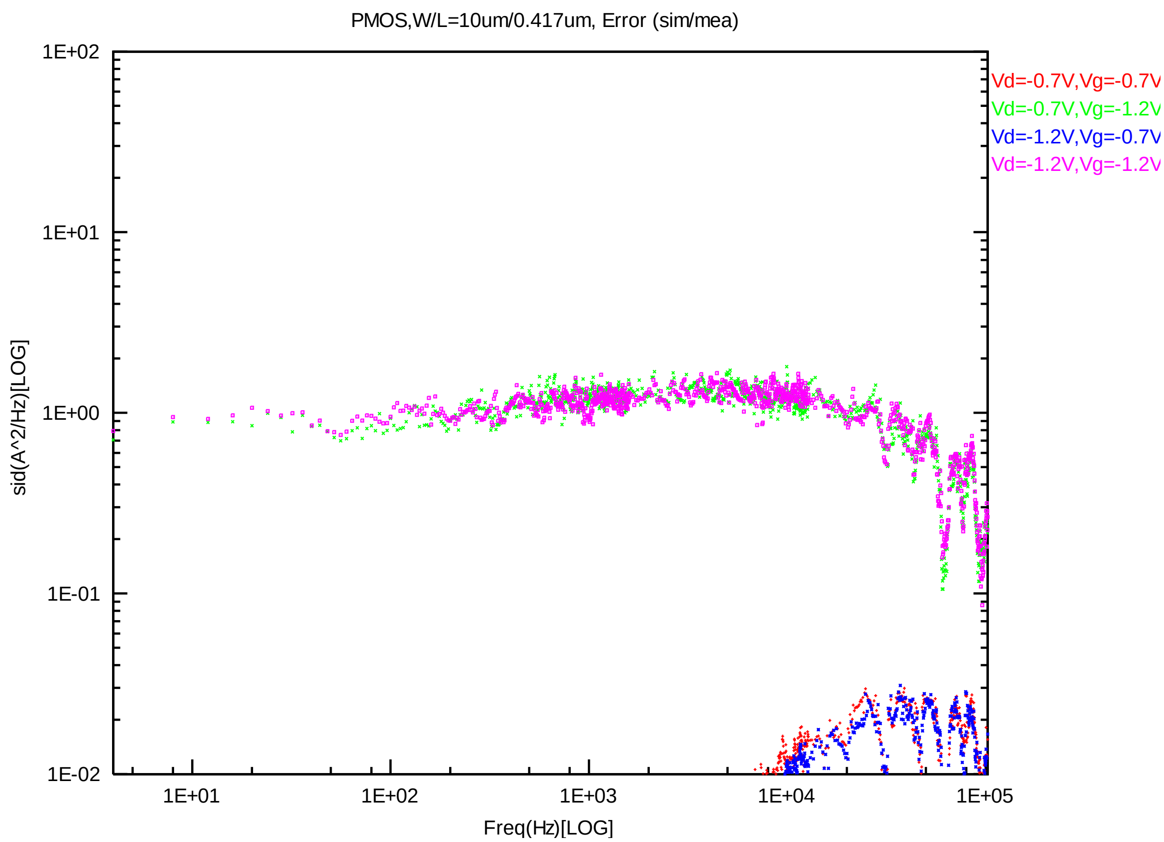

Figure 8-27 PMOS W/L=10um/0.417um error distribution plot

PMOS, W=10um L=0.246um

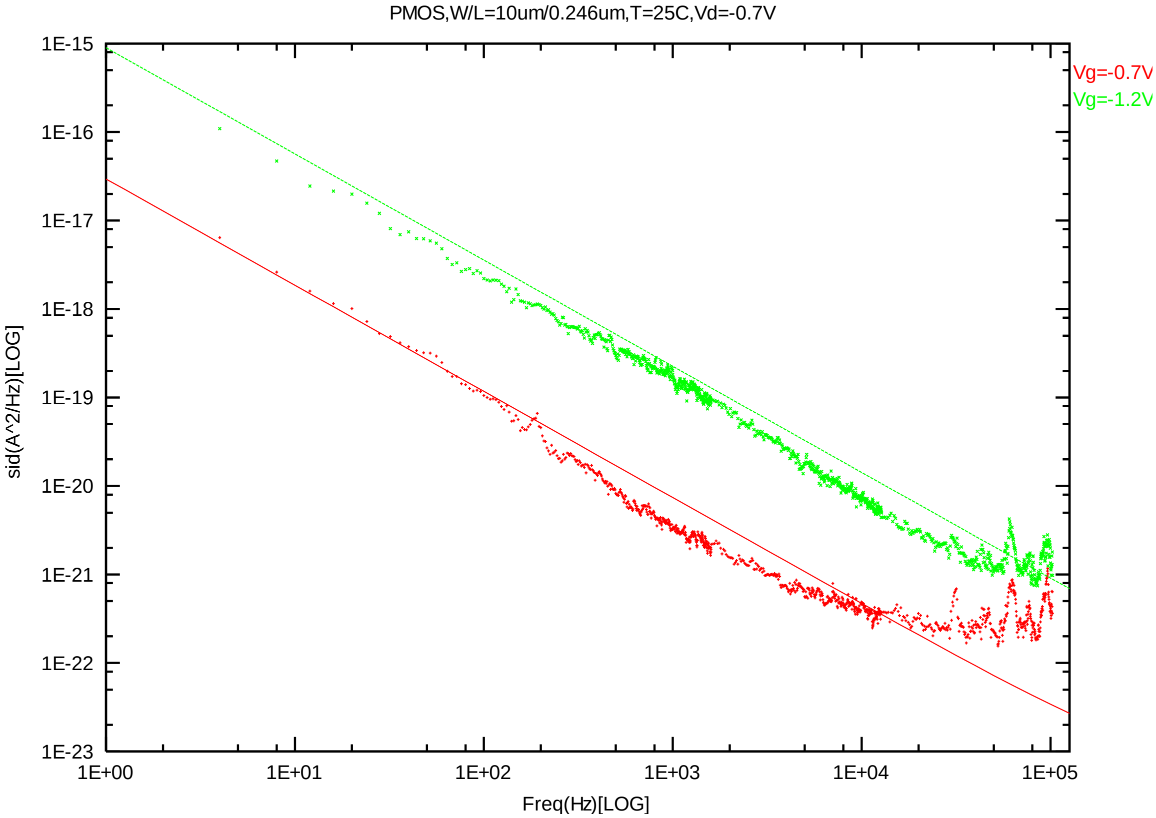

Figure 8-28 PMOS W/L=10um/0.246um @ Vd=-0.7V

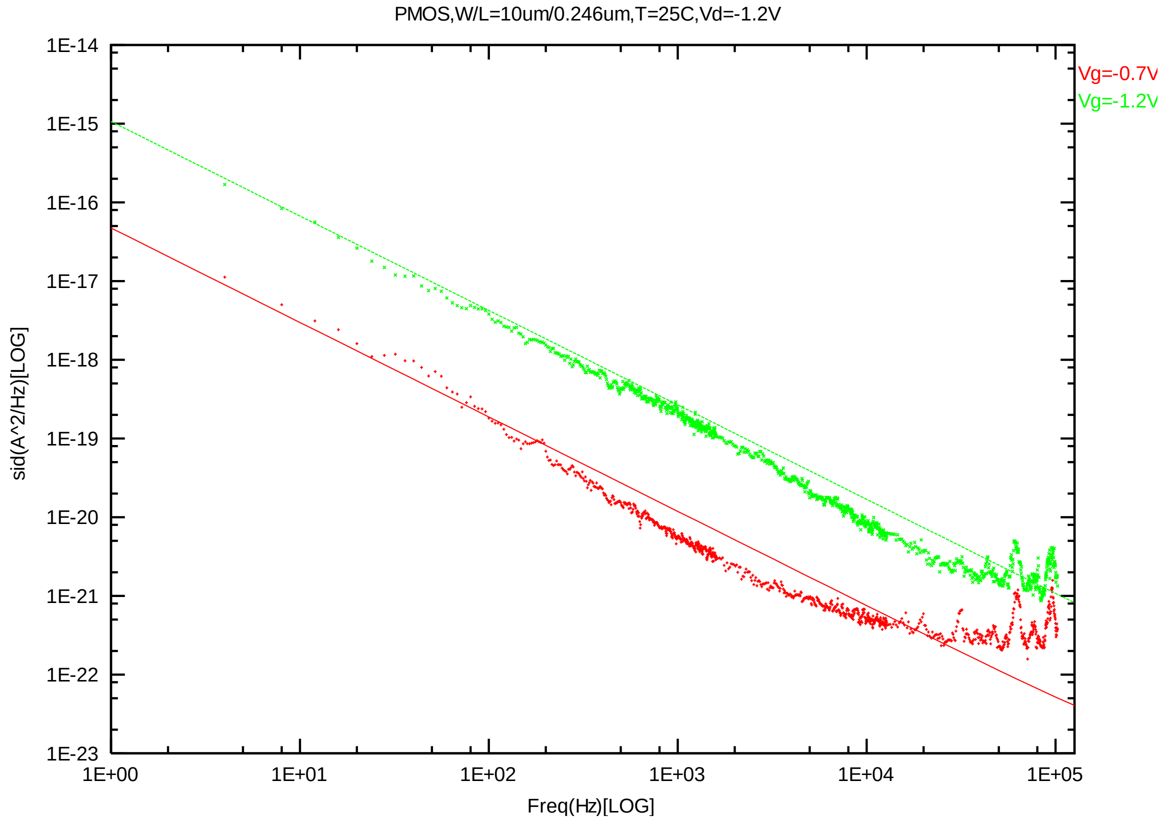

Figure 8-29 PMOS W/L=10um/0.246um @ Vd=-1.2V

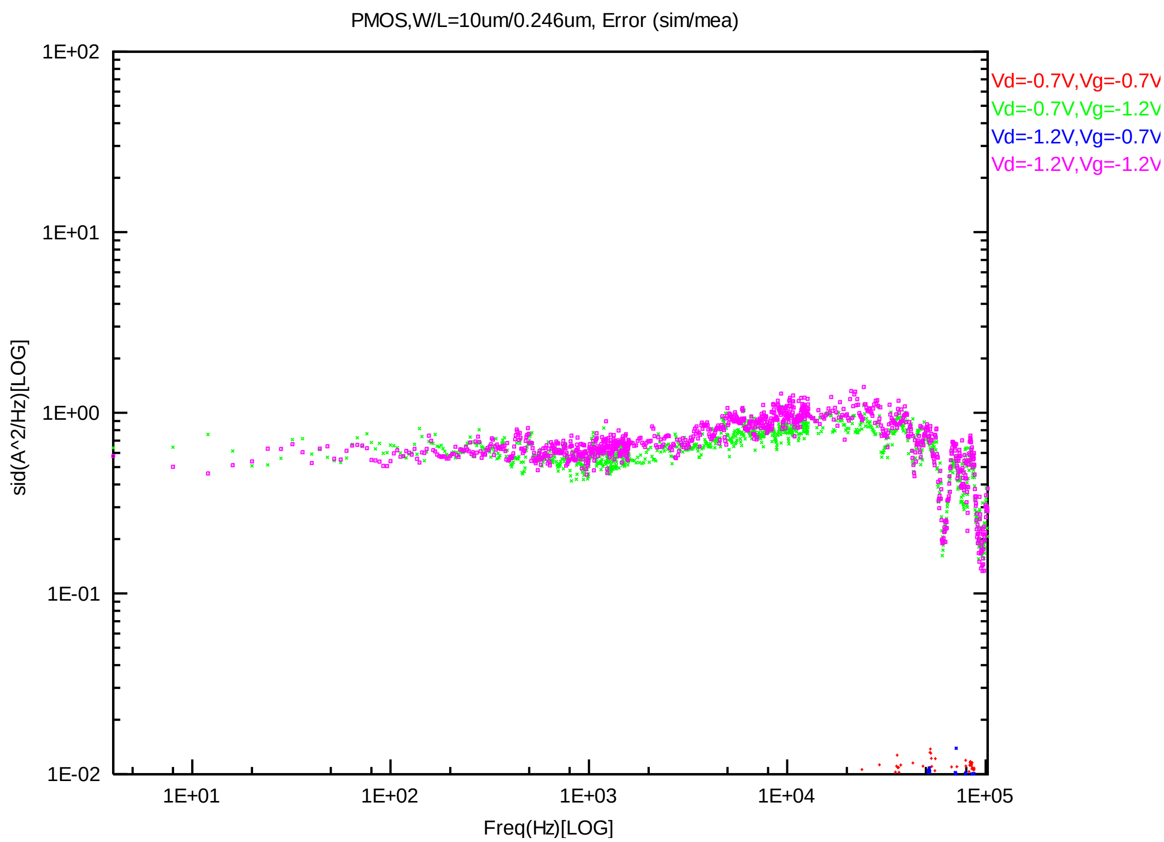

Figure 8-30 PMOS W/L=10um/0.246um error distribution plot

8.2 Trend Plots at Frequency=100Hz

A. NMOS

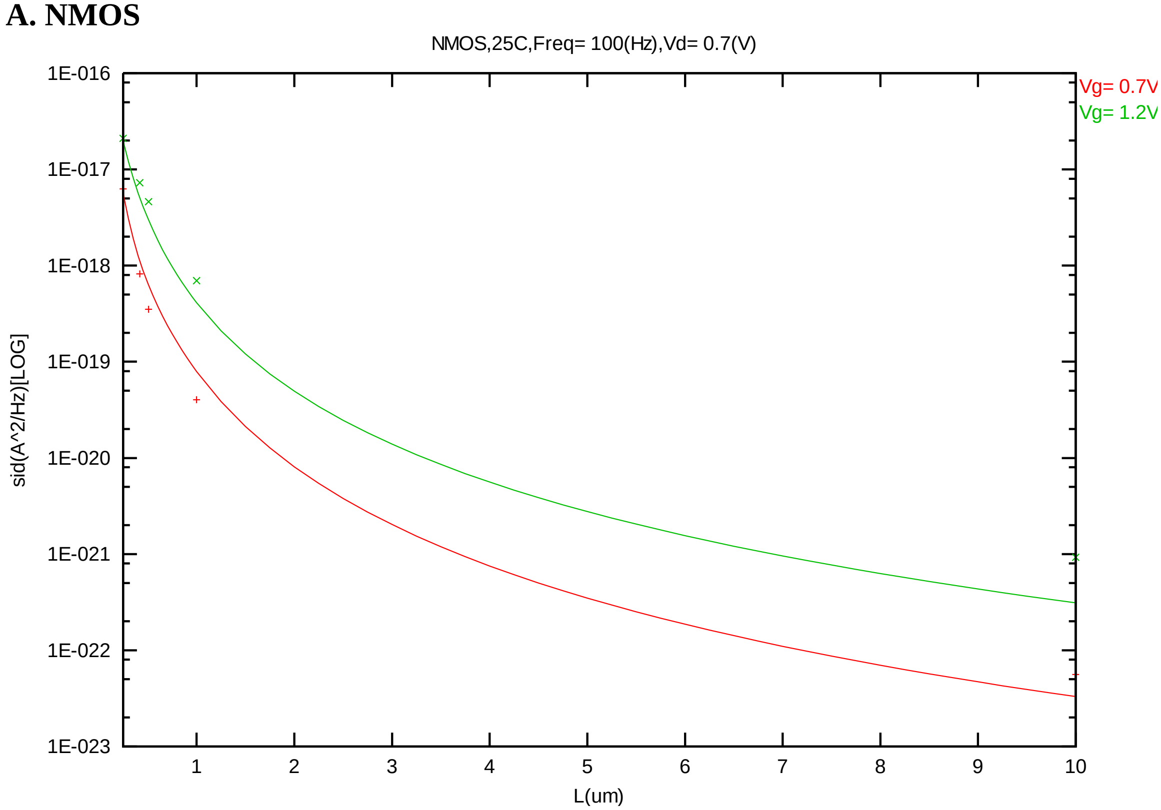

Figure 8-31 Noise vs. Length for NMOS @ Vd=0.7V

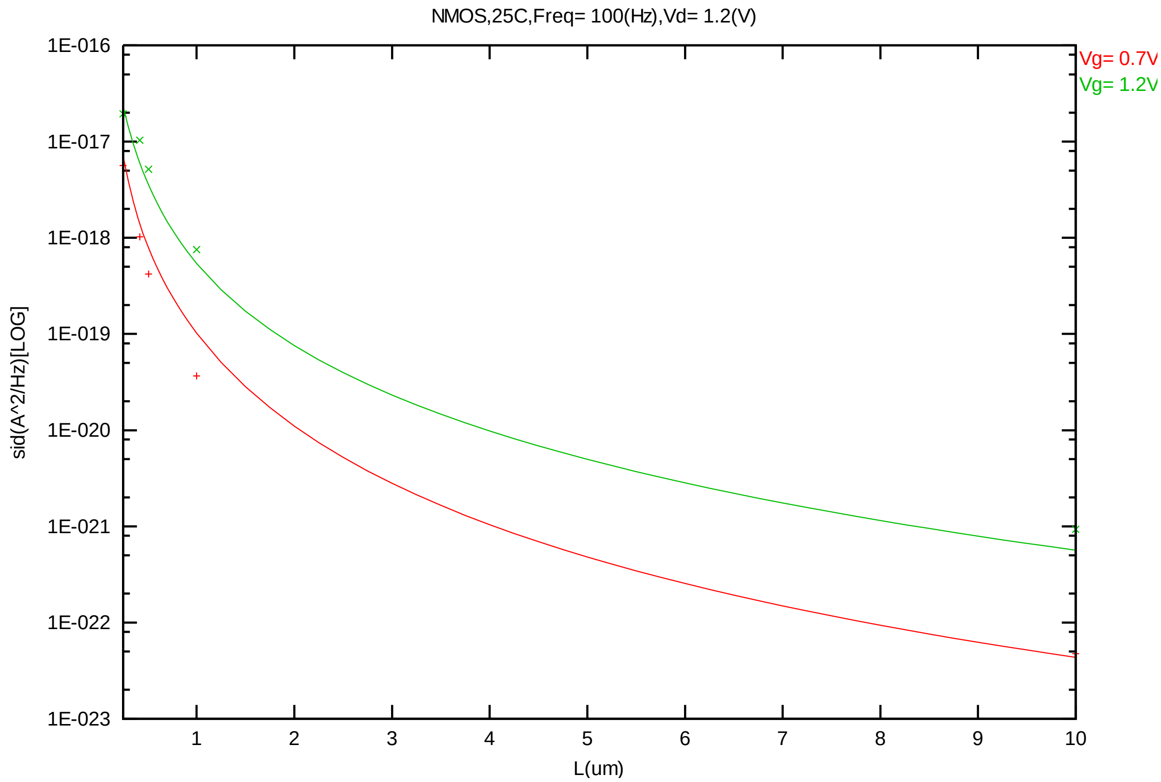

Figure 8-32 Noise vs. Length for NMOS @ Vd=1.2V

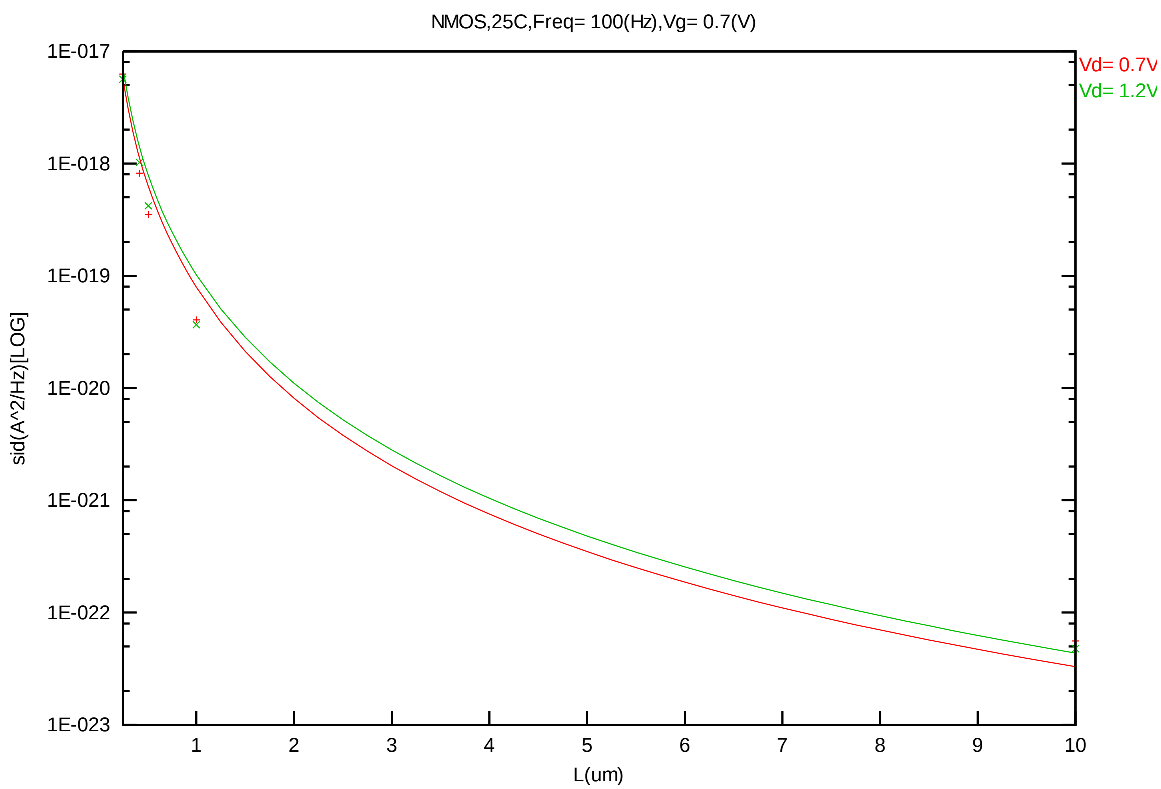

Figure 8-33 Noise vs. Length for NMOS @ Vg=0.7V

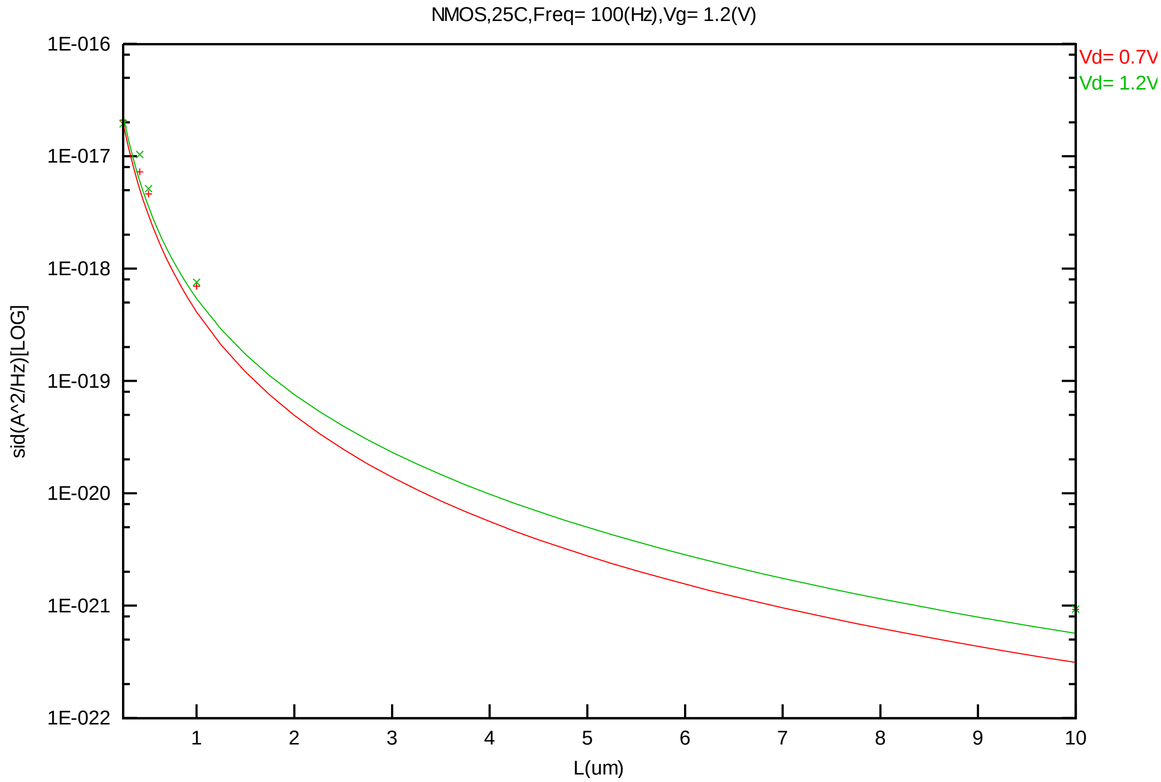

Figure 8-34 Noise vs. Length for NMOS @ Vg=1.2V

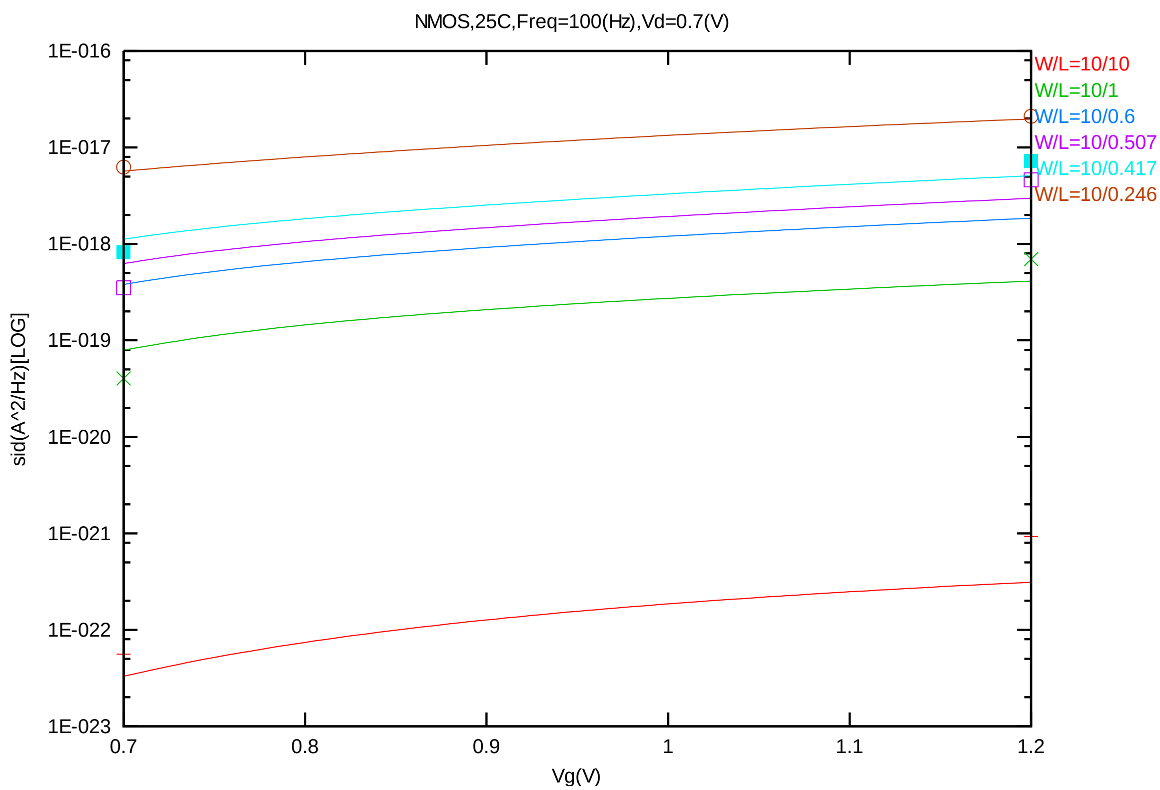

Figure 8-35 Noise vs. Vg for NMOS @ Vd=0.7V

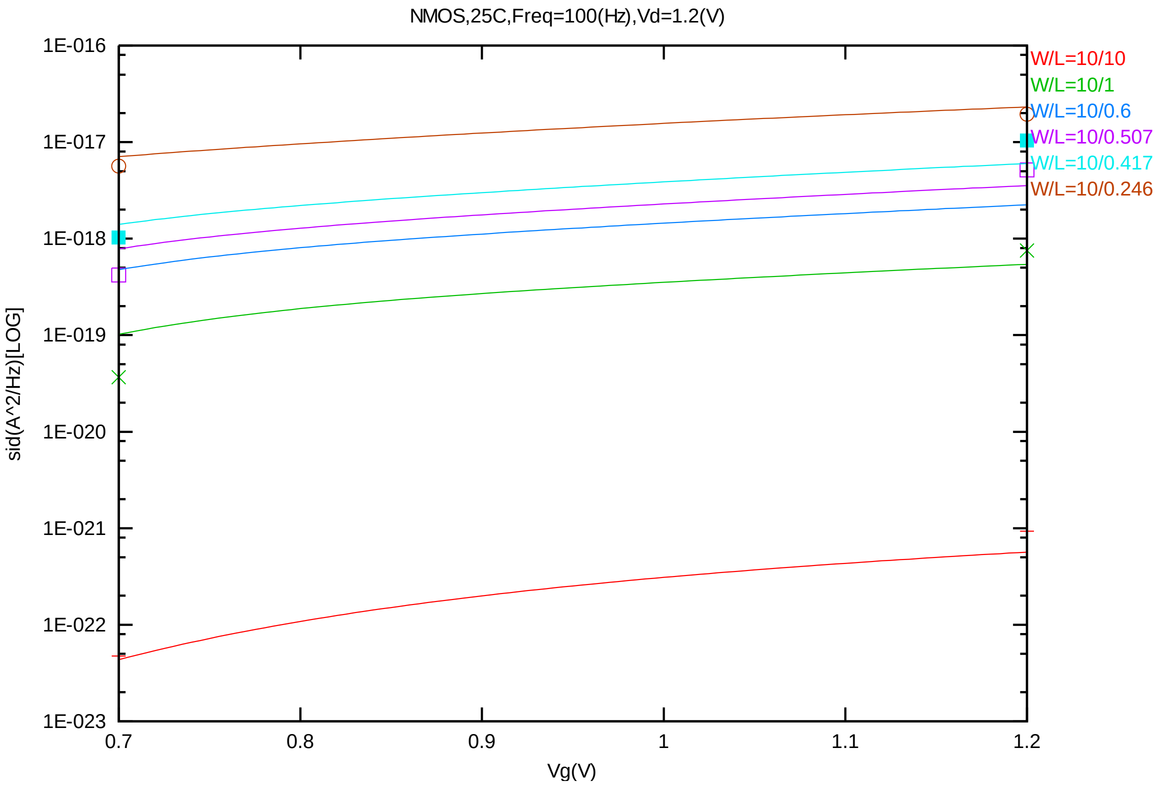

Figure 8-36 Noise vs. Vg for NMOS @ Vd=1.2V

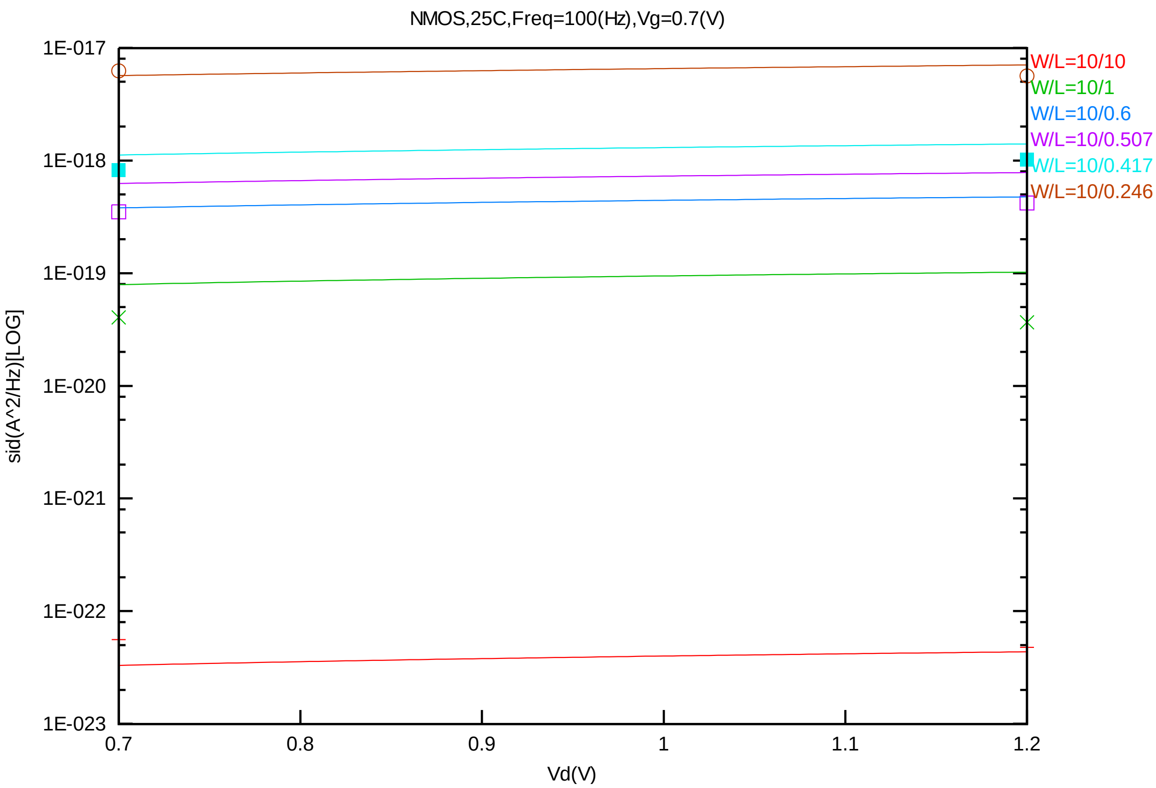

Figure 8-37 Noise vs. Vd for NMOS @ Vg=0.7V

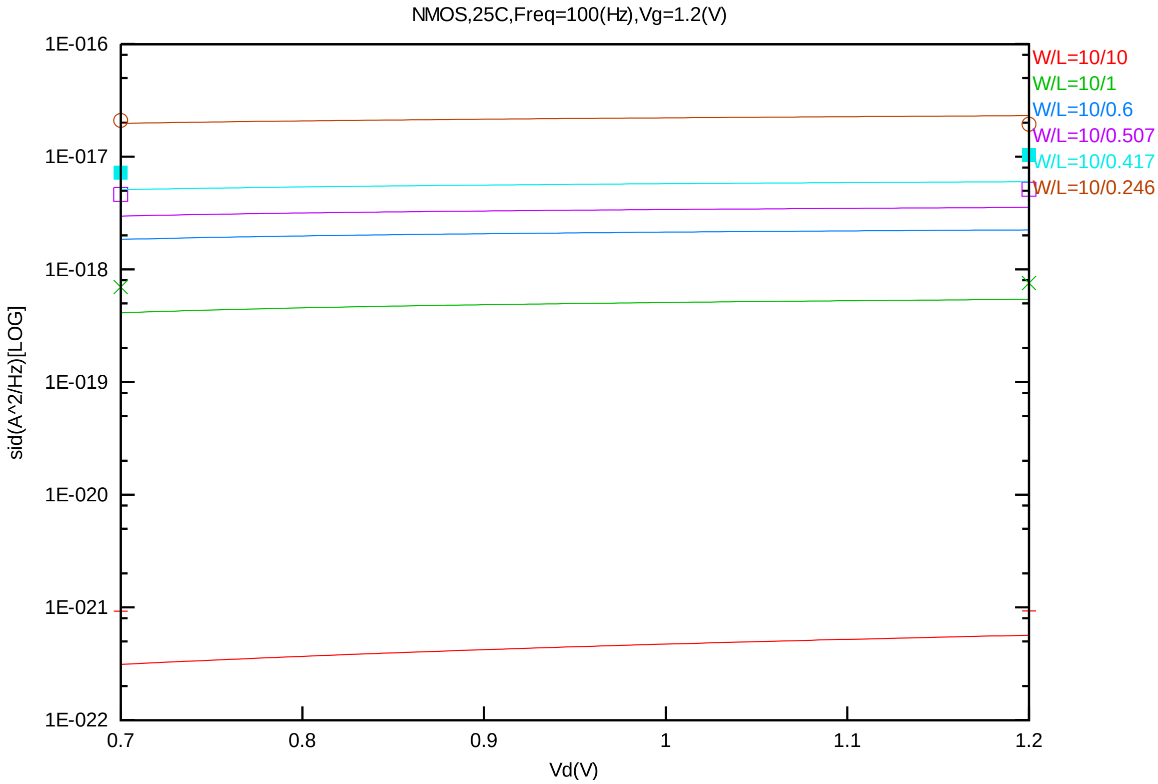

Figure 8-38 Noise vs. Vd for NMOS @ Vg=1.2V

B. PMOS

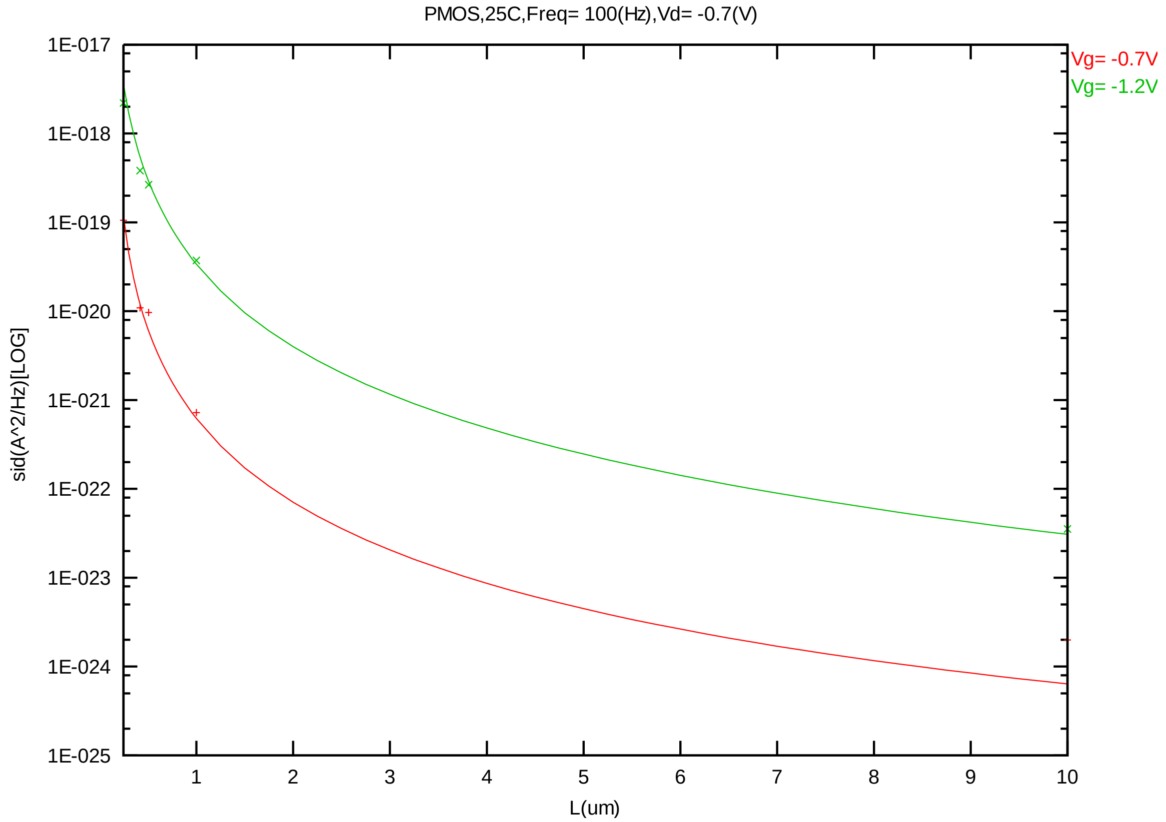

Figure 8-39 Noise vs. Length for PMOS @ Vd=-0.7V

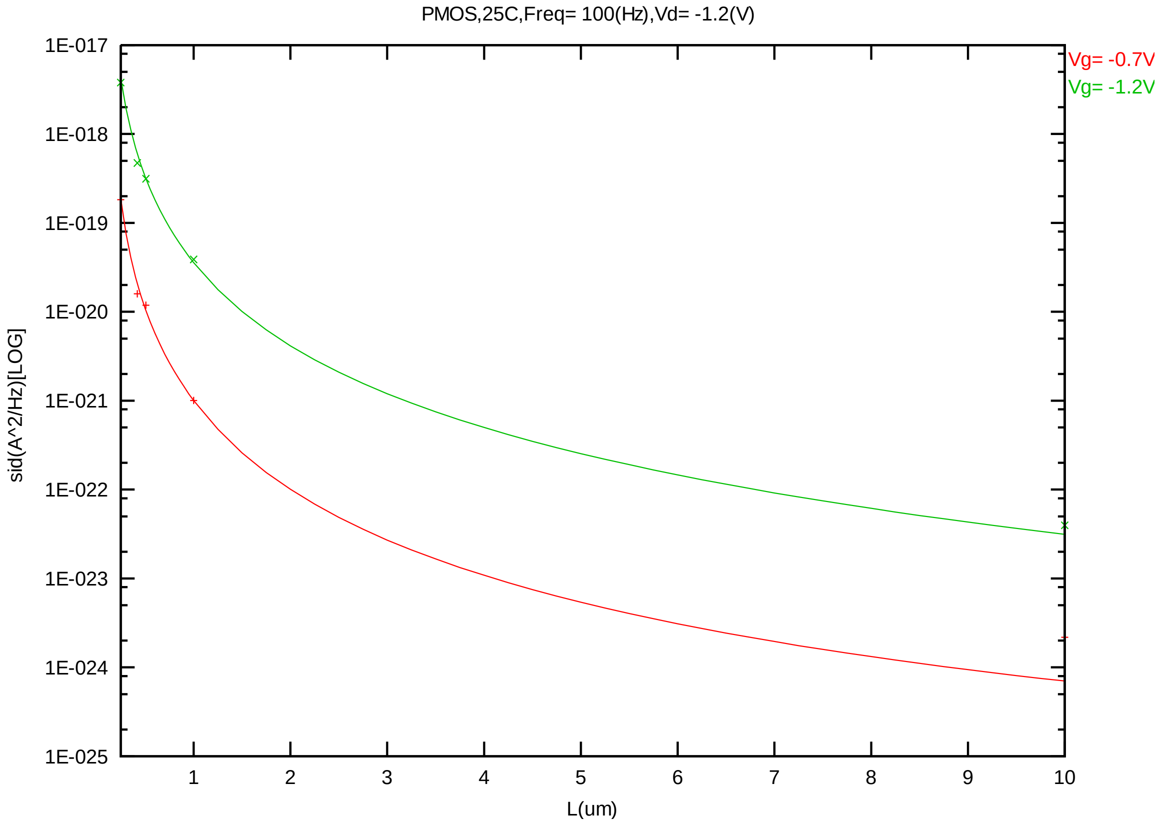

Figure 8-40 Noise vs. Length for PMOS @ Vd=-1.2V

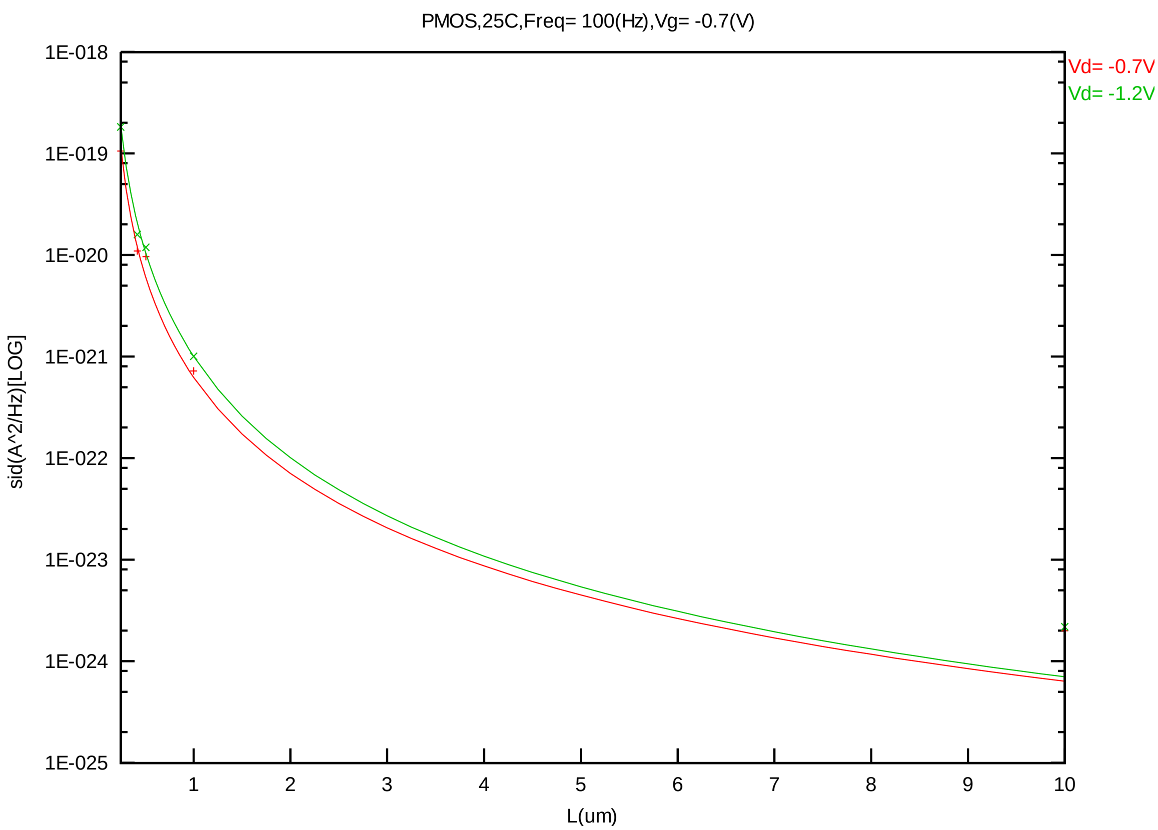

Figure 8-41 Noise vs. Length for PMOS @ Vg=-0.7V

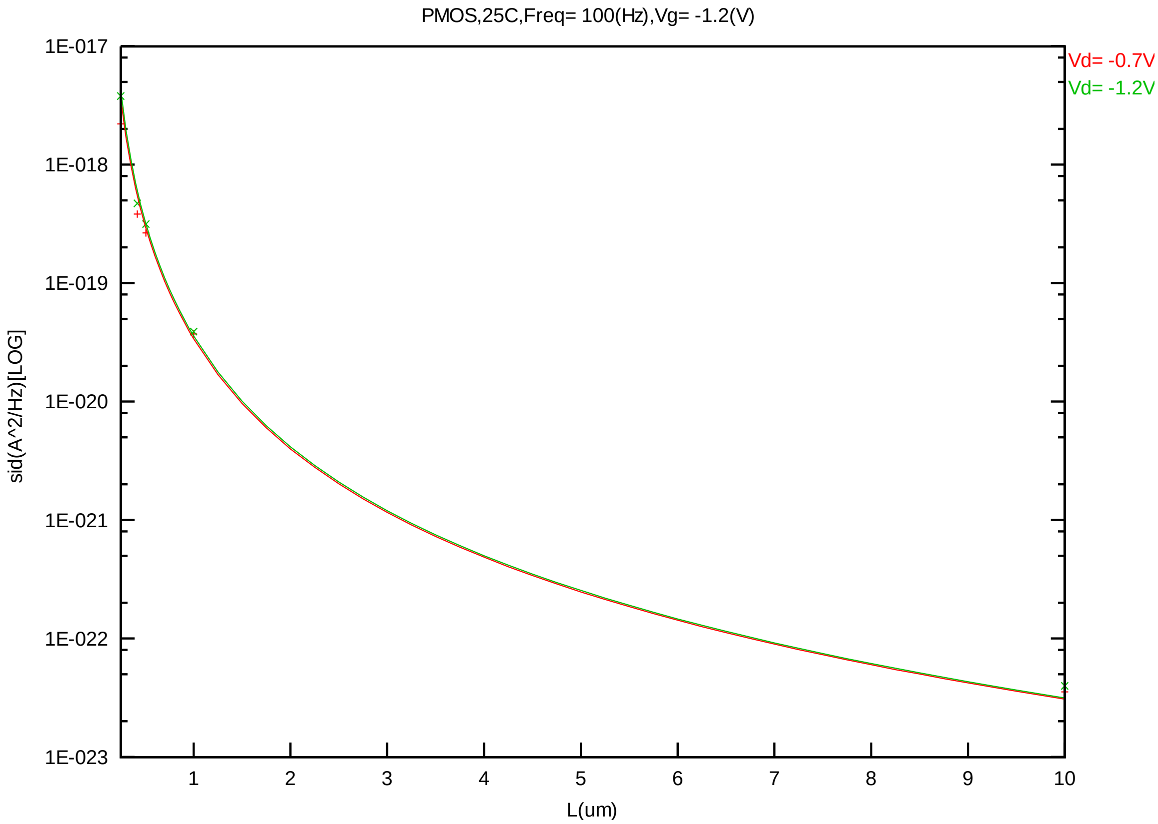

Figure 8-42 Noise vs. Length for PMOS @ Vg=-1.2V

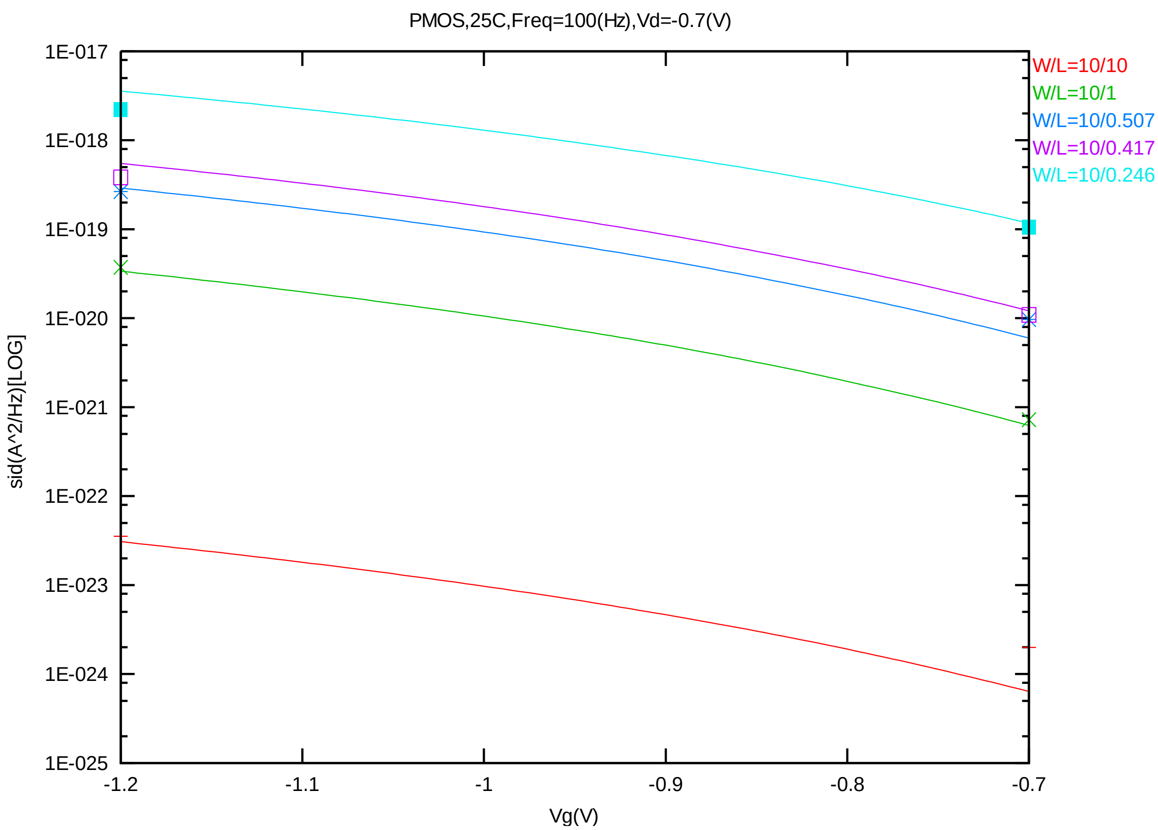

Figure 8-43 Noise vs. Vg for PMOS @ Vd=-0.7V

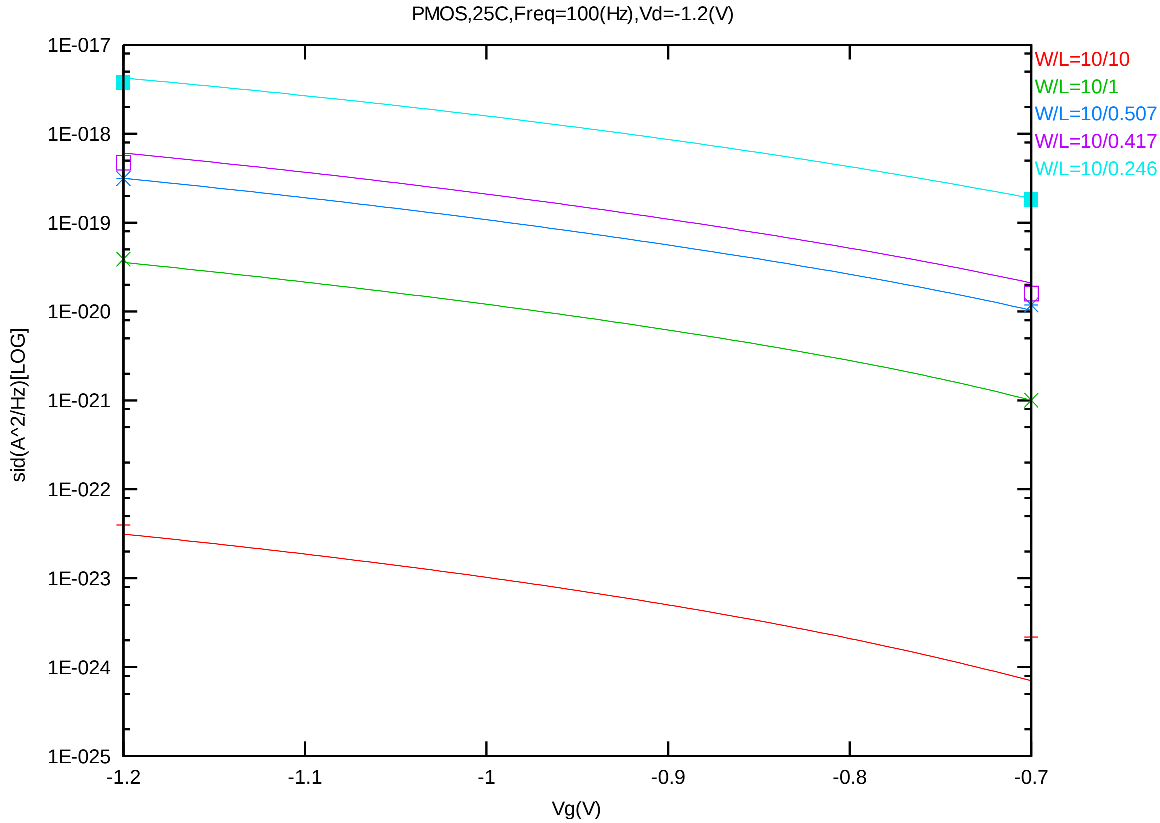

Figure 8-44 Noise vs. Vg for PMOS @ Vd=-1.2V

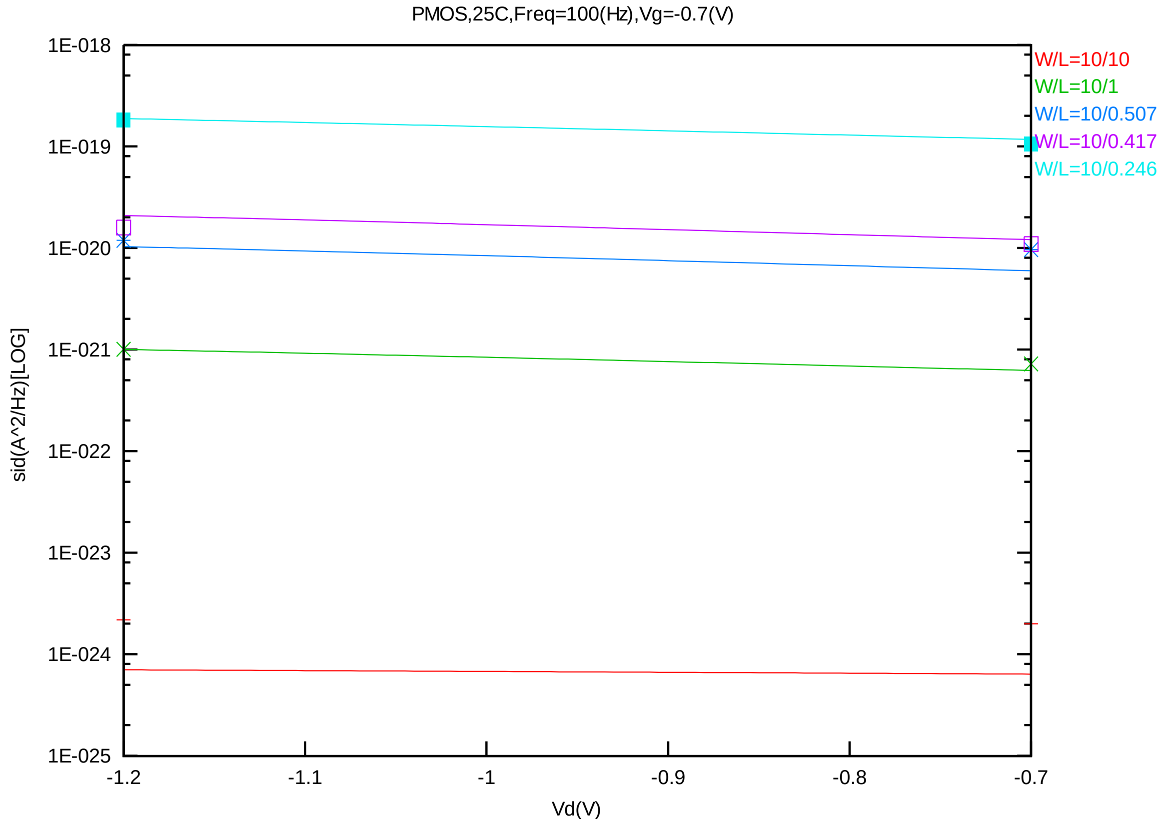

Figure 8-45 Noise vs. Vd for PMOS @ Vg=-0.7V

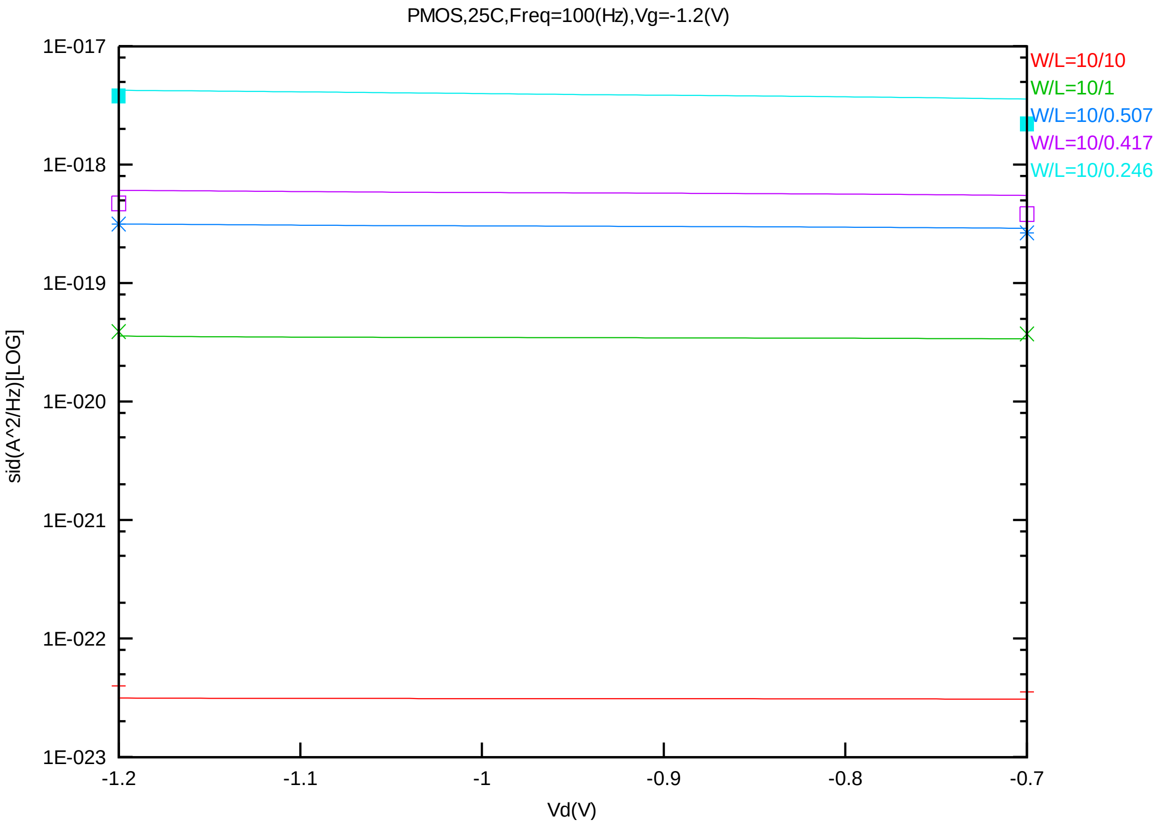

Figure 8-46 Noise vs. Vd for PMOS @ Vg=-1.2V

9 Appendix

9.1 Wafer Data vs. Corner Plots

Threshold voltage in linear region VTLIN

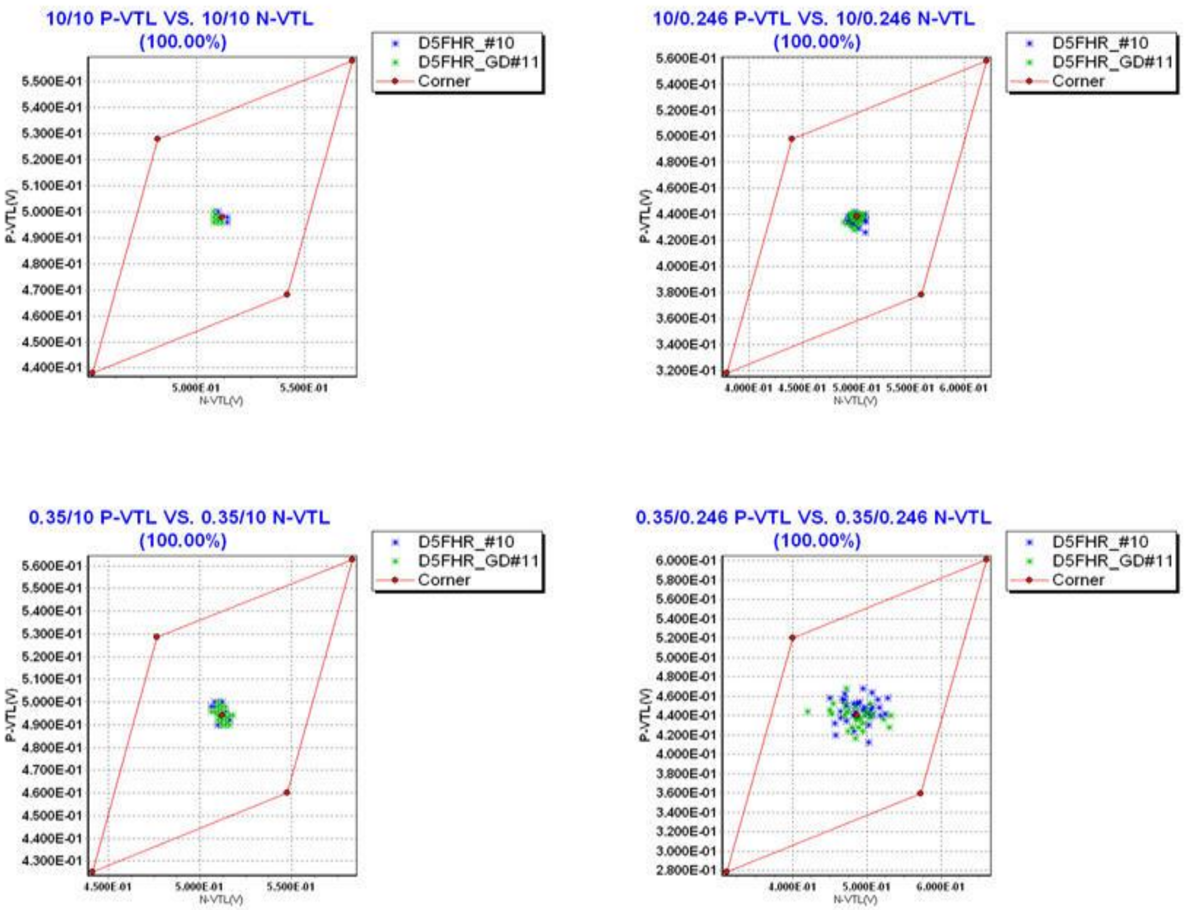

Figure 9-1 Corner plots of threshold voltage at low drain bias

Threshold voltage in linear region VTSAT

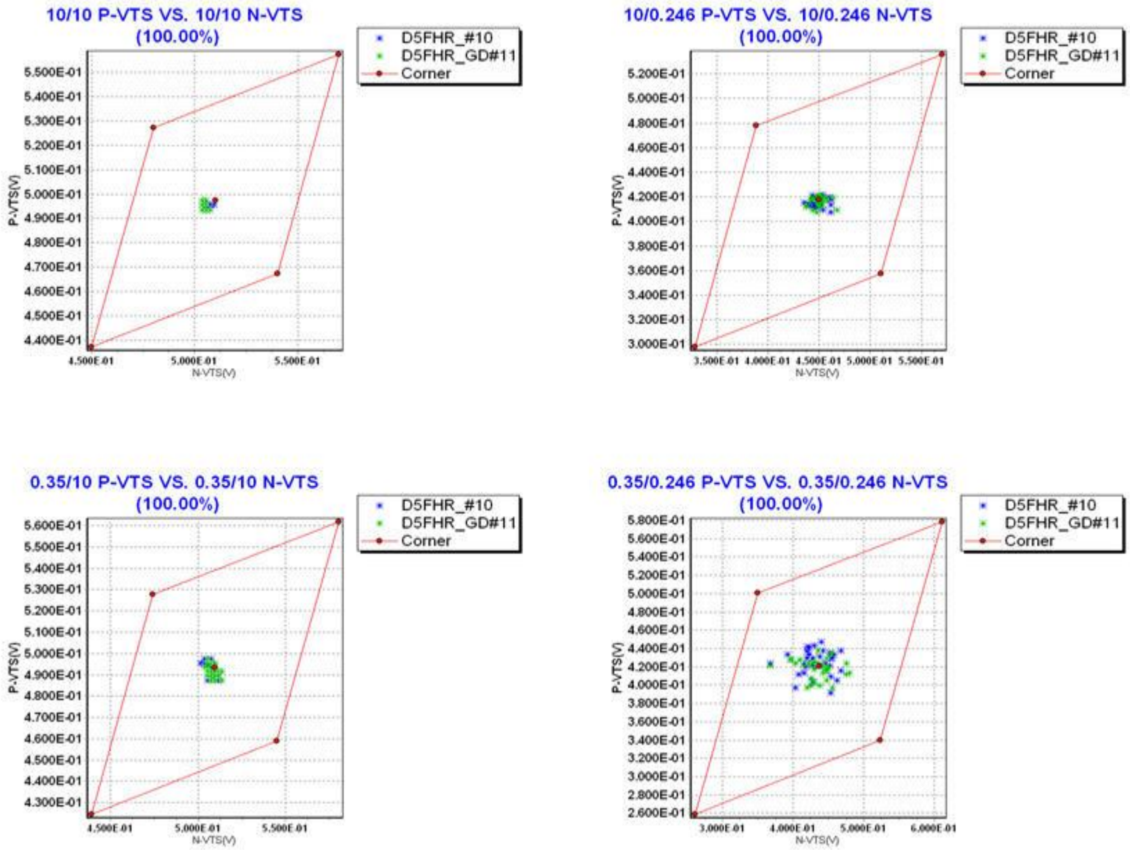

Figure 9-2 Corner plots of threshold voltage at high drain bias

Saturation current I_{ON}

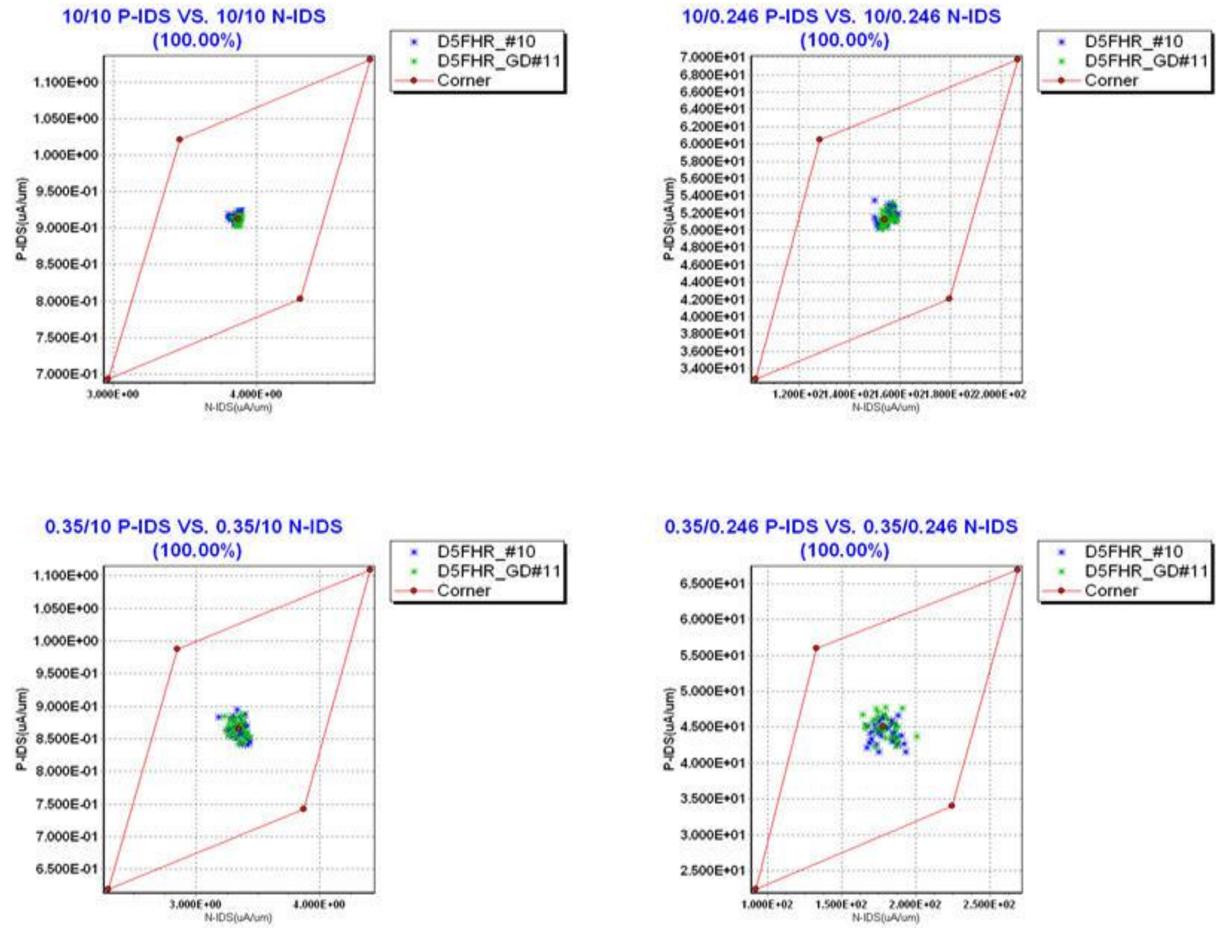

Figure 9-3 Corner plots of on-current

Off state current I_{OFF}

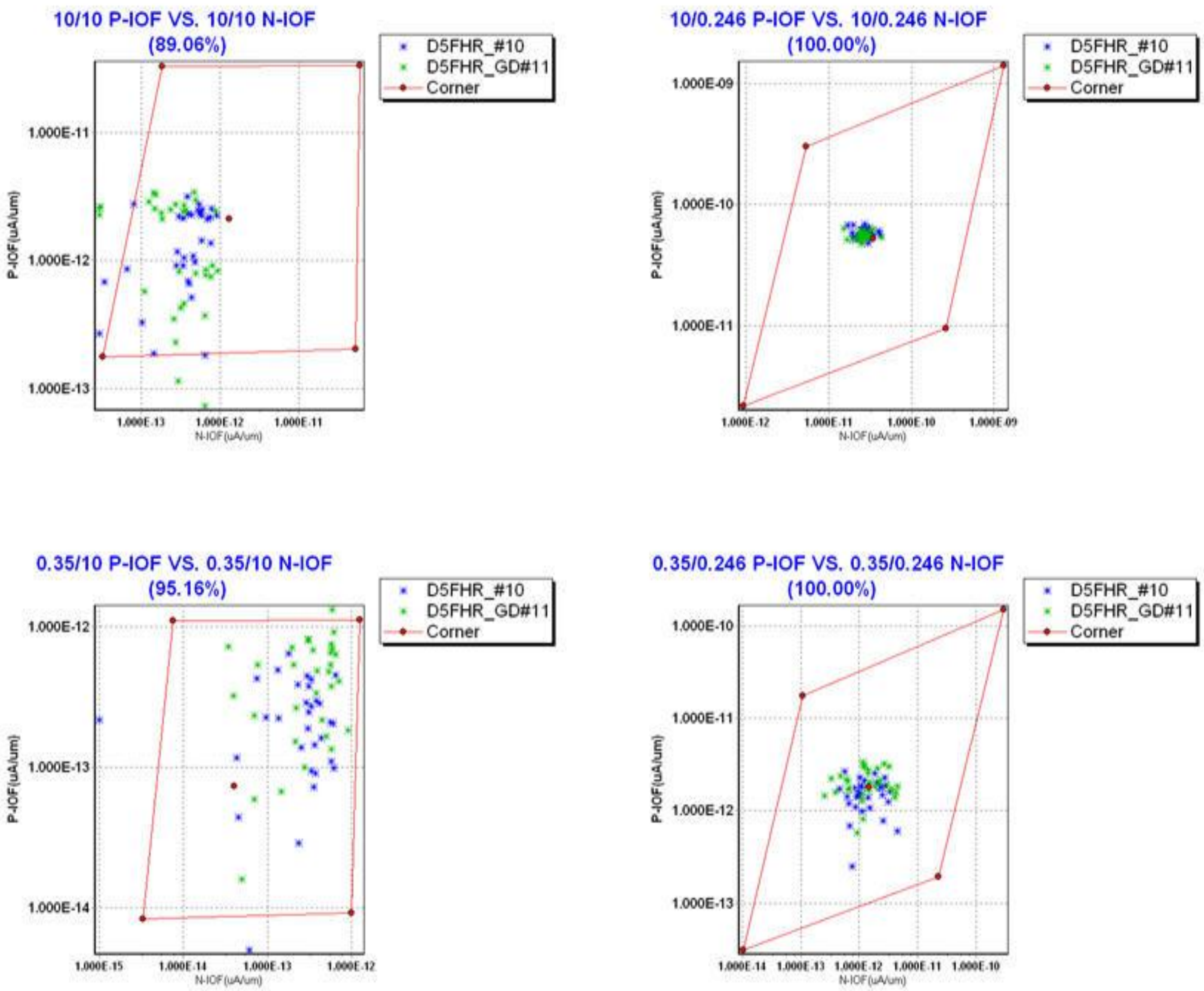

Figure 9-4 Corner plots of off-current

9.2 HSPICE Warning Message

Table 9-1 Hspice warning message

NO.	Warning Message	Explanation
1.	-	-

9.3 ELDO Warning Message

Table 9-2 Eldo warning message

NO.	Warning Message	Explanation
1.	-	-

9.4 SPECTRE Warning Message

Table 9-3 Spectre warning message

NO.	Warning Message	Explanation
1.	-	-